

Virtual compact specialness of hyperbolic one-relator groups

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Abstract

We prove that every word-hyperbolic one-relator group is virtually compact special. Combined with the work of Kielak–Linton, this resolves Wise’s virtual free-by-cyclic conjecture for hyperbolic one-relator groups: every such group is virtually free-by-cyclic, with the free kernel allowed to have infinite rank.

1 Introduction

One-relator presentations impose a single equation on a free group, but their geometric behavior ranges from free groups to groups with distorted cyclic subgroups. Hyperbolicity rules out the latter behavior. The purpose of this paper is to show that it also forces a strong finite-cover geometry.

Theorem 1.1. *Let X be finite, let $r \in F(X)$, and suppose that*

$$G = F(X)/\langle\langle r \rangle\rangle$$

is word-hyperbolic. Then G is virtually compact special. More precisely, there are a finite-index subgroup $H \leq G$, a finite connected nonpositively curved cube complex C , and a finite simplicial graph Λ such that

$$\pi_1(C) \cong H \quad \text{and} \quad C \longrightarrow S_\Lambda$$

is a cubical local isometry to the Salvetti complex of the right-angled Artin group defined by Λ .

The compactness assertion is part of the conclusion. An abstract embedding of a finite-index subgroup in a right-angled Artin group would not by itself supply the model required in Theorem 1.1.

The inductive study of one-relator groups begins with Magnus’s Freiheitssatz: omitting a generator that occurs in a cyclically reduced relator leaves a subgroup freely generated by the remaining generators [44]. Magnus subsequently solved the word problem for one-relator groups [45]. These results established the importance of reducing a one-relator problem to simpler presentations while retaining control of distinguished free subgroups. The Magnus–Moldavanskii hierarchy develops this approach through splittings over those subgroups [39, Sections 1.1–1.3]. The geometric question is whether the splittings have the quasiconvexity needed to construct finite special covers.

A second line of inquiry concerns finite quotients. A group is *residually finite* if each nonidentity element survives in some finite quotient. Baumslag’s 1967 Conjecture A asks for this property for one-relator groups with nontrivial torsion [6]. His paper contrasts this question with non-Hopfian one-relator examples, which cannot be residually finite. Wise’s quasiconvex hierarchy theory proved the stronger virtual compact specialness conclusion for

one-relator groups with torsion [66, Corollary 19.2]. Theorem 1.1 extends the compact-special conclusion, and hence residual finiteness, to every hyperbolic one-relator group.

The cubical approach explains why this strengthening matters. Haglund–Wise’s special cube complexes connect nonpositive curvature with embeddings in right-angled Artin groups and, in the hyperbolic case, separation of quasiconvex subgroups in finite quotients [28]. Wise’s hierarchy theorem makes quasiconvex splittings a way to obtain this structure [66, Theorem 13.1]; Agol’s theorem supplies it for hyperbolic groups acting properly and cocompactly on CAT(0) cube complexes [2, Theorem 1.1]. Thus the geometric conclusion controls both linear representations and finite-quotient information. Section 18 explains these consequences, including conjugacy separation and virtual retractions, as well as the fibering conclusions below.

In the torsion-free direction, Linton proved hyperbolicity and virtual specialness when all two-generator subgroups of a one-relator group are free [39, Theorem 7.2 and Section 1]. His subsequent work identifies primitive extension groups as the terminal cases in the general hyperbolic problem and specifies the quasiconvex towers through which a terminal result propagates [40]. The remaining task is therefore a geometric construction for those primitive groups. Here we construct virtually compact special quotients, recover that structure using selective covers of the fillings, and then propagate it up the truncated tower. Wise singled out hyperbolic one-relator groups in his broader question about cocompact cubulation of hyperbolic groups with hierarchies [65, Problem 13.15]. Theorem 1.1 establishes the compact-special conclusion for this one-relator class.

We use *free-by-cyclic* in the subgroup-closed sense: a group H fits into an exact sequence

$$1 \longrightarrow K \longrightarrow H \longrightarrow Q \longrightarrow 1$$

with K free of possibly infinite rank and $Q \leq \mathbb{Z}$, possibly trivial. Wise’s Conjecture 17.8 asks whether every hyperbolic one-relator group is virtually free-by-cyclic, as recalled by Kielak–Linton [37, Section 1.1]. Their Corollary 1.3 identifies virtual compact specialness as a sufficient additional hypothesis. Combining that corollary with Theorem 1.1 gives a positive answer. Corollary 18.1 also records the Baumslag–Solitar-free consequence from the companion hyperbolicity theorem and a stronger finite-index embedding into a free-by- $\mathbb{Z}$ group whose free kernel is finitely generated.

In the proof we use the hyperbolicity and Magnus-subgroup results of [52] at two specific points: embedded primitive extension groups are themselves hyperbolic, and their actual Magnus rows have the required hyperbolic geometry. The criterion in that paper excludes every Baumslag–Solitar subgroup $BS(m, n)$ with $mn \neq 0$. No virtual-specialness conclusion is imported from it.

The recovery step is a filling criterion proved in Section 3. Its input is a coherent, locally indicable hyperbolic group of cohomological dimension at most two, with a finite classifying complex acyclic over its Hughes-free division field, and arbitrarily deep virtually compact special fillings along specified peripherals. We replace these by *selective* fillings that retain any prescribed finite collection of kernel factors. Their covers have a quasiconvex cubical hierarchy. This geometric control realizes the finite coefficient data of a chain contraction in ordinary Novikov completions for both signs of one character. The resulting finite-index subgroup maps onto $\mathbb{Z}$ with a *finite-rank* free kernel, and hence is virtually compact special.

Before constructing the primitive fillings, we use this criterion to prove a second result. It will also supply the specialness of the periodic end groups encountered in that construction.

Theorem 1.2. *Let F be a finitely generated free group and let $\phi : F \rightarrow F$ be injective. If its ascending mapping torus*

$$F *_\phi = \langle F, t \mid t^{-1}ft = \phi(f), f \in F \rangle$$

is word-hyperbolic, then it is virtually compact special.

Hagen and Wise cubulate hyperbolic free-by-cyclic groups with finite-rank free kernel [27, Theorem A]. Their irreducible theorem also covers expanding, fundamental-group-injective train-track maps whose transition matrices are irreducible and whose mapping-torus groups are hyperbolic [26, Theorem B]. They posed the cubulation problem for hyperbolic ascending HNN extensions defined by general monomorphisms in [27, Problem C]. Mutanguha constructs expanding relative immersions for injective nonsurjective endomorphisms and proves the Baumslag–Solitar criterion for arbitrary injective ascending tori [51, Theorems 4.5 and 6.7]. He also records Kielak’s question whether hyperbolic ascending tori of injective nonsurjective free-group endomorphisms are virtually free-by-cyclic [51, Overview, p. 3, Question (Virtual fibering)]. The proof below retains arbitrary injective input and establishes the compact-special conclusion. Linton proved that every hyperbolic free-by-cyclic group is virtually compact special, allowing its free kernel to have infinite rank [41, Corollary 1.5]. The canonical kernel of a strict ascending torus is presented here as an increasing union of free groups; freeness of that union is not an input to our argument.

After Theorem 1.2 is proved, the ascending group’s field acyclicity and virtual RFRS property give a stronger fibering conclusion: it has a finite-index subgroup mapping onto $\mathbb{Z}$ with a finite-rank free kernel (Corollary 4.10). The resulting character is not identified with the canonical ascending height. For general one-relator groups we retain the possibly infinite-rank kernel in Corollary 18.1.

Kielak’s virtual-fibering method realizes division-field coefficients in ordinary Novikov completions along an RFRS filtration [36, Section 4.2 and Theorem 5.2]. Fisher’s later work relates this method to cohomological dimension of kernels [23]. Here the original group is not yet known to be virtually RFRS: the selective switch supplies the required finite coefficient data before the two-sided Novikov argument. Once compact specialness is established, Kielak–Linton’s theorems give the general one-relator fibering and ambient-embedding consequences [37].

The remaining construction provides the primitive group’s good fillings. Its Magnus splitting has a height coordinate. We call the vertex and edge spaces at specified heights its *rows*; these are the actual spaces in the developed Bass–Serre tree, not merely their quotient labels. Long paths with noncyclic stabilizers obstruct an acylindrical hierarchy. Section 5 controls them by a joint injection into one finite-dimensional homology space. Keeping this injection under subgroup restriction, rather than only its rank inequality, distinguishes coincident contributions from different path orbits.

Expanding trajectories then identify periodic ends and join them along bilateral trajectories (Section 8). Homology prevents circuits and rank growth in the resulting incidence trees. We close the corresponding quasiconvex groups under cyclic intersections, fold rows, and enlarge shared ends until they form a malnormal peripheral system. There are two termination arguments: horizontal joins with no homological growth stabilize because their increasing union is finitely generated, whereas each genuine enlargement at a shared line end strictly increases the injected homology dimension. The coefficient constructions in Sections 6 and 7 preserve the joint injections through these operations. The first retains prescribed subgroup supports in scalar chain equations; the second localizes semilinear affiliated coefficients to finitely many periodic cosets.

We fill finite-index subgroups of the peripheral *height kernels*, retaining the height direction. Exact intersection transfer (Section 14) makes the resulting splitting acylindrical. The compatible kernels of Section 15 also give exact link covers for the filled rows. Binary labels on these links have sparse changes; the changes specify horizontal wall attachments. Within each filled vertex row, panels join through midpoint slots in the central pieces and then through satellite and outer fans. This produces quasiconvex trees carrying binary cocycles

(Sections 16 and 17). Keeping central and satellite panels separate lets the labels control departures introduced by later attachments, including at root extensions.

Three boundary configurations require different panel tests. The resulting separators satisfy Bergeron–Wise’s boundary criterion [7, Theorem 1.4]; Agol’s theorem then gives virtual compact specialness of the filled rows. The transferred hierarchy assembles the good quotients in Theorem 17.19. Finally Theorem 17.20 applies the filling criterion to the unfilled primitive group, and the truncated one-relator tower proves Theorem 1.1.

2 Background, conventions, and the reduction

We write VCS for *virtually compact special*, QC for quasiconvex, and fg for finitely generated. RFRS means residually finite rationally solvable. An ascending mapping torus always uses an injective endomorphism unless explicitly described as a presentation torus before passing to its stable image. All subgroup occurrences in a Bass–Serre tree include their actual conjugating frames. Thus two occurrences of isomorphic subgroups need not be the same subgroup or the same peripheral object.

2.1 Cubulation and hierarchy results

We use the following standard results in their compact forms.

Proposition 2.1 (Cubulation and quasiconvex hierarchies). *1. A word-hyperbolic group acting properly and cocompactly on a CAT(0) cube complex is virtually compact special.*

2. A hyperbolic group with a finite graph-of-groups decomposition whose edge groups are quasiconvex in the ambient group and whose vertex groups are hyperbolic VCS is VCS. The same applies to a finite hierarchy of such decompositions.

3. A compact special cube complex admits a cubical local isometry to a Salvetti complex. Its defining graph can be taken finite.

These are the virtual special theorem [2, Theorem 1.1], the quasiconvex hierarchy theorem [66, Theorem 13.1], and the special-complex characterization [28, Proposition 3.10 and Theorem 4.2], respectively. For the last assertion, only finitely many hyperplanes and crossing relations occur in a finite complex. Passing to a further special cover, when needed, provides the oriented two-sided convention.

The hierarchy statement with VCS terminal groups follows also by cutting a compact special cover of each terminal model along its finitely many two-sided hyperplane families. The cutting traces are convex, hence quasiconvex, and the resulting hierarchy ends in trivial groups. Finite-index supergroups are allowed in the hyperbolic hierarchy class. In any finite splitting used here, quasiconvexity of an edge group is asserted in the ambient group, not merely in an adjacent vertex group. Finitely generated vertex groups are then undistorted: in a tree of Cayley spaces, replace excursions from a target vertex through incident edge cosets by paths in those cosets of comparable length.

We use residual finiteness, virtual RFRS, quasiconvex subgroup separability and virtual retractions for hyperbolic VCS groups [1, 28, 29, 66]. The malnormal special quotient theorem supplies designated finite-index normal subgroups $\dot{P}_i \trianglelefteq P_i$ for a finite almost malnormal quasiconvex peripheral family. Normal finite-index kernels $N_i \trianglelefteq P_i$ contained in $\dot{P}_i$ give hyperbolic VCS quotients [3, Theorem 2.7 and Corollary 2.8]. We retain this containment condition at every use; it is not replaced by finite avoidance. Additional finite avoidance conditions can be imposed by further intersections. Exact peripheral embeddings follow from the relative Dehn filling theorem [55, Theorem 1.1].

For relative hyperbolicity we use peripheralization of finite almost malnormal quasiconvex families and the uniform relative Dehn filling theorem [11, 53, 55]. For rows that are only relatively finitely generated, the fine-graph to relative-presentation comparison is [24, Proposition 3.47]; its hypotheses and exact infinite vertex stabilizers are checked at the constructions below. In particular, bounded isolated peripheral components in sufficiently long fillings have labels from a fixed finite list in the *original* peripheral groups. Later arguments explain the additional tests needed for exact subgroup-intersection transfer; that stronger conclusion is not assumed as part of the ordinary filling theorem.

2.2 The primitive extension reduction

Here is the shape of the groups that remain after the reduction. For $k \geq 1$, write $a_i = t^{-i}at^i$. Their presentations have the forms

$$\langle a, t \mid \text{pr}_{p/q}(x, y) \rangle, \quad \langle a, t \mid \text{pr}_{p/q}(xy, z) \rangle,$$

where p, q are relatively prime positive integers. The notation $\text{pr}_{p/q}(v, w)$ means substitution of v, w into the positive primitive word of slope p/q in two formal letters. Primitivity means membership in a free basis before substitution. The words x, y use the overlapping windows $a_0, \dots, a_{k-1}$ and $a_1, \dots, a_k$, respectively, and each lies outside their common subgroup $F(a_1, \dots, a_{k-1})$. The second form also uses a nontrivial word z in that common subgroup. These shapes are only a preview: the slot and primitive exceptional-property conditions in Definition 5.1, including its malnormality and exceptional-factorization qualifications, are part of the definition.

Proposition 2.2 (Reduction). *To prove Theorem 1.1, it suffices to prove that every torsion-free hyperbolic primitive extension group in the two forms of [40, Section 5] is VCS.*

Proof. The trivial relator gives a free group. Wise’s virtual compact specialness theorem handles the torsion case [66, Corollary 19.2]. For a torsion-free group, use the maximal factored one-relator tower and primitive extension reduction of [40, Theorem 4.8, Corollary 4.9, Lemma 5.2 and Theorem 5.6]. The primitive terminal groups have finite one-relator presentations and embed in G .

A hyperbolic group contains no $\text{BS}(m, n)$ with $mn \neq 0$: cyclic subgroups generated by infinite-order elements are undistorted and have virtually cyclic commensurators, whereas the indicated Baumslag–Solitar groups violate these alternatives. The embedded primitive groups are therefore BS-free. Theorem 1.1 of [52] makes each of them hyperbolic; no general assertion that finitely generated subgroups of hyperbolic groups are hyperbolic is being made.

With these terminal groups hyperbolic, the tower truncated at its primitive extension complex has the required acylindricity and quasiconvex inclusions by [40, Corollary 4.9]. Only this truncated tower is used; no acylindricity is asserted for the unused remainder of the maximal tower below it. If each primitive terminal group is VCS, Proposition 2.1 propagates VCS through these upward steps. If the tower reaches an elementary terminal group without a primitive obstruction, the same argument starts there. This proves the reduction with the compact conclusion. $\square$

The precise primitive presentations and their actual refined rows are given in Section 5. They have two generators and one nontrivial relator, so their standard aspherical complexes have Euler characteristic zero. Torsion-free one-relator groups are locally indicable [13, 30]; see also the direct proof in [31, Theorem 3.1 and Corollary 3.2]. They are coherent [34, Theorem 1.1(1)]. For the nontrivial non-power presentations here, Lyndon’s asphericity theorem [43] is also supplied by [5, Corollary 6.5]: specialize to the free group on its Cayley

tree and a singleton non-power relator. The root quotient is trivial, so the resulting exact augmented sequence is the cellular sequence of the simply connected presentation cover; that cover is acyclic and hence contractible. The rational Linnell division fields are available, with Hughes-free subgroup compatibility, by [35, 33].

For such a primitive group, the universal-cover chain complex is acyclic over its division field. The Fox 2-boundary is a nonzero column, hence injective over the field; degree-zero homology vanishes because the group is nontrivial, and Euler characteristic zero then gives degree-one acyclicity. These are the finiteness and field hypotheses used by the filling criterion.

2.3 Coefficient and tree conventions

For a subgroup U of the torsion-free ambient group, D_U denotes its rational Hughes-free Linnell field in the affiliated operator algebra $\mathcal{U}(U)$. Inclusions use the compatible subgroup-field embeddings. Distinct subgroup cosets are independent over the corresponding field. We compute chains with coefficients on the left and differentials given by right matrices; the Fox rule is

$$d(rs) = d(r) + r d(s), \quad \partial d(r) = r - 1.$$

Changing a subgroup frame by s conjugates group symbols and right-multiplies their coefficients by s^{-1} . This convention preserves their ambient homology images. The notation $H_*(V; D_U)$ for $V \leq U$ means scalar extension from the intrinsic subgroup field. We use the Bass–Serre homology and cohomology sequences, including infinite quotient graphs: direct sums occur in homology and products in cohomology [59, 14].

All matrix traces in the affiliated-operator argument are compatible *unnormalized* extensions of the group trace. This ensures equality of traces for rectangular products in the two possible orders. Matrix dimensions remain finite, so this convention does not change tightness in the measure topology.

Proof of Theorem 1.1, assuming the later constructions. Theorem 17.20 proves the primitive case required by Proposition 2.2. Apply that proposition and then the Salvetti characterization in Proposition 2.1. This gives a finite connected special cover, its finite-index fundamental group, and the stated local isometry. Theorem 1.2 is established in Section 4. $\square$

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3 A filling criterion and progressive characters

This section separates the geometric and algebraic ingredients of the filling criterion. The geometric ingredient preserves a prescribed finite part of a filling kernel in a virtually compact special quotient of a finite-index subgroup. The algebraic ingredient uses these quotients to resolve the coefficient problems left by an RFRS chain.

We specify the division-ring hypotheses, rather than assume that local indicability alone supplies them. For a group J , write $D(J)$ for a division ring containing $\mathbb{Q}J$. Throughout this section the rings for subgroups are compatible division closures inside the ambient ring; distinct subgroup cosets are linearly independent over the corresponding subgroup

division ring; and finite-index coset representatives span the ambient division ring over the subgroup division ring. We also use the following character consequence of Hughes-freeness. If $N = \ker \chi$, where χ is a real character of a finitely generated subgroup J , then

$$D(J) = \text{Ore}(D(N) * (J/N)). \quad (3.1)$$

Here the crossed product is embedded using coset representatives. Its quotient J/N is finitely generated free abelian, so the crossed product is a Noetherian skew Laurent domain. These are the subgroup-field properties of the Hughes-free Linnell fields used in our applications; see [32, 38] for their origins and [35, Corollaries 6.2–6.3] and [33, Section 2.3 and Proposition 2.2] for the locally indicable affiliated realization and the normal-subgroup crossed-product assertions. Arbitrary subgroup coset independence is justified separately at the start of Section 6. We refer to this collection of properties as the *subgroup-field hypotheses*.

Theorem 3.1 (Filling criterion). *Let G_* be a torsion-free hyperbolic group satisfying the following conditions.*

- (i) *G_* is locally indicable, coherent, of type F , and has integral cohomological dimension at most two.*
- (ii) *G_* has a division ring $D(G_*)$ satisfying the subgroup-field hypotheses, and the chains of a finite classifying space become acyclic over $D(G_*)$.*
- (iii) *There is a finite malnormal family $\{D_j\}$ of quasiconvex hyperbolic virtually compact special subgroups such that arbitrarily deep peripheral fillings*

$$G_* \longrightarrow Q_*$$

are hyperbolic and virtually compact special, and their inherited peripheral images are hyperbolic and virtually compact special.

In (iii), arbitrarily deep means that the peripheral kernels can avoid any prescribed finite subset of $\bigcup_j (D_j \setminus \{1\})$; the inherited peripheral structure is the one furnished by relative Dehn filling.

Then a finite-index subgroup of G_ is an extension of a finite-rank free group by $\mathbb{Z}$. Consequently G_* is virtually compact special.*

The last conclusion uses cubulation of hyperbolic free-by-cyclic groups [27] and Agol's theorem [2]. We prove the other conclusions below. In particular, the original peripheral images in Theorem 3.1 need not be finite.

3.1 Cofinite fillings and selective quotients

Discard trivial members of the peripheral family. If no members remain, the good filling is the identity, so hypothesis (iii) already says that G_* is VCS. The same conclusion follows directly from (iii) if a member equals G_* . In these cases the RFRS argument at the end of the section gives the stronger finite-rank free-by- $\mathbb{Z}$ conclusion without a geometric filling construction. Until then, assume all peripherals are nontrivial proper subgroups.

By malnormal quasiconvex peripheralization, the pair $(G_*, \{D_j\})$ is relatively hyperbolic [11, 53]. Suppose a sufficiently deep good filling as in Theorem 3.1(iii) has been chosen, written $q : G_* \rightarrow Q$. Choose this first filling to preserve every nonidentity element of the total prescribed original finite list. Its peripheral images $P_j = q(D_j)$ form an almost malnormal quasiconvex family in the hyperbolic VCS group Q . The malnormal special quotient theorem supplies designated finite-index normal subgroups $\hat{P}_j \trianglelefteq P_j$ [3, Theorem 2.7 and Corollary 2.8].

Each P_j is residually finite. Choose finite-index normal subgroups $L_j \trianglelefteq P_j$ avoiding the finitely many nontrivial images that must survive, as well as the additional relative-filling exclusions. Put $M_j = \dot{P}_j \cap L_j$ and $R_j = (q|_{D_j})^{-1}(M_j)$. Thus M_j meets the designated containment condition, and the composite quotient is exactly the filling by the normal finite-index R_j . It is hyperbolic VCS by the special quotient theorem, with exact peripheral images by relative filling. A finite P_j allows $M_j = 1$. Thus we may, and do, choose a cofinite good filling

$$K = \langle\langle R_j : j \rangle\rangle_{G_*}, \quad Q_* = G_*/K, \quad \bar{D}_j = D_j/R_j. \quad (3.2)$$

Every depth requirement below can be imposed before this choice.

For sufficiently deep fillings, the Cohen–Lyndon theorem [63, Definition 3.13 and Theorem 5.1] identifies K with a free product

$$K = \ast_{a \in \mathcal{A}} R_a, \quad (3.3)$$

where $\mathcal{A} = \coprod_j Q_*/\bar{D}_j$ is the set of peripheral apex images, and R_a is a suitably conjugated copy of R_j . The action of G_* permutes the conjugacy classes of these factors. There are two coset sets here. Original peripheral cosets G_*/D_j carry the conjugate rotation subgroups, whereas $Q_*/\bar{D}_j$ indexes their images after filling. All original apices over one quotient apex form a K -orbit, and their rotation groups are K -conjugates. Thus, for $K \leq H$, retaining an H -invariant set of quotient apices means retaining its full inverse image among original cosets. Killing rotations over the complementary original cosets has the same normal closure in H as killing the corresponding factors in (3.3).

Lemma 3.2 (Uniform selective hyperbolicity). *The depth in (3.2) can be chosen so that the following holds for every finite-index preimage $H \leq G_*$ containing K . For any H -invariant subset of quotient peripheral apices, kill the rotations over its full inverse image among original cosets, and let $K_- \triangleleft H$ be their normal closure. Then H/K_- is hyperbolic. The required depth is independent of the index of H and of the chosen invariant subset.*

Proof. Use a proper Cayley space of G_* . Replace each peripheral coset by a uniformly thickened hull of its limit set. The hulls are uniformly strongly quasiconvex and remain uniformly close to the corresponding cosets. Malnormality and quasiconvexity bound the coarse overlap of any two distinct hulls, at every fixed thickening.

Coulon’s geometric small-cancellation theorem [17, Theorem 6.11] applies to an invariant family of such hulls when every nonidentity rotation has sufficiently large infimal translation length. Its constants depend on the hyperbolicity and overlap bounds, not on the number of orbits. Rescale the original space first to meet those bounds, fix the cone radius, and then make the filling deep enough to meet the translation-length bound.

This last requirement is finite avoidance. Elements of bounded infimal displacement have representatives in finitely many conjugacy classes in a proper Cayley space. A nontrivial such class meets a malnormal peripheral group in at most one peripheral conjugacy class. Normality of R_j then reduces the requirement to excluding finitely many elements of D_j . Equivalently, one can use quasi-axes near the peripheral hulls and the uniform comparison between stable and minimum displacement.

Now restrict the action to H and retain only the hulls to be filled. All overlap bounds persist. There are finitely many H -orbits, and each rotation subgroup acts cocompactly on its hull because R_j has finite index in D_j . Coulon’s theorem therefore gives a proper cocompact action of H/K_- on the hyperbolic quotient cone-off. Thus H/K_- is hyperbolic, with the same initial depth requirement for every finite-index preimage and every invariant selection. $\square$

Proposition 3.3 (Selective-cover quotient). *Choose (3.2) sufficiently deep. Let H_{given} be a finite-index preimage in G_* , and let $\mathcal{B} \subset \mathcal{A}$ be finite. There is a further finite-index preimage $H \leq H_{\text{given}}$, normal in H_{given} , such that, on retaining all factors indexed by the H -translates of $\mathcal{B}$ and killing the other factors, the group*

$$P = H/K_- \quad (3.4)$$

*is hyperbolic and virtually compact special. The natural map to P is injective on $*_{a \in \mathcal{B}} R_a$.*

The injectivity follows at once from (3.3), since the retained index set is H -invariant and $K \subset H$. Hyperbolicity follows from Lemma 3.2. The remainder of the proof constructs a quasiconvex hierarchy for P . Its terminal groups will be finite free products of retained factors. To obtain the hierarchy, we construct a cocompact action on a CAT(0) cube complex whose hyperplanes project isomorphically to hyperplanes in a special model for the original quotient. This controls the cutting stabilizers. The vertex stabilizers require a different argument: they are generated by whole retained factors, and may be infinite. First we remove walls near the retained apices, then construct sheets over the resulting chambers, and finally identify both kinds of stabilizer before cutting the hierarchy.

3.2 Removing walls near the retained apices

Let X be the simply connected cone presentation complex for G_* : start with a Cayley presentation complex and cone the connected finite-generator Cayley graph of each peripheral coset. Set

$$Z = X/K, \quad \tilde{X} = X/K_-, \quad \pi : \tilde{X} \longrightarrow Z, \quad B = K/K_-. \quad (3.5)$$

The complex Z is locally finite and Q_* -cocompact; the links at its apices are finite because the filling is cofinite. Both quotient complexes are simply connected. Indeed, lift a cellular loop to X ; its endpoint discrepancy is a product of rotations. A rotation fixing an apex is accounted for by a path to that apex followed by the rotated return path, which projects to a null backtrack. Removing these discrepancies leaves a loop in the simply connected complex X . This argument applies to either quotient.

The map π is an ordinary B -cover away from the retained apex images, branched at those images with the full retained-factor stabilizers. One may subdivide the cone cells and remove small open apex cones to see ordinary covering charts. Restoring an apex permits sheets to switch precisely by its stabilizer. Connectedness of the apex links also permits paths through an apex to be replaced by paths in its punctured star.

Pass first to a torsion-free finite-index subgroup $V \leq Q_*$ with a compact special cubical model. Write C for its CAT(0) universal cover. There is a continuous V -equivariant coarse equivalence

$$f : Z \longrightarrow C. \quad (3.6)$$

For example, define it on finitely many cell representatives and extend equivariantly, using contractibility of C . The orbit metrics make it a quasi-isometry. We will use a fixed coarse inverse on vertices.

We record a control observation used repeatedly below. For any fixed radius, loops in a ball of that radius in Z fill in a uniformly bounded larger ball. Translate the ball into one of finitely many bounded subcomplexes; local finiteness gives a finite subcomplex there, and its finitely generated fundamental group is killed in a larger finite subcomplex because Z is simply connected. Consequently a sufficiently fine mesh of a path or homotopy in C can be returned to Z , including relative to a prescribed boundary path, with a fixed error in its f -image. Connect nearby inverse representatives and fill the resulting small polygons by this

observation. The error depends on the mesh scale and (3.6), not on the diameter or duration of the homotopy.

Fix a large radius R , whose final size will be specified after the coarse tolerances below. Let $\mathcal{W}_0$ be the finite set of hyperplanes meeting the R -neighborhood of $f(\mathcal{B})$, and put $m = |\mathcal{W}_0|$. Choose a finite-index subgroup $\bar{H} \leq V \cap (H_{\text{given}}/K)$ so that distinct $\bar{H}$ -translates of each $W \in \mathcal{W}_0$ have distance greater than $2m + \dim C$. We use combinatorial distances for this separation; fixed comparisons with CAT(0) distances are absorbed in R .

Here is why such a subgroup exists. Hyperplane stabilizers are quasiconvex and separable, and act cocompactly on their hyperplanes. For a fixed distance bound only finitely many double cosets of a hyperplane stabilizer can give translates at that distance. Separability excludes representatives of the double cosets not equal to the stabilizer itself, in a finite-index subgroup containing the stabilizer. Intersect these finitely many subgroups, and take a normal core in H_{given}/K . Passing to a subgroup preserves the separation. We henceforth take H to be its preimage.

Delete the hyperplanes in $\bar{H}\mathcal{W}_0$. The remaining wall pocset defines a CAT(0) cube complex Y , with median projection [58, Lecture 2, Sections 2 and 5.2]

$$p : C \longrightarrow Y. \quad (3.7)$$

It is useful to include the elementary details of this projection.

Lemma 3.4 (Bounded wall collapse). *The cubing Y is proper, the projection p is a quasi-isometry, and every cube of Y lifts to a cube of C . For a vertex $v \in Y$, let $E_v \subset C$ be the intersection of the closed mid-halfspaces selected by its retained-wall orientations. For a cube $\sigma \subset Y$, put*

$$E_\sigma = \bigcap_{v \in \sigma^{(0)}} E_v. \quad (3.8)$$

These sets are nonempty bounded convex sets. A point mapping into an open cube σ is within $\dim C$ of E_σ . Every retained branch image is either R -deep in a vertex chamber or R -far from it, after decreasing R by a fixed dimensional margin. Every E_σ with $\dim \sigma > 0$ is R -far from all retained branch images, with the same convention.

Proof. Vertices of Y are almost-principal orientations of the remaining walls. Starting at a vertex of C , intersect the finitely many combinatorial halfspaces whose orientations must change. This intersection is nonempty by finite Helly [57, Theorem 2.2]. Gate the starting vertex to it. Each required unchanged halfspace contains the starting vertex and meets this intersection, again by Helly; convexity of the gate [57, Lemma 2.11 and Theorem 8.2] therefore keeps the gate in every unchanged halfspace. Hence every vertex lifts.

A combinatorial geodesic cannot cross more than m consecutive omitted walls. Otherwise two of those walls would belong to the translates of one member of $\mathcal{W}_0$, contrary to their imposed separation. In particular vertex fibers have diameter at most m , and a geodesic between lifts crosses at most m omitted walls between successive retained walls. This gives the quasi-isometry inequalities and properness of Y .

For a cube of Y , constrain all its fixed orientations in C . The resulting convex combinatorial subcomplex contains lifts of every corner. Thus all four sectors for each pair of its variable walls occur there. These walls cross pairwise in that subcomplex and therefore cross in a common cube. This lifts the given cube and proves nonemptiness of (3.8). The closed mid-halfspaces are CAT(0)-convex. Looking cube by cube, their intersections lie within a dimensional bound of the bounded combinatorial fibers, so they are bounded. In a cube mapping onto σ , move its variable coordinates to their middle values to obtain the claimed open-cube distance estimate.

No retained hyperplane comes within R of a retained branch image, because all translates of the walls near $\mathcal{B}$ were omitted. The ball about that image therefore has a single retained-wall orientation. It lies in the corresponding E_v and is disjoint from the other chambers and from every E_σ with $\dim \sigma > 0$. This proves the last assertions. $\square$

3.3 Sheets and a cubical branched model

We give the sheet construction with uniform collars. This avoids any assumption that the branch geometry is locally a cubical vertex star. Choose constants

$$r_0 < r < D < R \tag{3.9}$$

in this order. The coarse return-of-homotopies observation following (3.6) has a fixed error at the chosen mesh scale. Choose r_0 larger than this error and $\dim C$, and choose r to absorb the short endpoint corrections when a path is projected to a chamber. Choose D larger than $2r + 10r_0 + \dim C$, with room for the same coarse-return error; this also accommodates the close-map lifts used below. Finally choose R much larger than these constants and make the sparse cover choice above. All these tolerances are fixed before H , m , or the chamber diameters are known. Projection homotopies of exterior collar points remain in a fixed-width collar, regardless of how far the chamber extends along its boundary.

A *near point* for E_σ is a point $z \in \tilde{X}$ with $d(f\pi(z), E_\sigma) \leq r_0$. Two near points determine the same *sheet* if they are joined by a path whose $f\pi$ -image lies in the r -neighborhood of E_σ .

Lemma 3.5 (Sheet stability and stabilizers). *With the choices above, the following hold.*

- (i) *Enlarging the path tolerance from r to D does not change the equivalence relation on near points. Points at any fixed intermediate distance can be assigned consistently to sheets by connecting them to near points.*
- (ii) *B acts transitively on the sheets over each E_σ . If $\dim \sigma > 0$, this action is free. The transitions from a cube to one of its positive-dimensional faces are bijections on sheets.*
- (iii) *The stabilizer of a vertex sheet is generated by the stabilizers of the branch lifts accessible in that sheet. These are whole conjugates of retained factors in (3.3). There are finitely many orbits of such accessible branch lifts under the sheet stabilizer.*

Proof. We first prove the collar assertion. Project an excursion of a path towards the convex set $E = E_\sigma$, using nearest-point projection in C . Leave portions already within the small collar unchanged. For an excursion that goes beyond the larger tolerance, interpolate to full projection away from its endpoints, retaining the endpoints by short connecting pieces. Only finitely many excursions cross the separated collar thresholds: a continuous path on a compact interval is uniformly continuous.

The homotopy of an exterior point to its nearest point stays outside the interior of E until its endpoint, which is on the boundary. Thus the sweep avoids all branch images deep inside E . It also stays in the prescribed fixed neighborhood of E , so avoids the branch images far outside it. Discretize this homotopy back in Z relative to the original path, using the control observation following (3.6). Its uniformly bounded additional errors still avoid the branch images because R was chosen last. This homotopy therefore lifts through the ordinary covering away from those images. Its replacement path lies in the r -neighborhood after the uniform endpoint corrections. This proves (i). The same argument assigns points in larger fixed collars to sheets, independently of the chosen short return paths, and respects inclusions $E_\sigma \subset E_\tau$ for faces $\tau \subset \sigma$.

Paths within a fixed neighborhood of E can be contracted downstairs in a controlled enlargement. Indeed project to E , contract there by convexity, and use the same discretization.

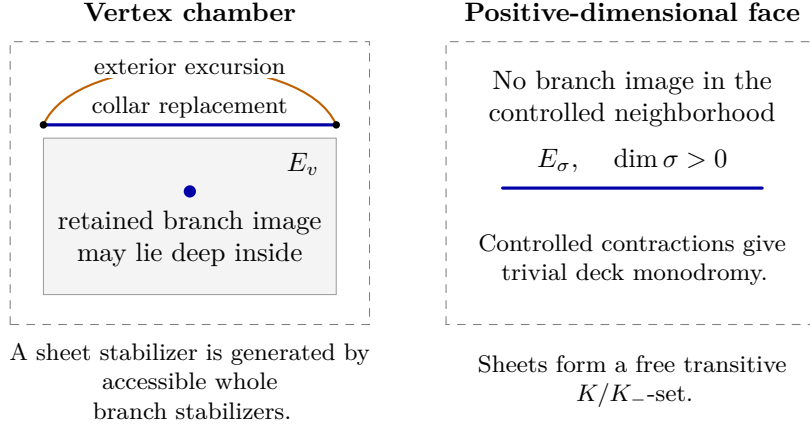


Figure 1: Schematic coarse neighborhoods for selective cubulation. The closed mid-halfspace chamber E_v may contain retained branch images; only the exterior collar used to shorten excursions must avoid them. Every positive-dimensional face chamber E_σ has a branch-free controlled neighborhood. Consequently sheet transitions between positive-dimensional faces are equivariant bijections, whereas vertex sheet stabilizers may be nontrivial. The drawn regions and distances are schematic: E_v is not asserted to be the literal fiber $p^{-1}(v)$, and the collar constants do not depend on chamber diameter.

If E has no branch points nearby, these contractions lift through the covering. Hence a deck transformation stabilizing one of its sheets is trivial. Coarse convex connectivity downstairs and path lifting also show that the deck action on sheets is transitive: connect two reference images within the allowed collar, lift the path, and adjust its endpoint by a deck transformation. This additionally shows that any prescribed point of E has a nearby reference lift in each sheet.

For a positive-dimensional σ , Lemma 3.4 excludes branch points from all the neighborhoods just used. Its sheet set is therefore a free transitive B -set. Inclusion $E_\sigma \subset E_\tau$ induces a B -equivariant map of sheet sets. When both dimensions are positive, this is a map between nonempty B -torsors and hence is bijective. This proves (ii).

For a vertex chamber, let a deck transformation stabilize a sheet. Choose a path from a reference point to its translate within the allowed collar. Replace passages through branch lifts by link paths. Such paths exist because the links are connected; even a long path in a lifted link has image in the bounded star of its branch image. Contract the projected loop in a controlled enlargement of E . A cellular version of this contraction uses a finite subcomplex and may include the entire small cone star of each apex that it visits. By van Kampen, after deleting these apex cones the loop is a product of conjugated loops in their links.

Lift this factorization. The discrepancy of a link loop is an element of the full stabilizer of the corresponding accessible branch lift. All the paths to these lifts stay in the enlarged collar, so (i) assigns them to the original sheet. Conversely every such branch stabilizer preserves the sheet. We have proved generation by these whole stabilizers, not merely an assertion about individual elements of them.

There are finitely many branch images in the controlled neighborhood: E is bounded, f is a quasi-isometry, and Z is locally finite. Two accessible lifts of the same branch image lying in the sheet differ by a deck transformation preserving that sheet. Consequently the accessible lifts have finitely many orbits under its stabilizer. This proves (iii). $\square$

Figure 1 summarizes the distinction between vertex chambers and positive-dimensional face chambers in Lemma 3.5.

Construct $\tilde{Y}$ by taking one copy of each cube of Y per sheet of E_σ , gluing along the face transitions. These transitions are compatible by construction, so this is a cubical complex, with

$$q : \tilde{Y} \longrightarrow Y \quad (3.10)$$

and an action of P , inducing the $\tilde{H}$ -action downstairs.

Lemma 3.6 (Cubical branched model). *The complex $\tilde{Y}$ is $CAT(0)$, the P -action is cocompact, and every hyperplane maps isomorphically under q to a hyperplane of Y .*

Proof. First consider links. Two adjacent lifted edge germs project to distinct germs, since they occur on a cube projecting isomorphically to a base cube. Hence a clique projects to a clique with distinct germs. An individual lifted edge determines a unique lift of every positive-dimensional containing cube by the positive-face bijections of Lemma 3.5. The base flag condition and these bijections therefore place all germs of the clique on its unique lifted cube. Links are simplicial and flag.

We now establish simple connectivity; local flagness alone would not suffice. There is a useful canonical-lifting rule. Suppose b maps a path, disc, or homotopy into $\tilde{X}$, and a maps the same parameter space into C , with

$$d(a, f\pi b) \leq 2r + 10r_0. \quad (3.11)$$

By the choice of D , if pa lies in the open cube σ , the open-cube estimate in Lemma 3.4 assigns b to a sheet of E_σ . Use the corresponding copy of σ to lift pa .

This lift is continuous. Near a parameter mapping into an open face, local finiteness downstairs confines nearby values to cofaces of that face. Local path connectivity of the parameter space and the collar assignment in Lemma 3.5 make their sheet restrictions agree on the face. Although there can be infinitely many coface sheets upstairs, their points approach this same face lift: distance to the face is bounded by the corresponding coordinate distance inside a unit cube, uniformly in dimension. Thus the rule is continuous in the cubical path metric. In particular it applies to $a = f\pi b$.

Conversely, let α be an edge loop in $\tilde{Y}$, viewed as a loop based at an edge midpoint. For each of its edge occurrences choose $c_e \in E_e$ projecting to that midpoint, using the cube-lifting assertion of Lemma 3.4. Choose a nearby reference point $z_e \in \tilde{X}$ in its assigned edge sheet. Consecutive reference points belong to the same vertex sheet, so join them by paths over the r -neighborhood of the relevant E_v . These paths concatenate to a loop b in $\tilde{X}$.

Project the image $f\pi b$ piecewise to the convex sets E_v , adjusting its endpoints to the chosen c_e 's. Nearest-point projection and bounded endpoint interpolation give a homotopy satisfying (3.11). Its canonical lift ends in paths lying in closed half-stars of the prescribed lifted vertices. To see this precisely, a point of E_v has all outside-wall orientations equal to those of v , and every variable coordinate is at most halfway from that corner. Face consistency fixes the assigned lifted corner throughout each path piece. Straight-line contraction towards that corner, compatible on faces, replaces the piece by the two required half-edges. The resulting loop is α .

Since $\tilde{X}$ is simply connected, b bounds a disc there. The canonical lift for $a = f\pi b$ maps this disc into $\tilde{Y}$, proving that α is nullhomotopic. Every lifted vertex is also connected to the canonical image: choose a reference point close to a lift of a base vertex in E_v and use the same bounded interpolation. Thus $\tilde{Y}$ is connected and simply connected. All cells have bounded finite shapes and embedded faces; after subdivision, metric-open cell stars allow ordinary fine-mesh simplicial approximation of the compact discs just used. Hence this argument proves the usual combinatorial simple connectivity as well. Together with flag links it proves the $CAT(0)$ conclusion by [58, Lecture 1, Section 1.2].

The action is cocompact because $Y/\bar{H}$ has finitely many cube orbits and B is transitive on every sheet set. Finally, positive-dimensional face transitions make the restriction of q to a hyperplane an ordinary covering of a base hyperplane. Base hyperplanes are $\text{CAT}(0)$, hence simply connected. Each such covering is therefore an isomorphism. $\square$

Completion of Proposition 3.3. Take the finitely many hyperplane orbit families of the compact special cover of C , omitting families with no surviving members. Within each family hyperplanes are two-sided and disjoint. These properties pass to Y : remaining walls do not acquire new crossings, by Lemma 3.4. They also pass to $\bar{Y}$.

Cut successively along the lifts of each family, within the convex components left by previous cuts. Disjoint two-sided tracks in a simply connected component give a tree splitting. The resulting graph-of-groups quotients are finite. Indeed, subdivide along the midplanes. A chamber and its incident side pieces are specified by cube-side flags up to the group action. At a cell only the hyperplanes cutting that cell can vary the choices of sides, so the number of choices is bounded in terms of the dimension. Branching at original vertices introduces no additional midplane choices.

A cutting trace is a convex part of a lifted hyperplane. It contains a dual-edge midpoint in its chamber: in a cube meeting the trace, fix the dual coordinate at its middle and choose the remaining corner coordinates towards the chamber's defining sides. Thus the trace determines its containing hyperplane. A deck transformation preserving the trace preserves this hyperplane, acts trivially on it by Lemma 3.6, and is trivial because positive-cell sheet actions are free. Hence the trace stabilizer injects into $\bar{H}$.

The trace projects to a convex subcomplex of the corresponding base hyperplane. Each earlier cutting hyperplane meeting the lifted hyperplane has a unique corresponding base trace; thus its side condition projects to the same convex halfspace condition downstairs. The stabilizer acts cocompactly on the trace by the finite flag description. Its image is consequently undistorted in Q_* . The quotient map $P \rightarrow \bar{H}$ is Lipschitz in word metrics, so this also proves undistortion of the trace stabilizer in the hyperbolic group P . All edge groups of the hierarchy are therefore quasiconvex in P .

After all families are cut, each component has one original cubical vertex. Its stabilizer projects trivially to $\bar{H}$, since $\bar{H}$ is torsion-free and acts properly on Y . It is exactly the deck stabilizer of the corresponding vertex sheet. Lemma 3.5(iii) says that this group is generated by whole accessible retained factors, with finitely many factor orbits. Apply Kurosh to its subgroup of

$$B = \ast_{a \text{ retained}} R_a.$$

Each accessible whole factor is one of its Kurosh factors, up to subgroup conjugacy. Killing these factors kills the entire sheet stabilizer by the proved generation statement. Hence there can be no remaining free summand or partial-factor summand. The vertex stabilizer is a finite free product of retained factors and is hyperbolic and virtually compact special.

The hierarchy has finitely many levels and finite quotients. Working upwards from its terminal groups shows that all vertex groups are finitely generated. The globally quasiconvex edge groups give quasiconvex vertex groups successively: replace excursions of an ambient path outside a vertex space by paths in the attaching edge cosets, using their ambient undistortion. The quasiconvex hierarchy theorem [66] therefore makes P virtually compact special. The normal-core choice made above establishes the asserted normality of H , and any later normal core can be accommodated by repeating the same sparse-cover construction. $\square$

3.4 Novikov realization along quotient chains

The geometric construction now supplies quotients that retain any prescribed finite set of filling-kernel factors. Our algebraic target is to realize any finite list of rational expressions, after finite-index restriction, in the ordinary Novikov rings for both signs of one real character. Applied to the entries of a division-ring chain contraction, this will give the two acyclicity statements needed for fibering. We first descend through an RFRS chain of the quotient until the remaining coefficients lie in $D(K)$. A selective quotient then makes their finitely many subgroup supports detectable by a second RFRS chain. That descent ends with ordinary group-ring coefficients; recombination recovers the original expressions for one character and its negative.

For a real character $\chi : J \rightarrow \mathbb{R}$, write

$$\widehat{\mathbb{Q}J}^\chi = \left\{ \sum_{g \in J} a_g g : \#\{g : a_g \neq 0, \chi(g) < c\} < \infty \text{ for every } c \in \mathbb{R} \right\}. \quad (3.12)$$

Our completion convention is towards positive height. The subgroup-field hypotheses embed $D(J)$ and this ring in the same skew series division ring over $D(\ker \chi)$. Indeed, (3.1) embeds the rational field there; a nonzero series has a leading term, and its positive remainder has a positive minimum valuation, so geometric inversion is defined. This remains valid when $\chi(J)$ is dense in $\mathbb{R}$, because the support condition is finite below each threshold.

Definition 3.7. Let $K \triangleleft J$, with J/K virtually RFRS. We say that (J, K) has the *selective realization property* if, for every finite-index preimage $J' \leq J$ and every finite set $T \subset K$, there are a finite-index preimage $H \leq J'$, normal in J' , a subgroup $U \leq K$ containing T , and a homomorphism from H to a virtually RFRS group that is injective on U .

Proposition 3.3 supplies this property in Theorem 3.1: a finite set in K lies in a finite subproduct of (3.3), and its selective quotient is virtually compact special, hence virtually RFRS [1]. More explicitly, let $\rho : H \rightarrow L$ be the selective homomorphism and let S be a finite transversal for $H \backslash J'$. Normality of H makes

$$h \mapsto (\rho(s^{-1}hs))_{s \in S} \quad (h \in H)$$

a homomorphism on the common domain H . Its s -coordinate is injective on sUs^{-1} . These are the conjugate support groups arising in the right-coset coordinates for restriction of scalars. Finite products and subgroups of RFRS groups are RFRS, so the product target remains virtually RFRS.

Lemma 3.8 (Controlled linearization). *A rational expression over a coefficient ring in a division ring has a representation $pA^{-1}q$, where A is a square polynomial matrix and p, q are constant selectors. The entries of A^{-1} can be given rational expressions using sums and products of the original subexpressions and their original inverses. In particular linearization introduces neither a larger nested inverse depth nor new bottom denominators on a maximal-depth inverse chain.*

Proof. A polynomial f is the upper-right entry of the inverse of $\begin{pmatrix} 1 & -f \\ 0 & 1 \end{pmatrix}$. Use block diagonals for sums. If $E = pA^{-1}q$ and $F = rB^{-1}s$, use

$$\begin{pmatrix} A & -qr \\ 0 & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & A^{-1}qrB^{-1} \\ 0 & B^{-1} \end{pmatrix}$$

for the product. For E^{-1} , border A to obtain

$$M = \begin{pmatrix} 0 & p \\ q & A \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} -E^{-1} & E^{-1}pA^{-1} \\ A^{-1}qE^{-1} & A^{-1} - A^{-1}qE^{-1}pA^{-1} \end{pmatrix}. \quad (3.13)$$

The displayed formulas prove the assertion inductively. In (3.13) the new inverse is exactly the parent inverse already present in the original expression. $\square$

The data at a descent stage are finitely many square polynomial matrices invertible over the current division ring, together with chosen rational expressions for all entries of their inverses. Linearization puts a finite list of scalar expressions in this form. Truncating a character expansion records finitely many coefficient entries. Solving those coefficient problems means realizing their chosen rational values in both ordinary Novikov rings for one later character. The next lemma proves that this reduction terminates; the following lemma explains how to recover the original inverses.

Lemma 3.9 (Finite descent of inverse depth). *Let J/K have a normal RFRS chain, and pull it back to*

$$J = H_0 \geq H_1 \geq \cdots, \quad \bigcap_i H_i = K.$$

Let $N_i \triangleleft H_0$ be the preimage of the rational abelianization kernel of H_i/K , so $N_i \leq H_{i+1}$. At each stage expand finitely many specified rational expressions for both signs of a character with kernel N_i , retain finitely many coefficients, restrict those scalar coefficient problems to H_{i+1} , and use Lemma 3.8. After finitely many stages the resulting coefficient problems have inverse depth zero over $D(K)$.

Proof. Inverse depth is measured relative to $D(K)$: coefficient-field elements and group monomials have depth zero, sums and products take the maximum, and inversion adds one. At the bottom of an inverse chain, evaluate the inverse-free operand as a finite polynomial before taking slices.

For one character, write an operand with leading monomial L as $L(1 + T)$, with T of strictly positive valuation. Its inverse is $\sum_{n \geq 0} (-T)^n L^{-1}$. Every specified coefficient is a finite sum of products of operand coefficients, monomial transports, and the inverse of the leading coefficient. Thus depth does not increase.

There is a more precise invariant. If a coefficient retains the old maximum depth $d > 0$, each bottom denominator on a length- d inverse chain is a height slice of an old bottom polynomial on a length- d chain, up to common monomial transport and conjugation. For sums and products the assertion follows by tracing a maximal-depth chain into one factor. For an inverse, coefficients of its operand alone have depth at most $d - 1$. A chain of length d must therefore pass through the inverse of the leading coefficient. Apply the same induction to that coefficient of the operand. At the bottom this is simply a slice of an inverse-free polynomial. Cancellations remove terms but introduce no new support degree in that bottom polynomial.

Choose one right normalizing monomial for each slice. Replacing support elements g, h by gu^{-1}, hu^{-1} leaves gh^{-1} unchanged. A later whole conjugation conjugates this ratio. Consequently, while depth d persists, the nonidentity differences of its bottom support degrees modulo K lie among conjugates of a fixed finite list, chosen at the beginning of this depth block.

Restriction to the next subgroup does not mix the coefficients: for $c \in D(N_i)$ and a representative t of $H_{i+1} \setminus H_i$,

$$tc = (tct^{-1})t, \quad tct^{-1} \in D(N_i) \subset D(H_{i+1}). \quad (3.14)$$

Thus the coefficient operator is diagonal by conjugation on these coset coordinates. Controlled linearization also preserves the invariant by Lemma 3.8.

The bottom support differences at stage i belong to H_i/K . Since these subgroups are normal and have trivial intersection, eventually they exclude every element of the fixed finite list and all its conjugates. Every bottom operand on a surviving maximal-depth chain is then a monomial over $D(K)$. Inverting it uses only a monomial inverse and a coefficient-field inverse, so all such chains shorten. Start a new finite list after this depth drop and repeat. There are finitely many depth drops.

The qualification about maximal depth is essential to this argument. New denominator supports can appear after the depth drops; the construction expressly starts a fresh finite list at that point. It never asserts that one support list controls all later depths. $\square$

A given RFRS chain may be replaced by one normal in its initial group. Indeed, intersect its finitely many conjugates at each stage; the rational abelianization kernel of a subgroup maps into that kernel in every containing conjugate. The RFRS containment therefore persists under these cores. In the finitely generated setting of Proposition 3.11, characters with kernel exactly N_i exist by embedding the finite-rank free abelian group H_i/N_i in $\mathbb{R}$. A zero character is permitted at a stationary stage. Depth zero over $D(K)$ still allows coefficients in that division ring: it need not mean a polynomial over $\mathbb{Q}K$. Removing precisely those remaining coefficients will require the selective realization property.

Lemma 3.10 (Recombination by perturbation). *Suppose a finite stage of the preceding procedure used a character χ_i on H_i . For each of its polynomial matrices A , fix finite homogeneous truncations B_+, B_- of A^{-1} such that $I - AB_{\pm}$ and $I - B_{\pm}A$ have strictly positive weights for $\pm\chi_i$, respectively. Suppose all scalar coefficients of the matrices A, B_+, B_- have been realized, after restriction to a finite-index subgroup F , in both ordinary Novikov rings for a character ψ of F , compatibly with their rational-field values. Then its original matrix inverse problems are realized for both signs of*

$$\chi_i|_F + \epsilon\psi \quad (3.15)$$

for every sufficiently small positive ϵ .

Proof. Let A be one polynomial matrix and $B = A^{-1}$, with its specified rational expressions. The chosen truncations B_+ and B_- have coefficients in $D(N_i)$, and both errors

$$I - AB_{\pm}, \quad I - B_{\pm}A \quad (3.16)$$

have strictly positive weights in their respective directions.

First we check the support needed to compare the two characters. This is necessary even when a later finite-index cut does not contain N_i . For right-coset coordinates t_j of $F \setminus H_i$, the entry from t_j to t_k of a homogeneous term of degree u is supported in the single height

$$u + \chi_i(t_j) - \chi_i(t_k). \quad (3.17)$$

Its coefficient lies in one coset of $D(F \cap N_i)$: use the finite-index field decomposition on the intersection of its original homogeneous coset with the selected matrix coset. Put $U = F \cap N_i$ and first remove this common monomial shift. Choose a finitely generated subgroup of U supporting a rational expression for the remaining coefficient, and expand there for the restriction of ψ . Uniqueness of rational inverses in the ambient skew-series ring identifies this intrinsic expansion with the ambient one. At each ψ -height an ordinary Novikov realization has only finitely many group terms. Subgroup-coset independence applied to that finite coefficient sum excludes terms outside U . Multiplying back by the common shift proves

preservation of the original homogeneous support. The same argument applies across the selective change of quotient, where the support subgroup embeds.

On $F \cap N_i$, the new character in (3.15) is $\epsilon\psi$. The internal series expansion is therefore unchanged, with only its grading scaled; it represents the same rational entry. Every finite old-height slice remains summable. For each sign, subtract the row-minus-column offsets in (3.17) when measuring an error entry. There are finitely many positive old-height slices, and their $\pm\psi$ -valuations are bounded below. A common sufficiently small positive ϵ therefore keeps all adjusted error valuations strictly positive.

Offsets telescope in matrix products. Hence powers of either error in (3.16) have valuations tending to positive infinity, even in the unadjusted coordinates. Geometric inversion provides both a left and a right inverse for A in the relevant ordinary Novikov ring; they coincide. All identities are exact because the rational field and the ordinary series were embedded in the same skew series division ring. The same choice of ϵ works for the finite lists for both signs. $\square$

Proposition 3.11 (Progressive characters). *Let J be finitely generated with the subgroup-field hypotheses, and let $K \triangleleft J$ satisfy the selective realization property of Definition 3.7. Every finite family of rational expressions over $\mathbb{Q}J$ can, after restriction of scalars to a finite-index subgroup F , be realized compatibly in the ordinary Novikov rings for both signs of a single real character of F .*

In particular, if a finite complex of finitely generated free $\mathbb{Q}J$ -modules is acyclic over $D(J)$, then its restriction is acyclic over both these Novikov rings. If its degree-zero ordinary homology is nonzero, the character is nonzero.

Proof. Take the preimage H_0 of a finite-index RFRS subgroup of J/K . Restriction of scalars is right multiplication on right-coset coordinates, using a normal core when necessary. The finite-index field decomposition makes this an exact finite matrix operation. Linearize the finitely many expressions as in Lemma 3.8, retaining the specified expressions for all needed inverse entries.

Pull back a normal RFRS chain from H_0/K . At stage i , use (3.1) to expand each inverse matrix for both signs of a character χ_i with kernel N_i . Truncate sufficiently far to obtain the positive errors (3.16). Collect the finitely many coefficients of the matrix and its truncations; they lie in $D(N_i)$. Restrict to the next group by (3.14), and linearize these scalar problems. Lemma 3.9 terminates this procedure with finitely many unresolved coefficients in $D(K)$.

Choose a finite set of elements of K supporting rational expressions for all those coefficients. Apply the selective realization property, including its finite product of conjugate maps, to obtain a normal finite-index preimage H and a map injective on the required support subgroups. Pass to the preimage H_p of an RFRS subgroup of its image.

This cut can mix coefficient entries, so we describe it explicitly. For a scalar $c \in D(U_0)$, group the cosets of H_p into orbits of right multiplication by U_0 . An orbit has representatives gu_j , with $u_j \in U_0$. The entries of right multiplication by c in these coordinates come from decomposing $u_j c$ over

$$D(U_0 \cap g^{-1}H_p g)$$

and then conjugating by g . They therefore admit finite rational expressions entirely in these subgroup fields. Changing to one common transversal only multiplies entries by monomials, which does not affect summability.

Start the depth argument afresh with these new expressions. Use an RFRS chain in the new quotient and ordinary rational group polynomials. Work separately inside each support subgroup, choosing its own monomial transporters when taking a height coefficient. Later coefficients remain in conjugates of that support subgroup. Once a coefficient lies in the current character kernel, restriction to the next stage only conjugates it, because that whole

kernel is contained in the next subgroup. Thus the maximal-depth ancestry argument applies exactly as before. The new quotient map is injective on each support subgroup, so its normal RFRS chain separates the finite nonidentity support lists. This time the terminal problems are ordinary group-ring polynomials, with no unresolved division-field coefficients. They are realized for the zero character if necessary.

Now reconstruct the inverse problems backwards, through both finite chains and the selective switch, using Lemma 3.10. At each step use both signs' residual lists and choose one sufficiently small positive perturbation. The result is a single character on a common finite-index subgroup and compatible realizations of all original expressions for both of its signs.

For the chain-complex assertion, include the entries of a $D(J)$ -chain contraction and all the inverse operands used in their expressions. Their exact identities persist under the common series embeddings, giving contractions over the two ordinary Novikov rings. Restriction of scalars does not change the underlying chain complex or its ordinary homology. If the final character were zero, its Novikov rings would be $\mathbb{Q}F$, so these contractions would contradict the assumed nonzero ordinary degree-zero homology. $\square$

Corollary 3.12 (Open Novikov regions for RFRS groups). *Let J be virtually RFRS with the subgroup-field hypotheses. A finite family of bounded complexes of finitely generated free $\mathbb{Q}J$ -modules, with polynomial differential matrices and acyclic extension to $D(J)$, admits simultaneous ordinary Novikov contractions, after passage to a finite-index subgroup F , on a nonempty antipodally invariant open subset of $\text{Hom}(F, \mathbb{R})$. Here the character space has the topology of pointwise convergence. This property can be imposed after any prescribed further finite-index restriction.*

Proof. We use the descent above with $K = 1$, so there is no selective switch. For finitely generated J , the underlying realization is also supplied by [36, Theorem 5.2]. To retain the present group generality, we first justify the ambient characters and the character-field expansion without finite generation.

Choose a finite-index RFRS subgroup H_0 and a normal residual chain (H_i) . The embedding $H_0 \hookrightarrow \prod_i H_0/H_i$ into a countable product of finite groups gives $|H_0| \leq |\mathbb{R}|$. If N_i is the rational abelianization kernel of H_i , the rational hull of the torsion-free abelian group H_i/N_i therefore has dimension at most $\dim_{\mathbb{Q}} \mathbb{R}$. A rational-linear injection into $\mathbb{R}$ supplies a character of the whole group H_i with kernel exactly N_i . The quotient need not be free abelian or finitely generated. The RFRS containment $N_i \leq H_{i+1}$ still holds.

For any real character χ of a subgroup H , put $N = \ker \chi$. Coset independence embeds $D(N) * (H/N)$ in $D(H)$. This crossed product is the directed union of its subrings over finitely generated subgroups of H/N . Each such subgroup is free abelian of finite rank, and its crossed product is an iterated skew Laurent Ore domain. Every pair of elements lies in one such subring, where its left and right Ore equations can be solved. The union is therefore Ore as well. Its fraction division ring embeds in $D(H)$, since $D(N)$, the group lifts and all inverses of nonzero elements lie there. Conversely, the fraction ring contains $\mathbb{Q}H$, so it contains its division closure $D(H)$. The fraction ring is thus exactly $D(H)$, proving (3.1) in this generality.

The same skew-series realization applies: a nonempty support finite below every real threshold has a least term, and each strictly positive remainder has an individual positive minimum valuation. These facts justify geometric inversion even when the character image is dense or not finitely generated.

Start with the finitely many entries of division-ring chain contractions. The descent in Lemma 3.9 uses only finite rational expressions, finite retained coefficient lists and finite-index transversals. Normal residuality removes its finite support-difference lists, so it terminates

with coefficients over $D(1) = \mathbb{Q}$. Lemma 3.10 then applies backwards: there are finitely many old-height slices and lower bounds for their later-character valuations, so one positive perturbation works for both signs at each stage. This gives simultaneous ordinary Novikov contractions.

For openness, linearize the finitely many contraction entries over the ordinary group ring and truncate their realized inverse matrices to finite group-ring matrices. The left and right inverse errors have finitely supported strictly positive valuations. Their finitely many strict evaluation inequalities define an open neighborhood in the pointwise topology, and geometric inversion supplies the contractions throughout it. Intersect the neighborhood for the positive sign with the negative of the neighborhood for the negative sign, and then take the union with its negative. This is the required nonempty antipodally invariant open set.

On restriction to any deeper finite-index subgroup, a Novikov series remains summable in its finite coset coordinates, with only fixed monomial offsets. Repeat the finite truncation argument there to obtain openness in the larger character space. $\square$

3.5 From two-sided Novikov acyclicity to a free kernel

Lemma 3.13 (The degree-one Novikov implication). *Let J be finitely presented, and let $\chi : J \rightarrow \mathbb{R}$ be nonzero. If degree-one homology vanishes over $\widehat{\mathbb{Q}J}^\chi$, then χ belongs to the Bieri–Neumann–Strebel (BNS) invariant. If the vanishing holds for both signs, some nonzero rational character has finitely generated kernel.*

Proof. We include the rational-coefficient argument in the positive-completion convention of [36, Theorem 3.11]; Sikorav established the integral-coefficient criterion [60]. Choose a symmetric generating set containing a generator t with $\chi(t) > 0$. In the positive completion the formal ray

$$c = \sum_{j \geq 0} [t^j, t^{j+1}]$$

has boundary $-[1]$. For each oriented generator x , the chain

$$[1, x] + xc - c$$

is a cycle, hence bounds a completed rational two-chain in the finite presentation complex.

Truncate that two-chain sufficiently high. Since relators have bounded height offsets, its boundary cancels every low term of the displayed cycle. Truncate both rays at a common sufficiently large exponent k . The resulting finite one-chain can be made to have arbitrarily high support and boundary

$$[xt^k] - [t^k].$$

The component of its edge support containing one endpoint must contain the other: the total boundary coefficient in a finite connected component is zero. Thus t^k and xt^k are joined by an actual path at uniformly high height, for every generator x .

Translate these finitely many paths to replace each edge $g \rightarrow gx$ by a path from gt^k to gxt^k . Choose the height threshold so this replacement raises the minimum height by a fixed positive amount. Iterating pushes any finite path above height zero. Its original nonnegative endpoints are connected to their pushed copies by positive t -rays, so the nonnegative Cayley graph is connected. This is the BNS condition.

The finite paths and their strict height inequalities persist in a neighborhood of χ . Applying the same construction to $-\chi$ gives both BNS conditions on a nearby nonzero rational line. The two-sided BNS criterion [9, Theorem B1], also stated in [36, Theorem 3.10], says that the kernel of its integral representative is finitely generated. $\square$

Lemma 3.14 (A finitely presented kernel in dimension two). *Suppose $\text{cd } J \leq 2$ and J maps onto $\mathbb{Z}$ with finitely presented kernel M . Then M is free.*

Proof. We have $\text{cd } M \leq 2$. If $\text{cd } M = 2$, finite presentation gives a finite-length resolution of $\mathbb{Z}$ by finitely generated projective $\mathbb{Z}M$ -modules, with the last module in degree two. Its dual has nonzero degree-two cohomology: otherwise its last map would split, and dualizing back would shorten the original projective resolution. Thus

$$H^2(M; \mathbb{Z}M) \neq 0.$$

As an M -module, $\mathbb{Z}J$ is a direct sum of copies of $\mathbb{Z}M$ indexed by $J/M = \mathbb{Z}$. Finite generation of the resolution gives the corresponding direct-sum description of $H^2(M; \mathbb{Z}J)$. The generator of J/M shifts the summands, with its conjugation twist. Its coinvariants are therefore nonzero. The Lyndon–Hochschild–Serre cohomology spectral sequence has only columns zero and one for this quotient, so

$$H^3(J; \mathbb{Z}J) \cong H^1(\mathbb{Z}; H^2(M; \mathbb{Z}J)) \neq 0,$$

contrary to $\text{cd } J \leq 2$; see [14, 8]. It follows that $\text{cd } M \leq 1$, and Stallings–Swan [61, 64] makes M free. $\square$

Proof of Theorem 3.1. First suppose G_* is already VCS, as happens when the peripheral family is empty after discarding trivial members or contains G_* itself. Then G_* is virtually RFRS, so Corollary 3.12 applies directly to its finite classifying-space chains. This gives two-sided Novikov contractions on a finite-index subgroup F , without using the filling construction.

In the remaining case choose the cofinite good filling (3.2). Proposition 3.3 gives the selective realization property, and Proposition 3.11 applied to the same chains again supplies two-sided Novikov contractions on a finite-index subgroup F .

In either case the restricted group-ring chain complex has $H_0 \cong \mathbb{Q}$: it is the cellular complex of the same connected universal cover, now with the F -action. The character cannot be zero, since its Novikov ring would then be $\mathbb{Q}F$, and a contraction would contradict this ordinary H_0 . Lemma 3.13 therefore gives a nonzero rational character on F with finitely generated kernel. Rescaling its image gives an epimorphism $F \rightarrow \mathbb{Z}$.

Coherence makes this kernel finitely presented. Lemma 3.14 makes it free, and finite generation makes its rank finite. The extension splits by choosing a lift of a generator of $\mathbb{Z}$, so the subgroup is finite-rank free-by-cyclic. It is hyperbolic, hence admits a proper cocompact cubulation by [27]; the elementary rank-zero case is immediate. Agol’s theorem [2] now gives virtual compact specialness, which passes to the finite-index overgroup G_* . $\square$

4 Hyperbolic ascending tori

For an injective endomorphism $\phi : F \rightarrow F$, write

$$J_\phi = \langle F, t \mid t^{-1}gt = \phi(g) \ (g \in F) \rangle.$$

Its height homomorphism sends t to 1 and F to 0. Changing the sign of height, or replacing the stable letter by its inverse, will not affect any assertion below. A *free splitting tree* means a minimal simplicial tree with trivial edge stabilizers and finite quotient; its vertex stabilizers are free factors. A path map of trees is *alignment preserving* if it maps each segment monotonically onto a segment, allowing constant subsegments.

Theorem 4.1. *Let F be a free group of finite rank and let $\phi : F \rightarrow F$ be injective. If J_ϕ is word-hyperbolic, then J_ϕ is virtually compact special.*

We prove the theorem by induction on $\text{rk } F$. The rank-zero group is infinite cyclic. Rank one gives $\text{BS}(1, n)$, with $n \neq 0$, and is incompatible with hyperbolicity. When ϕ is an automorphism, the conclusion follows from the cubulation theorem for hyperbolic free-by-cyclic groups [27] and Agol’s theorem [2]. For a nonsurjective injection, we either reduce rank or construct a noncontracting map of a free splitting tree. In the expanding case, its associated real-tree action gives a finite family of point stabilizers with nontrivial height-zero part. They are virtually ascending tori of smaller rank and will serve as peripherals. Filling their height-zero parts makes the base virtually free; a finite graph cover then separates the expanding map into irreducible train-track blocks. We remove any failure of fundamental-group injectivity in those blocks before applying Hagen–Wise. This supplies the good fillings for Theorem 3.1. The nonexpanding case is handled directly by a quasiconvex hierarchy.

The ascending result is also an input to the filling construction in Section 10. After proving it, we recover its stronger virtual finite-rank free-by- $\mathbb{Z}$ consequence in Corollary 4.10, including the induction branches that do not directly use the filling criterion.

4.1 Finiteness hypotheses and elementary changes of model

Lemma 4.2. *An ascending torus of an injective finite-rank free-group endomorphism is torsion-free, locally indicable, coherent, residually finite, and has a finite aspherical complex of dimension at most two. It has the Hughes-free subgroup-field setup used in Theorem 3.1, and its finite cellular chain complex becomes acyclic over its rational Linnell field.*

Proof. The height kernel is an increasing union of conjugates of F . Consequently a finitely generated subgroup contained in that kernel lies in a free group and, if nontrivial, maps onto $\mathbb{Z}$. A subgroup not contained in the kernel has a nonzero height homomorphism. This proves local indicability and also torsion-freeness. Coherence is the theorem of Feighn–Handel, which includes injective endomorphisms [22, Theorem 1.1]. Residual finiteness is due to Borisov–Sapir [10].

Represent ϕ by a based map of a finite rose. Its mapping torus is a finite aspherical two-complex: the associated graph of spaces has contractible vertex and edge universal covers and injective attaching homomorphisms. The usual tree-of-spaces construction therefore gives a contractible universal cover.

The height kernel belongs to Linnell’s class generated by free groups under directed unions; its extension by $\mathbb{Z}$ belongs to the same class under elementary-amenable extensions. Linnell’s theorem, together with the locally indicable Hughes-free field setup, gives the rational subgroup fields, their compatible inclusions and coset independence [38, 32]; the locally indicable affiliated realization and compatible rational subfields are given by [35, Corollaries 6.2–6.3] and [33, Section 2.3 and Proposition 2.2]. In particular the field embeds in the height skew-series division ring with coefficients in the kernel field, with either order on height.

For completeness, take a basis $x_1, \dots, x_r$ of F and the relators $t^{-1}x_it\phi(x_i)^{-1}$. The horizontal Fox matrix is

$$M = t^{-1}I - \left(\frac{\partial \phi(x_i)}{\partial x_j} \right)_{i,j}.$$

After removing the invertible initial factor t^{-1} , its initial height term is I . Geometric-series inversion therefore makes M invertible in a height skew-series division ring. A square matrix over a division subring that is invertible over an extension division ring is already invertible over the subring, so M is invertible over the Linnell field. Thus the second boundary is

injective after base change. The zeroth homology vanishes because $t - 1$ is invertible in that field. The cellular Euler characteristic is $1 - (r + 1) + r = 0$, so the first homology vanishes as well. $\square$

We will use the following elementary observation twice. It applies even when a graph map is not injective on fundamental groups.

Lemma 4.3. *Suppose based maps $f : X \rightarrow X$ and $g : Y \rightarrow Y$ of connected finite graphs admit based maps $q : X \rightarrow Y$ and $i : Y \rightarrow X$ such that*

$$qf = gq, \quad ig = fi, \quad iq = f^m, \quad qi = g^m$$

for some $m \geq 1$. Their mapping-torus groups are isomorphic, by an isomorphism preserving height.

Proof. The commuting maps induce homomorphisms of the mapping-torus presentations, sending stable letter to stable letter. Their composites are induced by f^m and g^m . In a mapping-torus presentation the map induced by the defining endomorphism is conjugation by the stable letter; hence both composites are inner automorphisms. The two homomorphisms are therefore isomorphisms. The same argument works up to based homotopy, with the corresponding change of stable-letter marking. $\square$

4.2 An expanding relative tree model

Expanding relative immersions for injective, nonsurjective free-group endomorphisms were constructed by Mutanguha [51, Proposition 4.1 and Theorem 4.5]. The following form is tailored to the induction on rank.

Lemma 4.4. *Let $\phi : F \rightarrow F$ be injective and nonsurjective, with $\text{rk } F \geq 2$. After replacing F by an iterated image, one of the following holds.*

- (i) *A finite-index height subgroup of J_ϕ is an ascending torus of an injection of a free group of rank strictly smaller than $\text{rk } F$.*
- (ii) *There is a nontrivial free splitting tree T and a ϕ -equivariant embedding $f : T \rightarrow T$ that is a homothety of scale $\lambda \geq 1$. The edge-transition matrix is irreducible, every vertex stabilizer is a proper free factor, and the quotient graph is finite.*

Proof. We compare successive iterated-image core graphs using the lengths of their arcs between branch vertices. Collapsing the arcs whose lengths stay bounded gives two trees and two maps: substitution identifies the collapsed trees, while their original inclusion gives a possibly collapsing map back. Their composition yields the required self-map. Its transition diagram will either lose an edge orbit, allowing rank reduction, or supply a recurrent irreducible part.

Fix the unit-edge Cayley tree S of F . Put $F_n = \phi^n(F)$ and let $S_n \subset S$ be its convex minimal subtree. Each S_n is leafless. Suppress the vertices of valence two and call the resulting edges *natural arcs*. The finite core graph $F_n \setminus S_n$ has rank $r = \text{rk } F$, and thus at most $3r - 3$ natural edges.

Substitution by the fixed injection ϕ is a quasi-isometric embedding of S : its image group is finitely generated and undistorted in F . Restricting to S_n and straightening to the hull S_{n+1} gives ϕ -equivariant quasi-isometries

$$S_n \longrightarrow S_{n+1}$$

with constants independent of n . Indeed the boundary of the target hull is the image of the source boundary, and straightened axes remain within the uniform quasigeodesic-stability bound. Branch points are centers of tripods with three unbounded legs. These quasi-isometries therefore give uniformly controlled branch-point correspondences in both directions; the two composites move branch points a uniformly bounded distance.

Choose a subsequence of pairs

$$U_k = S_{n_k}, \quad V_k = S_{n_k+1}$$

on which the two finite reduced core-graph types are fixed and each of their natural-edge lengths is either bounded or tends to infinity. There is one bound B for all bounded coordinates, and the minimum ℓ_k of all unbounded coordinates tends to infinity. Collapse in U_k and V_k the components of their respective bounded-arc forests. These forests need not have bounded-diameter components.

For sufficiently large k , the branch-point correspondence induces a ϕ -equivariant simplicial isomorphism between the two collapsed trees. Here are the details needed when a collapsed component is unbounded. Adjacent branch points joined by a bounded arc have images at a uniformly bounded distance. Their images consequently lie in the same bounded-arc component, since that distance is less than ℓ_k . Apply this to chains of bounded arcs and then apply the coarse inverse; the bounded errors of the two composites show that the resulting maps on components are inverse bijections. To verify adjacency, let $[v, w]$ be a long natural source arc and choose the corresponding target branch points v', w' . For every branch point z of $[v', w']$, quasigeodesic stability puts its corresponding source branch point within a uniform distance of $[v, w]$. The projection of a branch point to the interior of a branch-free natural arc is impossible: an off-arc branch point would create a branch point where its joining arc meets the interior. Its projection is therefore v or w . It follows that z is uniformly close to v' or w' . If $[v', w']$ crossed two long target arcs, a branch point between them would be at distance at least ℓ_k from both endpoints, a contradiction. It crosses at least one long arc because the endpoint short components are distinct. Thus it crosses exactly one. The reverse correspondence gives the same conclusion in the opposite direction, proving adjacency.

The collapsed tree is nontrivial for some such subsequence. Otherwise all natural arcs have uniformly bounded lengths along an infinite subsequence. There would then be only finitely many unbased labelled core graphs, so two nested groups F_n, F_{n+j} would be conjugate. A finitely generated subgroup of a free group cannot be conjugate to a proper subgroup of itself. To see this last fact directly, if $gAg^{-1} \leq A$, the nested translates $g^{-i}S_A$ all contain S_A . Fix a point on an axis in S_A . Since A acts cocompactly on S_A , that point is a uniformly bounded distance from every coset $g^{-i}A$. Only finitely many cosets can meet a fixed finite ball, so $g^j \in A$ for some $j > 0$. The nested conjugacy chain is now periodic and all its inclusions are equalities. Hence $F_n = F_{n+j}$, which by injectivity implies $F = \phi^j(F)$ and contradicts nonsurjectivity.

Inclusion $V_k \subset U_k$ maps bounded-arc components to bounded-arc components. In fact a bounded natural arc of V_k is subdivided by branch points of U_k into arcs of length at most $B < \ell_k$. Fix one sufficiently large k , and write $n = n_k$ and T_U, T_V for the collapsed trees. Their actions are the actual actions of F_n and F_{n+1} , respectively. The maps just constructed are

$$T_U \xrightarrow{\Phi_k} T_V \xrightarrow{I_k} T_U.$$

Here Φ_k is a simplicial isomorphism equivariant for $\phi : F_n \xrightarrow{\sim} F_{n+1}$, and I_k is alignment preserving and equivariant for the inclusion $F_{n+1} \leq F_n$. Hence $I_k\Phi_k$ is a ϕ -equivariant alignment-preserving self-map of the single F_n -tree T_U . The subsequence was used to classify natural-arc lengths; no identification of the markings at different k is needed. Replacing the base by F_n does not change the ascending torus: the base F is conjugate into F_n by a power

of the stable letter, and ascending normal form gives the converse inclusion. Rename this base F .

If an iterate of the whole tree misses an edge orbit, its connected image lies in one component of the complement of that orbit. The image group $\phi^m(F)$ stabilizes that component. Its stabilizer C is a proper free factor, as follows by collapsing every complementary component in the original free splitting. Consequently

$$\phi^m(F) \leq C, \quad \phi^m(C) \leq C.$$

The ascending torus of $\phi^m|_C$ embeds by normal form in the height- m subgroup, and contains its whole base after conjugating by the stable letter. It is therefore that subgroup. This is alternative (i).

Otherwise choose a source strongly connected component in the finite edge-transition diagram. If it were nonrecurrent, it would consist of one vertex with no self-loop and no incoming arrow; that edge orbit would already be missing from the image of the whole tree. The chosen component is therefore recurrent. No incoming arrows means that each edge outside it maps entirely through edges outside it. Their union is an invariant forest in the tree, so collapsing it gives a well-defined self-map. Each remaining edge has a transition inside the recurrent component and is not collapsed. Alignment preservation also survives: lift a segment through the collapsed vertex components, apply alignment preservation before collapse, and delete the collapsed subsegments. No reduced path can acquire backtracking in this operation. The transition matrix is now irreducible. Its Perron eigenvector assigns positive lengths to the finitely many edge types, and linear parametrization makes the map multiply all path lengths by the Perron eigenvalue $\lambda \geq 1$. Alignment preservation and noncollapse make it an embedding. All vertex stabilizers are proper free factors because the resulting free splitting is nontrivial. This proves alternative (ii). $\square$

4.3 Point stabilizers and their quasiconvexity

Assume alternative (ii) of Lemma 4.4. Over each vertex b of the ascending Bass–Serre tree $\mathcal{B}$ place a copy T_b of T . Orient $\mathcal{B}$ toward its distinguished end, so every vertex has one parent and any two upward rays eventually agree. The attaching map to the parent is the homothety f . Rescale the metric in each copy according to its height so that these maps are isometric embeddings. Their convex directed union is a real tree $\mathcal{T}$; it need not be complete. The torus acts on it by homotheties, whose scales are powers of λ determined by height.

Lemma 4.5. *The height-zero subgroup of J_ϕ has trivial stabilizers of nondegenerate arcs of $\mathcal{T}$. There are finitely many J_ϕ -orbits of points with nontrivial height-zero stabilizer. For each such point x , the full stabilizer A_x has a finite-index subgroup that is an ascending torus on a proper free factor of F . If $\lambda > 1$, the family of these point stabilizers is malnormal.*

Proof. For a height-zero element and a compact arc, choose a sufficiently high ancestor copy containing both endpoints and whose base group contains the element. A nontrivial element fixing that arc would fix an edge of the free splitting, which is impossible.

Suppose a nontrivial base element fixes x . Its own copy T_b is a complete simplicial tree, since its finitely many edge lengths have a positive minimum. It is thus a closed convex subtree of $\mathcal{T}$. If x were outside it, the element would fix the segment from x to its nearest point in T_b , contradicting the preceding paragraph. Hence $x \in T_b$ and is a vertex with nontrivial vertex stabilizer.

Let $C_{b,x}$ be the stabilizer of x in the base group at b , and consider the vertices b for which $C_{b,x} \neq 1$. They form a connected parent-closed subtree of $\mathcal{B}$: nontrivial elements remain nontrivial on passing to a parent, and upward rays coalesce. There are only finitely many

types of pairs (b, x) modulo J_ϕ , because the free splitting has finite quotient. This proves finiteness of point orbits. On an upward ray two pair types repeat, yielding an element $s \in A_x$ of positive height that takes one vertex b to an ancestor. Its positive translates are cofinal along that ray. Every height-zero element of A_x belongs to a sufficiently high base group and fixes x there. Consequently $\ker(h|_{A_x})$ is the increasing union of the corresponding translates of $C_{b,x}$. The subgroup with heights in $h(s)\mathbb{Z}$ is therefore the ascending torus based on $C_{b,x}$, with stable letter s and the appropriate orientation. It has finite index in A_x . The group $C_{b,x}$ is a proper free factor, so has smaller rank.

Finally suppose $\lambda > 1$. An element fixing two distinct points has scale one, hence height zero. It must then be trivial by the arc assertion. Thus distinct point stabilizers intersect trivially, including conjugate stabilizers of distinct points. This is precisely malnormality for a list of orbit representatives. $\square$

If $\lambda = 1$, irreducibility and Perron eigenvalue one imply that the transition matrix is a cyclic permutation matrix. The maps take edges to single edges. The union tree is simplicial and its torus quotient is finite. Its edge stabilizers inject into the height group and are therefore cyclic or trivial. Vertex stabilizers either inject into height or are those in Lemma 4.5. Cyclic edge groups are quasiconvex in the hyperbolic torus. The vertex groups are finitely generated and quasiconvex as well: replace excursions of an ambient path outside a vertex space by paths in its incident edge cosets, using their ambient undistortion. Induction applies to the smaller-rank ascending tori, and the quasiconvex hierarchy theorem [66] proves Theorem 4.1 in this case.

We henceforth assume $\lambda > 1$.

Lemma 4.6. *Each subgroup A_x in Lemma 4.5 is quasiconvex in J_ϕ . Consequently the finite malnormal family of these subgroups is a relatively hyperbolic peripheral structure. If the inductive hypothesis holds below rank $\text{rk } F$, every member is hyperbolic and virtually compact special.*

Proof. Realize the free splitting as a finite graph with a bouquet at each vertex carrying its vertex free factor; its other edges realize the edges of the splitting. Represent f by a map of this graph that sends bouquets into bouquets and the other edges along their prescribed tree paths, with bouquet words between successive edges. Such a representative can be constructed equivariantly on universal covers: choose the prescribed stabilizer map on each vertex fiber and join the finitely many representative edge endpoints along their tree paths. The finitely many bouquet legs in these edge images have a uniform length bound.

Use this representative to form the ordinary tree of graph spaces for the torus. Its one-skeleton X is a proper cocompact model for the torus word metric. In every row with $C_{b,x} \neq 1$, let W_b be the convex bouquet lift representing x . Let Y_x be the union of these lifts and the upward strips joining them. It is connected. Finiteness of the pair types in the proof of Lemma 4.5 shows that A_x acts cocompactly on Y_x .

Project each retained horizontal row to its convex subtree W_b . These projections are 1-Lipschitz within rows. They agree, up to a uniform error, across an upward step. Indeed take the last relative edge on the path toward the source bouquet. Its image is noncollapsed; alignment preservation prevents relative backtracking. The target gate can differ from the image of the source gate only by a terminal bouquet leg in the image of that one edge. This leg has the uniform bound just specified. Thus the projections are uniformly Lipschitz across all vertical edges between retained rows, for the intrinsic metric of Y_x .

It remains to treat a component of omitted rows. It attaches to the retained subtree of $\mathcal{B}$ by one step $d \rightarrow b$ with $C_{d,x} = 1$. The gates to W_b of the entire immediate attaching image have uniformly bounded diameter. To verify this, use the embedding of relative trees. If

that image misses x , its connected image has one gate. If it meets x at an interior point of a relative edge, only the image of that one edge can meet the bouquet, and its bouquet legs have bounded length. If it meets x at a vertex, that vertex has trivial stabilizer: otherwise its injected stabilizer would give $C_{d,x} \neq 1$. Such a vertex has uniformly finite valence, and only its point image and the finitely many bounded initial edge legs can meet W_b . In each case the actual attaching image is connected in the horizontal tree. Every gate from that image to a bouquet that it meets lies in their intersection, so the preceding bounds apply to the whole image, not merely its vertices near the attachment.

Map the entire omitted component to a point within bounded distance of these gates. The resulting map on the vertices of X is a coarse Lipschitz retraction to Y_x : horizontal edges, retained vertical edges, and the boundary of every omitted component have just been checked. Therefore inclusion of Y_x with its intrinsic path metric is a quasi-isometric embedding. Cocompactness of the A_x -action proves that A_x is undistorted, hence quasiconvex in the hyperbolic group J_ϕ .

A finite malnormal family of quasiconvex subgroups of a hyperbolic group is a relatively hyperbolic peripheral structure [11]. Each A_x is itself hyperbolic. Its finite-index ascending torus has smaller-rank base by Lemma 4.5; induction and finite-index invariance give the final assertion. $\square$

4.4 Deep fillings with train-track finite covers

Lemma 4.7. *Assume the inductive hypothesis below rank $\text{rk } F$ and use the peripheral family in Lemma 4.6. For every prescribed finite avoidance set there is a peripheral filling*

$$\bar{J} = J_\phi / \langle\langle N_x \rangle\rangle$$

that is hyperbolic, has finite-by-cyclic peripheral images, and has a finite-index subgroup equal to the mapping-torus group of an expanding train-track map $p : V \rightarrow V$ of a finite connected graph. After passing to a height power, the edge-transition components of p are disjoint primitive irreducible blocks with connected invariant supports.

Proof. Choose a finite-index normal subgroup $H \triangleleft J_\phi$ and put

$$N_x = H \cap \ker h \cap A_x.$$

Residual finiteness permits H to avoid any prescribed finite set of nontrivial elements, simultaneously for all the finitely many peripherals. The image of $\ker(h|_{A_x})$ in A_x/N_x is finite, whereas its height image is cyclic. Thus A_x/N_x is finite-by-cyclic. For sufficiently deep avoidance, the relative filling theorem [55] gives exact peripheral reduction and a relatively hyperbolic quotient with these peripheral images. As the peripheral images are hyperbolic, $\bar{J}$ is hyperbolic.

Let C_v be the vertex groups of the base free splitting. Define $\bar{F}$ by replacing them by $C_v/(C_v \cap H)$, keeping the underlying graph and trivial edge groups. Equivalently, quotient F by the normal closure of all its vertex kernels. The induced tree is denoted T^{fin} ; it is the Bass–Serre tree of this graph of finite groups. Every N_x is exhausted by the vertex kernels along its upward union description in Lemma 4.5. Consequently $\bar{J}$ has the presentation torus of the induced endomorphism of $\bar{F}$. This is a statement about presentations and does not assert that this endomorphism is injective. The relative tree map descends to an equivariant edge-path map of T^{fin} .

We impose one additional finite avoidance condition that makes every iterate of every edge reduced in T^{fin} . There are finitely many seed turns, namely the internal turns of the images of representative edges. All turns in later iterates are translates of derivatives of these

seed turns. Here the derivative sends an edge germ to the initial germ of its image. Before filling their two directions are distinct, because f is an embedding.

For each seed turn whose directions ever enter the same oriented edge orbit, choose the first such derivative. Its two directions differ by a nonidentity multiplier in their vertex stabilizer. Add that multiplier to the finite avoidance set for H . At subsequent derivatives the two directions stay in the same oriented edge orbit, and their multipliers are obtained from the first one by ϕ and changes of vertex frame. Since ϕ is conjugation in J_ϕ and H is normal there, none of these multipliers belongs to H . Directions in different oriented edge orbits cannot identify under vertex reduction; directions in the same orbit identify precisely when their multiplier is in the vertex kernel. Thus no derivative of a seed turn folds. This proves the assertion for all edge iterates at once.

Put $D = F/(F \cap H)$. Conjugation by the stable letter induces an automorphism σ of this finite group: its induced injection into itself is necessarily surjective. The natural map $\bar{F} \rightarrow D$ is injective on every finite vertex group. Its kernel acts freely on T^{fin} and has a finite connected graph quotient V . The descended map gives a train-track map $p : V \rightarrow V$, satisfying

$$p(dz) = \sigma(d)p(z) \quad (d \in D).$$

All edges expand, since their relative lengths are multiplied by $\lambda > 1$. With compatible basepoint and stable-letter markings, its mapping torus is a finite cover of the presentation torus for $\bar{F}$: the map on the finite fiber is the permutation σ . Equivalently it is the preimage of the stable cyclic subgroup under $\bar{J} \rightarrow D \rtimes_\sigma \mathbb{Z}$, and has index $|D|$. This argument remains valid without injectivity of the endomorphism of $\bar{F}$ or of p_* .

Finally consider reachability of unoriented edges in V . If α reaches β , irreducibility downstairs gives a path of transitions from β to $d\alpha$ for some $d \in D$. The composite gives an occurrence $\alpha \rightarrow d\alpha$ in an iterate of length m . Repeated substitution follows the permutation

$$D \longrightarrow D, \quad s \longmapsto \sigma^m(s)d.$$

Its orbit of the identity closes. The resulting closed chain of occurrences passes through β , so β reaches α . Every transition is therefore contained in a strongly connected component, with no arrows between distinct components. Take a power to split their periods, making all blocks primitive. A sufficiently long iterate of any one edge contains all edges of its block and contains no edges outside it; hence each block support is connected. It is invariant and has a periodic vertex. Taking a further power fixes all periodic vertices. This gives the claimed form. $\square$

4.5 Cubulating the filled groups

The next lemma treats the possible failure of fundamental-group injectivity in the preceding finite graph model.

Lemma 4.8. *Let $p : V_0 \rightarrow V_0$ be a based train-track map of a finite connected graph, fixing its base vertex. Suppose all edges expand, the transition matrix is primitive, and its mapping-torus group is hyperbolic. Then that group is virtually compact special, whether or not p_* is injective.*

Proof. The free groups $P_m = p_*^m(\pi_1 V_0)$ have nonincreasing ranks. Choose $m \geq 1$ after these ranks stabilize. The surjection $p_* : P_m \rightarrow P_{m+1}$ is between free groups of the same finite rank, so is injective by Hopficity. Thus its restriction to P_m is an injective endomorphism.

Let $\rho : \hat{V} \rightarrow V_0$ be the based cover corresponding to P_m . Lift p^m to $q : V_0 \rightarrow \hat{V}$ and lift p to $\hat{p} : \hat{V} \rightarrow \hat{V}$, with both maps fixing the chosen basepoint as appropriate. Uniqueness of

based lifts gives $\widehat{p}q = qp$. The image graph $W = q(V_0)$ is finite and connected, and carries exactly P_m : its inclusion into the cover is injective on fundamental groups, while the image of q_* already projects onto all of P_m . The restriction $p_W = \widehat{p}|_W$ is thus injective on fundamental groups. It is a train-track map and every edge expands, because projection takes its edge iterates to the corresponding reduced expanding iterates of p .

The maps $q : V_0 \rightarrow W$ and $\rho|_W : W \rightarrow V_0$ intertwine p and p_W . Their composites are p^m and p_W^m : the latter equality again follows from uniqueness of based lifts. Lemma 4.3 identifies their mapping-torus groups.

We check irreducibility of p_W , since mere restriction to a finite image graph does not by itself establish it. Given edges e', f' of W , choose edges e, f of V_0 whose q -images contain them. Primitivity supplies a fixed k such that the image under p^k of each edge contains f or its reverse. Consequently every path $qp^k(a)$, for an edge a of V_0 , contains $q(f)$ or its reverse. Their lengths have a common finite bound B . For $n \geq k$ the reduced path $qp^n(e)$ is a concatenation of these bounded chunks. Reduction causes no cancellation, since its projection is the train-track path $p^{n+m}(e)$. The path $p_W^n(e')$ is a subpath of $qp^n(e)$ and its length tends to infinity. Once its length exceeds $2B$, it contains an entire chunk and hence an occurrence of f' . Thus every edge reaches every other edge.

The map p_W satisfies exactly the hypotheses of [26, Theorem B]: it is a fundamental-group-injective train-track map of a finite graph, all edges expand, its transition matrix is irreducible, and its mapping-torus group is hyperbolic. That theorem gives a free cocompact cubulation. Agol's theorem [2] gives virtual compact specialness. $\square$

Lemma 4.9. *The fillings in Lemma 4.7 are virtually compact special.*

Proof. Use the finite-index graph torus supplied there, and pass to the height power for which all its blocks are primitive and its periodic vertices are fixed. At every vertex of V make a separate copy in each incident block, retain one central vertex, and join the central vertex to its block copies by connector intervals. Collapsing each connector star recovers V , so this operation is a homotopy equivalence. The map acts on each block as before, sends central vertices according to p , and sends each connector linearly to its corresponding connector.

The eventual image of this graph map is connected and contains all blocks, since each block map is primitive. Its remaining intervals and central vertices are periodic. Restriction to the eventual image does not change the mapping-torus group: inclusion and a sufficiently high power of the map satisfy Lemma 4.3. Connector orbits now give annuli joining the point tori and the block tori. This is a finite graph-of-groups decomposition over infinite cyclic groups. Each edge injection is detected by height, so is injective.

The total group is hyperbolic, as a finite-index subgroup of $\bar{J}$. Its cyclic edge groups are quasiconvex, and the finitely generated vertex groups are consequently quasiconvex. In particular every block torus is hyperbolic. Each block has a fixed vertex and satisfies the other hypotheses of Lemma 4.8, so is virtually compact special. The point tori are infinite cyclic. The quasiconvex hierarchy theorem [66] now gives virtual compact specialness of the total group, and therefore of its finite-index supergroup $\bar{J}$. $\square$

Proof of Theorem 4.1. The base cases and the automorphism case were disposed of at the start. Apply Lemma 4.4. Its rank-reduction alternative is settled by induction and finite-index invariance. In the tree-model alternative, the case $\lambda = 1$ was settled following Lemma 4.5. For $\lambda > 1$, Lemma 4.6 supplies a finite malnormal quasiconvex family of hyperbolic virtually compact special subgroups. Lemmas 4.7 and 4.9 produce arbitrarily deep fillings to hyperbolic virtually compact special groups, with their inherited finite-by-cyclic peripheral images. Finite-by-cyclic groups are virtually cyclic and hence virtually compact special. Finally Lemma 4.2 verifies the coherence, local-indicability, finiteness, dimension and field-acyclicity assumptions

of Theorem 3.1. That criterion applies and proves that J_ϕ is virtually compact special. The induction is complete. $\square$

Corollary 4.10. *Let F be a free group of finite rank and let $\phi : F \rightarrow F$ be injective. If J_ϕ is word-hyperbolic, it has a finite-index subgroup H fitting into an exact sequence*

$$1 \longrightarrow F_d \longrightarrow H \longrightarrow \mathbb{Z} \longrightarrow 1$$

for some finite $d \geq 0$, where F_d is the rank- d free group. Thus $H \cong F_d \rtimes_\alpha \mathbb{Z}$ for an automorphism α of F_d .

Proof. Theorem 4.1 makes J_ϕ virtually RFRS. Lemma 4.2 supplies a finite classifying complex whose chains are acyclic over the subgroup division field. Corollary 3.12 therefore gives two-sided ordinary Novikov contractions on a finite-index subgroup H . The restricted group-ring complex is still the cellular complex of a connected universal cover, so its ordinary H_0 is $\mathbb{Q}$. Its character is nonzero: a zero character would make the Novikov ring $\mathbb{Q}H$ and the contraction would contradict this H_0 .

Lemma 3.13 gives an epimorphism $H \rightarrow \mathbb{Z}$ with finitely generated kernel. Coherence and cohomological dimension at most two pass to H by Lemma 4.2. The kernel is therefore finitely presented, and Lemma 3.14 makes it free. Its rank is finite by finite generation. A lift of $1 \in \mathbb{Z}$ splits the sequence. This proof applies after every branch of the induction establishing Theorem 4.1. $\square$

The epimorphism in this corollary need not be the restriction of the canonical height homomorphism of the original ascending presentation.

5 Actual splittings and row homology

We now work with a torsion-free hyperbolic primitive extension group. The filling criterion of Section 3 and the ascending mapping-torus conclusion of Section 4 are available. An ascending mapping torus always allows an injective, possibly nonsurjective, endomorphism. Quasiconvexity of a subgroup of a row means quasiconvexity in that row unless the ambient group is specified. Our immediate goal is to control the stabilizers of paths in the actual splitting. The row homology will provide a common finite-dimensional space into which their first homology groups inject, and hence the bound used to construct the periodic peripheral groups. We need two outputs. The first is a simultaneous homology injection for path stabilizers, including after restriction to a subgroup. The second is a quasiconvexity criterion for finitely generated periodic subgroups whose height kernels have positive finite-dimensional homology. The first controls which periodic groups can coexist; the second will keep their later joins quasiconvex in the ambient group.

5.1 Primitive presentations and actual trees

Definition 5.1 (Primitive data). Fix $k \geq 1$, set $a_i = t^{-i}at^i$, and put

$$B_0 = \{a_1, \dots, a_{k-1}\}, \quad P_- = F(a_0, \dots, a_{k-1}), \quad P_+ = F(a_1, \dots, a_k).$$

An empty B_0 is allowed. For relatively prime positive integers p, q , write $\text{pr}_{p/q}(v, w)$ for the positive primitive word of slope p/q in two slots. The two presentations are

$$G = \langle a, t \mid R \rangle, \quad R = \text{pr}_{p/q}(x, y) \quad \text{or} \quad R = \text{pr}_{p/q}(xy, z), \quad (5.1)$$

where

$$x \in P_- \setminus F(B_0), \quad y \in P_+ \setminus F(B_0), \quad 1 \neq z \in F(B_0)$$

(the last choice is needed only in the second form). We assume that G is torsion-free and hyperbolic and that the word has Linton's primitive exceptional property [40, Sections 3.1 and 5]. Explicitly, for the first form the family $\{\langle x \rangle, \langle y \rangle\}$ is malnormal in the free group on $a_0, \dots, a_k$; if $p = q = 1$, the same word must not admit a factorization $x'y'$, with x', y' on the respective sides outside $F(B_0)$, whose cyclic family is nonmalnormal. In the second form $\langle z \rangle$ is malnormal; if p/q is an integer, the displayed word must not admit such a factorization $x'y'$. Inversions can be absorbed into the slots.

Let

$$h : G \longrightarrow \mathbb{Z}, \quad h(a) = 0, \quad h(t) = 1, \quad L = \ker h.$$

The orientation can be reversed without changing any argument. Write $H = \langle a_0, \dots, a_k \mid R \rangle$. The coarse Magnus splitting is the HNN extension with vertex group H , identifying P_- with P_+ by a shift of subscripts. Denote its Bass–Serre tree by T_0 .

The refinement below is the graph-of-groups decomposition of [40, Corollaries 3.10–3.11 and Section 5], with the edge embeddings kept explicit for later intersection arguments.

Lemma 5.2 (Refinement). *There is an actual graph-of-groups decomposition*

$$H = P_- *_I Q *_J P_+, \quad I = F(B_0, x), \quad J = F(B_0, y). \quad (5.2)$$

For nonzero integers d, n , its middle group is

$$\begin{aligned} Q &= F(B_0, u), \quad x = u^d, \quad y = u^n, \quad R = \text{pr}_{p/q}(x, y), \\ Q &= F(B_0, x) *_{\langle z \rangle = \langle u^d \rangle} \langle u \rangle, \quad y = x^{-1}u^n, \quad R = \text{pr}_{p/q}(xy, z). \end{aligned}$$

The second group has the same root-extension description with $F(B_0, y)$ as base. Refine the vertices of T_0 by this splitting and collapse the original shift edges, whose maps onto their P -endpoints are isomorphisms. Denote the resulting tree by T .

Proof. A subgroup generated by a free factor $F(B_0)$ and one element outside it is freely generated by that factor and the new element. Kurosh's theorem preserves $F(B_0)$ as a free factor; the rank increases by exactly one, and Hopficity identifies the displayed generating tuple with a free basis. Thus the edge maps from I, J into their P -sides are injective.

Amalgam normal form in $F(a_0, \dots, a_k) = P_- *_F(B_0) P_+$ shows that the two slots x, y , or xy, z , freely generate before imposing the relation. Kill their primitive relator by changing a basis of this abstract rank-two free group. Its quotient is infinite cyclic, with generator u , and the two slots have nonzero images. This yields exactly (5.2) and its displayed equations. The substitution $y = x^{-1}u^n$ gives the alternative free base in the second case.

Both extreme letters remain essential in the cyclically reduced relator. In the first form, nonzero powers of x and y stay outside the common free factor, which is root-closed, so alternating syllables retain their extreme letters. In the second form, expand the positive word on xy, z : the x, y factors alternate the two sides, separated where needed by nontrivial z -powers. Normal form again retains both extremes. With inverted slots, conjugating a nontrivial common-factor element by an extreme-letter syllable stays outside the common factor by malnormality, giving the same conclusion. The two alphabet omissions are therefore genuine Magnus omissions. The Freiheitssatz and HNN normal form justify the coarse splitting, and the injective edge maps justify its refinement and collapse. $\square$

A row is a vertex or an edge midpoint ξ of T , together with its stabilizer X_ξ . Here is an explicit height convention. Give the coarse row $t^i H$ height i . In its replacement put P_- , Q , and P_+ at heights $i + \frac{1}{2}$, i , and $i - \frac{1}{2}$, respectively. The shift edge joins its P_+ to the P_- in the replacement of $t^{i-1} H$, at the same height, so height descends after its collapse. Extend

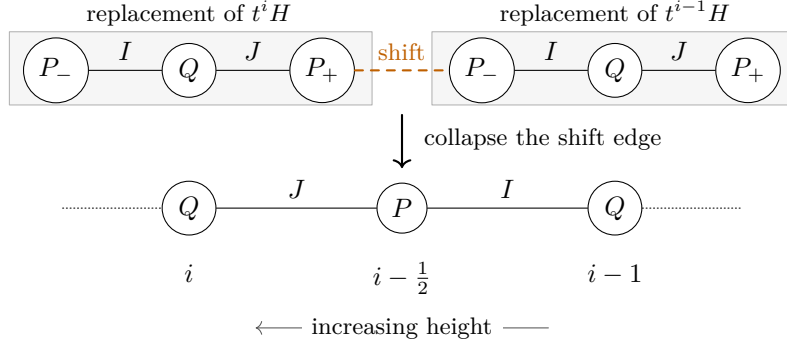


Figure 2: A local part of the actual row refinement. Each coarse row is replaced by $P_- *_I Q *_J P_+$. The original shift edge has isomorphisms at its two P -endpoints and can be collapsed; the I - and J -edges remain. The heights shown below agree with $h(t) = 1$. Each displayed group is in the actual conjugate frame of its coarse row. The picture shows a local path, not the whole Bass-Serre tree; the quotient by L is a line, while the actual tree can branch.

affinely across the remaining edges and equivariantly under G , with $h(gx) = h(x) + h(g)$. Thus the I - and J -edge midpoints have heights $i + \frac{1}{4}$ and $i - \frac{1}{4}$. A *level* consists of all rows of one type at one height and is a single L -orbit. A *slot* at a vertex specifies its upward or downward incident edge type. All actions preserve these orientations. We use the same terminology for T_0 when explicitly indicated. The quotient graph of groups $L \backslash T_0$ is the infinite line obtained by unfolding the coarse HNN splitting, and its fundamental group is L . The actual tree T_0 , and likewise T , can branch. Figure 2 shows one part of the refinement and the resulting height convention.

Lemma 5.3 (Row geometry and overlaps). *All coarse and refined rows are hyperbolic, and their incident maps are quasi-isometric embeddings. Refined rows are quasiconvex in their containing coarse row H . The coarse Magnus sides satisfy:*

- (i) *their ordinary intersection is $F(B_0)$, or $F(B_0) * \langle w \rangle$ with w outside the common factor on each side;*
- (ii) *a conjugate intersection can be noncyclic only in the aligned opposite-slot double coset; distinct same-slot cosets intersect at most cyclically;*
- (iii) *the overlap and its shifted copy have full-row-rank Fox matrices in their free sides. They are strongly inert there: for every finitely generated subgroup U of a free side P , with overlap $O \leq P$,*

$$\sum_{UgO \in U \backslash P/O} \max\{\text{rk}(U \cap gOg^{-1}) - 1, 0\} \leq \max\{\text{rk } U - 1, 0\}.$$

Proposition 5.7 proves the simultaneous subgroup-homology injection behind this inequality from the stated matrix property.

In a nonfree root row Q , z is not a proper power in the free base. Distinct cyclic incidences at a free-base vertex of its root splitting consequently have trivial intersection.

Proof. The one-relator group H embeds in G and contains no Baumslag-Solitar subgroup, so it is hyperbolic by [52, Theorem 1.1]. Its Magnus sides are quasiconvex [40, Theorem 4.7]. Collins's ordinary-intersection theorem gives (i) [15, Theorem 2], and his conjugate-intersection theorem gives (ii), including the case of equal Magnus subgroups [16, Theorem 2]; see also [52, Section 8.3]. The common alphabet is a connected Magnus subgraph.

For (iii), the Fox matrix keeps the identity columns for B_0 . If w occurs, its derivative in the missing extreme letter is nonzero: in a reduced word in a free group the relevant prefix vertices are distinct, so those signed terms cannot cancel. The matrix is triangular with a nonzero final pivot. This supplies the matrix input for the induced-module restriction argument below; the simultaneous rank bounds will follow from that argument.

Each refined edge is finitely generated in a quasiconvex free P -group, and hence is quasiconvex in H . In a finite splitting over ambient quasiconvex edge groups, the vertex groups are also undistorted: replace successive excursions of a geodesic outside a chosen vertex space by paths in the attaching edge cosets. Ambient edge undistortion bounds their lengths linearly. This proves quasiconvexity of the refined vertex groups in H .

If $|d| = 1$, the root row is free. Otherwise, an equality $z = v^m$ with $|m| > 1$ would, by amalgam normal form, embed $\langle v, u \mid v^m = u^d \rangle$ into Q . Its central common power, together with a lift of an infinite-order element of $C_{|m|} * C_{|d|}$, generates $\mathbb{Z}^2$. This contradicts hyperbolicity. Thus $\langle z \rangle$ is maximal cyclic and malnormal in the free base. $\square$

5.2 Coefficients and level homology

Notation 5.4. For $U \leq G$, let D_U be the rational Linnell division field inside the affiliated-operator algebra $\mathcal{U}(U)$. Local indicability and the strong Atiyah theorem give these compatible Hughes-free fields [13, 30, 35]. We use natural subgroup embeddings, independence of distinct cosets over subgroup fields, and rational skew-Laurent extension over the kernel of a cyclic quotient [33, Section 2.3 and Proposition 2.2]. Coset independence also holds for affiliated coefficients: clear finitely many coefficients to bounded subgroup operators and use their disjoint Fourier supports.

For $V \leq U$, $H_*(V; D_U)$ uses right group multiplication on coefficients and left vector spaces; equivalently compute over D_V and extend scalars. Put

$$\beta(V) = \dim_{D_V} H_1(V; D_V).$$

This dimension is unchanged by overfield extension. For a nontrivial free group of finite rank r , it is $r - 1$; for an infinitely generated free group it is infinite; for the trivial group it is zero.

Coefficients precede chain vectors and boundary matrices act on the right. Write $d(v)$ for the Fox one-chain of a word v , so that

$$d(vw) = d(v) + v d(w), \quad \partial d(v) = v - 1, \quad \Delta(v) = (v - 1)^{-1} d(v) \quad (v \neq 1).$$

Thus $\partial \Delta(v) = 1$, and $\Delta(v) - \Delta(w)$ is a cycle. Intrinsic expressions are taken modulo two-boundaries. A subgroup *detects* a noncyclic subgroup in ambient homology if some such difference from the latter subgroup has nonzero class.

Changing an orbit representative by s conjugates the group symbols by s and right-multiplies their coefficients by s^{-1} . Conjugation of symbols and coefficients acts on ambient homology by left multiplication by s , so this convention preserves ambient classes. Differentiating svs^{-1} gives the same verification: connecting-path terms cancel for a cycle.

Every noncyclic subgroup of a free group detects noncommensurable elements. Indeed the support realization of Section 6 places the normalized Fox chains near their respective cyclic axes. Those axes have bounded overlap when they are noncommensurable. Equality would give a finitely supported chain with boundary the mass at the identity, contrary to the zero total augmentation of a finite boundary.

The height-kernel presentation is

$$L = \langle a_i \ (i \in \mathbb{Z}) \mid R_i \ (i \in \mathbb{Z}) \rangle, \tag{5.3}$$

where R_i is the shifted relator on $a_i, \dots, a_{i+k}$. It comes from the infinite cyclic cover of the aspherical one-relator complex [43], after collapsing its stable-letter tree.

Lemma 5.5 (Level basis). *Over D_L , the two-boundary in (5.3) is injective. Every interval of k consecutive a -columns is a basis of $C_1/\text{im } \partial_2$. The corresponding quotient for any refined row, after extension to D_L , maps isomorphically to the global quotient. The incident edge-to-row maps are isomorphisms on these quotients, and*

$$\dim_{D_L} H_1(L; D_L) = k - 1.$$

Proof. Adjoin u_i at each index. In the first form replace R_i by $x_i = u_i^d$, $y_i = u_i^n$; in the second form use $u_i^d = z_i$, $x_i y_i = u_i^n$. These are the elementary expansions and substitutions supplied by the primitive change of basis. For $m \neq 0$, the u_i -derivative of u_i^m is nonzero and hence is a unit over D_L . Eliminate the u_i -column. The resulting relation has nonzero entries at both extreme a -columns, from the missing-letter derivatives of x_i, y_i in their free sides.

In any finite dependence between shifted two-boundaries, an outermost nonzero coefficient has an extreme entry which cannot cancel. Hence the two-boundary is injective. Conversely either extreme column can be eliminated using that shifted relation. Repeated elimination moves any finite chain into a prescribed k -letter interval. A dependence within that interval would trim from both ends of a finite filling and force every relator coefficient to vanish. This proves the basis assertion.

For P -rows this is the alphabet basis. The free tuples (B_0, x) and (B_0, y) give triangular changes of basis with nonzero extreme pivots. In a root row, first quotient by its root relation using the unit u -entry, and then make this triangular change. The first form uses the two power pivots instead. This proves the other row assertions. Conjugate rows add connecting Fox paths, whose coefficients cancel on cycles.

The boundary of the resulting k -dimensional quotient is onto D_L , because $v - 1$ is a unit for $1 \neq v \in L$. Its kernel has dimension $k - 1$. $\square$

Proposition 5.6 (Restriction to row orbits). *For $U \leq L$ and an actual level $\mathcal{L}$, inclusion and the above coefficient transport give an injection*

$$\bigoplus_{\xi \in U \setminus \mathcal{L}} H_1(U \cap X_\xi; D_U) \longrightarrow H_1(U; D_U). \quad (5.4)$$

For U in a vertex row, the analogous sum over one incident slot injects. The same assertions hold for coarse rows. The assertions hold in the induced chain modules before passing to an overfield.

Proof. Use

$$E_U = D_U \otimes_{\mathbb{Q}U} \mathbb{Q}L,$$

embedded in D_L by coset independence. Its restriction to a row, the double-coset decomposition, and Shapiro's lemma give precisely the summands of (5.4) [14, 59]. Thus we must check that a row cycle bounding over E_U already bounds in the row over the same module. For a free row, this is the column-independence argument.

For a root row keep the added columns u_j , so that no division inside E_U is needed. A Q -chain at index i , with free base (B_0, x) , uses its common alphabet, the word x_i , and u_i , but not the other extreme letter. Trim a finite filling from its outermost index $j \neq i$. There the relation $x_j y_j = u_j^n$ has a nonzero extreme-letter entry not otherwise used at that end. Its coefficient is zero in D_L , and therefore in E_U . The u_j -column next kills the coefficient of $u_j^d = z_j$. Repeat from both ends. At the remaining index i , the unused extreme letter kills the $x_i y_i = u_i^n$ coefficient. What remains is exactly the row's root relation, with its original

coefficient in E_U . Substitution for x_i is injective by its nonzero missing-letter entry. Hence it is already a row filling. The first form and the coarse H -row use the same paired-column or extreme-column trimming.

For a single slot replace L by the vertex group in this argument. At a free vertex the attaching free tuple has full Fox rank; at a root vertex the root column gives the extra pivot. The overfield is used only to infer that a coefficient multiplying a nonzero entry vanishes, never to divide within E_U . This also proves the assertion about induced modules. $\square$

Proposition 5.7 (Path inertia). *In T and T_0 , every path with noncyclic pointwise stabilizer is monotone. For each fixed monotone shape in a fixed absolute level range, the first homologies of its stabilizers, indexed modulo L , inject jointly into $H_1(L; D_L)$. A path containing an edge has finitely generated free stabilizer, and a fixed finite shape has finitely many G -orbits with nontrivial stabilizer, or finitely many L -orbits when its absolute level range is fixed. In particular*

$$\sum_{\gamma \bmod L} \max\{\text{rk Stab}(\gamma) - 1, 0\} \leq k - 1. \quad (5.5)$$

The same joint-injection statement holds for continuations of any fixed additional shape inside any subgroup U of a current path stabilizer, with the intersection summands indexed by U -orbits. It holds after restriction of the actual induced matrices and double-coset decompositions, not just as a numerical inequality.

Proof. Apply the slot rule with U an incident edge group. The identity edge orbit already accounts for all of $H_1(U; D_U)$. Every different same-slot intersection has zero first field homology, so, being free, is trivial or cyclic. A reversal of a reduced path passes through two such edges. This proves monotonicity in T . The coarse argument is the same, or follows from the equal-Magnus-subgroup intersection theorem. Inserting collapsed shift edges at successive P -passages takes a long monotone refined path to a long monotone coarse path.

Begin at a free side and extend a monotone path one step in either direction. Only the aligned opposite-slot double coset can have a noncyclic intersection, by Lemma 5.3(ii). Its overlap $O \leq P$ has a full-row-rank Fox matrix. For arbitrary $U \leq P$, perform this calculation over $D_U \otimes_{\mathbb{Q}U} \mathbb{Q}P$. Injectivity on cycles, followed by its double-coset decomposition, gives the simultaneous injection for all $U \cap sOs^{-1}$. Other continuation double cosets are cyclic and have zero first homology. Repeat for each current stabilizer at the next step. This proves both the global injection and the asserted arbitrary-subgroup version. The intervening refined steps use Proposition 5.6.

This keeps the subgroup matrices throughout: restriction to a smaller subgroup merely decomposes the same induced modules again. If an independent finite family of the resulting cycles acquired an ambient relation, the trimming proof of Proposition 5.6 would reduce it to a relation at the preceding step. This supplies the restriction property used later for different double-coset orbit summands.

At each step, intersect quasiconvex subgroups in a hyperbolic row. Their intersections are finitely generated and quasiconvex, and only finitely many double cosets have nontrivial intersection. Iterating over a fixed shape gives the claimed finite lists. Each path group lies in a free edge group and is free. The joint injection and Lemma 5.5 prove the budget. $\square$

We will use more than the numerical budget. Inside a later path group, a predecessor already contributing its full first homology excludes a second noncyclic predecessor orbit. The induced-module statement also retains the distinction between different double cosets after restriction to a subgroup. These are the two forms of path inertia used in the trajectory construction of Section 9. If $k = 1$, the budget is zero and no path containing an edge has a noncyclic stabilizer; the next lemma then gives the acylindrical case.

Lemma 5.8 (Cyclic-path dictionary). *Fix a finite edge-containing path shape. In a reduced tree path fixed pointwise by a nontrivial element, the number of pairwise disjoint subsegments of that shape having cyclic stabilizer is bounded. Monotonicity is not required.*

Proof. The fixed shape has finitely many nontrivial stabilizer G -orbits; record traversal directions as well. Too many such subsegments give an element s taking an earlier one to a later one with direction preserved. Both cyclic stabilizers contain the common nontrivial element c . Thus s commensurates $\langle c \rangle$. The commensurator is cyclic in a torsion-free hyperbolic group. It is elliptic in this tree, since it contains a nonzero power fixing the first segment. But taking an oriented edge forward to one pointing the same way on a geodesic is tree-hyperbolic: the intervening segment and its successive translates concatenate without backtracking into an axis. This contradiction gives the bound. $\square$

5.3 Subtree gates and finite homology in a row

The path-homology estimate is now available. We turn to the second output of this section: recognizing when a periodic subgroup is quasiconvex in G . First we give a geometric criterion for a subgroup carried by a finite-quotient subtree. Finite row homology will then supply its local hypotheses, and the period rule will supply the subtree.

Lemma 5.9 (Subtree gate criterion). *Let a torsion-free hyperbolic group J act cocompactly on a tree of hyperbolic Cayley spaces with uniformly quasi-isometrically embedded edge maps and hyperbolic total space. We use the standard Cayley realization with uniformly proper vertex inclusions and isometric product strips over open edges. For the finite splittings here, properness is uniform: normalize a vertex to the identity and take the maximum intrinsic length of the finitely many elements in each ambient word ball, over the finitely many vertex types. Let $A \leq J$ preserve a connected subtree Y , with $A \backslash Y$ finite. Suppose its vertex and edge stabilizers are finitely generated and quasiconvex in the respective rows. Either of the following suffices for A to be quasiconvex in J :*

- (i) *ambient vertex rows are quasiconvex in J ;*
- (ii) *every edge leaving Y intersects A trivially or cyclically.*

This applies in particular to G acting on T or T_0 .

Proof. At every included vertex v , take a connected quasiconvex neighborhood of an orbit of $A_v = A \cap J_v$. Connect these neighborhoods across included edges by strips over edge-intersection orbits. The attachment offsets have finitely many types modulo A . The resulting connected path space W is proper and A -cobounded, and is quasi-isometric to A . We construct a coarse Lipschitz retraction from the total space to W . Here an intersection orbit means the orbit of the intersection of the two stabilizers through a chosen attachment point. The literal cosets need not meet: fixed bounded connecting paths are included in the attachment data. All statements about gates and intersections below allow these bounded offsets.

Inside an included vertex take coarse nearest-point projection to its orbit neighborhood. These projections agree across an included edge through their gates to its exact intersection subgroup. Indeed a geodesic in the attaching hull to a point of the intersection passes close to the vertex gate. It stays near the attaching hull, placing that gate in a bounded-neighborhood overlap. For two fixed subgroup cosets such an overlap is within bounded distance of their intersection orbit: properness leaves finitely many possible discrepancies of a close pair, and pairs with one fixed discrepancy form an orbit of the intersection subgroup. First-entry gates in the edge thus coarsely agree with both vertex gates. Quasi-isometric edge maps preserve

this description by quasigeodesic stability. There are finitely many included attachment types, so the bounds are uniform and are measured in the intrinsic path metric of W .

Each omitted tree component attaches across one edge. In case (i) the vertex subgroup at that edge is quasiconvex in J . Ambient projection to its orbit extends the gate throughout the omitted component, agreeing at the boundary up to bounded error. Its target metric is also intrinsic to the subgroup orbit. This proves case (i).

For case (ii), consider an omitted attaching hull in the included boundary row. Its projection to the vertex-subgroup neighborhood has bounded diameter, or is within bounded distance of an orbit of their cyclic intersection. The alternatives are uniform. To see this, write the vertex subgroup as M and an attaching coset as gE , where E belongs to the finite list of incident edge types. A projection of sufficiently large diameter forces a close pair $p \in M$, $q \in gE$, by thin quadrilaterals. Translate by $p^{-1} \in M$. The pair becomes $(1, s)$, with s in a fixed finite intrinsic word ball, and the attaching coset becomes sE . There are therefore only finitely many such configurations modulo M . In each one, pairs at a fixed bounded discrepancy form an orbit of $M \cap sEs^{-1}$; the finitely many discrepancies give the bounded connecting offsets described above. Hence the projection lies uniformly near the intersection orbit. Trivial intersections give bounded projections, and the unbounded cases have finitely many cyclic types.

A bounded projection extends by a constant. In an unbounded case choose a generator c of the cyclic intersection and its periodic axis coordinate. Let f be the gate coordinate on the attaching hull. Projection first to the quasiconvex M -orbit and then to this axis agrees, up to a uniform error, with direct intrinsic projection to the axis: the first projection lies near the cyclic intersection orbit, and thin triangles compare the two gates. The boundary row itself may be distorted in J , so its intrinsic Lipschitz estimate does not suffice. We claim in the ambient metric that

$$|f(x) - f(y)| \leq C d_J(x, y) + C. \quad (5.6)$$

The inclusion of the boundary vertex row has a Cannon–Thurston extension [50, Corollary 3.11], since the splitting has finite quotient, hyperbolic vertex and edge groups, and quasi-isometrically embedded edge maps. Its periodic-axis endpoints map to the two distinct endpoints of c in ∂J . For each normalized cyclic configuration there exist M_0, R_0 such that, whenever the intrinsic gate coordinates of x, y lie on opposite sides of the origin with margins at least M_0 , an ambient geodesic between them meets $B_J(1, R_0)$. Otherwise choose pairs with both margins at least n and ambient geodesics missing $B_J(1, n)$. Thin triangles make the two sequences converge to the opposite intrinsic axis endpoints: their Gromov products with those endpoints tend to infinity even when their off-axis legs do also. Cannon–Thurston continuity gives the two distinct ambient endpoints of c as their limits. But avoidance of $B_J(1, n)$ makes their mutual ambient Gromov products tend to infinity, forcing the same limit, a contradiction. Taking maxima over the finite list makes M_0, R_0 uniform.

Consequently an ambient geodesic from x to y passes uniformly near every intervening c^j , outside fixed endpoint margins. Indeed recenter by c^{-j} and apply the preceding assertion. The constants are independent of j by cyclic translation. The cyclic orbit is undistorted in J . Choose a uniformly spaced subcollection of those translates with disjoint, positively separated fixed-radius balls. A geodesic meeting all those balls has length at least a constant times their number. This proves (5.6).

On the discrete attaching hull enlarge the constant so that f is Lipschitz (or replace it within bounded distance by a Lipschitz function). Extend it over the outside component by the metric inf-envelope

$$\tilde{f}(z) = \inf_x \{f(x) + C d_J(x, z)\}.$$

Round its values to integers and send them to the same periodic orbit. This gives a coarse Lipschitz extension in the intrinsic target metric, agreeing with the attaching gate up to bounded error. Only the cyclic coordinate has been estimated in the ambient metric; undistortion of the boundary row was not assumed.

Combine these extensions with the compatible included-vertex gates. Across every ordinary or strip edge the image distance in W is uniformly bounded, giving a global coarse Lipschitz retraction. Since W is quasi-isometric to A , the retraction proves ambient undistortion, and hence quasiconvexity in J . $\square$

Lemma 5.10 (Finite homology in a refined row). *Every subgroup M of a refined row with $\beta(M) < \infty$ is finitely generated and quasiconvex in that row. If $\beta(M) = 0$, then M is trivial or cyclic.*

Proof. For a free row this follows from the free-group rank formula. Consider a nonfree root row

$$Q = F(B_0, x) *_{\langle z \rangle = \langle u^d \rangle} \langle u \rangle$$

and restrict its cyclic splitting to M . All edge first homologies vanish. The tree Mayer–Vietoris sequence injects the sum of vertex first homologies into $H_1(M; D_M)$. Thus all free-base vertex stabilizers are finitely generated, and only finitely many orbits of them are noncyclic.

At a cyclic free-base stabilizer, at most one incident edge has nontrivial intersection, and its maximal-cyclic property implies that this whole vertex stabilizer fixes that edge. At a noncyclic free-base stabilizer, quasiconvexity in the free base gives finitely many orbits of nontrivial edge intersections. Root-side vertices have valence $|d|$. Hence each component of nontrivially stabilized edges has finite quotient and finitely generated stabilizer: a path between root-side vertices must pass through a noncyclic free-base vertex, while cyclic free-base vertices are leaves. There are only finitely many noncyclic vertex orbits and finitely many relevant incidences at each.

Collapse these components and split over the remaining trivial edges. Bass–Serre normal form expresses M as a free product of their nontrivial stabilizers, remaining nontrivial vertex stabilizers, and a free group. The case $M = 1$ is immediate. For $M \neq 1$, with s nontrivial factors and free rank r , the field homology dimension is

$$\sum_{j=1}^s \beta(M_j) + s + r - 1.$$

Infinitely many factors or free generators give infinite dimension, as follows either by finite subproducts or the degree-zero part of the tree sequence. Finite $\beta(M)$ therefore gives a finite graph-of-groups core and finite generation.

If $\beta(M) = 0$, no free-base stabilizer is noncyclic. Each nontrivial-edge component is then a finite star whose cyclic leaves collapse into its root-side cyclic group. The free-product formula leaves only the trivial or cyclic case.

The free-base and cyclic vertices of the root splitting are quasiconvex in Q : the cyclic edge is ambient quasiconvex, and the excursion argument of Lemma 5.3 applies. The vertex groups of the finite M -core are quasiconvex there, as are the edge groups. Lemma 5.9(i) proves quasiconvexity of M in Q . $\square$

5.4 Periodicity and the homology budget

Proposition 5.11 (Period rule). *Suppose $A \leq G$, $h(A) \neq 0$, and*

$$S = A \cap L, \quad 0 < \beta(S) < \infty.$$

At every actual level $\mathcal{L}$, inclusion induces an isomorphism

$$\bigoplus_{\xi \in S \setminus \mathcal{L}} H_1(S \cap X_\xi; D_S) \longrightarrow H_1(S; D_S).$$

Both incident edge-level maps to a vertex level are isomorphisms. Every level sum therefore has dimension $q_X = \beta(S)$. The convex hull Y of the rows with noncyclic $S \cap X_\xi$ contains no edge with trivial S -stabilizer. If A is also finitely generated, then $A \setminus Y$ is finite and A is quasiconvex in G .

Proof. Use D_S -coefficients. Let V_j be the summed vertex homology at the j -th vertex level and E_j the sum at the intervening edge level. Proposition 5.6 makes these finite dimensional and both maps $E_j \rightarrow V_j, V_{j+1}$ injective. The cokernel C of the edge-to-vertex first-homology map in the tree sequence injects into $H_1(S; D_S)$.

For an interval $[a, b]$ set

$$C[a, b] = \left(\bigoplus_{j=a}^b V_j \right) / \text{im} \left(\bigoplus_{j=a}^{b-1} E_j \right).$$

It injects into C . Indeed any exterior finite-support relation trims at its outermost edge, since the map to that outer vertex is injective, leaving only the relations already imposed. The same trimming makes the internal edge map injective. Thus

$$\dim C[a, b] = \dim V_a + \sum_{j=a}^{b-1} (\dim V_{j+1} - \dim E_j).$$

Each summand is nonnegative. A positive-height element of A periodically identifies these data, semilinearly over D_S . A positive summand would recur indefinitely and contradict $\dim C \leq \beta(S) < \infty$. Expanding intervals in the other direction proves surjectivity onto V_j as well. All adjacent maps are isomorphisms and each level maps isomorphically to C .

The remaining part of the tree sequence is

$$K_0 = \ker \left(\bigoplus_{\text{edge orbits}} H_0(S_e; D_S) \longrightarrow \bigoplus_{\text{vertex orbits}} H_0(S_v; D_S) \right).$$

It is a quotient of $H_1(S; D_S)$ and hence finite dimensional. A nonzero vector in K_0 has finite level support. Sufficiently spaced height translates would give linearly independent vectors with disjoint supports. Thus $K_0 = 0$, and $H_1(S; D_S) = C$.

Take the actual path between rows with noncyclic stabilizer. Its edge chain with coefficient 1 has boundary supported at those endpoints, whose H_0 's vanish. If the path contained a trivially stabilized edge, its orbit coefficient would be a nonzero signed sum of distinct group monomials: distinct actual edges with trivial stabilizer have distinct transporters, and a tree path repeats no edge. It would therefore yield a nonzero vector in K_0 , a contradiction. This proves the hull assertion.

By Lemma 5.10, all row intersections are finitely generated and quasiconvex; every noncyclic one contributes at least one dimension. The level bound and height periodicity give finitely many A -orbits of such rows. When A is finitely generated, join a base vertex to its translates under a finite generating set and to representatives of these finitely many row orbits. All A -translates of these paths form an invariant tree of finite quotient containing Y . Hence $A \setminus Y$ is finite.

At a boundary edge, the A -stabilizer equals the S -stabilizer, since row groups have height zero. A noncyclic edge intersection would put both endpoints in Y , so every leaving intersection is at most cyclic. Apply Lemma 5.9(ii). The finite-generation assumption has been used only for this last cocompactness and quasiconvexity conclusion, not for the preceding homology or nontrivial-edge assertions. $\square$

6 Support and descent for division-ring coefficients

The closure construction requires two forms of algebraic coefficient control. Disjoint subgroup support patterns must give independent expressions. We also need descent from intersecting subgroup-field spans when the subgroup intersection is separable in both groups. Theorem 6.1 supplies these assertions. Section 7 supplies the separate localization argument for coefficients subject to height translation.

Throughout this section, J is a locally indicable group and D_J is its Hughes-free division ring over $\mathbb{Q}$. For $A \leq J$, write D_A for the division subring generated by $\mathbb{Q}A$. We use the subgroup inheritance and crossed-product properties of these fields: if $H \leq J$ is finitely generated and $\chi: H \rightarrow \mathbb{Z}$ has kernel K , then, choosing $t \in H$ with $\chi(t) = 1$,

$$D_H = D_K(t; \text{Ad } t) \hookrightarrow D_K((t; \text{Ad } t)). \quad (6.1)$$

The left side is the Ore division ring of skew Laurent polynomials, and the right side consists of series bounded below in degree. Distinct subgroup cosets are independent over the corresponding subgroup field. The normal-subgroup crossed-product assertion is [33, Section 2.3 and Proposition 2.2]; the compatible rational affiliated realization is [35, Corollaries 6.2–6.3], extending the Hughes-free framework of [32].

For arbitrary, possibly nonnormal A , coset independence follows directly in that realization. If $\sum_i d_i g_i = 0$ with $d_i \in \mathcal{U}(A)$ and distinct right cosets Ag_i , put $b = (1 + \sum_i d_i d_i^*)^{-1/2}$. This is injective and all bd_i are bounded. The bounded terms $bd_i g_i$ have disjoint Fourier supports, so each is zero and hence each $d_i = 0$. Left cosets use a common right regularizer. This argument does not attribute nonnormal coset independence to the normal-subgroup proposition.

Our arguments concern algebraic rational expressions and do not assign Fourier coefficients to an arbitrary affiliated operator.

6.1 Permitted support patterns

A *support pattern* is a subset of J of the form

$$\mathcal{P} = A_1 g_1 A_2 g_2 \cdots A_m g_m, \quad A_i \leq J, \quad g_i \in J.$$

A product $d_1 g_1 \cdots d_m g_m$, where $d_i \in D_{A_i}$, is *permitted on* $\mathcal{P}$. For $S \subseteq J$, let $\mathcal{V}(S)$ be the $\mathbb{Q}$ -span of all such products whose support pattern is contained in S . These are permitted algebraic supports; no choice of a scalar-function realization is involved in the definition. In particular, $\mathcal{V}(S)g = \mathcal{V}(Sg)$.

Theorem 6.1 (Support and coset descent). *The following assertions hold.*

(i) *If $S \cap T = \emptyset$, then $\mathcal{V}(S) \cap \mathcal{V}(T) = 0$.*

(ii) *There is a $\mathbb{Q}$ -linear map*

$$\mathcal{F}: D_J \longrightarrow \mathbb{Q}^J$$

with $\mathcal{F}(1) = \delta_1$, commuting with every right group translation, such that $v \in \mathcal{V}(S)$ implies $\text{supp } \mathcal{F}(v) \subseteq S$.

(iii) *Let $A, B \leq J$ and $C = A \cap B$. If $At \cap Bs$ is empty, then $D_{At} \cap D_{Bs} = 0$. Otherwise, for any $x \in At \cap Bs$,*

$$D_{At} \cap D_{Bs} = D_C x. \quad (6.2)$$

If, in addition, C is separable in both A and B , then

$$\left(\sum_{g \in J} D_A g \right) \cap \left(\sum_{g \in J} D_B g \right) = \sum_{g \in J} D_C g, \quad (6.3)$$

where each sum denotes its algebraic, finite-support span.

Applied entrywise to a finite row v , the map in (ii) satisfies

$$\mathcal{F}(vR) = \mathcal{F}(v)R \quad \text{for every compatible finite matrix } R \text{ over } \mathbb{Q}J.$$

Indeed each matrix entry acts by a finite sum of right group translations. Thus the same scalar-function map preserves the Fox boundary and divergence equations used in the closure construction. This consequence uses linearity and right equivariance; it requires no multiplicativity for division-ring coefficients.

We first make the complexity argument precise. The two descent arguments will use the same coefficient extraction, but have different conclusions: disjoint supports give vanishing, while intersecting single cosets give a smaller field.

6.2 A hereditary complexity and coefficient extraction

For predecessors of coefficient descent by rational-expression complexity, see Dicks–Herbera–Sánchez [18, Sections 3–6, especially Theorem 6.3] and Jaikin-Zapirain–López-Álvarez [35, Proposition 5.1]. Here we combine an explicit ordinal complexity with bookkeeping for the prescribed subgroup support patterns.

The difficulty in an induction is inversion. A coefficient of an inverse series is a finite sum of products, but neither the lengths of those products nor the number of their displayed terms has a uniform bound. We therefore use two ordinal layers: the first makes simplifying an inverse operand outweigh any finite increase in product length, and the second makes simplifying a term outweigh any finite increase in the number of terms. We define them on the displayed expressions, retaining terms even when their values cancel.

A displayed rational expression can be normalized recursively as a finite sum of terms

$$qP_1^{-1} \cdots P_s^{-1}g, \quad q \in \mathbb{Q} \setminus \{0\}, \quad g \in J. \quad (6.4)$$

Each P_i is a displayed sum in the same form, has nonzero value, and has at least one top term with final offset 1. To obtain this normalization before taking an inverse, choose a displayed top offset g , factor it to the right, and move the resulting monomial through other inverse factors by conjugation. For example, $gP^{-1} = (gPg^{-1})^{-1}g$. These operations preserve the number and nesting of inverse occurrences. They do not identify equal displayed terms or cancel terms against one another.

The *hereditary offsets* of an expression are all final group elements appearing at any depth of its displayed expression tree. For an inverse occurrence P^{-1} , let $H(P)$ be the subgroup generated by the hereditary offsets of P . Assign the term (6.4) the pattern

$$H(P_1) \cdots H(P_s)g. \quad (6.5)$$

When the original factors have prescribed subgroup-field patterns, normalize them separately before multiplying. Every resulting term pattern is then contained in its original prescribed pattern. Indeed all offsets used to represent an element of D_A can be chosen in A ; moving a monomial through a later factor conjugates that factor's subgroup and moves the same monomial to the right.

Write $\oplus$ for natural, or Hessenberg, addition of ordinals: add the coefficients of equal powers in Cantor normal form. Define the *path weight* and *complexity* recursively by

$$w(qP_1^{-1} \cdots P_s^{-1}g) = \bigoplus_{i=1}^s \omega^{c(P_i)}, \quad (6.6)$$

$$c(E) = \bigoplus_{T \text{ a displayed term of } E} \omega^{w(T)}. \quad (6.7)$$

An empty sum has complexity zero. Every complexity here is a finite hereditary ordinal below ε_0 . A monomial has weight zero; a displayed sum of r monomials has complexity r . Conjugating an expression, changing a final monomial offset, or performing the preceding normalization preserves these complexities. When an equality is under consideration, its joint complexity counts all displayed terms on both sides, including terms that cancel upon evaluation.

We shall use two elementary properties of this measure. Replacing an occurrence of ω^α in a path weight by finitely many ω^{β_i} , with every $\beta_i < \alpha$, strictly decreases that path weight. Replacing a displayed term of weight η by finitely many terms of weights less than η strictly decreases the outer complexity. Both statements follow directly from Cantor normal form; unaffected summands may be retained by strict monotonicity of natural addition. There is no bound on the finite number of replacement terms. Also, the complexity of any inverse operand in a displayed expression is strictly smaller than the complexity of the expression containing it.

Relative to a character χ , call an inverse occurrence P^{-1} , or its normalized operand P , *nonhomogeneous* when χ is nonzero on some hereditary offset of P , equivalently on $H(P)$. This refers to the displayed expression: for example, $1 + t - t$ is nonhomogeneous when $\chi(t) \neq 0$, although its value is constant. When all hereditary offsets are killed by χ , the operand can instead be retained unchanged in degree zero.

Lemma 6.2 (Coefficient extraction invariant). *Let finitely many normalized rational expressions have hereditary offsets in a finitely generated subgroup H , and let $\chi: H \rightarrow \mathbb{Z}$ be a character with kernel K . Expand in the positive skew Laurent field in (6.1), writing powers of t on the right. Every coefficient has a finite normalized rational expression in D_K , with the following properties.*

- (a) *If a summand is permitted on S , the expression for its coefficient of degree n has term patterns contained in*

$$(S \cap \chi^{-1}(n))t^{-n}.$$

Here S can be replaced by its intersection with H .

- (b) *Coefficient extraction does not increase the complexity of an arbitrary top expression. If an inverse occurrence is nonhomogeneous, the decrease is strict. In particular, for a normalized operand P whose hereditary offsets are not all killed by χ , every coefficient expression has complexity strictly less than $c(P)$.*
- (c) *For a collection of top expressions, assume at least one represented top offset equals 1 and that χ is nonzero on the subgroup generated by all their hereditary offsets. The joint complexity of the coefficient expressions at each degree is strictly smaller than the original joint complexity.*
- (d) *If the input is represented in one field coset $D_A a$, with all inverse operands internal to A , a nonzero coefficient of degree n is represented in one coset of $D_{A \cap K}$. Its permitted group coset is $(Aa \cap \chi^{-1}(n))t^{-n}$, and its inverse operands remain internal to $A \cap K$.*

Proof. We construct the coefficients by induction on the expression trees. An inverse operand whose hereditary offsets are all killed by χ is retained unchanged in degree zero. For any other inverse operand P , first use a transporter inside $H(P)$: if $\chi(H(P)) = d\mathbb{Z}$, choose $a \in H(P)$ with $\chi(a) = d > 0$. The cyclic-field property for this subgroup gives the expansion of P in powers of a . Its leading nonzero coefficient is itself a rational expression. Factoring this coefficient and the leading power gives the inverse expansion by the geometric series $(1 + R)^{-1} = \sum_{j \geq 0} (-R)^j$, where R has positive valuation. A fixed coefficient of that expansion is a finite sum of finite products of coefficients of P and inverses of its leading coefficient,

with the conjugations imposed by the skew multiplication. All series in this construction are bounded below, so coefficient multiplication and inverse expansion involve only finitely many terms in each fixed degree.

For completeness, the complexity comparison can be proved at the same time by structural induction. A normalized nonhomogeneous operand P has a top offset 1. If one of its inverse occurrences is nonhomogeneous, the induction hypothesis decreases the complexities of that occurrence's operand coefficients. If all its inverse occurrences are homogeneous, every term contributes to just one degree, and the nonzero character distinguishes at least two top offsets. Thus every degree omits at least one term. These are the two cases proving that every coefficient of P has complexity less than $c(P)$. Consequently, in expanding P^{-1} , the inverse of its leading coefficient has operand complexity less than $c(P)$. Every inverse operand appearing in one of the other coefficients also has complexity less than $c(P)$. Any coefficient term of P^{-1} therefore has path weight less than $\omega^{c(P)}$. This replaces the original inverse's entire path contribution by finitely many lower contributions. The two natural-sum comparisons above prove strict decrease even after arbitrary finite multiplication and distribution of the coefficient expressions. They also justify the nonhomogeneous-inverse case used in the preceding structural induction.

This proves (b). For (c), retain the tags recording each original term. If some inverse occurrence is nonhomogeneous, every resulting term coming from that occurrence has smaller path weight, while unchanged terms contribute at most once at any degree. If all inverse occurrences are homogeneous, the top offset 1, together with nontriviality on all hereditary offsets, forces another top degree. Every coefficient then omits a top term. This proves (c), also for a joint collection whose individual expressions need not have a top offset 1.

It remains to check that this construction respects the designated supports rather than just the field identities. In the local expansion at P^{-1} , all transporters lie in $H(P)$, and its coefficient of height n is supported on the corresponding coset of $H(P) \cap K$. To compare with the global variable, write $a^m = k_m t^{dm}$, with $k_m \in K$. A local coefficient then acquires the final monomial offset k_m . Moving powers of t to the right in a product conjugates its subsequent kernel factors. Multiplying the corresponding group elements shows that every resulting term pattern is contained in the original product pattern's total-degree slice, followed by t^{-n} . The new offsets are monomials; their word lengths play no role in (6.6)–(6.7). Conjugations and final-offset changes preserve the measure. This proves (a) and its compatibility with (b) and (c). For (d), all the local transporters before the final offset lie in A . When intermediate global powers are moved to the right and the resulting monomials are normalized, each conjugating prefix is consequently an element of A . It preserves $A \cap K$, and the final offset lies in the displayed single coset. This also follows by first expanding the whole D_A -factor using an A -transporter and only then converting its coefficients to the global variable. $\square$

6.3 Disjoint supports and the scalar-function map

Proof of Theorem 6.1(i)–(ii). Suppose $v \in \mathcal{V}(S) \cap \mathcal{V}(T)$, with $S \cap T$ empty. Choose finite rational representations of its two values, retaining the prescribed pattern for each top summand. We induct on their joint complexity. Right-multiply the equality by the inverse of one represented top offset. This translates both allowed sets and preserves their disjointness and the joint complexity, while putting one top offset at 1. Let H be the finitely generated subgroup of all hereditary offsets in the shifted equality.

If $H = 1$, all nonzero displayed terms have pattern $\{1\}$. Disjointness forces one side to have no such term, and the equality therefore has value zero. Otherwise local indicability gives an epimorphism $\chi: H \twoheadrightarrow \mathbb{Z}$. Apply Lemma 6.2. At each degree the two coefficient expressions have strictly smaller joint complexity, and their allowed supports are disjoint subsets of K . The induction hypothesis makes each side's coefficient zero. Injectivity of (6.1)

makes both original values zero as well. Well-foundedness of the ordinal measure completes the induction.

Set $W = \mathcal{V}(J \setminus \{1\})$. Part (i) gives $W \cap \mathbb{Q}1 = 0$. Define a linear functional on $W + \mathbb{Q}1$ by $\ell(w + q) = q$, and extend it to a $\mathbb{Q}$ -linear functional on D_J . Put

$$\mathcal{F}(v)(g) = \ell(vg^{-1}). \quad (6.8)$$

Then $\mathcal{F}(1) = \delta_1$. If $g \notin S$ and $v \in \mathcal{V}(S)$, then $vg^{-1} \in \mathcal{V}(Sg^{-1}) \subseteq W$, so this coefficient vanishes. Finally,

$$\mathcal{F}(vh)(g) = \ell(vhg^{-1}) = \mathcal{F}(v)(gh^{-1}),$$

which is precisely equivariance for right group multiplication. This proves all support restrictions simultaneously. No multiplicativity of $\mathcal{F}$ is asserted or needed. $\square$

6.4 Rational descent from a smaller coefficient field

We isolate the rational-series step, including its handedness.

Lemma 6.3 (Rational descent). *Let $E \subseteq F$ be division rings and let conjugation by u induce an automorphism of F preserving E . If an element of $F(u)$ has a bounded-below Laurent expansion*

$$S = \sum_n u^n f_n, \quad f_n \in E,$$

then $S \in E(u)$.

Proof. Choose a nonzero right Laurent-polynomial denominator $P = \sum_{j \in I} u^j p_j$, with I finite and $p_j \in F$, such that SP is a Laurent polynomial. For all sufficiently large n , its coefficient of u^n , with coefficients written on the right, is zero:

$$\sum_{j \in I} (u^{-j} f_{n-j} u^j) p_j = 0. \quad (6.9)$$

Every parenthesized coefficient belongs to E . Choose a basis $(b_\gamma)_\gamma$ of F as a *left* vector space over E , and expand the finitely many coefficients as $p_j = \sum_\gamma e_{j\gamma} b_\gamma$. Only finitely many basis vectors are used. Linear independence makes every coordinate tuple $(e_{j\gamma})_{j \in I}$ satisfy all equations (6.9) simultaneously. Since $P \neq 0$, at least one such tuple is nonzero. Fix it and put $P' = \sum_{j \in I} u^j e_{j\gamma} \neq 0$.

The product SP' has coefficients in E , has a lower degree bound, and vanishes in every sufficiently high degree. It is therefore a Laurent polynomial over E , and $S = (SP')(P')^{-1} \in E(u)$. There is no finite-subsystem argument here: one fixed coordinate tuple satisfies the entire family of recurrence equations. $\square$

6.5 Intersections of field cosets

Proof of Theorem 6.1(iii), single cosets. The empty-intersection assertion follows from part (i). Suppose $At \cap Bs = Cx$ and $0 \neq v \in D_{At} \cap D_{Bs}$. We induct on the joint complexity of expressions for v in these two field cosets. Choose all inverse operands on the first side inside A , and all those on the second inside B . Right-shift by one represented top offset as in the preceding proof, and let H be the subgroup generated by the new hereditary offsets.

Intersect the two permitted cosets with H . Each nonempty intersection is a coset of $A \cap H$, respectively $B \cap H$. All represented inverse operands remain internal to these subgroups: their offsets lie in the original subgroup and, by definition, in H . Likewise, ratios of represented top offsets on a fixed side lie in its intersected subgroup. Thus this restriction neither changes

the expression nor loses its single-coset form. The restricted cosets have a common point by part (i), since their common field value is nonzero. We may accordingly work entirely in H , temporarily reusing A, B, C for their intersections with H . If $H = 1$, the assertion is immediate. Otherwise choose $\chi: H \rightarrow \mathbb{Z}$, with kernel K .

For each degree n , the two coefficient expressions lie in single cosets of the fields of $A \cap K$ and $B \cap K$, allowing the final offsets in K described in Lemma 6.2. Their joint complexity has strictly decreased. The induction hypothesis therefore says that their common value is zero if the two group cosets do not meet, and otherwise lies in the field of $A \cap B \cap K$ times a point of their intersection. Returning to the original degree, the coefficient is thus permitted precisely on the corresponding slice of the common coset Cx , with its subgroup-field coefficient in $D_{C \cap K}$.

Choose a common point x inside H and divide v by it on the right. The resulting rational series has coefficients in the common kernel field and group support restricted to C . If $\chi(C) = 0$, it has only degree zero and already belongs to $D_{C \cap K} = D_C$. If $\chi(C) = d\mathbb{Z}$, with $d > 0$, choose $u \in C$ of height d , and put $E = D_{C \cap K}$, $F = D_K$. Then it has the form

$$vx^{-1} = \sum_m u^m f_m, \quad f_m \in E, \quad (6.10)$$

and conjugation by u preserves E .

We must also check that (6.10) is rational over $F(u)$, not just over $F(t)$. The subgroup $H_d = \chi^{-1}(d\mathbb{Z})$ has index d in H , and its field is $F(u)$. The finite crossed decomposition gives

$$D_H = \bigoplus_{r=0}^{d-1} D_{H_d} t^r.$$

The Laurent series of the r -th summand uses only degrees congruent to r modulo d . Since (6.10) has only zero-congruence degrees, injectivity of Laurent expansion makes all other summands vanish. Hence $vx^{-1} \in F(u)$. Lemma 6.3 now gives $vx^{-1} \in E(u) = D_C$. This proves the required inclusion; the reverse one follows from $C \leq A, B$. Undoing the initial common right shift proves (6.2) in the original frame. If a restricted ambient subgroup was used, its common subgroup is contained in the original C , so this conclusion also gives the original claim. $\square$

6.6 Finite spans and separability

Proof of Theorem 6.1(iii), finite spans. One inclusion in (6.3) is immediate. For the other, write a common element as

$$v = \sum_{i=1}^r a_i t_i = \sum_{j=1}^s b_j s_j, \quad a_i \in D_A, \quad b_j \in D_B.$$

Merge repeated cosets on each side, so the original At_i 's are pairwise disjoint, as are the Bs_j 's. Every nonempty overlap is $At_i \cap Bs_j = Cx_{ij}$.

For a fixed i , two distinct overlaps Cx_{ij} and Cx_{ik} have $x_{ij}x_{ik}^{-1} \in A \setminus C$. Separability supplies a finite-index subgroup $A_0 \leq A$ containing C and excluding all these finitely many differences. Hence no right A_0 -coset inside At_i meets two distinct original B -cosets. To see this independently of representatives, a common A_0 -coset meeting both overlaps would imply $cx_{ij}(c'x_{ik})^{-1} \in A_0$ for some $c, c' \in C$, and therefore $x_{ij}x_{ik}^{-1} \in A_0$, a contradiction. Choose $B_0 \leq B$ symmetrically, of finite index and containing C .

These two choices already give the simultaneous refinement required. Since $A_0 \cap B = C$, the intersection of a refined A_0 -coset with its unique possible original B -coset is one C -coset.

As $C \leq B_0$, that entire intersection lies in just one refined B_0 -coset. The symmetric argument applies on the other side. Thus the bipartite intersection graph of all refined cosets is a matching, with possible isolated vertices. Also $A_0 \cap B_0 = C$.

We use the finite-index field decompositions

$$D_A = \bigoplus_{a \in A_0 \setminus A} D_{A_0} a, \quad D_B = \bigoplus_{b \in B_0 \setminus B} D_{B_0} b.$$

For clarity, these do not require the chosen subgroups to be normal. One may first pass to a finite-index normal core: its finite crossed span in the overfield is a finite-dimensional domain over its division ring, hence a division ring, and so equals the larger subgroup field. Grouping the resulting summands by the original subgroup cosets gives the displayed formulas.

Refine the two expressions for v by these decompositions. Group the terms of their difference according to the connected components of the refined intersection graph. Distinct components have disjoint unions of permitted supports. Part (i) therefore makes the value of each component zero. Isolated terms vanish. Each remaining component is a single pair, giving an equality between one D_{A_0} -coset and one D_{B_0} -coset. By (6.2), their common value belongs to $D_C x$ for a point of their group-coset overlap. Adding the finitely many pairs proves (6.3). $\square$

7 Semilinear localization of affiliated coefficients

Write $\mathcal{N}(J)$ for the group von Neumann algebra of a discrete group J , with its canonical trace, and write $\mathcal{U}(J)$ for its algebra of affiliated operators. The division fields used here are embedded in these affiliated algebras. Our target is the following localization statement for a finite semilinear coefficient system.

Lemma 7.1 (Semilinear finite-dimensional support). *Let G be a torsion-free word-hyperbolic group, let $h : G \rightarrow \mathbb{Z}$ be a homomorphism, and put $L = \ker h$. Let $D \leq G$ be quasiconvex and put $S = D \cap L$. Choose $g \in G$ and $g_D \in D$ such that*

$$h(g) = h(g_D) \neq 0.$$

Suppose that the division fields in use are embedded in the affiliated group algebras, and that, for an integer $r \geq 1$, a row $\lambda \in D_L^{1 \times r}$ satisfies

$$g \lambda g_D^{-1} = \lambda B, \quad B \in \mathrm{GL}_r(D_S). \quad (7.1)$$

Then

$$\lambda \in \sum_{xS \in \mathcal{P}} x \mathcal{U}(S)^{1 \times r}, \quad \mathcal{P} = \{xS : x \in L, g^m \in x D x^{-1} \text{ for some } m \in \mathbb{Z} \setminus \{0\}\}. \quad (7.2)$$

The set $\mathcal{P}$ is finite. In fact, the same assertion holds for any affiliated row λ and any affiliated invertible matrix B satisfying (7.1).

Here is the normalization behind the proof. Put $\alpha = \mathrm{Ad} g_D$, and first consider the case in which the map $w \mapsto \lambda w$ on affiliated S -columns has zero kernel. We seek a positive invertible affiliated matrix P over S such that

$$P = B \alpha(P) B^*.$$

Then

$$Z = \lambda P^{1/2}, \quad u = P^{-1/2} B \alpha(P)^{1/2}$$

satisfy $uu^* = u^*u = 1$ and $gZg_D^{-1} = Zu$. Thus changing the positive form makes the coefficient transport unitary. It follows that $gZZ^*g^{-1} = ZZ^*$. Hyperbolicity and the nonzero height of g will force this invariant affiliated operator to be scalar, so every entry of Z is bounded. These entries generate a closed right S -module in $\ell^2(L)$ of von Neumann dimension at most r . Its invariance under height translation excludes infinite coset orbits from the support, and quasiconvexity of D leaves only finitely many periodic cosets.

The invariant-form strategy parallels the bounded unitarization argument of Sz.-Nagy and Dixmier [19, Section 5, Theorem 6]. The positive mean constructed below supplies P . Its inputs need not have a common operator-norm bound, so we average bounded resolvent expressions and control the spectral tails of the inputs and their inverses. When the column kernel is nonzero, the proof first represents its complement by a projection over $\mathcal{N}(S)$ and carries out the same construction between the corresponding corners.

The affiliated conclusion in (7.2) is sufficient for Proposition 10.5. There the Bass–Serre homology injection is first established over a subgroup division field. Extending scalars to its affiliated algebra preserves that injection and its nonzero classes: choosing a vector-space basis identifies the scalar extension with a direct sum of copies of the nonzero affiliated algebra. Injectivity into ambient homology still requires the separate comparison using actual row chains and coset independence in that proposition.

7.1 Measure estimates and a projection inequality

Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra with a faithful normal finite trace. The trace need not have total mass one. On matrices we use the compatible *unnormalized* traces

$$\tau_r((a_{ij})) = \sum_{j=1}^r \tau(a_{jj}).$$

In particular, for a bounded rectangular matrix a , $\tau_r(a^*a) = \tau_k(aa^*)$, with the appropriate source and target matrix sizes. Corners carry the restriction of these traces. Products and sums of affiliated operators mean their closed affiliated products and sums.

A family $\mathcal{A}$ of affiliated operators is *tight* if

$$\lim_{K \rightarrow \infty} \sup_{a \in \mathcal{A}} \tau(1_{(K, \infty)}(|a|)) = 0.$$

For rectangular operators the trace is taken in the source corner; the nonzero spectral distributions of $|a|$ and $|a^*|$ agree with the compatible traces. We use convergence in measure: $a_j \rightarrow 0$ if, for every $\varepsilon, \eta > 0$, eventually there is a projection q with $\tau(1-q) < \eta$ and $\|a_j q\| < \varepsilon$.

Lemma 7.2 (Measure calculus). *The affiliated algebra of a finite von Neumann algebra is complete for convergence in measure. Products of tight families are tight. If $a_j \rightarrow 0$ in measure and b_j, c_j range over tight families, then $b_j a_j c_j \rightarrow 0$ in measure. These assertions also hold uniformly over an additional index.*

Positive square roots are continuous in measure on positive affiliated operators, and inversion is continuous in measure on positive affiliated units. A uniformly norm-bounded family that tends to zero in measure tends to zero in $L^1(\tau)$; the same assertion holds uniformly.

Proof. The complete topological $*$ -algebra assertion and the singular-number estimates used below are the standard measurable-operator calculus; see [21, Sections 1.4–1.5, Lemma 2.5 and Lemma 3.1]. For clarity, the estimates that provide all the uniformity we need are

$$\mu_{s+t}(ab) \leq \mu_s(a)\mu_t(b), \quad \mu_{s+t}(a+b) \leq \mu_s(a) + \mu_t(b), \quad s, t > 0,$$

where

$$\mu_s(a) = \inf\{\|aq\| : q \text{ a projection, } \tau(1 - q) \leq s\}.$$

Their proofs cut the two operators to bounded parts and add the trace codimensions; polar decomposition identifies the trace codimensions of the source and range cuts. Tightness says precisely that $\sup_{a \in \mathcal{A}} \mu_s(a) < \infty$ for every $s > 0$. Convergence in measure says that $\mu_s(a_j) \rightarrow 0$ for every $s > 0$. The two displayed estimates therefore prove the product and uniform assertions.

We recall how to obtain the stated functional calculus without any norm bounds on the original operators. For positive $a_j \rightarrow a$, the resolvent identity and the product estimate give

$$(1 + a_j)^{-1} - (1 + a)^{-1} = (1 + a_j)^{-1}(a - a_j)(1 + a)^{-1} \rightarrow 0$$

in measure. On bounded self-adjoint contractions, polynomial approximation gives continuity of continuous functional calculus in measure. Apply this to the positive contractions $q_j = (1 + a_j)^{-1}$, with limit $q = (1 + a)^{-1}$. Here is the endpoint control needed for unbounded scalar functions. If q has no spectral atom at 1, then for every $\eta > 0$ one can choose $b < 1$ with $\mu_{\eta/2}(q) < b$. The sum inequality gives $\mu_\eta(q_j) \leq \mu_{\eta/2}(q) + \mu_{\eta/2}(q_j - q)$, so the spectral mass of q_j sufficiently near 1 is at most η , for all large j . Applying the same argument to $1 - q_j$ controls the endpoint 0 when q has no atom there. The latter condition always holds; the former holds when a is a positive unit. Cut off the functions $(q^{-1} - 1)^{1/2}$, and, in the unit case, $(q^{-1} - 1)^{-1}$, at these endpoints. The clipped functions are continuous on $[0, 1]$, and the cutoff errors have arbitrarily small support trace. This proves square-root and inverse continuity. The same proof is uniform whenever the required tail bounds are uniform.

Finally, if $\|z_j\| \leq C$, then

$$\|z_j\|_1 \leq \varepsilon \tau(1) + C \tau(1_{(\varepsilon, \infty)}(|z_j|)).$$

First use convergence in measure and then let ε tend to zero. $\square$

A *positive affiliated unit* in a corner is a positive affiliated operator whose support is the identity of that corner; its inverse is allowed to be unbounded. For two such units define

$$F(P, X) = P^{1/2}(P + X)^{-1}P^{1/2} = (1 + P^{-1/2}XP^{-1/2})^{-1}. \quad (7.3)$$

This is a bounded positive contraction.

Lemma 7.3 (Spectral-projection comparison). *Let T, X be positive affiliated units in $\mathcal{M}$, and let $t > 0$. For $p_+ = 1_{(t, \infty)}(T)$ and $p_- = 1_{(0, t)}(T)$,*

$$\tau(p_+ F(T, X)) \geq \tau(p_+ F(t1, X)), \quad (7.4)$$

$$\tau(p_- F(T, X)) \leq \tau(p_- F(t1, X)). \quad (7.5)$$

Proof. First assume that T, X and their inverses are bounded. Put $p = p_+$ and $q = 1 - p$, and decompose with respect to $p + q$:

$$T = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \quad X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}.$$

Here $A \geq tp$ and $B \leq tq$. The positive Schur complements

$$K_B = X_{11} - X_{12}(B + X_{22})^{-1}X_{21}, \quad K_t = X_{11} - X_{12}(tq + X_{22})^{-1}X_{21}$$

satisfy $K_B \leq K_t$. Block inversion and cyclicity of the trace give

$$\tau(pF(T, X)) = \tau(A(A + K_B)^{-1}).$$

For positive invertible A, K in $p\mathcal{M}p$, set

$$\psi(A, K) = \tau(A(A + K)^{-1}).$$

This function is increasing in A , because

$$\psi(A, K) = \tau(p) - \tau(K^{1/2}(A + K)^{-1}K^{1/2}),$$

and is decreasing in K , by its defining expression with $A^{1/2}$ on the two sides of the inverse. Consequently

$$\psi(A, K_B) \geq \psi(tp, K_B) \geq \psi(tp, K_t) = \tau(pF(t1, X)),$$

which proves (7.4). For $p = p_-$, the comparisons $A \leq tp$, $B \geq tq$, and $K_B \geq K_t$ reverse both inequalities, giving (7.5). The cases $p = 0$ or $q = 0$ follow from the same argument with the empty block omitted.

For affiliated units, clip T and X spectrally into $[1/R, R]$, with $1/R < t < R$. The high and low spectral projections of the clipped T at t are the original projections. The clipped operators converge in measure to T, X , and their inverses converge to the corresponding inverses. Lemma 7.2 and (7.3) show that the resulting positive contractions converge in L^1 . Both inequalities pass to the limit. $\square$

7.2 An affiliated positive mean

Fix a shift-invariant mean m on $\ell^\infty(\mathbb{Z})$. Such a mean is obtained as a weak-* cluster point of the averaging functionals on the intervals $[-N, N]$: translating an interval by a fixed integer changes its average by a quantity tending to zero. For a bounded sequence $Y_n \in \mathcal{M}$, define $\mathfrak{M}_n Y_n \in \mathcal{M}$ by duality with the predual:

$$\varphi(\mathfrak{M}_n Y_n) = m(n \mapsto \varphi(Y_n)), \quad \varphi \in \mathcal{M}_*.$$

This mean is positive and unital, commutes with fixed bounded left and right multiplications, and is shift invariant. It also satisfies

$$\|\mathfrak{M}_n Y_n\|_1 \leq \sup_n \|Y_n\|_1. \quad (7.6)$$

Indeed, pair with an arbitrary element of the unit ball of $\mathcal{M}$, using the dual characterization of the L^1 -norm. No interchange of a nonuniform monotone limit with this mean will be used.

For finitely many bounded positive invertible inputs and a weighted average, the equation below defines the induced operator mean associated with the binary harmonic mean [56, Proposition 6.2]. We prove the affiliated version needed here, under tightness of the inputs and their inverses and with the fixed shift-invariant mean m .

Proposition 7.4 (Affiliated positive mean). *Let X_n , $n \in \mathbb{Z}$, be positive affiliated units in a finite von Neumann algebra. Suppose that both $\{X_n\}$ and $\{X_n^{-1}\}$ are tight. There is a unique positive affiliated unit P satisfying*

$$\mathfrak{M}_n F(P, X_n) = \frac{1}{2} 1. \quad (7.7)$$

The construction is covariant under congruence by affiliated invertible operators, including affiliated invertible maps between corners with their restricted traces.

Proof. We divide the argument into bounded construction, uniform tails, comparison, and passage to the limit.

For an integer $R \geq 1$, let $X_{n,R}$ be the spectral clipping of X_n into $[R^{-1}, R]$. Define

$$H(P, X) = (P^{-1} + X^{-1})^{-1}, \quad \Phi_R(P) = 2\mathfrak{M}_n H(P, X_{n,R}).$$

Inversion reverses positive order, so Φ_R is order preserving. It maps the interval $[R^{-1}1, R1]$ into itself. Starting from $P^{(0)} = R1$, the iterates $P^{(j+1)} = \Phi_R(P^{(j)})$ decrease and remain in that interval. Their strong limit P_R lies in the same interval, and normality of the finite trace gives convergence in L^1 .

This convergence passes through Φ_R uniformly in n . In fact,

$$H(P, X) = X - X(P + X)^{-1}X,$$

and the resolvent identity gives, on the fixed interval in question,

$$\|H(P, X) - H(Q, X)\|_1 \leq C_R \|P - Q\|_1$$

with a constant independent of $X = X_{n,R}$. Equation (7.6) therefore shows that $P_R = \Phi_R(P_R)$. The identity

$$H(P, X) = P^{1/2}(1 - F(P, X))P^{1/2} \quad (7.8)$$

and bounded invertibility of P_R imply

$$\mathfrak{M}_n F(P_R, X_{n,R}) = \frac{1}{2} 1. \quad (7.9)$$

The clipped families $\{X_{n,R} : n \in \mathbb{Z}, R \geq 1\}$ and their inverses are tight: above any threshold $K > 1$, clipping can only decrease the relevant spectral tail. Consequently

$$\begin{aligned} \sup_{n,R} \|1 - F(t1, X_{n,R})\|_1 &\longrightarrow 0 & (t \longrightarrow \infty), \\ \sup_{n,R} \|F(s1, X_{n,R})\|_1 &\longrightarrow 0 & (s \longrightarrow 0). \end{aligned}$$

For example, cutting $X_{n,R}$ at a fixed K bounds the first quantity by

$$\frac{K}{t} \tau(1) + \sup_{n,R} \tau(1_{(K,\infty)}(X_{n,R})).$$

For the second, cut $X_{n,R}^{-1}$ at K ; the analogous bound is $sK\tau(1)$ plus its tail trace.

Apply Lemma 7.3 to $p = 1_{(t,\infty)}(P_R)$ and use (7.9). It gives

$$\frac{1}{2} \tau(p) \geq \tau(p) - \sup_n \|1 - F(t1, X_{n,R})\|_1.$$

For $q = 1_{(0,s)}(P_R)$, the low-projection inequality gives

$$\frac{1}{2} \tau(q) \leq \sup_n \|F(s1, X_{n,R})\|_1.$$

Thus both $\{P_R\}$ and $\{P_R^{-1}\}$ are tight, uniformly in R .

We next remove clipping from the equation. Put

$$q_{n,R} = 1_{(0,R^{-1}) \cup (R,\infty)}(X_n).$$

Two-sided tightness implies $\sup_n \tau(q_{n,R}) \rightarrow 0$. The resolvent identity expresses

$$\begin{aligned} F(P_R, X_n) - F(P_R, X_{n,R}) &= P_R^{1/2}(P_R + X_n)^{-1} \\ &\quad \cdot (X_{n,R} - X_n)(P_R + X_{n,R})^{-1} P_R^{1/2}. \end{aligned}$$

The middle factor is supported on $q_{n,R}$. Polar decomposition shows that the support trace of a product cannot exceed the support trace of any one of its factors. The displayed difference, being a difference of positive contractions, has norm at most two. Hence

$$\sup_n \|F(P_R, X_n) - F(P_R, X_{n,R})\|_1 \leq 2 \sup_n \tau(q_{n,R}) \longrightarrow 0.$$

It follows that

$$E_R := \mathfrak{M}_n F(P_R, X_n) - \frac{1}{2} 1 \quad \text{satisfies} \quad \|E_R\|_1 \longrightarrow 0. \quad (7.10)$$

We record precisely the congruence rule. If C is an affiliated invertible operator, set $P' = CPC^*$ and $X' = CXC^*$. Then

$$F(P', X') = UF(P, X)U^*, \quad U = (P')^{-1/2}CP^{1/2}. \quad (7.11)$$

The affiliated algebra identities $UU^* = U^*U = 1$ show that U is a bounded unitary. The formula follows by taking the inverse of $C(P + X)C^*$. For maps between corners the same calculation makes U a partial unitary between their identities. Crucially, U depends on P, C but not on X ; errors in a mean equation are therefore conjugated by one fixed unitary and keep their L^1 -norm.

We now prove that P_R is Cauchy in measure, rather than appealing to any compactness assertion for tight families. Given R, S , put

$$C = P_R^{-1/2}, \quad T = CP_S C, \quad X'_n = CX_n C.$$

The equations with errors E_R, E_S , after this congruence, become

$$\mathfrak{M}_n F(1, X'_n) = \frac{1}{2} 1 + E_R, \quad \mathfrak{M}_n F(T, X'_n) = \frac{1}{2} 1 + UE_S U^*$$

for a unitary U independent of n . The first congruence has unitary equal to 1. By the previously established tightness and Lemma 7.2, the families X'_n and $(X'_n)^{-1}$, with both R and n varying, are tight.

Fix $d, \varepsilon > 0$. If $p = 1_{(1+d, \infty)}(T)$ has $\tau(p) \geq \varepsilon$, then (7.4) gives

$$\|E_R\|_1 + \|E_S\|_1 \geq \tau(p \mathfrak{M}_n \delta_d(X'_n)), \quad \delta_d(x) = \frac{dx}{(1+d+x)(1+x)}. \quad (7.12)$$

Choose $0 < a < b < \infty$, independently of R, n , such that

$$\tau(1 - 1_{[a,b]}(X'_n)) < \varepsilon/2.$$

On $[a, b]$, the function δ_d has a positive minimum c . With $q_n = 1_{[a,b]}(X'_n)$, functional calculus gives

$$\tau(p \delta_d(X'_n)) \geq c \tau(p q_n) \geq c(\tau(p) - \tau(1 - q_n)) \geq c\varepsilon/2.$$

These inequalities do not require p and q_n to commute. They hold for every n , so positivity of the scalar mean gives the same lower bound in (7.12). This contradicts (7.10) for sufficiently large R, S .

For the low projection $p = 1_{(0, 1-d)}(T)$, $0 < d < 1$, use (7.5) and the positive scalar function

$$\frac{1}{1+x} - \frac{1-d}{1-d+x} = \frac{dx}{(1+x)(1-d+x)}.$$

The identical argument bounds its trace away from zero whenever $\tau(p) \geq \varepsilon$. Thus $T - 1 \rightarrow 0$ in measure as $R, S \rightarrow \infty$, uniformly over these pairs. The high and low estimates also imply $T^{-1} - 1 \rightarrow 0$ in measure. Since

$$P_S - P_R = P_R^{1/2}(T - 1)P_R^{1/2}, \quad P_S^{-1} - P_R^{-1} = P_R^{-1/2}(T^{-1} - 1)P_R^{-1/2},$$

tightness and Lemma 7.2 show that both (P_R) and (P_R^{-1}) are Cauchy in measure.

Completeness gives limits P, V . Positivity is closed under convergence in measure, and continuity of multiplication gives $PV = VP = 1$. Hence P is a positive affiliated unit and $V = P^{-1}$. Square-root continuity and the uniform product estimate show that

$$P_R^{-1/2} X_n P_R^{-1/2} - P^{-1/2} X_n P^{-1/2} \longrightarrow 0$$

in measure uniformly in n . The resolvent identity in (7.3) gives uniform measure convergence of $F(P_R, X_n)$ to $F(P, X_n)$. These are contractions, so convergence is uniform in L^1 . Equations (7.6) and (7.10) prove (7.7).

Finally, if P, Q are two exact solutions, their two-element families and their inverses are tight. Repeat the same congruence comparison with zero errors. Any nonzero high or low relative spectral projection would give a strictly positive lower bound in (7.12). Therefore $P^{-1/2} Q P^{-1/2} = 1$, proving uniqueness. The congruence formula then proves the asserted covariance: a congruent solution solves the congruent equation, and it is the unique such solution. $\square$

7.3 Proof of the support lemma

Proof of Lemma 7.1. If $\lambda = 0$, the conclusion is immediate. Assume otherwise. The automorphism $\alpha = \text{Ad } g_D$ preserves L, S , their group von Neumann algebras, and their traces. On affiliated rows over L , let

$$\mathcal{T}(\xi) = g \xi g_D^{-1}.$$

This is well defined because $h(g) = h(g_D)$. It is right α -semilinear: $\mathcal{T}(\xi w) = \mathcal{T}(\xi) \alpha(w)$.

We first identify the column kernel with a projection in $M_r(\mathcal{N}(S))$. Choose the positive affiliated unit

$$v = (1 + \lambda \lambda^*)^{-1/2} \in \mathcal{U}(L), \quad a = v \lambda.$$

The row a is bounded. Let

$$H = \mathbb{E}_S(a^* a) \in M_r(\mathcal{N}(S)), \quad e = \text{supp } H,$$

where $\mathbb{E}_S : \mathcal{N}(L) \rightarrow \mathcal{N}(S)$ is the trace-preserving group conditional expectation, applied entrywise. Concretely, if $P_S : \ell^2(L) \rightarrow \ell^2(S)$ is orthogonal projection, then $\mathbb{E}_S(a) = P_S a|_{\ell^2(S)}$. Compression commutes with right S -translations and is positive and normal. Elements of $\mathcal{N}(S)$ reduce $\ell^2(S)$, giving bimodularity; the vector trace at δ_1 gives trace preservation and faithfulness. This is also the tracial expectation of [4, Theorem 9.1.2 and Remark 9.1.3]. Neither normality nor finite index of S is required. For a bounded S -column w , compatible traces give

$$\tau((aw)^* aw) = \tau(w^* H w).$$

Faithfulness shows that $aw = 0$ if and only if $ew = 0$. For an affiliated column, multiply on the right by increasing spectral cuts for $|w|$; the cut columns are bounded and converge in measure to w . The same equivalence follows by measure continuity. Consequently

$$\ker(\lambda : \mathcal{U}(S)^r \rightarrow \mathcal{U}(L)) = (1 - e)\mathcal{U}(S)^r, \quad \lambda e = \lambda. \quad (7.13)$$

Here $\lambda e = \lambda$ also follows directly from $a(1 - e) = 0$. In particular $e \neq 0$.

The semilinear relation controls the transported row $g^n \lambda g_D^{-n}$. To apply the positive mean, we need control of the coefficient maps themselves on the complement of the column kernel. The following quantitative consequence of the kernel description will transfer that control:

$$\left. \begin{array}{l} w_j = e w_j, \\ \{\lambda w_j\} \text{ is tight} \end{array} \right\} \implies \{w_j\} \text{ is tight.} \quad (7.14)$$

The matrices w_j here may have any fixed finite number of columns. To prove it, suppose that $p = 1_{(R,\infty)}(ww^*) \leq e$ has $\tau_r(p) \geq \varepsilon$. Since $\text{supp } H = e$, choose $\delta > 0$ such that

$$\tau_r(e - 1_{[\delta,\infty)}(H)) < \varepsilon/2.$$

Then, writing $q = 1_{[\delta,\infty)}(H)$,

$$\tau(apa^*) = \tau_r(pH) \geq \delta \tau_r(pq) \geq \delta \varepsilon/2 =: c > 0.$$

Also $0 \leq apa^* \leq \|a\|^2 1$. Thus apa^* has spectral mass at least $c/(2\|a\|^2)$ above $c/2$, using the canonical trace $\tau(1) = 1$ in $\mathcal{N}(L)$. Positive order gives

$$aww^*a^* \geq Rapa^*.$$

Monotonicity of spectral distribution under positive order therefore forces the same positive amount of spectral mass above $Rc/2$ for aww^*a^* . This contradicts tightness of aw as $R \rightarrow \infty$. Since $aw = v(\lambda w)$, tightness of λw implies tightness of aw by Lemma 7.2. This proves (7.14).

Define the full invertible cocycle C_n , $n \in \mathbb{Z}$, by

$$C_0 = 1, \quad C_1 = B, \quad C_{n+m} = C_n \alpha^n(C_m).$$

Iterating (7.1) gives

$$\mathcal{T}^n(\lambda) = g^n \lambda g_D^{-n} = \lambda C_n.$$

Semilinearity and (7.13) imply

$$C_n \alpha^n((1-e)\mathcal{U}(S)^r) = (1-e)\mathcal{U}(S)^r.$$

Hence the induced maps on the quotient columns have representatives

$$b_n = e C_n \alpha^n(e) : \alpha^n(e)\mathcal{U}(S)^r \longrightarrow e\mathcal{U}(S)^r. \quad (7.15)$$

Their inverses are

$$d_n = \alpha^n(e) C_n^{-1} e = \alpha^n(b_{-n}), \quad b_n d_n = e, \quad d_n b_n = \alpha^n(e).$$

Compression does not spoil the cocycle law. Indeed the discarded summand $\alpha^n(1-e)$ is carried by C_n into the kernel and is killed by e . It follows that

$$b_{n+m} = b_n \alpha^n(b_m). \quad (7.16)$$

Moreover,

$$\lambda b_n = \lambda C_n \alpha^n(e) = \mathcal{T}^n(\lambda). \quad (7.17)$$

The spectral distributions of the rows on the right of (7.17) are independent of n : their absolute squares are $\alpha^n(\lambda^* \lambda)$. Thus they form a tight family. Since $eb_n = b_n$, (7.14) shows that $\{b_n\}$ is tight. Trace invariance of α and the formula for d_n show that $\{d_n\}$ is tight as well. In the fixed corner $eM_r(\mathcal{U}(S))e$, put

$$X_n = b_n b_n^*, \quad X_n^{-1} = d_n^* d_n.$$

Both families are tight, by the product estimate.

Apply Proposition 7.4 in this corner to obtain the unique positive affiliated unit P satisfying

$$\mathfrak{M}_n F(P, X_n) = \frac{1}{2} e.$$

Equation (7.16) gives

$$X_{n+1} = b_1 \alpha(X_n) b_1^*.$$

The automorphism α commutes with the weak-* mean. Its application moves the equation to the corner $\alpha(e)$; congruence by the affiliated invertible map $b_1 : \alpha(e) \rightarrow e$ then gives an equation for $b_1 \alpha(P) b_1^*$ with inputs X_{n+1} . Shift invariance and uniqueness yield

$$P = b_1 \alpha(P) b_1^*. \quad (7.18)$$

Set

$$Z = \lambda P^{1/2}, \quad u = P^{-1/2} b_1 \alpha(P)^{1/2}.$$

Inverses of P here and below are taken in the corner e . Equation (7.18) gives $uu^* = e$ and $u^*u = \alpha(e)$, so u is a bounded partial unitary with entries in $\mathcal{N}(S)$. Equations (7.17) and (7.18) imply

$$gZg_D^{-1} = Zu, \quad gZZ^*g^{-1} = Zuu^*Z^* = ZZ^*.$$

Every conjugation-invariant positive affiliated operator in $\mathcal{U}(L)$ under this g is scalar. To see this, a nonidentity $l \in L$ with finite orbit under conjugation by g would commute with a nonzero power g^m . The centralizer of an infinite-order element in a hyperbolic group is virtually cyclic; in the present torsion-free setting its height-zero subgroup is trivial, since it contains g^m of nonzero height. Thus no such l exists. Each spectral projection of an invariant positive affiliated operator belongs to $\mathcal{N}(L)$ and has square-summable Fourier coefficients constant on these conjugacy orbits. All coefficients off the identity vanish. Each spectral projection is therefore scalar, proving the assertion. The resulting scalar is finite because the affiliated operator is densely defined.

Applying this to ZZ^* , we obtain $ZZ^* = c1$, $c \geq 0$. It follows that every entry Z_j is bounded, because $Z_j Z_j^* \leq c1$. Let $\mathcal{H} \subseteq \ell^2(L)$ be the closed right S -module generated by $Z_1, \dots, Z_r$. The map

$$\ell^2(S)^r \longrightarrow \ell^2(L), \quad (\xi_1, \dots, \xi_r) \longmapsto \sum_{j=1}^r Z_j \xi_j$$

is bounded and right S -equivariant, with range closure $\mathcal{H}$. Its polar decomposition shows that the right S -dimension of $\mathcal{H}$ is at most r [42, Definitions 1.8 and 1.10, Theorem 1.9(6)]. The unitary $\mathcal{T} : \xi \mapsto g\xi g_D^{-1}$ on $\ell^2(L)$ preserves $\mathcal{H}$: the formula $\mathcal{T}(Z) = Zu$ gives one inclusion, and $\mathcal{T}(Z)u^* = Z$ gives the other, using right semilinearity.

Decompose $\ell^2(L)$ along the left cosets xS , which are the natural summands for its right S -action. If Π is the orthogonal projection onto $\mathcal{H}$, its diagonal trace on xS is

$$t_{xS} = \langle \Pi \delta_x, \delta_x \rangle \geq 0.$$

It is independent of the representative x , and the module dimension formula, valid also for infinite subgroup index, is

$$\sum_{xS \in L/S} t_{xS} = \dim_S \mathcal{H} \leq r.$$

The unitary $\mathcal{T}$ permutes these cosets by

$$xS \longmapsto gxg_D^{-1}S.$$

It commutes with Π , so the diagonal traces are constant on its permutation orbits. Every infinite orbit therefore has trace zero. If $t_{xS} = 0$, positivity and right S -equivariance of Π give $\Pi \delta_{xs} = 0$ for every $s \in S$; hence $\mathcal{H}$ has zero Fourier component on that coset. It follows that the coefficients of Z are supported on periodic cosets.

A coset is periodic precisely when, for some $m \neq 0$,

$$g^m x g_D^{-m} S = xS.$$

This implies $x^{-1}g^m x \in D$. Conversely, if $x^{-1}g^m x \in D$, then $x^{-1}g^m x g_D^{-m}$ has height zero and lies in D , hence lies in S ; this proves the reverse implication. Thus the periodic cosets are exactly $\mathcal{P}$.

Finally, $\mathcal{P}$ is finite by the ambient quasiconvexity of D . For $xS \in \mathcal{P}$, left multiplication by a nonzero power of g stabilizes the uniformly quasiconvex coset xD . Both endpoints g^+, g^- therefore lie in its limit set. A geodesic joining these endpoints is contained in a uniformly bounded neighborhood of xD , with a bound depending only on the hyperbolicity and quasiconvexity constants. Fix a point on this geodesic. Every such xD meets one fixed finite ball about that point, so there are finitely many such cosets xD . Since $x \in L$,

$$xD \cap L = xS,$$

and distinct xS in $\mathcal{P}$ correspond to distinct xD . This proves finiteness.

Choose one representative $x \in L$ for each coset in $\mathcal{P}$. Fourier projection, or equivalently the group conditional expectation, now gives for each bounded entry

$$Z_j = \sum_{xS \in \mathcal{P}} x \mathbb{E}_S(x^{-1} Z_j), \quad \mathbb{E}_S(x^{-1} Z_j) \in \mathcal{N}(S).$$

This is an equality first in $\ell^2(L)$, hence also in $\mathcal{N}(L)$. Multiplying the row on the right by $P^{-1/2}$ recovers λ , because $\lambda e = \lambda$. The coefficients become affiliated S -coefficients, proving (7.2). $\square$

8 Expansion gates

We show that a sufficiently long carried arc expands in one direction, and that enough expansion determines a unique continuation in that direction. These path lemmas will be applied to the ordinary rows and, later, to relative rows. The relative statement is conditional on the full package below; passing to relative rows does not automatically preserve that package. We retain the ambient torsion-free hyperbolic group, its height homomorphism, and the cyclic noncommensuration property established above. In particular, a nonzero-height element cannot commensurate a nontrivial height-zero cyclic subgroup.

8.1 Carried arcs and bounded comparisons

For the ordinary application, sample successive coarse Magnus edges of T_0 . The short window is the sampled edge itself, its group H_i is the corresponding free Magnus group, and R_i is its free Cayley tree. There are no peripheral groups in this application. The row geometry and path-inertia results supply quasiconvex window groups, finite orbit lists for each bounded shape, and the cyclic intersection bound for competing continuations. We formulate the argument more generally because Section 13 will use relative free-product trees in place of these Cayley trees.

Definition 8.1 (The expansion package). Consider oriented monotone paths in a tree of groups with height. Sample at successive occurrences of one edge type, and number these occurrences by integers. At position i fix a short window W_i , of a fixed radius s ; the window may consist of the sampled edge itself. Write $H_i = \text{Stab}(W_i)$, with pointwise stabilizers throughout. In the relative case fix an invariant system of height-zero peripheral groups. A group is *nonparabolic* if it is nontrivial and is not contained in any one of these actual peripheral groups. The required package consists of the following properties.

- (E1) The total tree of ordinary or relative Cayley spaces is hyperbolic, and all adjacent edge maps are uniformly quasi-isometric embeddings.
- (E2) Nonparabolically stabilized windows of each fixed shape have finitely many orbits. The same holds for their nonparabolically stabilized extensions, including configurations with a specified common subwindow. Their stabilizers, with intrinsic relative metrics, embed quasi-isometrically in every row of the window; constants are uniform for each bounded shape.
- (E3) A stabilizer of a window containing a short window is a free product of representatives of its nontrivial peripheral intersections and a free group, with a finite relative decomposition. The peripheral systems restrict exactly under further window restrictions. Every window group under consideration has a loxodromic element in the relevant short-window free-product trees.
- (E4) Two distinct continuations through the same short window, with their terminal positions at the same height, have stabilizers intersecting in an at most cyclic group.

For H_i use the Bass tree R_i of the stated free-product decomposition, with trivial edge stabilizers and finite quotient. Its elliptic subgroups are exactly the prescribed restricted peripheral subgroups. In the ordinary application R_i is the horizontal free Cayley tree, and the deep-window groups under consideration are noncyclic. Relative trees may have vertices of infinite valence.

A *trajectory* is one of these monotone paths through the row tree; it may be finite, one-sided infinite, or bilateral. The arcs carried along it lie in the corresponding trees R_i .

The choices in this definition, including the comparisons constructed next, are equivariant under translation of a path together with its marked positions. For a group acting on R_i , its *core* means its minimal invariant subtree; its quotient will be called the *core graph*. These are different objects: the subtree is generally infinite, while the core graphs below have finitely many edges. An edge of R_i will also be called a *regular edge*, to emphasize its trivial stabilizer. An arc is *carried* by a window if it lies on a bi-infinite line in that window group's core. For an infinite path, carrying means carrying by every relevant finite subwindow.

Lemma 8.2 (Bounded comparisons). *Assume the expansion package. For a fixed bounded pair of positions i, j , let K stabilize the segment spanned by $W_i \cup W_j$. There are K -equivariant coarse quasi-isometries between the K -cores in R_i, R_j . They have coarse inverses, preserve the line endspaces of every further window, and are coarsely compatible on every fixed finite diagram of positions. The constants depend on the bounded diagram, not on a further window carrying the arc.*

Proof. The K -action on either core is cocompact. To see this, take the finite relative generating set supplied by (E3), paths from a base vertex to its translates by those generators, and paths to fixed vertices of the finitely many peripheral factors. The translates of the resulting finite subtree contain the minimal subtree. Conversely, a segment in the core reads as a bounded-cost word in relative generators: there are finitely many edge representatives, and changes between their lifts are accounted for by the vertex stabilizers. Exact restriction of the elliptic systems identifies these vertex stabilizers with the allowed peripheral moves. Thus the orbit metrics on both cores are quasi-isometric to the same intrinsic relative Cayley metric of K .

For each representative vertex choose an image fixed by its stabilizer in the other tree; such a vertex exists by exact ellipticity. Extend these choices equivariantly and then over edges. Finite quotient gives bounded choices, coarse inverses, and the asserted quasi-isometries. For a fixed finite diagram, first compare the maps and their composites on the core of the common

window spanned by that diagram. They are equivariant for its group and agree within bounded distance on a finite set of orbit representatives. Property (E2) makes this bound uniform for the bounded shape of the diagram. A further window has its core inside this common core, so restriction inherits the same bound; no coboundedness constant depending on that further window is introduced. This proves compatibility and preservation of the boundary points of further window cores.

We use the following precise convention for arcs. Join the images of the endpoints by a geodesic and delete a fixed bounded segment at each end. The remaining arc lies in every transferred line containing the original arc. The trimming constant follows from stability of quasi-geodesics in a tree. Under immediate inverse comparison, the transferred arc is contained in a bounded enlargement of the original arc; in particular its length is at most the original length plus a uniform additive constant. $\square$

8.2 The corridor estimate and phase uniqueness

Lemma 8.3 (Flaring). *Under the expansion package there are integers $b, R > 0$ with the following property. An arc of length at least R carried across the positions $[-b, b]$ has trimmed image of length at least 100 times its original length in one of the comparisons $0 \rightarrow -b$ and $0 \rightarrow b$.*

Proof. We establish the corridor estimate with its dependence on the window length made explicit. Fix m . Work with the stabilizer of the comparison window through $[-m, m]$, including its short tails. Its core is cobounded, so the endpoints of a carried arc at position 0 can be approximated by two orbit points. The error may depend on m but not on the arc. Translate one fixed section through this window by the corresponding two group elements. These sections form the two sides of a quadrilateral and cross each sampled fiber exactly once. Their total length, and all endpoint approximation errors, are bounded in terms of m , uniformly over window types.

Let l_j be the distance between the untrimmed comparison images at position j , with l_0 the original length. Use the fixed comparisons through the short windows, rather than comparisons chosen afresh for a larger carrying window. By Lemma 8.2, the horizontal width between the sections is comparable to l_j with uniform multiplicative constants and an additive error depending on m .

For $d \leq m$, fill the loop formed by the two sections and horizontal paths in the rows $-d, d$ by a linear-area Rips disc in the hyperbolic total space [12, Proposition III.H.2.7]. Here the Rips parameter and the linear isoperimetric constant are fixed. We explain why its area bounds the sum of the intervening fiber widths. Extend projection to the underlying tree across the edges of the triangulation by bounded paths and across triangles by tripod fillings. A transverse cut through an intervening tree edge has two boundary crossings, one on each section. A component of its preimage joins those crossings: preimage tracks pair boundary crossings, and these are the only two. A small perturbation arranges transversality without altering the boundary projection.

Label consecutive track crossings by points of the relevant actual fiber within bounded total-space distance of the triangle vertices. Such points exist because a bounded path whose projected endpoints lie on opposite sides crosses the fiber. Two resulting nearby fiber points have uniformly bounded intrinsic distance. Indeed a bounded total-space path between them has a bounded projected edge word; successively remove its backtracks, using the uniform quasi-isometric edge embeddings. This bounds the intrinsic fiber displacement by a constant depending only on the fixed Rips parameter. Each triangle meets only a uniformly bounded number of sampled cuts. Summing over tracks therefore costs only a fixed multiple of the

number of triangles. The horizontal boundary lengths and the bounded section lengths give

$$\sum_{-d < j < d} l_j \leq C(l_{-d} + l_d) + C_m, \quad 1 \leq d \leq m, \quad (8.1)$$

where C is independent of m .

Put $P_d = \sum_{|j| \leq d} l_j$. Equation (8.1) implies

$$P_d \geq (1 + C^{-1})P_{d-1} - C_m/C.$$

With m fixed, iteration shows that the two moving ends at m dominate l_0 by a factor tending exponentially to infinity with m , apart from an additive error depending on m . First choose $m = b$ to make this factor exceed the desired constant, and then choose the threshold for l_0 to absorb the additive error and the two trimming errors. Increasing that threshold to an integer gives R . $\square$

Choose R still larger so that every bounded-comparison additive error and trimming loss used below is small compared with R . Comparisons send vertices to vertices, and we take integral trimming lengths. An oriented arc of length $2R$ has an *expanding event of sign ε* if its trimmed image under the εb comparison has length at least $3(2R)$. After such an event, expansion continues in that sign, with factor at least 100 at each subsequent b -step for which the arc is carried: immediate inverse comparison is too short to be the expanding direction in Lemma 8.3. By partitioning a long expanded arc into length- $2R$ pieces, the triangle inequality and the uniform upper Lipschitz bound show that arbitrarily many disjoint expanding events occur far along an infinite expanding trajectory.

The next lemma isolates the cyclic obstruction used in phase uniqueness. It also supplies the infinite-intersection assertion needed later: a rational line surviving all finite windows will have a common nonzero translation. The finite datum in the proof is an actual pair of continuation windows with a marked edge, not an isomorphism type of free groups.

Lemma 8.4 (Cyclic indices along a trajectory). *Assume the expansion package. Fix a bounded sampling step, and let W_j be the successive short windows, with sampled edges e_j , along an interval of a monotone trajectory. The interval may be finite, one-sided infinite, or bilateral. Put*

$$H_j = \text{Stab}(W_j), \quad K_j = \text{Stab}(\text{Hull}(W_j \cup W_{j+1})).$$

Let $C = \langle c_0 \rangle \leq L$ be a maximal cyclic subgroup of G . Suppose $C \cap K_j$ is nontrivial and loxodromic in the short-window trees for every adjacent pair. Write

$$C \cap H_j = \langle c_0^{q_j} \rangle, \quad C \cap K_j = \langle c_0^{m_j} \rangle, \quad q_j, m_j > 0.$$

There are uniform constants B, E such that both indices m_j/q_j and m_j/q_{j+1} are at most B , and at most E of these indices differ from one over the whole interval. In particular all partial products of their ratios lie between B^{-E} and B^E . For an infinite interval some nonidentity element of C belongs to every K_j .

Proof. Let $V_j = \text{Hull}(W_j \cup W_{j+1})$ and set $r_j = c_0^{q_j}$. The axis of r_j in R_j lies in the K_j -core, since a nonzero power of r_j belongs to K_j . That core has uniformly finitely many edge orbits for this bounded window shape. Among successive translates of a marked oriented axis edge by r_j , two therefore belong to the same K_j -orbit after uniformly many steps. An oriented edge has trivial stabilizer, so the element identifying them is the corresponding power of r_j . This bounds $[C \cap H_j : C \cap K_j] = m_j/q_j$. The argument in R_{j+1} bounds the other index.

Suppose $m_j/q_j > 1$. Then r_j fixes W_j but not V_j : a height-zero element preserving the finite monotone window setwise fixes it pointwise. Consequently V_j and $r_j V_j$ are distinct

actual continuations through W_j , with their terminal positions at the same height. Their stabilizers K_j and $r_j K_j r_j^{-1}$ both contain $C \cap K_j$. By (E4) their intersection is cyclic, and its maximal cyclic overgroup in G is exactly this same C .

Mark an oriented regular edge on their common axis in R_j . There are only finitely many orbits of the resulting actual marked pairs. Indeed (E2) gives finitely many continuation-window orbits; their finite core edge quotients give finitely many orbits after adding an edge marking. Normalize the short window and that edge. Since its stabilizer is trivial, only finitely many actual lifts of each continuation remain. Hence only finitely many pairs remain, with their actual subgroup embeddings retained. The same argument applies to a nonunit index at the W_{j+1} -end, recording which end is marked.

If one marked-pair type occurred at two different heights, its transporter $g \in G$ would conjugate the two cyclic intersection groups just described. Both have maximal cyclic overgroup C as an actual subgroup of G , so g would commensurate C . But equivariance of height gives $h(g) = h(e_{j'}) - h(e_j) \neq 0$, which is impossible. A station has only the two adjacent comparison directions. The finite list thus bounds the total number of nonunit indices on the whole interval, including when the interval is all of $\mathbb{Z}$. This proves the bound E and the partial-product assertion.

Except at these finitely many indices, $q_j = m_j = q_{j+1}$. Thus only finitely many different integers q_j, m_j occur, even on an infinite interval. A common multiple M gives $1 \neq c_0^M \in K_j$ for every j . $\square$

Corollary 8.5 (Persistence of a rational line). *Suppose a bilateral trajectory carries, in R_0 , the whole axis of a loxodromic element $c \in H_0$ through every finite subwindow containing position zero. Some nonzero power of c fixes the entire trajectory pointwise.*

Proof. For each finite window, its core contains this axis and has finite edge quotient. Repetition of a marked axis edge under powers of c therefore puts a nonzero power of c in its stabilizer. This conclusion initially allows the power to depend on the window. Equivariant comparisons transfer the axis to any other station as the axis of that same actual element, after taking a power lying in the comparison group. Thus the maximal cyclic overgroup C of c meets every adjacent comparison group nontrivially and loxodromically. Lemma 8.4, applied over all $\mathbb{Z}$, supplies one common nonzero power. The comparison windows cover the trajectory, so that power fixes it pointwise. $\square$

Lemma 8.6 (Phase uniqueness). *There is a uniform depth N such that two trajectories carrying the same expanding event of length $2R$, and agreeing through depth N in its expanding direction including the short comparison windows, cannot subsequently diverge in that direction while still carrying the arc. One or both trajectories may be finite. The same conclusion holds for arcs carried by all finite subwindows of an infinite trajectory.*

Proof. The obstruction to late divergence is a long periodic overlap. We pull its periods back to the original event. Adjacent primitive periods can change by a bounded index, but Lemma 8.4 bounds the number of such changes. Exponential expansion will therefore exceed the total loss in this pullback.

Suppose there is a first divergence. At the last available common b -station before it, the expanded arc lies in the cores of two bounded continuation groups. The continuation windows need extend only far enough to exhibit the first divergent edges; the distance from the station to these edges is bounded. No comparison beyond the divergent edges is required. Their intersection is at most cyclic by (E4).

We record the finite-state overlap argument. For an oriented edge in the intersection of two cores, record its orbit in each core. There are uniformly finitely many such pairs. If two edges have the same pair of records, the two elements carrying one edge to the other

coincide, because an oriented edge has trivial stabilizer. Their common value belongs to the intersection of the two groups. Thus the overlap has uniformly finitely many edge orbits under that intersection. If the intersection is trivial, the overlap has uniformly bounded diameter. A nontrivial cyclic intersection can also be elliptic in a relative tree. In that case project one of its fixed vertices to the invariant overlap; the projection is fixed too. A radial segment from this point contains at most one edge of each cyclic orbit, since the action preserves distance from the fixed point. The finite edge-orbit bound therefore bounds the radius of the overlap. Thus a sufficiently long overlap forces the cyclic intersection to be loxodromic. It then runs along its axis apart from bounded initial and terminal pieces: an off-axis hanging segment cannot contain a repeated oriented-edge state. Its period is uniformly bounded.

Let r_j be the primitive translation of that axis in the short-window group at the last common station j , oriented with the arc. The arc contains exponentially many complete periods. Pull these periods back through the preceding comparison group. If sufficiently many remain, repetition in its finite-edge-quotient core gives

$$r_j^a = r_{j-b}^{a'} \quad \text{as actual group elements,} \quad 1 \leq a, a' \leq C',$$

where each side is the least positive power of the corresponding primitive root lying in the comparison group. Exact ellipticity ensures that the common element remains loxodromic on the other side. Repetition bounds the indices by the uniform edge-orbit count. Equivariance of the coarse inverse on the axis implies that if p periods remain on the later side, there are at least $(a'/a)p - O(1)$ periods on the earlier side. Removing partial blocks and end margins costs only a uniformly bounded number of periods.

Every equality above identifies powers as actual elements of G . All the roots reached in the pullback therefore lie in one maximal cyclic subgroup $C \leq L$. Lemma 8.4 applies on every interval already reached, with bounds independent of its length. In particular the product of the ratios a'/a , on any part of the pullback and in either direction, lies between B^{-E} and B^E .

Let t be the number of backward steps to the initial event and p_t the number of terminal periods. The recurrence just proved gives, after t steps,

$$p_0 \geq B^{-E} p_t - C_0 B^E t$$

for a uniform trimming constant C_0 ; the same lower bound with fewer losses holds at every intermediate station. The terminal period has uniformly bounded length, while the event has expanded by at least 100^t . Hence $p_t \geq c_1 100^t - C_1$, with $c_1 > 0$ uniform. For sufficiently large t , all intermediate period counts exceed the finite repetition threshold required to take the next backward step, and the displayed lower bound exceeds $2R$. But each initial period has length at least one, whereas the initial event has length $2R$. This is the contradiction. Choosing the common depth N to allow this many steps proves the lemma. All terminal constants came from the bounded windows exhibiting the first divergence, so the proof is uniform for finite trajectories and for finite-window carrying of infinite trajectories. $\square$

Lemma 8.7 (Changing the sampling offset). *For an ordinary bilateral trajectory, an expanding event of a given sign produces events of that sign at every sufficiently distant row in that direction, not merely on its stride- b subsequence.*

Proof. There are only finitely many offsets of the stride- b sampling. Fix one. If at a sufficiently distant row on this offset there were no events of the given sign on the transferred lines, partitioning long arcs into length- $2R$ pieces bounds their transfer in that sign. Lemma 8.3 then forces expansion by at least 100 in the opposite sign, and this expansion persists.

Start with a long arc already expanding on the original subsequence, transport it forward m steps and then across the bounded offset, and transport it backward m steps on the

offset subsequence. The resulting trimmed arc lies, up to bounded error, inside the direct bounded-offset transfer of the starting arc. This follows from Lemma 8.2 on the common finite diagram, and stability of lines under the comparisons. Choose m large enough that the two opposed exponential expansions exceed the upper Lipschitz constants of the bounded-offset comparisons. With this m fixed, move the initial row far enough along the original expanding trajectory to dominate every additive error for that finite diagram. The length comparison is then impossible. Taking the maximum over the finitely many offsets proves the assertion. $\square$

9 Actual trajectory clusters

Outside the acylindrical case, we use expansion to construct a G -invariant system of periodic quasiconvex groups, finite modulo G , whose height kernels contain all sufficiently long noncyclic path stabilizers. We work in the coarse actual tree T_0 . The ordinary expansion package follows from the actual splitting and path-inertia statements of Section 5. In particular the total ordinary space is quasi-isometric to G , the horizontal trees are locally finite free Cayley trees, and Proposition 5.7 supplies joint homology injection and its subgroup-wise version. The sum of reduced ranks for paths of a fixed shape is at most $k - 1$. The induced-coefficient content of that lemma, not merely this numerical bound, will be used below.

There are three steps. We first obtain periodic end stabilizers from branching surviving subtrees. We then connect positive and negative ends when they support a common bilateral trajectory. Finally, we show that each connected component of this switch graph is a tree and that its stabilizer captures every sufficiently long path group seen by that component.

9.1 Surviving cores and periodic ends

For an infinite monotone path all of whose relevant finite windows have noncyclic stabilizer, intersect their decreasing cores in a fixed horizontal tree. Call the intersection the *surviving subtree*. We only consider an event or a branch when it belongs to this intersection. Local finiteness implies that any nonempty surviving subtree is a union of full lines: extending a finite surviving arc in each finite-window core and taking subsequences gives extensions in both directions. The same argument preserves a specified surviving arc. For a bilateral path, Lemma 8.2 transfers the endspace of its surviving subtree between rows.

This is a geometric intersection of subtrees, not yet the core of the intersection of their groups. A surviving line need not initially come with a common nontrivial translation. A trajectory or tail is called *branching* when its surviving subtree has a branch vertex, equivalently three distinct ends. Comparisons make this condition independent of the sampled row. The next approximation lemma supplies the group-theoretic control needed alongside these geometric data.

Lemma 9.1 (Decreasing core graphs). *Let K_n be a decreasing sequence of finitely generated subgroups of one free row, with uniformly bounded ranks, and put $M = \bigcap_n K_n$. Then M is a finite-rank free group. Its core graph embeds in that of K_n for all sufficiently large n , and its first field homology injects there.*

Proof. Choose a finite subgraph of the core graph of M , and choose actual horizontal lifts of its vertices and edges. An identification of two chosen vertices in a K_n -quotient has one possible transporter, because a horizontal vertex has trivial stabilizer. If it does not belong to M , it fails to belong to some K_n and to every later group. There are only finitely many candidate identifications, so the chosen finite subgraph eventually embeds. An infinite-rank core graph would have finite subgraphs of arbitrarily large rank, contradicting the uniform rank bound on K_n . Thus M has finite rank. Apply the same argument to its entire finite

core graph. An embedded connected subgraph gives a free factor, after choosing a connecting path to the basepoint, and consequently an injection on first field homology. $\square$

Lemma 9.2 (Branch states and tail groups). *There are finitely many orbits of branching bilateral trajectories and of branching one-sided tails. Every branching bilateral trajectory is periodic under a nonzero-height element; every branching one-sided tail is eventually periodic.*

For a monotone ray from an edge e to an end z , let

$$M_{z,e} = \text{Stab}([e, z))$$

be its pointwise stabilizer. This is a finite-rank free group. Its homology injects in sufficiently long finite-prefix homologies, and the finite-path restriction rule also holds on restricting within a later tail group, with the same common-level and orbit conventions. If some $M_{z,e}$ is noncyclic, let Y_z be the union of all monotone rays to z having noncyclic stabilizer, and let $D_z = \text{Stab}_G(z)$. Then:

- (i) Y_z is connected, D_z has a height translator with axis in Y_z , and tail ranks along such an axis are constant;*
- (ii) $D_z \cap L$ is transitive on the Y_z -edges of each fixed level;*
- (iii) D_z is quasiconvex in G and virtually an ascending mapping torus of a finite-rank free group; in particular it is virtually compact special.*

Proof. We first produce translators using a finite set of trajectory orbits. The core-graph lemma then gives the tail stabilizers and their restriction maps. These inputs establish transitivity along levels and finally ambient quasiconvexity.

In a noncyclic finite-rank free core, the number of branch-vertex orbits is at most twice the reduced rank. At any finite depth, count pairs consisting of a path and a branch vertex of its core, modulo the horizontal free vertex action. Joint path inertia bounds this count uniformly by $2(k-1)$. A specified horizontal vertex has trivial stabilizer, so marking it removes all ambiguity in the transporter. Normalize the sampled edge to a fixed representative edge at the chosen level and a surviving branch vertex to one horizontal vertex. More than $2(k-1)$ distinct branching infinite paths in these coordinates would already be distinct at some common finite depth. Thus there are finitely many L -orbits of branching bilateral trajectories with a sampled edge at this level, and finitely many L -orbits of branching tails starting there. These unmarked trajectory orbits are the states; branch markings only bound their number and are not part of a proposed canonical shift of vertices.

Choose a height-one element t . For a positive trajectory, move the sampled position to its next edge and translate the whole picture by t^{-1} to restore its height. On bilateral states this gives a bijection, with inverse obtained from the preceding edge and t . On positive tail states first delete the initial segment, then translate by t^{-1} ; this is a map of the finite state set. Both operations are well defined on L -orbits since L is normal, and preserve branching because comparisons preserve the three ends of a surviving tripod. A repeated bilateral state gives an element of nonzero height preserving the same actual trajectory. A repeated tail state gives an element taking a later tail to a proper subray of itself, hence translating its eventual axis. Negative tails use the reversed operation. This proves periodicity and eventual periodicity without choosing a transported branch vertex.

The tail group $M_{z,e}$ is the intersection of its decreasing finite-prefix groups. Their ranks are bounded by path inertia, so Lemma 9.1 proves finite rank and the asserted homology injection into sufficiently long prefixes. For a finite family in a prescribed backward restriction sum, choose these depths simultaneously and compose with Proposition 5.7. When the restriction takes place within a later tail group $M_{z,f}$, use the subgroup-wise version of that

lemma: intersection with the finite segment from an earlier edge to f is exactly the earlier tail group. This proves the precise inherited restriction rule used below.

Rays to a common end coalesce; their union Y_z is consequently connected. A noncyclic tail group gives a branch in the surviving subtree. Eventual periodicity of that tail produces a height translator in D_z . Its axis has a tail in Y_z , and invariance under its powers puts the whole axis in Y_z . Tail groups increase as the starting edge moves towards z . The inherited inertia gives injective first-homology maps for these inclusions. Periodicity makes their dimensions repeat, so those dimensions, and hence their ranks, are constant along the axis.

Compare an arbitrary Y_z -edge at a given level with the axis edge at that level. Their rays meet at a later axis edge f . Inside $M_{z,f}$, the axis predecessor already contributes its full first-homology dimension. The joint restriction rule within $M_{z,f}$ therefore excludes a second noncyclic predecessor orbit. A transporter realizing the common orbit belongs to $M_{z,f} \leq D_z \cap L$. This proves (ii).

A height-zero element preserving z fixes a tail of every ray to z . Consequently $D_z \cap L$ is the increasing union of tail stabilizers along the axis. After passing to a finite-index subgroup to select a height period, a translator together with one finite-rank tail group gives the ascending mapping-torus presentation. At a vertex of Y_z , the stabilizer equals the stabilizer of its parent edge towards z . It is quasiconvex in the vertex row by the preceding finite-rank assertion and Lemma 5.10. An edge leaving Y_z has at most cyclic intersection with D_z , since a noncyclic such intersection would create a second outgoing continuation, contrary to the reversal restriction. The action on Y_z has finite quotient by (ii) and periodicity. Lemma 5.9 now gives ambient quasiconvexity of D_z . It is therefore hyperbolic, and Theorem 4.1 makes it virtually compact special. $\square$

9.2 Active colors and switches

Color an expanding event on an actual bilateral trajectory by its expanding end and its sign. Lemma 8.6 says that sufficiently large fixed-window data determine this color uniquely. Such colors are called *active*. A *switch trajectory* is a bilateral trajectory whose surviving subtree supports events of both signs at some row.

Lemma 9.3 (Periodic switches and noncyclic active tails). *Every switch trajectory is periodic. For an active color z occurring at an edge e , the tail group $M_{z,e}$ is noncyclic. There are finitely many orbits of active colors and of switch trajectories. A switch trajectory has both signs at every row.*

Proof. Every vertex of a surviving subtree lies on a full line and admits a centered event in at least one sign, by flaring. If both signs occur at a row, connect their centers in the surviving tree. There are opposite centered events at distance at most one: along the connecting vertex path the available sign changes, or a vertex admits both signs. The fixed-window data of such a nearby pair have finitely many orbits, and phase uniqueness determines both ends from these data.

Suppose first that a switch has both signs at infinitely many rows of a stride- b subsequence. A repeated nearby-pair state determines the same two ends and hence the same bilateral path. The change of height gives a translator, proving periodicity. Otherwise, persistence of its initial positive and negative events gives purely positive rows far in the positive direction and purely negative rows far in the negative direction.

At a pure row, a surviving line lies in a bounded neighborhood of the orbit of the corresponding tail group. Indeed, its centered events have only finitely many marked types. An element transporting one occurrence of a type to another fixes the marked row and preserves the uniquely determined expanding end. It therefore belongs to the tail group.

Choose one representative for each occurring type. Every vertex of the line lies within a fixed distance of a translate of one of those finitely many representatives. This proves the assertion and puts the two endpoints of the line in the limit set of the tail group.

This limit-set conclusion transfers back to an earlier row. Intersect the later tail group with the finite connecting-path group. The result is the earlier tail group, expressed in the later frame. Both groups are finitely generated subgroups of a free row, hence quasiconvex, and their limit-set intersection is exactly the limit set of their intersection. The equivariant comparison transfers this intersection of endspaces back to the earlier row. Apply the argument on both sides of the switch. The endpoints of a central surviving line then lie in the limit set of the bilateral pointwise stabilizer.

If the surviving subtree is not already branching, it is a line; the last conclusion makes it a rational cyclic axis. A nontrivial translation of that axis fixes every window. Every finite-window group is noncyclic, so its core branches somewhere on this axis. Translate the branch point into a fixed translation period. Local finiteness and nesting leave a branch direction present at every depth. This gives a surviving branch, a contradiction. The trajectory is thus branching and periodic by Lemma 9.2.

Now consider an active event on a periodic trajectory. Expansion produces arbitrarily many disjoint events of the same sign. Translate them back by periods to one central row. Repeated marked event types at different horizontal positions give a nontrivial element of $M_{z,e}$. This group cannot be cyclic: its tail groups are nested and commensurable if cyclic, and the period translator would commensurate a nontrivial horizontal cyclic subgroup.

For completeness, a nonperiodic single-sign event requires a separate argument. Far along its expanding stride- b subsequence the rows are pure in that sign; otherwise the nearby-pair argument would make the trajectory periodic. The pure-row limit-set argument and transfer back put the endpoints of every surviving line in the limit set of $M_{z,e}$, so this group is nontrivial. If it were cyclic, the line would be rational. That rational line belongs to every finite-window core by the definition of the surviving subtree. Corollary 8.5 therefore supplies one nontrivial translation fixing the whole bilateral trajectory. In particular this conclusion uses the index bound on both directions together, rather than choosing a different power for each finite window. The preceding branch-in-a-period argument again produces a surviving branch, and hence periodicity, a contradiction. Therefore every active tail group is noncyclic.

A nearby pair determines a switch, so there are finitely many switch orbits. A single fixed-window event determines its color, giving finitely many color orbits. Finally Lemma 8.7 supplies both signs at all sufficiently far appropriate rows on a periodic switch. Translation by periods brings those signs to any prescribed row. $\square$

Form the actual bipartite *switch graph* $\mathcal{C}$: its vertices are active positive and negative ends, and each switch trajectory joins its two ends. Retain isolated active vertices. A pair of ends determines at most one bilateral path in the tree, so there are no parallel edges. The graph has finitely many vertex and edge orbits by Lemma 9.3.

Lemma 9.4 (The switch graph seen by a long window). *There is a uniform depth such that the following holds for every centered monotone window W whose two endpoints are at least that far from its sampling center and whose pointwise stabilizer is noncyclic. The colors realized in the central core of $\text{Stab}(W)$ are exactly the active ends z with $W \subset Y_z$. These vertices, together with the switch trajectories traversing W , form a nonempty connected tree, denoted $\mathcal{C}(W)$. If it is an isolated vertex locally, that vertex is isolated in $\mathcal{C}$.*

Proof. First justify the passage from a finite window to bilateral witnesses. A local state consists of a centered event, or a pair of events with centers at distance at most one, together with a fixed-size path neighborhood large enough for phase uniqueness. There are finitely

many normalized states. More generally, fixing an actual regular horizontal marking, there are finitely many carried extensions of any prescribed finite depth: the relevant paths have finitely many orbits and their cores have finite edge quotients, while the marking has trivial stabilizer. Thus the extension sets form a finitely branching inverse system.

If a given state occurs in arbitrarily deep window cores, choose compatible extensions successively. The resulting bilateral path carries its specified event or pair in every finite subwindow. Local finiteness of the horizontal tree preserves the full lines carrying those arcs. States that do not persist in this way have a maximum possible depth. Taking a maximum over the finitely many short states gives a uniform depth beyond which every state in question has a bilateral witness. The witness is initially required to agree only on the fixed-size neighborhood recorded in the state. Lemma 8.6, applied to the witness and the finite window, then forces its expanding ray to agree throughout the corresponding side of W . This last step is what permits use of a fixed-size state for an arbitrarily long W .

Every vertex in the finite-window core admits a centered event. Assign its expanding color using a bilateral witness. Phase uniqueness makes the assignment unambiguous and equivariant under $\text{Stab}(W)$. Adjacent same-sign labels agree: use a common bilateral witness for the pair. Adjacent opposite-sign labels give a switch through W . Connectedness of the core therefore connects all recorded colors in the switch graph. A recorded color either participates in a switch or is the unique color and is preserved by the noncyclic window group. In the switch case, periodicity and Lemma 9.3 imply $W \subset Y_z$. In the unique-color case, the ray to that end extends W monotonically; the window group fixes that end and W , and hence its entire continuing tail. The tail stabilizer is noncyclic, so again $W \subset Y_z$.

Conversely, suppose $W \subset Y_z$. Take events far along an expanding trajectory for z , and use periods and Lemma 8.7 to bring them to the required levels. By the transitivity in Lemma 9.2, match the starting edge on the side away from z . The path towards z from that edge is unique. Taking the original event far enough along its expanding part ensures that its comparison subwindow lies on this ray in Y_z ; it therefore realizes the desired event in the central core of W . The same argument applied to switch paths shows that a locally isolated occurrence has no global switch incidence.

It remains to exclude a cycle. At the center, use the convex cores of the tail groups $M_{z,e}$ for the colors just obtained, with one fixed thickening. Nearby paired events are uniformly close to the corresponding tail cores. To verify uniformity, normalize their marked fixed-window data; there are finitely many types, each determining the actual end, and the tail group acts coboundedly on its core. Every switch through W has such nearby paired events by periodicity and Lemma 8.7. Thus every switch edge gives an intersection of the two thickened cores.

For sufficiently deep W , the thickenings of distinct same-sign cores are disjoint. Otherwise normalize a pair at bounded distance near one horizontal origin. There are only finitely many such actual lifts: use finite color orbits, edge transitivity on Y_z , and coboundedness of each tail core. Increasing depths then give arbitrarily long agreement of the corresponding rays, forcing equality of their ends by phase uniqueness. This contradicts distinctness. Finally, the intersection graph of subtrees of a tree is chordal, by convexity and the Helly property. With no same-sign intersections it is bipartite, hence a forest. The switch graph under consideration is a subgraph of this intersection graph; extra opposite-sign intersections cause no problem. It is connected by the preceding argument, and therefore is a tree. $\square$

9.3 Splitting marks and simultaneous free factors

We next pass from individual periodic ends to the full stabilizer of a connected component of switches. To see this group through finite windows, temporarily distinguish incidences at an end according to where their rays cross a chosen level. Widely separated levels cut the

component into the local trees already constructed. Moving the levels towards the respective ends joins those pieces. We will show that these joins neither create circuits nor increase the rank of their stabilizers. This is where the actual homology injections, rather than only the rank bound, enter.

Fix a nonisolated component of $\mathcal{C}$, let A be its full stabilizer, and put $S = A \cap L$. For integers j, l , split each positive vertex by also remembering the level- j edge of each incident trajectory, and split each negative vertex by remembering its level- l edge. Incidences with the same end and the same additional mark remain together. Denote the resulting graph by $\mathcal{C}_{j,l}$, retaining its S -action. Increasing j and decreasing l coarsens the graph: marks move towards the common end and eventually coincide. All these actions preserve the bipartition and have no inversions.

For j sufficiently smaller than l , its components are precisely the local trees $\mathcal{C}(W)$ for the paths across $[j, l]$ just considered. Their stabilizers in S are exactly

$$K(W) = \text{Stab}(W).$$

Indeed the two marks determine the segment W , and equivariance gives both inclusions. The split vertex stabilizers are $M_{z,e}$ and the split edge stabilizers are

$$M_\tau = \text{Stab}(\tau),$$

the pointwise stabilizers of the bilateral switch trajectories. The same finite-core approximation used for tails shows that M_τ is finitely generated free. It is noncyclic or trivial: a nontrivial cyclic M_τ would be normalized by a period translator of τ , contrary to cyclic noncommensuration.

Lemma 9.5 (Orbit control under coarsening). *The components of each $\mathcal{C}_{j,l}$ form one S -orbit. Their quotient vertex, edge, and incidence counts are finite and do not change as j increases and l decreases, including after the two levels cross. The corresponding vertex and edge ranks do not change either.*

Proof. Consider two incident switch trajectories sharing an end. Orient their period translators towards that end, so their heights are negative when the common end is negative, and take positive powers having equal heights. Far enough towards the shared end, any prescribed finite subwindows of the two trajectories lie on their common ray. Move the first subwindow there using its period translator and move back using the other translator. Both translators stabilize the original component because they preserve an edge in it. Their composite therefore belongs to A ; it has height zero, hence belongs to S , and identifies the relevant split components. Connectivity of the original component propagates this comparison along any finite path of switch incidences. Hence all split components lie in one S -orbit.

For a fixed end, $D_z \cap L$ is transitive on the eligible marked edges at a level by Lemma 9.2; their stabilizer ranks are constant at all levels. More explicitly, an A -orbit of ends contributes

$$[h(A) : h(D_z)]$$

S -orbits of marked ends, independently of the marking level. An A -orbit of switch trajectories contributes $[h(A) : h(\text{Stab}_A(\tau))]$ S -orbits of split edges. The marks on a switch are its uniquely determined edges at the two chosen levels, and its pointwise group M_τ does not change. Each split edge has exactly one incidence in each bipartition, so the same edge list also controls the two incidence lists. All these indices are finite because the stabilizers have height translators.

There are finitely many A -orbits: if a group element carries one end or switch in this component to another, it carries their entire connected components to one another and hence

belongs to A . Thus the finite global orbit lists restrict to finite A -lists. Because the split components form one S -orbit, the lists just counted are exactly the quotient lists in one component under its stabilizer. Their counts and ranks are therefore independent of the levels. They persist under coarsening, and every resulting component comes from components at a sufficiently separated pair of levels. $\square$

Lemma 9.6 (Simultaneous incident free factors). *At every vertex of every split action, the nontrivial incident edge groups, indexed by incidence-orbit representatives, are simultaneous free factors of its vertex group, up to separately chosen inner conjugations.*

Proof. Fix a marked end (z, e) and write $M = M_{z,e}$. Start a long path at e towards z , and compare at a center sufficiently far along it and far from both endpoints. By periodicity and the finite number of switch orbits, the core graphs of the nontrivial M_τ have uniformly bounded sizes at these centers.

In the core graph of the finite-path group K , include the core graphs for *all* switch lines through that path. They eventually embed jointly and disjointly modulo their K -orbits. Here is the identification test. An alleged overlap of chosen actual horizontal vertices has a unique transporter. Normalize at that vertex. Finite switch orbits and the uniform core-size bound leave only finitely many normalized switch lifts nearby. Distinct such bilateral paths have only a bounded common centered subpath. Once the comparison path is longer, any persistent identification must preserve the entire switch line. An element of K has height zero, so if it preserves that line it fixes it pointwise. Consequently its line stabilizer is exactly M_τ . These are precisely the identifications already made in the M_τ core graph. This proves both embedding and disjointness. The same finite test, applied to the whole finite collection of core representatives, is uniform.

Disjoint embedded connected core subgraphs in a free graph give simultaneous free factors of its fundamental group, after choosing connecting paths. Therefore the indicated groups are simultaneous free factors of K . Since $M \leq K$, apply the Kurosh subgroup theorem to this free-product decomposition. Within a K -orbit of switch lines, the desired incidences modulo M correspond to distinct double cosets

$$M \backslash K / M_\tau.$$

Their pointwise groups are already contained in M , so the Kurosh intersection is the entire conjugate edge group. This yields exactly the requested simultaneous free factors in M , with the independent conjugations arising from its choices of connecting paths. $\square$

Lemma 9.7 (A homology obstruction to a proper free quotient). *Let F_r map onto a nontrivial subgroup $B \leq L$, and use D_L coefficients. If this map has nontrivial kernel, then*

$$\dim_{D_L} H_1(B; D_L) \leq r - 2.$$

In particular, it cannot surject onto a group containing a subgroup whose ambient homology image has dimension $r - 1$.

Proof. Choose a shortest nontrivial freely reduced word in the kernel. Its proper prefix vertices in the quotient are distinct; a repeated one would give a shorter nontrivial kernel word. Its closed labelled path has no cyclic backtrack, since cyclic cancellation would again shorten the kernel word. At least one evaluated Fox derivative is nonzero. More explicitly, cancellation of a contribution to a generator derivative would require an opposite traversal of the same oriented labelled edge. This repeats prefix vertices unless it is the forbidden backtrack. Thus this relator supplies a nonzero 2-boundary over the injective group ring inside D_L . The 1-boundary for the r generators has rank one, since B is nontrivial. Consequently

its kernel has dimension $r - 1$, and the nonzero 2-boundary lowers first homology by at least one. Adding any remaining relations can only lower that dimension further. $\square$

Proposition 9.8 (The component trees). *Every component of the actual switch graph is a tree. For a nonisolated component with stabilizer A and $S = A \cap L$, there is an integer $r \geq 2$ such that S is a nested union of rank- r free groups and*

$$\dim_{D_L} H_1(S; D_L) = r - 1.$$

These approximants and S have injective ambient first-homology maps. Moreover A is finitely generated, quasiconvex, virtually compact special, and field-acyclic.

Proof. There are two steps to excluding circuits. Simultaneous incident free factors make the abstract incidence graph of groups free of a fixed rank r . Its natural quotient onto a component stabilizer cannot be proper, because an old path subgroup already contributes $r - 1$ independent ambient homology classes. We apply this first to finite coarsenings, then exhaust the unsplit component.

For separated marks, a split component is the tree $\mathcal{C}(W)$ and its stabilizer is the noncyclic free group $K(W)$. Put $r = \text{rk } K(W) \geq 2$. For arbitrary coarsened marks, form the abstract incidence graph of groups for a component and its stabilizer. Lemma 9.6 makes its fundamental group free: eliminate each glued free factor on one side of an edge, retaining the remaining free factors and the usual stable letters. Its rank is constant under coarsening, because the finite quotient incidence counts and all vertex and edge ranks are constant by Lemma 9.5. At separated marks this rank is r .

For a group acting without inversions on a connected graph, the incidence graph-of-groups fundamental group surjects onto the acting group, with kernel the ordinary fundamental group of the acted-on graph. This is seen by lifting the action to its universal covering tree. Apply this to a coarsened component. Its stabilizer contains an old $K(W)$, whose homology injects into $H_1(L; D_L)$ with dimension $r - 1$ by path inertia. Lemma 9.7 excludes a proper quotient of the abstract rank- r free group. Hence the surjection is an isomorphism and the coarsened component is itself a tree.

Every finite circuit in the unsplit component would lift to a sufficiently coarsened split graph: move its finitely many shared-end marks far enough to make the necessary marks coincide. Since no split graph has a circuit, the unsplit component is a tree. Track split components containing one chosen switch line. Their stabilizers increase. Every $s \in S$ sends that line to a line connected to it by a finite path in the original component; the finitely many required marks eventually coalesce. Therefore s belongs to one tracked stabilizer, proving that their union is S . Each is free of rank r , contains an old $K(W)$ with full ambient homology dimension, and hence has injective ambient homology of dimension $r - 1$. The maps between them, and their direct limit, preserve that injection.

The quotient of the component tree by A is finite. Its vertex groups are the end groups D_z , already finitely generated and quasiconvex. An edge stabilizer is the setwise stabilizer of a switch trajectory and fits into

$$1 \longrightarrow M_\tau \longrightarrow \text{Stab}_A(\tau) \xrightarrow{h} h(\text{Stab}_A(\tau)) \longrightarrow 1.$$

Its height image is nonzero by periodicity, and M_τ is finitely generated, including the possible trivial case. Thus edge stabilizers are finitely generated, and so is A .

The period rule, Proposition 5.11, now applies to A : its height image is nonzero and $0 < \beta(S) = r - 1 < \infty$. On the refined actual tree it supplies the finite-quotient hull required by Lemma 5.9, with quasiconvex row intersections and at most cyclic boundary intersections. Hence A is quasiconvex in G . The switch edge groups are intersections of the two quasiconvex

end groups, so they are quasiconvex in G and in A . The quasiconvex hierarchy theorem applied to this finite graph-of-groups decomposition, with its virtually compact special vertex groups, makes A virtually compact special.

Finally, every end vertex group is a finite extension of an ascending free-group mapping torus. Every switch edge group has a finite-rank bilateral kernel and a nonzero height translator; the kernel is trivial or noncyclic. These vertex and edge groups are field-acyclic. One can compute this directly by the height homology sequence: over the skew Laurent division field, translation minus identity on the finite-dimensional kernel homology is invertible, as is seen from its leading Laurent degree. The same computation covers a cyclic edge group with trivial kernel. Finite-index passage preserves the vanishing. Mayer–Vietoris on the switch-tree splitting now gives field acyclicity of A . $\square$

9.4 The initial captured system

Proposition 9.9 (Capture by periodic quasiconvex groups). *Unless G is already virtually compact special by the acylindrical case, there is an invariant system of actual objects, finite modulo G , with full stabilizers A and kernels $S = A \cap L$, having these properties:*

- (i) $h(A) \neq 0$; each A is quasiconvex, virtually compact special, and field-acyclic.
- (ii) Each S is a nested union of finite-rank free approximants of one constant rank at least two. Its field first-homology dimension is positive and finite, and its ambient homology detects every noncyclic subgroup.
- (iii) For representatives of the actual object orbits under L , the maps

$$\bigoplus H_1(S; D_L) \longrightarrow H_1(L; D_L)$$

are jointly injective. Distinct actual object stabilizers intersect in at most cyclic groups.

- (iv) There is a uniform length such that the pointwise stabilizer of every longer noncyclically stabilized actual path, in T_0 or in the refined tree T , is wholly contained in one of these kernels S .

Proof. Use the components of $\mathcal{C}$ as objects, taking D_z for an isolated color. Nonisolated components have the required finiteness, quasiconvexity, specialness, and acyclicity by Proposition 9.8. For an isolated color the increasing tail groups along a translating axis have constant positive reduced rank and exhaust $D_z \cap L$. On a sufficiently long path towards that end, the path group preserves the unique color by Lemma 9.4. It fixes the starting edge and the end, hence the whole ray, so it equals the tail group at the starting edge. This proves the corresponding approximant and homology assertions in the isolated case. Field acyclicity follows from the same ascending-torus calculation.

To obtain a simultaneous injection, choose finitely many free approximants from distinct L -orbits of actual objects. In each nonisolated component choose an older separated-window subgroup $K(W)$ inside the selected coarsened approximant. Both are free of the same rank, and $K(W)$ already has full ambient homology dimension. Its map to the approximant is therefore an isomorphism on first homology after extending to D_L . The separating marks can be taken arbitrarily far apart, so make these choices for all selected components at one common absolute level range. For an isolated end, use an earlier tail approximant at the required starting level; constant tail rank gives the same homology image, and a sufficiently long window realizes that tail group as above.

The resulting windows belong to distinct L -orbits: their equivariantly determined local switch trees lie in the selected distinct component orbits. Proposition 5.7 now injects their homologies jointly into $H_1(L; D_L)$. Replacing each window group by its approximant preserves

its ambient image and dimension. Passing to increasing unions, and testing any proposed relation in a finite collection of approximants, proves (iii)'s joint injection. Here the indexing is by actual object orbits, before any possible identification of their abstract subgroup kernels. Every noncyclic subgroup of one kernel contains a noncommensurable pair in a single free approximant. Such a pair is detected in the approximant's field homology, and the injected ambient map preserves that detection. This proves (ii).

We spell out the same-orbit intersection issue. Different L -orbit types cannot have a noncyclic common subgroup: its detected cycle would contradict their joint injection. For an L -translate of the same type, write $S = \bigcup_n F_n$ for the free approximants of rank r . Put $E = D_S \otimes_{\mathbb{Q}S} \mathbb{Q}L$. Coset independence embeds E in D_L . A nonzero E -valued cycle for the free group F_n remains a nonzero class over D_L , because its free resolution has no 2-chains. The ambient injection for F_n then prevents its image from being an E -valued boundary in L . Consequently $H_1(F_n; E) \rightarrow H_1(L; E)$ is injective. Decompose the induced module into double cosets and apply Shapiro's lemma; this gives

$$\bigoplus_{SgF_n \in S \backslash L / F_n} H_1(S \cap gF_n g^{-1}; D_S) \hookrightarrow H_1(S; D_S).$$

This is the induced-coefficient restriction behind the slot rule; a dimension estimate alone would not suffice. It applies to the coarsened free approximants because their ambient homology maps have already been proved injective. The identity double coset already has dimension $r - 1 = \beta(S)$, so every other summand has dimension zero. Its group is a subgroup of the finite-rank free group $gF_n g^{-1}$, and therefore is cyclic or trivial. For $g \notin S$, taking the increasing union gives $S \cap gSg^{-1}$ locally cyclic. A locally cyclic subgroup of a torsion-free hyperbolic group is cyclic: any nontrivial element puts the whole subgroup in its cyclic centralizer. This also shows $N_L(S) = S$, since S is noncyclic. In particular equal kernels do not create distinct objects hidden by a change of representatives.

Now let C be the intersection of two distinct full stabilizers A . Its subgroup $C \cap L$ is at most cyclic by what was just proved. If $h(C) = 0$, this proves the desired assertion. If $h(C) \neq 0$ and $C \cap L$ were nontrivial, a nonzero-height element of C would normalize a nontrivial height-zero cyclic group, contradicting cyclic noncommensuration. Thus $C \cap L = 1$ in this case, and C embeds in $\mathbb{Z}$, so is again at most cyclic.

Finally, a sufficiently long noncyclically stabilized coarse path has a deep centered window W . Its nonempty connected label graph $\mathcal{C}(W)$ lies in a single original component. The path stabilizer preserves this graph, hence that component, and has height zero because it fixes the path. It is therefore contained in the component kernel. The same argument includes an isolated color. A long noncyclic path in the refined tree is monotone by the reversal bound and projects, after reinserting collapsed shift edges, to a long monotone path of T_0 . This proves (iv).

If there are no active colors, Lemma 9.4 instead gives a uniform bound on lengths of noncyclically stabilized paths. The cyclic dictionary, Lemma 5.8, then bounds lengths of all nontrivially stabilized paths, so the refined action is acylindrical. The direct projection argument in Proposition 14.11, applicable with the already established hyperbolic rows and quasi-isometric edge maps, makes this a quasiconvex hierarchy. Its only potentially nonfree vertex is a hyperbolic root row, which itself splits over a quasiconvex cyclic subgroup with free and cyclic vertex groups. The hierarchy theorem therefore makes G virtually compact special in this case. $\square$

10 Cyclic closure of the periodic system

The captured periodic groups may still intersect cyclically. We first join these overlaps to obtain a malnormal system. The next two sections fold its actual subtrees into lines and enlarge groups when distinct lines share an end. Throughout, joint injection into $H_1(L; D_L)$ keeps the total dimension at most $k - 1$; the termination arguments must also control the operations that do not increase this dimension.

We retain the ambient torsion-free hyperbolic primitive extension group G , its height map $h : G \rightarrow \mathbb{Z}$, the height kernel L , and the actual tree T of Section 5. Coefficients and transport of orbit representatives have the conventions of that section. In particular, $\beta(V) = \dim_{D_V} H_1(V; D_V)$, and all dimensions below are unchanged on extension to an overfield. A noncyclic subgroup is *detected* when one of its normalized Fox differences $\Delta(a) - \Delta(b)$ has nonzero image in $H_1(L; D_L)$.

The actual-row inputs are Lemmas 5.5, 5.10, 5.8, and 5.9, together with Propositions 5.6, 5.7, and 5.11. We also use the support and coset descent assertions of Theorem 6.1, the semilinear localization assertion of Lemma 7.1, and the captured periodic system of Proposition 9.9. In particular, the field homology of L has dimension $k - 1$, the ambient two-boundary is injective, and sufficiently long noncyclic actual path stabilizers have already been captured. All statements below concern this actual tree, even when a new folded tree is constructed.

10.1 The closure input and its budget

Definition 10.1 (Admissible periodic system). An admissible system is a G -invariant set of objects with finitely many G -orbits, whose full stabilizers A satisfy the following conditions.

1. $h(A) \neq 0$; A is quasiconvex in G , virtually compact special, and acyclic over D_A .
2. Stabilizers of distinct objects intersect at most cyclically.
3. For $S = A \cap L$, one has $0 < \beta(S) < \infty$, every noncyclic subgroup of S is detected, and the natural map

$$\bigoplus_{d \in L \setminus \mathcal{D}} H_1(S_d; D_L) \longrightarrow H_1(L; D_L)$$

is injective. Representatives and coefficient transports are taken in the convention of Section 5.

4. There is a fixed length such that every noncyclic stabilizer of an actual path of that length lies wholly in one of the S_d .

The system supplied by Proposition 9.9 is admissible. Its *budget* is

$$\mathcal{B}(\mathcal{D}) = \sum_{d \in L \setminus \mathcal{D}} \beta(S_d) \leq k - 1.$$

This is a finite sum: each G -orbit splits into finitely many L -orbits because its stabilizer has nonzero height image.

We use the same closure operation after each end enlargement. Form a bipartite graph with the old objects as one class of vertices. The other class consists of maximal cyclic groups $C \leq G$ meeting at least two old stabilizers nontrivially; join C to precisely those objects it meets. The action at cyclic vertices is conjugation.

Lemma 10.2 (The cyclic incidence forest). *The cyclic incidence graph has finitely many G -orbits of edges and is a forest.*

Proof. Quasiconvex double-coset finiteness gives finitely many incidence-edge orbits. Suppose that a simple circuit has successive old vertices A_i and cyclic vertices C_i , with indices read cyclically. Choose $1 \neq c_i \in C_i \cap A_i \cap A_{i+1}$. Such a common power exists because both intersections with C_i are nontrivial. We will obtain a contradiction from one endpoint of the normalized path for c_i .

Use literal Fox chains based at the identity, with the same word for c_i in its two occurrences. Put

$$E_i = D_{A_i} \otimes_{\mathbb{Q}A_i} \mathbb{Q}G = \sum_{g \in G} D_{A_i} g.$$

The difference $\Delta(c_i) - \Delta(c_{i-1})$ is a cycle over E_i . Shapiro's lemma and the field acyclicity of the old group A_i give a finite two-chain b_i over this induced module with

$$\partial_2 b_i = \Delta(c_i) - \Delta(c_{i-1}), \quad \sum_i b_i = 0.$$

The second equality follows because the boundaries telescope and the ambient two-boundary is injective.

Apply the single scalar-function map $\mathcal{F}$ of Theorem 6.1(ii) to every coefficient. It commutes with the polynomial boundary matrices. Each coefficient of b_i belongs to a finite sum of $D_{A_i} g$, so its function support lies in a fixed neighborhood of A_i . The pointwise equality $\sum_i \mathcal{F}(b_i) = 0$ then confines the support of each summand to the union of the pairwise coarse intersections with the other A_j . For this finite family, each such coarse intersection lies in a bounded neighborhood of $A_i \cap A_j$. Indeed close pairs in the two groups have finitely many possible discrepancies, and pairs with one fixed discrepancy form one orbit under $A_i \cap A_j$. Choosing representatives of those finitely many orbits gives the claimed bound.

Old admissibility makes every $A_i \cap A_j$ cyclic or trivial. Consequently these function two-chains are supported, outside a bounded set, near finitely many periodic axes. Group together axes with the same endpoints. The remaining endpoint directions separate outside every sufficiently large ball: distinct maximal cyclic groups in a torsion-free hyperbolic group share no endpoint.

To see the flux on one of these directions explicitly, write $p_n = \mathcal{F}((c_i - 1)^{-1})(c_i^n)$. Support preservation and right equivariance give

$$p_{n-1} - p_n = \delta_{n,0}.$$

Thus the values on the two half-orbits are constants p_-, p_+ with $p_- - p_+ = 1$. Choose a half-orbit on which the constant is nonzero. The corresponding tail of $\mathcal{F}(\Delta(c_i))$ is a nonzero constant sum of translates of the chosen word path for c_i . Since $C_{i-1} \neq C_i$, the term $\Delta(c_{i-1})$ has no tail in this endpoint direction. No positivity or one-sided support of $\mathcal{F}((c_i - 1)^{-1})$ is needed.

Let R bound the propagation of the boundary matrices. Beyond a large ball, the distinct endpoint support clusters just described are more than $2R$ apart. Retain from $\mathcal{F}(b_i)$ the entire cluster in the chosen direction beyond that ball, obtaining a function two-chain b_i^+ . A selected coefficient and an omitted coefficient can contribute to the same boundary edge only near the cutoff ball. In particular all cancellations along the sides of the selected cluster remain intact. Therefore

$\partial_2 b_i^+ =$ the chosen nonzero constant half-ray of word paths + a finitely supported one-chain.

The correction is finite because the Cayley complex is locally finite and the boundary matrices have finite propagation. The half-ray has divergence of nonzero total mass, whereas the divergence of any finitely supported one-chain has total mass zero. This contradicts $\partial_1 \partial_2 b_i^+ = 0$ and rules out the circuit. $\square$

Lemma 10.3 (Component groups before the injection step). *Let A' be the full stabilizer of a component of the cyclic incidence forest, and put $S' = A' \cap L$. Then A' is finitely generated, quasiconvex in G , virtually compact special, and field-acyclic; $h(A') \neq 0$ and $0 < \beta(S') < \infty$. Stabilizers of different new components intersect at most cyclically. If e_+ and c_+ count, modulo S' , the incidences and cyclic vertices with nonzero height image, respectively, then*

$$\beta(S') = \sum_{v \in S' \setminus \text{Old}} \beta(S_v) + e_+ - c_+. \quad (10.1)$$

Here the old vertex groups are understood in their actual conjugate positions. No joint ambient injection for the new system is assumed in this lemma.

Proof. The finite quotient of the incidence forest makes A' a finite graph of groups with old vertex groups and maximal cyclic vertex groups. Its edge groups are nontrivial cyclic. The Bass–Serre homology sequence therefore gives field acyclicity, and the same finite graph of finitely generated groups gives finite generation. Also $h(A') \neq 0$, since A' contains an old group.

For $S' = A' \cap L$, each old vertex orbit still splits into finitely many orbits, because the old group has nonzero height image. The same holds for the nonhorizontal cyclic vertices and incidences. Horizontal cyclic stabilizers have zero H_0 and H_1 over their subgroup fields. At a nonhorizontal cyclic vertex or incidence the S' -stabilizer is trivial, and contributes one dimension in H_0 . The tree sequence, with direct sums even when the quotient is infinite, thus gives exactly

$$\beta(S') = \sum_{v \in S' \setminus \text{Old}} \beta(S_v) + e_+ - c_+.$$

It is positive and finite. The period rule and the subtree gate criterion now give quasiconvexity of A' . Its cyclic edge groups are ambient quasiconvex, so its finite splitting over old VCS vertex groups gives VCS by the quasiconvex hierarchy theorem [66].

We next control intersections; this must precede the new joint injection. For any two field-acyclic quasiconvex VCS groups B, C obtained here, their intersection V is quasiconvex and separable in each. The finite-span assertion of Theorem 6.1 gives

$$\left(\sum_g D_B g \right) \cap \left(\sum_g D_C g \right) = \sum_g D_V g.$$

A cycle over the induced D_V -module has fillings over both modules on the left. Ambient two-boundary injectivity makes those fillings equal, and the displayed equality puts their coefficients in the module on the right. Shapiro therefore gives $H_1(V; D_V) = 0$. The same induced injectivity gives $H_2(V; D_V) = 0$, and $H_0 = 0$ when $V \neq 1$. Thus every nontrivial such intersection is field-acyclic.

Apply this first to A' and an old group from another component. Their intersection has trivial stabilizers on the join tree: any nontrivial intersection with an old node or its cyclic neighbor would join that old group to the component. It is therefore free; its zero first field homology makes it at most cyclic.

Finally the intersection of two distinct new component groups has at most cyclic old-node stabilizers. At an old node, any nontrivial such stabilizer lies in its unique maximal cyclic germ, and hence in the corresponding incidence stabilizer. Collapse the resulting nontrivial-edge stars. The intersection is free, and its field acyclicity again makes it at most cyclic. This supplies the intersection hypothesis for the next join forest without using the joint injection to be proved next. $\square$

Lemma 10.4 (Detection under cyclic joins). *Every noncyclic subgroup of a new height kernel S' is detected in $H_1(S'; D_{S'})$. Its image will therefore be detected in ambient homology once the joint injection below is established.*

Proof. We first treat a horizontal piece and then the free-product joins. On the tree of S' , split along trivial-stabilizer edges (including positive-height joins), into free product pieces. In horizontal pieces edge and cyclic node H_0, H_1 vanish, leaving the direct sum of old vertex H_1 's. For a noncyclic subgroup contained in such a piece, if elliptic use old detection.

Otherwise take two independent hyperbolics x, y with axes strictly on opposite branches from a chosen old node v (take sufficiently separated translates/conjugates). This is possible because the tree action is acylindrical (successive turns through two distinct maximal cyclic nodes have trivial path stabilizer); elementary nonelliptic actions of subgroups there are virtually cyclic. Indeed end or line stabilizers have trivial kernel of tail translations (respectively, after preserving orientation on a line), and an all-elliptic subgroup has a fixed vertex or fixes tails towards an end. Independent hyperbolics give as widely separated axes as desired, by translating one by large powers of the other (their axis projections to the latter are bounded).

Choose a nonzero c in the first edge group along $[v, xv]$; the last has xcx^{-1} . Telescope the normalized paths $\Delta(c_j)$ of arbitrary edge-group elements along this segment with those two terminal choices. Each difference is an internal vertex cycle (cyclic vertices contribute zero); and the entire sum over $[v, xv]$ is

$$\Delta(xcx^{-1}) - \Delta(c) = (x - 1)(\Delta(c) - \Delta(x))$$

in affine chain coordinates modulo boundaries.

This writes $\Delta(x)$ as $\Delta(c)$ plus normalized cycle contributions at the internal nodes. Likewise for y with a noncommensurable incident cyclic element c' . Thus in the coordinates of the old summand orbit of v , the cycle $\Delta(x) - \Delta(y)$ contains $\Delta(c) - \Delta(c')$ at the identity coset; all internal pieces have coefficients supported in products $\langle x \rangle s S_v$ or $\langle y \rangle s S_v$ disjoint from S_v (no translated path interiors revisit that node). By disjoint support and bases over D_{S_v} , equality to zero would force $\Delta(c) = \Delta(c')$ in that old summand, impossible by induction (equality for generators collapses the entire affine cycle space of their subgroup).

For the trivial-edge free-product splitting, an elliptic noncyclic subgroup uses the injected vertex homology. Otherwise its action is nonelementary: trivial edge stabilizers make an elementary nonelliptic subgroup cyclic. Choose two hyperbolics x, y with axes on opposite sides of a separating edge, away from its midpoint q . The based segment chain $[q, hq]$ in free edge coordinates is a derivation, so vanishing of a normalized difference in homology would also kill this normalized path difference.

Read the oriented half-edge E from q towards the x -axis. Only E, xE from $\langle x \rangle E$ occur along $[q, xq]$, as first and reversed last: these are the axis hairs with distinct attachment points. No $\langle y \rangle E$ occurs on $[q, yq]$. The two contributions for x give coefficient -1 on normalization, which other pieces cannot cancel by disjoint support, since these edge stabilizers are trivial. Thus the normalized difference is nonzero, proving detection in $H_1(S')$. $\square$

Proposition 10.5 (Joint injection after a cyclic join). *For the new component system the map*

$$\bigoplus_{i \in L \setminus \mathcal{D}'} H_1(S'_i; D_L) \longrightarrow H_1(L; D_L)$$

is injective. Consequently a cyclic-join round preserves admissibility and does not decrease the budget. A round containing a positive-height cyclic join increases the budget by at least one.

Proof. Write $\mathcal{D}'$ for the new component system, and let

$$V = \ker \left(\bigoplus_{i \in L \setminus \mathcal{D}'} H_1(S'_i; D_L) \longrightarrow H_1(L; D_L) \right).$$

There are three trees in the argument: the current join forest computes these homologies, the next join forest receives localized coefficients, and one level of the actual tree T supplies independent chain columns. Lemma 10.3 makes the next forest available before the current joint injection has been established.

In the current Bass–Serre sequence, old joint injection makes the map from V to the sum of the positive-height-edge H_0 coordinates injective. The old vertex orbits summed here are exactly the old L -orbits. Choose a common positive height b for all these finitely many orbit coordinates. At a component i , height transport uses an element $g_i \in A'_i$ with $h(g_i) = b$ and acts on coefficients by conjugation followed by left multiplication by $t^b g_i^{-1}$. Inner transport on homology equals left multiplication, so this action is independent of the chosen lift and respects the tree sequence. At a positive-height edge we may instead choose its stabilizer element g of height b ; its coordinate action is

$$p \longmapsto t^b p g^{-1}.$$

Each one-dimensional coordinate is simple over the skew Laurent ring. Their finite direct sum is semisimple. If $V \neq 0$, it therefore contains a simple submodule whose nonzero projection onto one coordinate is an isomorphism. Normalize its generator to have coordinate 1. Its height translate is $t^b g^{-1}$ times itself.

Choose an intrinsic $D_{S'_i}$ -basis at every component, and let B_i be the matrix of transport by g_i , including its action on coefficients. A coefficient row λ_i of the chosen vector then satisfies

$$t^b \lambda_i g_i^{-1} B_i = t^b g^{-1} \lambda_i, \quad g \lambda_i g_i^{-1} = \lambda_i B_i^{-1}.$$

Lemma 7.1 localizes these rows to finitely many cosets whose transporters conjugate the contributing objects into one component of the next cyclic-join forest: their stabilizers share powers of g . Denote its group by A^* and its height kernel by $S^* = A^* \cap L$.

For each nonzero row choose one of its height-zero transporters x_i as the new representative. Any other contributing transporter x has $x x_i^{-1} \in S^*$: it sends the same object into the same next component and has height zero. Also $x_i S'_i x_i^{-1} \leq S^*$. Conjugating the basis cycles by x_i and replacing their coefficients by $\lambda_i x_i^{-1}$ therefore puts the sum over $\mathcal{U}(S^*)$, without changing its ambient class. Distinct selected indices remain distinct vertex orbits in the next tree, since an S^* -identification would already identify their old L -orbits. Edge first homology is zero in that tree, so its Bass–Serre sequence makes the resulting sum nonzero in $H_1(S^*; \mathcal{U}(S^*))$. Scalar extension is flat here because each intrinsic coefficient ring is a division ring.

Use the period rule to represent each contributing intrinsic basis at one fixed actual P -row level. This replacement differs from the old representative by a boundary over $D_{S'_i}$. After conjugation and extension it is still a boundary over $\mathcal{U}(S^*)$ in the induced complex, so the nonzero class just obtained is preserved. All x_i have height zero, and the period rule holds at every level; thus these representatives use one common interval of horizontal a -columns. Connecting frame chains cancel on cycles.

Their ambient class is zero by the definition of V . Independence of the columns at this actual free-row level makes their literal sum zero. This equality already holds in the restricted induced complex: for finitely many affiliated coefficients a common invertible affiliated multiplier makes them bounded, after which distinct S^* -cosets have disjoint Fourier supports. This is coset independence for $\mathcal{U}(S^*) \cdot L$. It contradicts the nonzero class in $H_1(S^*; \mathcal{U}(S^*))$, proving $V = 0$.

At a positive-height cyclic node of degree $m \geq 2$, the dimension formula in Lemma 10.3 adds $m - 1$ to the old total. Horizontal nodes add zero. Detection and capture are preserved by the preceding lemmas and containment of the old kernels. Hence admissibility is preserved, the budget does not decrease, and a positive-height join increases it by at least one. $\square$

Proposition 10.6 (Finite strict closure). *Iterating cyclic joins stops after finitely many rounds. The final system is admissible and strictly malnormal: stabilizers of distinct objects have trivial intersection. Each is the full stabilizer of its object. For each final object d , the union of all actual rows x with $X_x \cap S_d \neq 1$ is a connected subtree Z_d with finite D_d -quotient, where D_d is its object stabilizer and $S_d = D_d \cap L$.*

Proof. Every round containing a positive-height join raises the injected integer dimension, which is at most $k - 1$. There can therefore be only finitely many such rounds. We must separately show that an infinite sequence of horizontal joins cannot remain after them.

Follow a nested chain of component groups and put

$$A^\infty = \bigcup_n A_n, \quad S^\infty = A^\infty \cap L, \quad M_x^\infty = S^\infty \cap X_x.$$

Homology commutes with this increasing union, and joint ambient injection gives $0 < \beta(S^\infty) \leq k - 1$. Also $h(A^\infty) \neq 0$. The homology part of the period rule consequently applies before finite generation is known. Let Y be the convex hull of the rows with noncyclic M_x^∞ . Every edge of Y has nontrivial S^∞ -stabilizer.

Consider a finite run in Y on which the M_x^∞ are cyclic. Successive edge stabilizers are commensurable in each intervening cyclic M_x^∞ , so one nonidentity power $c \in S^\infty$ fixes the whole run. If a subblock of the fixed capture length has noncyclic *ambient* actual stabilizer K , then K lies in an originally captured kernel. Choose a stage n containing c on our chain and follow that captured object's descendant to the same stage. Both stage- n groups contain c , so they coincide or join at the next round. Thus $K \leq S^\infty$. The rows of this block would then have noncyclic M_x^∞ , a contradiction. All such subblocks therefore have cyclic ambient stabilizer, and the cyclic path dictionary bounds the length of the run uniformly.

We now build a finite quotient of Y . The level homology bound and height periodicity give finitely many A^∞ -orbits of rows with noncyclic M_x^∞ . Every M_x^∞ has finite beta by row restriction, hence is finitely generated and quasiconvex in X_x by Lemma 5.10. Quasiconvex double-coset finiteness, applied to the finitely many actual incident-edge types in X_x , gives finitely many incident orbits with nontrivial intersection. The uniform bound on intervening cyclic runs means that every row of the hull is reached within bounded distance of a noncyclic row. Induction over that distance, using the finite local orbit lists, proves that $A^\infty \backslash Y$ is finite. Its stabilizers are finitely generated, so A^∞ is finitely generated.

A finite generating set of A^∞ belongs to one stage; the chain equals its union from that stage onward. No further distinct object can merge into a stationary component: it would bring a noncyclic old group previously meeting that component group at most cyclically. Every component descends from one of finitely many initial orbit representatives, so stationarity is simultaneous. The final objects have trivial pairwise stabilizer intersections and have full stabilizers by construction.

For completeness, in this strict state let Y_d be the noncyclic hull for S_d . If an actual row x has $X_x \cap S_d \neq 1$, any nontrivial element of that intersection fixes the bridge from x to Y_d . A noncyclic ambient capture block on the part outside Y_d would belong to some $S_{d'}$; its common bridge element forces $d' = d$ by strictness, contradicting its position outside the hull. The cyclic dictionary therefore bounds the length of the bridge. Finite local orbit branching gives a finite quotient after adding all these rows. The union is connected, since every point of each bridge also has nontrivial intersection with S_d . This is Z_d . $\square$

11 Strict row folding

Cyclic closure has produced a strict system and its actual subtrees Z_d . We identify points at equal height within each Z_d , sending it to a line. The task is to prove that this quotient remains a tree and that its rows and incident-edge maps retain the homology injections needed to control path stabilizers.

11.1 Subtrees and horizontal fibers

Lemma 11.1 (Root closure and the enlarged actual subtrees). *In a strict system, each D_d is root-closed in G . Its subtree Z_d consists exactly of the actual rows with $M_x^d = X_x \cap S_d \neq 1$, and has finite D_d -quotient. At every actual level the sum of the homologies of the M_x^d , over S_d -orbits, maps isomorphically onto $H_1(S_d)$. At an actual vertex the analogous sum over one slot maps isomorphically onto $H_1(M_x^d)$.*

Proof. If $g^m \neq 1$ belongs to D_d , it also belongs to $gD_dg^{-1} = D_{gd}$. Strictness forces $gd = d$, hence $g \in D_d$. Thus D_d is root-closed. Proposition 10.6 constructed Z_d from all rows with nontrivial M_x^d and proved its finite quotient. The period rule identifies the sum at each actual level with $H_1(S_d)$. Its adjacent-level equality, restricted to the summands at one actual vertex, gives the stated local slot equality. $\square$

Equate points of each Z_d at the same height, and take the transitive quotient of T . It is a tree $\widehat{T}$, with the same maximal-orbit types as before (one orbit at each level under L), and each Z_d maps onto a monotone line ℓ_d .

Indeed its image has one point per height, and any circuit in the quotient lifts after adding connecting paths in the Z_d 's which project to backtracks.

At a vertex or mid-edge level the components before identification are described by the bipartite graphs Γ_w , with nodes the actual rows x and the d 's over w , and an edge when $x \in Z_d$. Here w denotes one resulting folded place (row position in $\widehat{T}$); $Y_w = \text{Stab}_G(w) \leq L$. The coefficient transport conventions remain in force after folding. In particular Y_w is always a subgroup of L .

Figure 3 shows the strict folding and its horizontal incidence data. The tree and homology assertions needed for this construction are proved below.

Lemma 11.2 (Raw horizontal jump vectors). *Fix an actual row group X at a given level, and represent its actual rows by frame cosets $pX \subset L$. A jump from pX to qX through one object d has a raw horizontal one-chain representative with divergence $q - p$. Its coefficient support is contained both in a finite union of $S_d r$ and in the union of the actual X -cosets incident to d . Root-row representatives are taken before eliminating the root relation.*

Proof. The support assertion uses Theorem 6.1. Fix an actual group X of the given level, using its one-chain generators in the horizontal Fox quotient of L ; rows now have frame cosets pX in L . We use raw vectors on those generators, i.e. before eliminating the root relation if there is one. For any jump $pX \rightarrow qX$ through d , take $1 \neq a \in S_d \cap pXp^{-1}$, $1 \neq b \in S_d \cap qXq^{-1}$. The difference $d(q) - d(p)$ (here the symbol on group elements denotes Fox chains) is represented by a raw vector obtained as

$$q\Delta_X(q^{-1}bq) - p\Delta_X(p^{-1}ap) - (\Delta(b) - \Delta(a)).$$

For the last parenthesis use an expression by cycles from the M_x^d in the current level, with coefficients over D_{S_d} , possible by the period isomorphism. In moving any such cycle from a conjugate row to its horizontal coordinates the frame chain terms cancel. In particular each may be written by differences of normalized elements just as in this formula, since

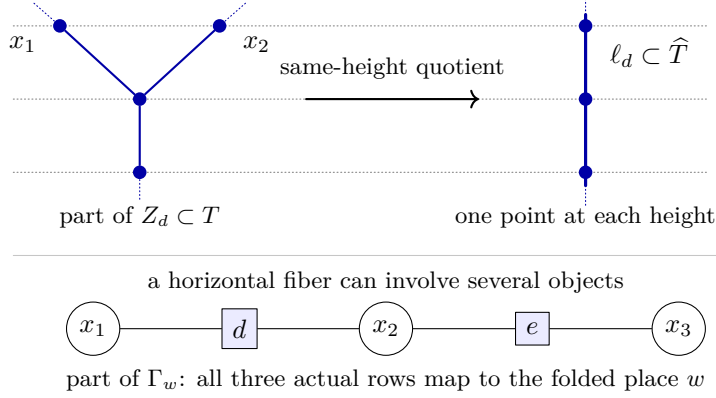


Figure 3: Strict folding is the transitive quotient generated by identifying same-height points within each Z_d . The upper panel shows the image of one subtree; horizontal guides indicate height. It does not identify arbitrary equal-height points of T . The lower panel records how two objects can make three actual rows equivalent at one height: an incidence x, d means $x \in Z_d$. Such incidence chains describe the horizontal fiber Γ_w , whose stabilizer and homology are analyzed below.

such differences span the intrinsic cycle space modulo 2-boundaries for a nontrivial group. This proves the representation assertion by the conjugation rule. All coefficients are thus supported (in the sense of Theorem 6.1) in finite unions of $S_d r$, lying also within the union of actual X -cosets incident to d . The divergence is literally $q - p$. $\square$

11.2 The root-cylinder correction

Lemma 11.3 (Isolation and splitting of root cylinders). *Suppose the actual row is*

$$X = B *_{\langle z \rangle = \langle u^a \rangle} \langle u \rangle, \quad a > 1,$$

where B is free and z is root-indivisible in B . Put $U = \langle u \rangle$, $\alpha = 1 + u + \cdots + u^{a-1}$, and $R = \alpha \mathbf{e}_u - d_B(z)$. Let $V = \sum_i V_i$ be a finite sum of raw vectors whose coefficients have the support patterns of Lemma 11.2, for distinct members of a strict quasiconvex system. If $V = PR$ in the division-field chain module (necessarily $P = V_u \alpha^{-1}$), then, after applying the simultaneous scalar-function map, P is supported within a bounded neighborhood of the corresponding finite union of peripheral neighborhoods. It can be split among those neighborhoods outside a bounded set, so that subtracting the corresponding multiples of R makes each V_i finitely supported without changing its divergence. The same assertion holds after restriction to a specified actual X -coset.

Proof. The subgroup U is maximal cyclic in X . Indeed, distinct edges at a B -vertex of the root Bass–Serre tree have trivially intersecting stabilizers, since z is root-indivisible in B . Each nontrivial u^m therefore has its own cyclic vertex as its only fixed vertex of that type, with stabilizer U .

The internal root relation generates the kernel of the row chain quotient. Thus

$$V = PR, \quad P = V_u \alpha^{-1}, \quad R = \alpha \mathbf{e}_u - d_B(z).$$

After applying the simultaneous scalar-function map, P is supported on U -cosets passing through a bounded neighborhood of the same subgroups. Their union is quasiconvex, since it consists of finitely many quasiconvex sets based at the identity. We show that P vanishes far from this union, working within each specified actual X -coset.

Choose a ball radius large enough to contain every translate used in the fixed words for u^a and z , and every right multiplication in R . If two contributing periodic U -quasigeodesics meet such a ball far from the quasiconvex union, thin quadrilaterals force them to fellow-travel for a distance tending to infinity on their way back to that union. Distinct cosets of the ambient maximal cyclic group have uniformly bounded coarse overlap, so these quasigeodesics belong to the same ambient cyclic coset. In a specified X -coset they are the same U -coset: the intersection of the ambient maximal cyclic group with X is U , by maximality in X .

Write $p_n = P(lu^n)$ in this isolated cylinder. Where $V_u = 0$, the u -equation says that a consecutive values sum to zero. Subtracting two consecutive such equations gives $p_n = p_{n-a}$. The B -equations can now be read on an oriented edge of the axis of z in the free Cayley tree of B . The sum of the translates of the based path $d_B(z)$ has nonzero signed count on that edge; any initial or terminal hairs cancel. Only coefficients with indices in the same $\langle z \rangle$ -coset contribute, and their indices lie in a fixed finite interval. The local periodicity makes them equal, so this nonzero integer multiple of $P(l)$ must vanish. Thus $P(l) = 0$.

After this support bound, write $P = \sum_i P_i + P_0$, with P_0 finitely supported and each P_i supported in the separated tail near the i th peripheral subgroup. Right multiplication by the finite polynomial matrix R enlarges supports by a fixed amount only. Since $PR = \sum_i V_i$, each $V_i - P_i R$ vanishes outside a bounded set. Moreover

$$\partial(P_i R) = P_i(\alpha(u - 1) - (z - 1)) = 0.$$

Consequently the correction preserves divergence. These are operations on scalar functions after projection; no multiplicativity of the scalar-function map for α^{-1} is asserted or needed. Its right-group equivariance is precisely what preserves the polynomial equations. Finally, all these right multiplications belong to X , so they preserve each left X -coset and commute with restriction to it. $\square$

11.3 The folded incidence trees

Lemma 11.4 (Tree structure and homology of a folded row). *For every folded place w , its incidence graph Γ_w is a tree. The Y_w -action is cocompact, with one orbit of actual row vertices; its vertex groups are the indicated X_x and S_d , and its incidence groups are M_x^d . The inclusion induces an isomorphism*

$$H_1(Y_w; D_L) \xrightarrow{\cong} H_1(L; D_L).$$

Proof. Sum these raw vectors around a putative simple circuit, fixing one frame per visited x . The sum is zero in the actual row chain quotient. Apply the simultaneous scalar function assertion, with all divergence and polynomial right products preserved. Outside bounded sets the supports of the various summands are disjoint, even in any fixed thickenings: they lie within bounded distance of the distinct D_d , with trivial intersections. For free X the sum was zero literally, forcing finite function support for each vector. But finite chains in one actual X -coset cannot have total divergence 1 or -1 there.

For a root row the raw sum around the circuit lies in the kernel of the row chain quotient and therefore is PR with $P = V_u \alpha^{-1}$. Lemma 11.3 corrects each circuit summand by a divergence-zero function boundary and makes it finitely supported. In any frame coset used at the end of a jump the remaining divergence has total mass 1 or -1 . A finitely supported one-chain in a Cayley graph has total divergence zero, giving the same contradiction as in the free case.

The stabilizer and cocompactness assertions in the statement now follow by equivariance. The Bass–Serre Mayer–Vietoris sequence on Γ_w gives the homology assertion: edge $H_0 = 0$ and the sum of edge H_1 ’s into the S_d -summands is exactly an isomorphism, leaving just the X inclusion. $\square$

Lemma 11.5 (Detection and separation at a folded level). *Every noncyclic subgroup of Y_w is detected in ambient height-kernel homology. Stabilizers of two distinct folded places at the same level intersect at most cyclically.*

Proof. We verify detection: every noncyclic $H \leq Y_w$ has some $\Delta(h) - \Delta(h')$ nonzero in ambient height-kernel homology. For an elliptic subgroup on Γ_w use the known S_d detection, or detection in X . The latter is immediate in a free group by disjoint support away from bounded sets for normalized paths of two noncommensurable elements and divergence 1.

In a root group use its own cyclic Bass tree. Elliptic noncyclic subgroups are in a free base with independent columns. Otherwise choose two hyperbolics with axes in different components off a pivot B -node (neither axis through the pivot), taking that node as base by conjugation. Such separated axes exist: the tree action has trivial long-segment stabilizers, so a nonelliptic noncyclic subgroup has independent hyperbolics (an all-elliptic subgroup without a fixed point fixes tails towards an end; an end-fixing subgroup has trivial kernel of translation on tails); translate independent axes to separate them as far as needed.

Write normalized paths from the base to its translates by using cyclic-amalgam normal forms and substituting $d_X(u) = \alpha^{-1}d_B(z)$. For each hyperbolic h , support at the pivot B -coset comes only from finitely many powers' translates of the used pieces, staying within bounded distance of the one adjacent cyclic stabilizer towards the axis (plus finite sets). Indeed U -steps spread just in the B -cosets incident to that cyclic node, using finitely many U -coset translates; the translated tree paths go out and back along the axis with disjoint hairs to $h^j B$.

The two incident cyclic stabilizers intersect trivially. Equality would force a function chain on that pivot coset with finite support and divergence of total mass one, impossible.

We next treat a subgroup that is nonelliptic on the folded incidence tree. If H itself is nonelliptic and noncyclic on Γ_w , similarly pick separated axes there off an X -node pivot (long segment stabilizers trivial by strictness). Conjugate to frame 1 there. Represent $d(h)$ raw as above on the path through d -nodes to hX , ending at frame h . Normalize by $(h - 1)^{-1}$. Restricting support to the pivot coset X , only finitely many $\langle h \rangle$ -translates contribute, namely jumps through a neighbor of the pivot, always the one d towards its axis. Thus pivot support stays near D_d . The two normalized paths have distinct such d 's and divergence 1.

These are enclosing support patterns for sums/products, hence remain usable before any scalar function projection. In particular in the root case a further factor α^{-1} on the right can hit the pivot coset only from inside that coset, and spreads pivot support just along U -cosets based near the corresponding D_d . If equal modulo boundaries, in the root case their difference is PR as above, with P -support thus satisfying the isolated-cylinder claim inside the pivot. Splitting P there at infinity separates two disjoint tails, allowing correction of the normalized paths by R to finite function support there while retaining divergence.

This is impossible, in the free case directly so without correction.

Finally compare two distinct folded places. For different components at this level, any common subgroup cycle has expressions in the level basis supported inside their respective actual frame cosets by these same path formulas (in an arbitrary starting frame p compute normalized jumps to hp and on normalization adjust by $d(p)$, which cancels in differences; moves changing only the frame within one coset just use an actual X -word). Eliminating u does not leave those cosets. Disjoint support in Theorem 6.1 makes the cycle zero in ambient homology. Detection proves the asserted separation. $\square$

Proposition 11.6 (Folded slot structure and restriction). *Fix a folded vertex v and one slot of incident folded edges f . Let K_f be the image of the endpoint map $\Gamma_f \rightarrow \Gamma_v$. Then K_f is a Y_f -cocompact subtree, every object vertex d of Γ_v occurs in exactly one such K_f with its full stabilizer, and any absent incidence has at most cyclic stabilizer. For each $H \leq Y_v$ the*

natural map

$$\bigoplus_{f \bmod H} H_1(H \cap Y_f; D_L) \longrightarrow H_1(H; D_L) \quad (11.1)$$

is injective.

Proof. Map each $\Gamma_f \rightarrow \Gamma_v$ by the endpoint map, fixing the d labels. Its image is a subtree K_f of finite quotient under Y_f . Each d of Γ_v occurs in exactly one of these subtrees (on the chosen slot, by its line) with full stabilizer. An incidence x, d can be absent there only with M_x^d cyclic, by the local period equality. The f 's with $x \in K_f$ form a single X_x -orbit. Moreover $\tilde{F} = X_x \cap Y_f$ there acts on the local tree of actual edges over x mapping to f , connected by the d 's incident to these actual edges. This is connected: a simple path between such edges in Γ_f projects back to x and has no repeated d , hence uses only jumps with that same x in Γ_v . Its quotient is finite with one actual edge-node orbit (X_x is transitive on the unfurled slot, and a transporter here necessarily fixes f ; incident object types there are finite as before). Denote the edge group of the actual slot by E ; these actual edge-nodes now have groups E up to conjugation. There are other vertex groups M_x^d and gluing groups M_e^d for actual edges e . Thus $\tilde{F}$ is finitely generated, and $H_1(E) \rightarrow H_1(\tilde{F})$ is an isomorphism after extension by the local period equality and nonzero incidence stabilizers again.

If X_x is free, the preceding isomorphism makes $\tilde{F}$ a rank- k free quasiconvex subgroup. Its Fox columns in X_x are independent, using the already known injection $H_1(E) \rightarrow H_1(X_x)$. Restricting the same induced modules to any subgroup of X_x and decomposing over its cosets therefore gives the joint homology injection, exactly as in the actual slot rule of Proposition 5.6.

For a root row, write its root splitting in the chosen slot as

$$X_x = E *_{\langle z \rangle = \langle u^a \rangle} \langle u \rangle.$$

The actual edges of this slot correspond to the E -nodes of the root Bass–Serre tree. Suppose first that a nonzero power in a root cyclic-node stabilizer belongs to some S_d . Root closure puts that whole cyclic stabilizer in S_d . Every adjacent root edge then has nontrivial stabilizer intersection with S_d , so all the E -nodes incident to this cyclic node join through the same d . By X_x -equivariance, the same holds at every cyclic node, with its corresponding object label. Connectedness of the root tree therefore joins all actual edges of the slot over x into one folded edge. In this case $\tilde{F} = X_x$.

Suppose instead that no root power belongs to any S_d , and fix $M = M_x^d$. Its stabilizers at root edges and cyclic nodes are trivial. The local period equality says that the first homologies of its E -node intersections sum isomorphically onto $H_1(M)$. The tree sequence has no remaining degree-zero contribution. Consequently, if an E -node intersection is nontrivial, there is only one such vertex-group orbit and no additional free graph factor: either a second nontrivial vertex group or a free factor would contribute extra first homology through the degree-zero edge terms. Thus M is contained in that E -node stabilizer. The fixed E -node is unique, since the intervening root edges have trivial M -stabilizer.

If M meets no E -node nontrivially, its root-tree action is free. It is a nontrivial finitely generated free group, and the same period equality gives $H_1(M) = 0$. Hence it is cyclic and acts hyperbolically on the root tree. Such an incidence meets no actual edge of this slot and gives no identification of slot edges. The other incidences meet exactly one actual edge, by the preceding paragraph. Thus no object joins distinct actual edges in the no-hit case, and $\tilde{F} = E$ with its actual inclusion.

Both root alternatives give quasiconvexity and the needed comparison after restriction to an arbitrary subgroup of X_x . When $\tilde{F} = X_x$, there is one folded-edge orbit over x and the comparison is the identity. When $\tilde{F} = E$, folded edges retain the actual E -cosets, and

the actual-slot induced-module injection of Proposition 5.6 applies to all their intersection summands simultaneously. These are the vertex-node comparisons used next.

Compare now the Bass–Serre Mayer–Vietoris sequence for H on Γ_v with the sum for $H \cap Y_f$ on K_f . At the d -vertices the summed rows map isomorphically, and likewise for incidence edge rows except for the absent ones, whose target stabilizers are at most cyclic (thus the H_1 map on edge rows is onto, and the H_0 map injects). At actual x -nodes the vertex H_1 map injects by the just-proved subgroup-wise statement. Chasing the sequence gives exactly (11.1): a putative kernel first has zero projected edge- H_0 coordinate, so lifts to vertex H_1 ; there subtract a lifted edge relation and use vertex injectivity. $\square$

Corollary 11.7 (Strict folded tree). *The quotient $\hat{T}$ is a tree with one L -orbit at each level, and every Z_d maps onto a monotone line ℓ_d . Its row groups and slot maps have the conclusions of Lemmas 11.4 and 11.5 and Proposition 11.6. Stabilizers of finite monotone paths in any fixed height range inject jointly on first field homology into L , over path-orbit representatives.*

Proof. The quotient-tree and orbit assertions were proved in the construction. Start the homology assertion at one folded edge, using Lemma 11.4, and extend a path one edge at a time at either endpoint. Apply (11.1) with H equal to the preceding path stabilizer. At every change of representative use the inverse monomial coefficient shift, so that each summand has its canonical ambient image. Iterating proves the asserted joint injection. $\square$

12 Enlargement at shared line ends

Distinct object stabilizers now intersect trivially, but their folded lines may still share an end. We replace such groups by full end stabilizers, prove that these remain quasiconvex and virtually compact special, and show that a genuine enlargement strictly increases the total injected homology dimension, which remains at most $k - 1$. This is the final step in constructing the terminal system.

12.1 End kernels and tail homology

Lemma 12.1 (End stabilizers). *Suppose two distinct folded lines share an end. Choose the positive orientation if possible, reversing all orientations otherwise. The set of distinct positive ends ξ of the ℓ_d has finitely many G -orbits. For $A_\xi = \text{Stab}_G(\xi)$ and $S_\xi = A_\xi \cap L$, one has $h(A_\xi) \neq 0$, and S_ξ is the increasing union of pointwise tail stabilizers on any chosen line ending at ξ . Distinct A_ξ intersect at most cyclically.*

Proof. If some folded lines of distinct d 's share an end, choose a sign with this property, say positive (otherwise reverse orientations). Consider the system of **distinct** positive ends ξ of the ℓ_d 's, with stabilizers A_ξ and kernels S_ξ . These have finite orbit set, and height translators. S_ξ is the increasing union of pointwise tail stabilizers along any such line ending there (height-zero translates preserving the end fix tails since height on the line is monotone). Different A_ξ 's meet at most cyclically: a common element preserves the connecting geodesic, hence has height zero (height has a minimum along it), and fixes the line pointwise; use distinct same-level places there. $\square$

Lemma 12.2 (Joint injection for tails and ends). *For representatives of the L -orbits of positive ends, the map*

$$\bigoplus_{\xi} H_1(S_\xi; D_L) \longrightarrow H_1(L; D_L)$$

is injective. The homology of every pointwise tail group K also injects into ambient homology and has finite dimension. In particular $0 < \beta(S_\xi) < \infty$.

Proof. Choose rays to representatives of the L -orbits of ends, starting at one common folded edge level w whose actual rows are free. For each ray let K be its pointwise stabilizer, and let K_n stabilize its length- n prefix, so $K_n \downarrow K$. We show that every finite-dimensional test space in the sum of these tail homologies injects into the sum for sufficiently long finite prefixes. Folded path inertia will then give its ambient injection. This finite-test argument is needed because the groups here form decreasing intersections.

Work on the fixed incidence tree Γ_w . At an object node d , a prefix stabilizer intersects S_d in the full group while the prefix lies on ℓ_d , and at most cyclically after it leaves that line. These actual intersections stabilize for each fixed d . To see this, consider a nontrivial cyclic intersection after departure. At an off-line place, a nonzero element of S_d fixing that place acts hyperbolically on its incidence tree: strictness forbids a fixed other object node, and enlargement of Z_d forbids a fixed actual row. If a power fixes a continuation f , its axis lies in K_f . The translate of K_f by the cyclic generator contains the same axis. But distinct K_f in one slot have no common object node, whereas that axis passes through object nodes. Hence the generator already fixes the continuation. There is no proper finite-index drop; a drop to the trivial group is permanent. The same reasoning gives stabilization at each fixed incidence edge.

In the Bass–Serre sequence for K on Γ_w , the incidence H_1 terms sum isomorphically onto the object-node terms: use the period rule at full nodes, and vanishing of cyclic first homology at the others. The remaining sequence has the actual-row homologies as an injected subspace, with quotient injecting into the incidence H_0 terms. Fix a finite-dimensional test space and finite cycle representatives. First consider their finitely supported H_0 projections. All their used incidence stabilizers have stabilized for large n . A transporter identifying two specified incidence lifts while matching length- n prefixes lies in a nested coset of one of these stabilizers. Once that stabilizer is fixed, persistent nonempty cosets coincide and identify the whole rays. Thus no new identification of the finite H_0 data remains for all large n , even when prefix orbits initially merge.

For the remaining actual-row part, use the free Cayley tree of each X_x in its actual frame. Represent the finitely many needed cycles inside finite connected subgraphs of the quotients by $K \cap X_x$. These subgraphs embed jointly and disjointly in the corresponding quotients for K_n once n is large. Indeed a proposed identification between two chosen regular frame vertices has a unique possible transporter. If it matched every prefix it would match the full rays, so each unwanted identification disappears at a finite stage. There are only finitely many pairs to test. Disjoint connected subgraphs of a graph give simultaneous free factors up to their basepoint conjugations, and hence joint first-homology injection, also after extension to D_L .

The tree sequences now give injection on the original finite test space into the finite-prefix homologies. Those inject jointly into $H_1(L; D_L)$ by Corollary 11.7. This proves both the individual tail assertion and the joint assertion after taking the increasing union over ray starting levels. That union is S_ξ . The ambient dimension is $k - 1$, so all these dimensions are finite. Finally S_ξ contains the kernel of an old line ending there; the just-proved injection preserves its positive homology. $\square$

12.2 Simultaneous boundary factors on a tail

Proposition 12.3 (The tail free-product decomposition). *Let K be the pointwise stabilizer of a positive ray contained in an old line, starting at an actual free-edge level w . An object vertex of Γ_w is called full if its line contains that entire tail. There are finitely many K -orbits*

of full vertices. With representatives $d_1, \dots, d_m$, one has

$$K \cong \left(\ast_{j=1}^m S_{d_j} \right) \ast F_r, \quad r < \infty. \quad (12.1)$$

The displayed embeddings of full factors are their actual subgroup embeddings, up to the usual independent choices of conjugating basepoint paths.

Proof. Call an object node of Γ_w full when its line contains the whole tail. At such a node the K -stabilizer is S_d . Each full node meets a noncyclic actual-row stabilizer, by its positive beta and the period rule. The actual-row homologies inject in the tree sequence used above. Thus there are finitely many K -orbits of noncyclic $K_x = K \cap X_x$, and each is a finite-rank free group. Quasiconvex double-coset finiteness inside X_x gives finitely many nontrivial incidence orbits at each. Consequently there are finitely many full-node orbits.

Remove the full nodes, and let B stabilize one of the remaining components. The sum of the boundary-incidence homologies is an isomorphism onto the removed full-node homologies. The tree sequence therefore bounds $\beta(B)$ by $\beta(K)$. We also need finite generation of B . Its remaining object-node stabilizers are cyclic or trivial. At a cyclic actual-row stabilizer, strictness and root closure imply that any nontrivial incident intersection is the whole stabilizer and occurs on at most one edge. Collapse these redundant cyclic-row leaves. Noncyclic actual-row stabilizers and their nontrivial incident edges have finite orbit lists as above; hence every remaining component of nontrivially stabilized edges has finitely generated stabilizer and finite core, or is a single cyclic group. The leftover trivial edges give a free product. Its free-product dimension formula and $\beta(B) < \infty$ leave only finitely many nontrivial factors and finite free rank. Thus B is finitely generated.

We claim that the boundary incidence groups at this component are *simultaneous* free factors of B . Choose an actual base vertex and take the paths to its translates by a finite generating set of B , together with all their B -translates. Their connected union $U \subset T$ is B -invariant and has bounded height range because $B \leq L$. Choose an actual free-edge level w' on the tail above that range by more than the actual capture length.

For an actual row y over w' , the group $B_y = B \cap X_y$ fixes y and preserves U , hence fixes the bridge from y to U . If B_y is noncyclic, the noncyclic ambient bridge stabilizer is captured in one $S_{d'}$, so $B_y \leq S_{d'}$ and $y \in Z_{d'}$. If B_y is cyclic and meets an incident $S_{d'}$ nontrivially, root closure again puts all of B_y there. Strictness makes the object unique. Thus all nontrivial incidences in the B -action on $\Gamma_{w'}$ form stars centered at objects, with redundant actual-row leaves. Collapsing those stars leaves trivial edge groups. Any remaining actual-row stabilizers are cyclic and contribute free factors of rank one.

Identify the surviving object groups in the original Γ_w . If $B \cap S_{d'}$ is noncyclic, it fixes both the selected tail place and the place of $\ell_{d'}$ at every original tail level. Same-level separation forces these places to agree, so d' was already full at the starting level. For any full d' , an element of $B \cap S_{d'}$ fixes d' and preserves the original component of B . If the connecting path crosses another full node, strictness makes this intersection trivial. Otherwise d' neighbors that component, and its intersection is precisely the unique boundary incidence stabilizer $M_x^{d'}$. This is free, being a subgroup of the free row X_x . Other surviving object groups are at most cyclic.

Every original boundary incidence is retained in this argument, including cyclic ones. Its actual object label d' is the same at w and w' . Two such labels become B -equivalent at the higher level exactly when they were already B -equivalent originally. The trivial-edge splitting just constructed therefore makes B free, with the entire original boundary family as simultaneous free factors, up to independent basepoint conjugations.

Return to the splitting of K obtained by removing the full nodes. Substitute these simultaneous factors for the boundary attachments on the B -sides of its Bass presentation.

This eliminates each boundary factor against its image in the corresponding S_d and gives, with the actual embeddings,

$$K \cong \left(\ast_{j=1}^m S_{d_j} \right) \ast F_r.$$

The trivial-edge H_0 terms give

$$\beta(K) = \sum_{j=1}^m \beta(S_{d_j}) + (m-1) + r. \quad (12.2)$$

Here $m \geq 1$, since the tail was chosen on an old line. Positive beta for the full factors and finite $\beta(K)$ give finite m, r . A cyclic free factor contributes one through r ; it is not lost through its zero intrinsic first field homology. $\square$

12.3 Virtually free ascending quotients

Filling the full factors in the tail decomposition (12.1) produces a finitely generated virtually free base. The induced ascending map may fail to be injective. We first remove this obstruction so that the ascending-torus theorem can be applied to the filled end groups.

Lemma 12.4 (Stabilizing a virtually free endomorphism). *Let V be a finitely generated virtually free group and $\phi : V \rightarrow V$ an endomorphism. The sequence $\ker \phi^n$ is eventually stationary. Its ascending torus is therefore isomorphic to the ascending torus of an injective endomorphism of a finitely generated virtually free group. In the injective case this torus has a finite-index subgroup that is an ascending torus of an injection of a finite-rank free group.*

Proof. Choose a finite quotient $q : V \rightarrow Q$ with torsion-free kernel, and put

$$F = \bigcap_{\psi: V \rightarrow Q} \ker \psi.$$

There are finitely many such maps because V is finitely generated. Thus F is a fully invariant finite-index normal subgroup contained in $\ker q$. It is torsion-free and virtually free, hence free of finite rank. The ranks of the free groups $\phi^n(F)$ form a nonincreasing sequence and eventually agree. For all sufficiently large n , the surjection $\phi : \phi^n(F) \rightarrow \phi^{n+1}(F)$ is consequently an isomorphism by Hopficity of finite-rank free groups. Hence the iterated kernels stabilize on F . They stabilize on all of V as well: once their intersections with F agree, the increasing kernels have a bounded finite quotient modulo that common intersection, bounded by $[V : F]$.

If $N = \ker \phi^n$ is the stable kernel, then $V/N \cong \phi^n(V)$ and the induced endomorphism is injective. In the ascending torus every element of N is already trivial, by iterating the stable-letter relation. Thus replacing V by V/N leaves that torus unchanged.

Now suppose ϕ is injective. Choose a fully invariant finite-index normal free subgroup $F_0 \leq V$ as above and replace it by

$$F_\infty = \bigcup_{j \geq 0} \phi^{-j}(F_0).$$

The terms increase, are normal, and have index at most $[V : F_0]$, so the union equals one of them. It is torsion-free by injectivity and therefore free of finite rank. Moreover $\phi^{-1}(F_\infty) = F_\infty$ and $\phi(F_\infty) \leq F_\infty$. The induced endomorphism of the finite group V/F_∞ is injective and hence an automorphism. Choose a positive power for which this finite action is trivial. Ascending normal forms then show that the subgroup generated by F_∞ and the corresponding stable-letter power has finite index and is the ascending torus of the induced injection on F_∞ . $\square$

Proposition 12.5 (Quasiconvex and virtually compact special end groups). *Each end stabilizer A_ξ is finitely generated, quasiconvex in G , field-acyclic, and virtually compact special. It is torsion-free, locally indicable, coherent, of type F , and of cohomological dimension at most two.*

Proof. Choose σ in the group of the original line with positive height p , and put $A'_\xi = A_\xi \cap h^{-1}(p\mathbb{Z})$. If K stabilizes a starting tail on that line, then $S_\xi = \bigcup_{n \geq 0} \sigma^n K \sigma^{-n}$. Thus A'_ξ is the ascending torus with base K and injection given by conjugation by σ^{-1} . The tail decomposition expresses K as finitely many full factors S_{d_j} and a finite-rank free group. Each S_{d_j} lies in the finitely generated group $D_{d_j} \cap A'_\xi$, which has finite index in D_{d_j} . These finitely many groups, the free generators, and σ generate A'_ξ . Therefore A_ξ is finitely generated.

Its height kernel has positive finite beta by tail injection, so the period rule and its actual-tree gate criterion make A_ξ quasiconvex in G . The ambient injective Fox two-boundary remains injective in subgroup-induced coefficients, giving $H_2 = 0$ over its division field. The cyclic height sequence gives the remaining field acyclicity: on finite-dimensional kernel homology, translation minus identity becomes invertible over the skew Laurent division extension, by its leading degree. Torsion-freeness, local indicability, coherence, and cohomological dimension at most two are inherited from G . Quasiconvexity supplies a finite classifying space [46, Theorem 5].

It remains to prove VCS. In A'_ξ take the peripheral occurrences $D_d \cap A'_\xi$ for lines ending at ξ . They form a strictly malnormal family of quasiconvex VCS groups. Every such occurrence can be moved, by a power of σ , to a full factor at the chosen starting tail, so there are finitely many conjugacy types. If one peripheral is all of A'_ξ , VCS is already known. Otherwise this is a proper relatively hyperbolic pair.

For each representative peripheral, choose a finite-index normal subgroup avoiding any prescribed finite set, and kill its intersection with S_d . Residual finiteness of the whole VCS peripheral permits these choices. The kernels are normal, height-zero, and equivariant on all conjugate occurrences. The peripheral images are finite-by-infinite-cyclic, so sufficiently deep relative filling gives a hyperbolic quotient [55].

The actual tail free-product decomposition identifies its filled base as

$$V = \left(\ast_{j=1}^m S_{d_j} / N_{d_j} \right) \ast F_r,$$

a finitely generated virtually free group. Conjugation by σ^{-1} takes each full line to another full line: its translated starting ray contains the original starting ray. Equivariance therefore takes its selected kernel into that of the corresponding factor. This supplies an endomorphism of V . All peripheral occurrences are reached by moving a sufficiently late full tail back to the starting level. Hence the entire filled quotient of A'_ξ is exactly the ascending torus presentation of that endomorphism; injectivity of the endomorphism is not assumed.

Lemma 12.4 replaces this presentation by an injective virtually free ascending torus and then gives a finite-index free ascending torus. It is hyperbolic, so Theorem 4.1 makes the filled group VCS. We have obtained arbitrarily deep such fillings, with hyperbolic VCS peripheral images. Together with the finiteness and field hypotheses proved above, these are precisely the inputs to Theorem 3.1. Thus A'_ξ , and hence A_ξ , is VCS. $\square$

12.4 Termination of both closure operations

Proposition 12.6 (Strict growth at a shared end). *Replacing the objects by distinct end stabilizers preserves all admissibility conditions. If some two distinct old lines share an end in the chosen orientation, this replacement strictly increases the budget. Subsequent cyclic closure does not decrease it.*

Proof. Tail injection gives the new joint homology injection. To check detection, choose a finitely generated noncyclic subgroup of any noncyclic subgroup of S_ξ . It lies in one sufficiently late tail group K . The free-product decomposition of K detects it, either in an old full factor or by the trivial-edge free-product argument used in Lemma 10.4. Tail injection preserves that nonzero cycle in L . Each new kernel contains its constituent old kernels, so actual noncyclic path stabilizers remain captured. The end-group and intersection lemmas supply the other admissibility conditions.

Fix a new end ξ . Old L -orbits over its L -orbit correspond exactly to S_ξ -orbits of old objects ending there. Their finite list of representatives is simultaneously full on a sufficiently late tail. If there are m such representatives, the tail free-product formula and tail injection give

$$\beta(S_\xi) \geq \sum_{j=1}^m \beta(S_{d_j}) + (m - 1).$$

This is strictly greater than the old sum when $m \geq 2$.

Suppose $m = 1$ and equality holds. Every later tail on the chosen original line contains its actual full factor S_d . Equality in (12.2) forces that tail group to have exactly this one full factor and no free part; hence it is S_d with its actual inclusion. Their increasing union gives $S_\xi = S_d$. Any second old object sharing this end would have its nontrivial kernel inside S_d , contradicting strictness. Thus a shared end forces strict growth even when the old objects belong to one S_ξ -orbit.

Sum over the new L -orbits. The budget increases by a positive integer whenever an end is genuinely shared. Cyclic closure then restores strictness without decreasing the budget, and allows a new folding on the same actual tree T . Testing both orientations at each strict state therefore terminates within the bound $k - 1$. $\square$

Theorem 12.7 (Terminal periodic system). *Starting with the system of Proposition 9.9, finitely many cyclic-closure and end-enlargement rounds produce an admissible strict malnormal quasiconvex virtually compact special system $\{D_d\}$, with $S_d = D_d \cap L$, and a folded tree $\hat{T}$ having all the conclusions of Corollary 11.7. The associated monotone lines ℓ_d of distinct objects share no end. Actual long noncyclic path stabilizers remain captured, the kernels still detect all noncyclic subgroups, and their joint homology budget is at most $k - 1$.*

Proof. Use Proposition 10.6 to reach a strict state. Whenever an end is shared, apply the preceding construction in the relevant orientation, then restore strictness and fold again. All foldings use the same actual tree T . Test both orientations each time strictness is restored. Proposition 12.6 gives a positive integral increase at every end-enlargement round, whereas all budgets are at most $k - 1$. There are consequently only finitely many such rounds, and each intervening cyclic closure is finite. The terminal state has exactly the asserted properties. $\square$

13 Relative geometry of terminal rows

The terminal folding gives row groups assembled from actual rows and height kernels. We first control their peripheral incidences and build their relative geometry. The resulting comparison and intersection lemmas will supply the finite-window hypotheses for the terminal path argument, before any filling is chosen.

13.1 Standing terminal data

We work with the terminal system supplied by Theorem 12.7. We retain the actual tree T , the folded tree $\hat{T}$, the height homomorphism $h : G \rightarrow \mathbb{Z}$, and $L = \ker h$. An *object* is

an actual label d , not merely a conjugacy class of subgroups. Its periodic group is D_d , its height kernel is $S_d = D_d \cap L$, its actual subtree is Z_d , and its monotone folded line is ℓ_d . The following previously established features will be used explicitly.

- (i) The groups D_d form an equivariant malnormal system with finitely many orbits of quasiconvex virtually compact special subgroups. Distinct actual objects have trivial stabilizer intersection, and no two distinct lines ℓ_d share an end. In particular the relevant cyclic intersections are root-closed in their row groups.
- (ii) At a folded place w , the fiber Γ_w is a bipartite tree with actual-row nodes x and object nodes d . Their stabilizers are X_x and S_d , and an incidence has stabilizer M_x^d . Endpoint maps send an adjacent edge fiber onto the invariant subtree $K_f \subset \Gamma_v$. On each slot an object node belongs to exactly one such image. All local data have finite quotient, and the actual intersections $\tilde{F}_{x,f} = X_x \cap Y_f$ are finitely generated and quasiconvex in X_x .
- (iii) There is a fixed actual capture length: the stabilizer of a sufficiently long actual segment, if noncyclic, lies wholly in one of the S_d . The enlarged Z_d 's contain every actual incidence with nontrivial intersection. The cyclic dictionary, the reversal bound, and folded homology inertia are available.
- (iv) For the relative free-product trees constructed below we use the expansion and phase-uniqueness conclusions of the expansion-gate argument, with their finite-window hypotheses. We will verify those hypotheses before invoking those conclusions.

Throughout this section a *nonparabolic* subgroup means a nontrivial subgroup not contained in any actual S_d . A folded place means either a vertex or an edge; its stabilizer is denoted Y_w .

Lemma 13.1 (Capture of a fixed monotone ray). *If $1 \neq c \in L$ fixes a monotone ray of $\hat{T}$ pointwise, then its end is an end of exactly one object line ℓ_d , and $c \in S_d$.*

Proof. Take a remote infinite subray W , high in the positive orientation or low in the negative one. Over W form the mapping cylinders of the place trees Γ and their endpoint maps. This is a simply connected tree of contractible spaces. Its actual vertical pieces are the connected components of the actual preimage of W ; its object vertical pieces are the intervals $\ell_d \cap W$. Construct a bipartite graph with these pieces as vertices and one edge for each incidence between them.

This incidence graph is a tree. There is a map from the cylinder complex to the graph: actual continuations and object intervals collapse to their vertex labels, place incidences map to the corresponding edges, and incidence strips have the same two labels at successive levels. Each prescribed vertex piece is connected. Thus a graph circuit could be lifted, using connecting paths inside these pieces, to a loop in the cylinder complex. Its image would be that circuit, contradicting simple connectivity.

The element c acts on this auxiliary tree. Every single-place image is connected and c -invariant. If c acted hyperbolically, its axis would lie in every such image. The axis passes through several object vertices, whose lines would consequently all contain W , contrary to the absence of shared line ends. Hence c fixes a vertex of the auxiliary tree.

A fixed object vertex gives $c \in S_d$ immediately. If an actual component is fixed, compare it with a bounded-height invariant subtree of actual T for $\langle c \rangle$, obtained from one generator path and its translates. For remote W these subtrees are disjoint and their long connecting bridge is fixed pointwise. The cyclic dictionary forces a capture-length subblock on that bridge to have noncyclic ambient stabilizer. Capture puts c in some S_d , and the fixed actual endpoint over W belongs to Z_d . Thus ℓ_d meets W in either case. Strictness makes this same d unique as W recedes, so its line has the given end and $c \in S_d$. $\square$

Lemma 13.2 (Bounded peripheral incidences). *There is a uniform constant C such that $Y_w \cap S_d \neq 1$ implies $d_{\hat{T}}(w, \ell_d) \leq C$. There are finitely many orbits of these incidences. If $w \notin \ell_d$, the intersection is cyclic.*

Proof. Suppose $Y_w \cap S_d \neq 1$, and follow the path from ℓ_d to w . At its departure vertex v , let f be the first edge off the line. The nontrivial intersection fixes d in Γ_v and its nearest point in K_f . These are adjacent: an intervening object node would contradict strictness. At their actual-row node x ,

$$S_d \cap Y_f = M_x^d \cap \tilde{F}_{x,f}.$$

This intersection is cyclic. Local cocompactness and quasiconvex double-coset finiteness in X_x give finitely many departure types (d, v, f) with nontrivial intersection.

At every later off-line vertex, a nonzero element of this cyclic group acts hyperbolically on the incidence tree. Its axis determines at most one preserved continuation in each slot. Moreover a cyclic generator fixing the vertex fixes a continuation whenever a nonzero power does, by the stabilization argument in Lemma 12.2. Thus the path cannot reverse slots after departure. Unbounded distance for one of the finitely many departure types would give an off-line monotone ray fixed by a common nonzero power. Lemma 13.1 would put its end on ℓ_d , which is impossible for a ray that has left that line in a tree. This gives the uniform distance bound. The same finite departure lists and unique continuations give finitely many incidence orbits within that bound. $\square$

13.2 The unfilled relative models

We now use these bounded incidence lists to construct the unfilled relative rows. The acting groups on adjacent places of $\hat{T}$ are denoted $Y_f \leq Y_v$, with conjugations implicit. Peripheral intersections mean all nontrivial $Y_z \cap S_d$, up to Y_z ; include also the intersections with the extra maximal cyclics when those are added. Recall the hypotheses reached for these arguments: finite intersection orbit lists; included S_d 's (line through the place) full, others cyclic; malnormality; the incidence tree Γ_v with quasiconvex M_x at its actual X -pieces, and adjacent image subtree $K_f \subset \Gamma_v$ of finite quotient, on which the S_d -nodes again have full groups and the X -intersections $\tilde{F}_x$ are finitely generated and quasiconvex in X_x . All groups here are torsion-free before filling.

Definition 13.3 (Exact relative embedding). Let J be hyperbolic relative to a finite list $\mathcal{P}$. An *exact relative embedding* of $B \leq J$ means that:

- (i) the nontrivial groups $B \cap p P p^{-1}$, for $P \in \mathcal{P}$, have a finite list of representatives up to B -conjugacy;
- (ii) B is finitely generated relative to this list; and
- (iii) its coned Cayley graph maps quasi-isometrically into the coned Cayley graph of J , with each peripheral apex sent to the corresponding actual peripheral coset.

Thus the apex map is injective on actual cosets. “Exact” refers to these induced intersections; it does not require that an induced subgroup equal the ambient peripheral.

Lemma 13.4 (Relative row geometry). *Under the local hypotheses just listed, each folded vertex row Y_v is hyperbolic relative to its finite induced peripheral system, and each adjacent edge map $Y_f \hookrightarrow Y_v$ is an exact relative embedding. The assertions remain valid after adjoining the external maximal cyclic intersections, provided their incidence lists satisfy the same local hypotheses.*

Proof. Each Y_v is hyperbolic relative first to just its included S_d 's: glue coned fine hyperbolic graphs for its actual X -nodes along the single apex points corresponding to their S_d -neighbors. In X these use the nontrivial malnormal quasiconvex incidence groups. This is a tree of graphs glued at points by Γ_v ; hyperbolicity and fineness follow piecewise, with finite edge orbit set and finite edge stabilizers.

The infinite vertex stabilizers of this glued graph are exactly the included object groups, in their actual coset positions; ordinary vertices have finite stabilizers. Thus [24, Proposition 3.47] gives a finite relative presentation with linear relative Dehn function. This comparison requires neither absolute finite generation of Y_v nor finite generation of an S_d . The same comparison applies to the edge-row model below with its exact induced apex stabilizers. A row equal to a single peripheral has the degenerate relative presentation with no ordinary generators or relators; a trivial row is immediate.

The edge group Y_f is relatively hyperbolic and cone-quasi-isometrically embedded with its induced peripheral structure for this list. Indeed use K_f , gluing the corresponding relative models of $\tilde{F}_x$ (with all nontrivial intersections with incident M 's) at the included points. A nontrivial S_d -intersection for $d \in \Gamma_v - K_f$ is witnessed adjacent to K_f , in the corresponding actual piece (project from d to K_f , using strictness).

The models at X embed relatively quasi-isometrically by ordinary quasiconvexity of $\tilde{F}_x$. One proof uses coarse nearest projection onto this quasiconvex orbit in X : projections of an incident M -coset have bounded diameter if the coset stays far away, otherwise lie boundedly near an orbit of the exact intersection. This follows by thin quadrilaterals and quasiconvexity, then ordinary properness (normalize a close pair, and pairs at bounded distance have only finitely many orbits under the intersection). Thus projections coarsely bound every cone jump in the induced cone metric. The intersection system itself is malnormal and quasiconvex there, finite up to the subgroup.

Summing distances over the point gluings preserves the relative embedding (each internal passage between distinct gluing apices costs positively). These models are equivalent to relative Cayley metrics by relative cocompactness or normal paths along the Bass tree.

External cyclic peripheral intersections to be added in Y_v are maximal loxodromic cyclics there, disjointly malnormal. Indeed root-closedness and disjointness are inherited (including for the extra family), these groups meet no included conjugate nontrivially, and loxodromic elementary closures are virtually cyclic, hence cyclic here.

Thus adding them still gives relative hyperbolicity by elementary-subgroup peripheral enlargement [54, Corollary 1.7]. For finitely many additions, iterate this statement over the distinct maximal cyclic types: a remaining type cannot become conjugate into an earlier one without being commensurable with it. The relative embedding above persists with the induced intersections, as we now prove separately.

Use coned Cayley graphs (ordinary T -edges for finite relative generators, radial edges to coset apices). An embedded subgroup's cone path is mapped with fixed representative offsets p_α for its peripheral list $B \cap p_\alpha P_i p_\alpha^{-1}$, via $b, bp_\alpha, bp_\alpha P_i$ at a jump. We can enlarge T by all offsets and ordinary letters needed, and also by a generator for each new loxodromic cyclic subgroup, whose translates now have uniform ordinary-edged quasigeodesic axis paths in the old coned graph. Embedded geodesics map to uniform quasigeodesics with no repeated old apices (actual apex map injective).

Recall the proper angle property at an old apex: for two incident ordinary vertices joined avoiding that apex in bounded length, the displacement belongs to a finite list inside the corresponding peripheral.

Project a new cyclic coset coarsely to the quasiconvex embedded orbit of B in this old graph. If two projections are far apart in the old metric, a B -comparison quasigeodesic between them, except for bounded ends, stays boundedly near that axis (coarse projection

and thin quadrilaterals).

Frame vertices b sufficiently inside this portion are in fact at finite-list ordinary displacement from axis frame points a . Indeed nearby apex turns on the comparison path have bounded angle: bypass via two short bridges to the axis a sufficiently large fixed distance before and after, avoiding the apex by quasigeodesicity, and the bounded axis subpath. The short bridge at b itself then has bounded angles at each jump, using one farther such bridge for a bypass and replacing occurrences of that apex along the comparison path by the bounded-angle detour. Take bridges geodesic so do not revisit the apex.

Proper angles give $b^{-1}a$ from a fixed finite list. Pairs with a given displacement form one orbit under the coset stabilizer in B . Normalizing one pair shows there are finitely many coset types with this large projection, and all such middle projections are at bounded relative distance of one another after coning the stabilizer if nontrivial (otherwise already bounded; finitely many pair orbits in the normalized types). End errors are bounded too by the old relative embedding.

Conversely every infinite cyclic intersection gives unbounded old projection. This gives exactly the finite induced new list (malnormal loxodromic cyclic subgroups also in B , by the old embedding), and the nearest projection now bounds each additional shortcut in the enlarged metric of B . Hence the assertions on relative hyperbolicity and embedding. Apply with $B = Y_f$. All these constructions use only finite orbit lists. "Full" relative maps of peripheral groups are not required. $\square$

Lemma 13.5 (Hyperbolicity of the partial metric). *The tree of relative row spaces before filling is hyperbolic. Its metric is quasi-isometric to the partial Cayley metric on G obtained by allowing arbitrary S_d -moves and, if present, arbitrary moves in the additional root cyclics.*

Proof. Spelling paths in the Bass presentation identifies the tree of relative row spaces, up to quasi-isometry, with the partial Cayley metric that permits arbitrary moves in the S_d for the finitely many G -orbit representatives. There are finitely many attaching frames and offsets, every local cone uses one such group, and each S_d is fully available on its line. The additional cyclic groups, if present, are fully available at their fixed places.

Use a finite linear relative presentation of G relative to the D_d and any additional cyclics. Inside D_d , the partial metric has bounded height fibers and is quasi-isometric to D_d/S_d , a line; inside an added cyclic it is bounded. A partial-metric loop therefore admits a relative filling with linearly many ordinary relative cells. Their peripheral labels belong to fixed finite lists, while the loop's own peripheral moves have bounded partial length. The total partial length on peripheral cluster boundaries is consequently linear in the original loop length.

Choose the filling with minimal ordinary relative area. A hole in a peripheral cluster can be filled entirely in that peripheral, by peripheral injectivity; doing so eliminates any ordinary cells inside. We may therefore treat clusters as discs. Each cluster boundary has a linear filling at a fixed Rips scale in its partial metric: push vertices by bounded moves to a lifted height line and fill there, or use the bounded metric for an extra cyclic. Combining these fillings gives a linear Rips filling for the partial Cayley graph, hence hyperbolicity. No finite generation of S_d is used. $\square$

13.3 Unfilled comparisons and intersections

In these coned Cayley graphs, a bounded path avoiding an apex gives a peripheral displacement from a finite list, by proper angles. Add the finitely many subgroup offsets and endpoint connectors as ordinary generators first. This convention lets the following comparison retain actual subgroup-frame coordinates.

Lemma 13.6 (Unfilled comparison paths). *Fix two exact relative embeddings into a row, and fixed ordinary endpoint connectors. Expanded intrinsic geodesics have the following comparison. Large-angle apices occur on both paths in the same order. Their corresponding inbound and outbound germs differ by peripheral elements from fixed finite lists. Between successive marked apices there are paired ordinary subgroup-frame checkpoints, at bounded successive path steps, whose discrepancies lie in a fixed finite subset of the row. The same conclusion holds in the central part of long paths tending to the endpoints of a common loxodromic element, with constants independent of remote endpoint costs.*

Proof. Expand the two intrinsic geodesics in the row, including their fixed ordinary offsets. They are uniform quasigeodesics with no repeated ambient apices. Hyperbolicity makes them fellow travel coarsely in order, after adding the fixed endpoint connectors. In particular, matched positions sufficiently far apart occur in the same order.

Choose a large angle threshold. A turn above this threshold on one path must also occur on the other. Otherwise two short bridges, placed a fixed distance before and after the turn, together with the other path, give a bounded bypass avoiding its apex. Near an endpoint use the prescribed endpoint connector instead. The same bypasses bound the angles between the respective inbound germs, and between the outbound germs, at a common marked apex. Proper angles give their finite peripheral discrepancy lists.

The marked apices occur in the same order. A reversed pair could only occur in a bounded portion, by ordered fellow travelling. If their orders were v, w and w, v , an earlier bridge to the second path, the segment there to w , and the first path back to a point after v would bypass v in bounded length. A later bridge gives the analogous bypass on the other path. Either contradicts the chosen large angle threshold.

Between successive marks all path turns have bounded angles. Pair ordinary subdivision points by short geodesic bridges, with bounded successive steps on each path. For a jump of a bridge at an apex z , another bridge a fixed distance away gives a bounded bypass using the two path segments. Any visit to z on those segments can be replaced by its bounded-angle detour. For a short chunk use an endpoint connector; at a marked endpoint use the already bounded germ comparison if that endpoint is z . Thus every bridge jump has bounded angle. Proper angles turn its bounded number of jumps into a finite list of ordinary group letters, yielding the checkpoint discrepancy list. Include offsets to return checkpoints to subgroup frames, while keeping the marked germ connectors.

All these constructions apply in long central portions of fellow-travelling paths without using their remote endpoint connectors. In particular they apply towards both endpoints of a common loxodromic element, with constants independent of those remote endpoint costs. Proper angles supply the finite lists; local compactness of the coned graph is not required. $\square$

Lemma 13.7 (Unfilled intersections). *If B, C have exact relative embeddings into a row J , then for every fixed $g \in J$, the subgroup $E = B \cap gCg^{-1}$ is relatively hyperbolic and has exact relative embeddings into J , B , and gCg^{-1} .*

Proof. Put $E = B \cap gCg^{-1}$. Take the finite path comparison for $x \in E$, using frames $1, x$ in B and g, xg in gC . All paired frame checkpoints have finite-list discrepancy in the actual group. For each discrepancy u , pairs $(b, d) \in B \times gC$ with $b^{-1}d = u$ form one orbit (hb_u, hd_u) under $h \in E$. Start and end use $(1, g)$ as their orbit representative. Small steps outside marked jumps therefore translate h by finite lists.

At a marked jump the shared actual apex has forms $ha = h'a'$ from the two successive checkpoints, where a, a' are apex states from a finite list (via b_u and offsets); thus $h^{-1}h'$ is a bounded ordinary move plus an actual E -stabilizer move at such an apex. This relatively generates E for a finite list, with cone lengths bounded by the comparison length, hence

quasigeodesically embedded. In more detail the number of these successive relative generators needed is linear in the coned distance in J , since the two subgroup comparison paths already have bounded embedding constants (for fixed end offsets); ordinary discrepancies and apex states above use finite sets independent of x .

We next identify the peripheral structure, rather than just a list of elliptic subgroups. All nontrivial peripheral intersections appear: powers in any such intersection force arbitrarily large-angle passages at its apex, by the proper angle test using a fixed path towards it and its translate, since comparison paths then have bounded relative length. The apex map is injective on actual cosets after taking orbit representatives. The domain coned graph is fine by mapping bounded avoiding paths to avoiding paths (add the finitely many ordinary generators/offsets first), and is hyperbolic by quasigeodesicity. Fineness follows from proper angles at the domain apices (bounded avoiding paths give only finitely many next directions) and finite degree at ordinary vertices.

This graph has finite edge stabilizers and finitely many edge orbits; its infinite vertex stabilizers are exactly the nontrivial induced peripheral intersections, since the groups are torsion-free. Proposition 3.47 of [24] therefore gives relative hyperbolicity in the finite-relative-presentation sense, without assuming absolute finite generation of E or its peripherals. The trivial and single-peripheral cases have their immediate relative presentations. Embedding holds likewise into B or gCg^{-1} with their structures. $\square$

Corollary 13.8 (Unfilled nonparabolic double cosets). *For fixed exact relative embeddings of B, C in a row J , the double cosets BgC having nonparabolic intersection $B \cap gCg^{-1}$ have representatives in a fixed finite subset of J .*

Proof. A nontrivial nonparabolic subgroup of a torsion-free relatively hyperbolic group contains a loxodromic element. Compare long paths in the two embedded groups tending to its two endpoints. The central comparison in Lemma 13.6 gives paired frame points $b \in B$ and $gc \in gC$ with $b^{-1}gc$ in a fixed finite list. This is a representative of BgC . The list is independent of g , because only central comparison constants are used, not the remote endpoint connectors. $\square$

14 Terminal paths and finite-path filling transfer

The unfilled relative models now supply the hypotheses for the expansion argument. We first obtain a terminal path bound, then fix the finite original tests that transfer path stabilizers exactly to a filling.

14.1 Relative free factors and the terminal path bound

Proposition 14.1 (Simultaneous relative factors). *For a sufficiently long monotone folded path W , its pointwise stabilizer H has a finite relative free-product decomposition*

$$H = \left(\ast_{j=1}^m (H \cap S_{d_j}) \right) \ast F_r.$$

The factors represent exactly the nontrivial peripheral intersections, up to H -conjugacy. Vertices of the corresponding relative Bass–Serre tree that have nontrivial stabilizer carry injective, equivariant labels by actual objects d . The trivial vertices can be chosen of finite valence.

Proof. Let $H = \text{Stab}(W)$. By bounded peripheral incidences, every nonzero $H \cap S_d$ has ℓ_d containing the sufficiently interior portion of W . Choose there two folded edge positions w, w' whose actual rows are free and whose height difference exceeds the actual capture length.

At an actual node x of Γ_w , put $H_x = H \cap X_x$. We show that all its nontrivial boundary incidences are simultaneous free factors by acting on $\Gamma_{w'}$. An actual-node stabilizer for H_x fixes a long bridge back to x in actual T . If noncyclic, it lies wholly in a captured S_d . If cyclic and meeting an incident S_d nontrivially, root closure again puts it wholly there. Strictness makes the incident object unique. Collapse the resulting nontrivial-edge stars, whose actual nodes are redundant leaves. The surviving splitting has trivial edge groups.

Every needed boundary incidence at x is represented by the same actual label d at w and w' , since its line contains the intervening interior window. The H_x -orbits of these labels cannot merge at the higher level without having been equal originally. Thus the corresponding groups are simultaneous free factors, including their specified embeddings up to attaching conjugations. As in Proposition 12.3, substitute these factors into the Bass presentation of H on Γ_w . This gives

$$H \cong \left(*_j (H \cap S_{d_j}) \right) * F_r$$

with all nontrivial peripheral intersections represented. Folded inertia bounds $\beta(H)$, and the trivial-edge free-product dimension formula then bounds both the number of nontrivial factors and r when $H \neq 1$; the trivial case is immediate. Thus the relative decomposition is finite.

Use its relative Bass–Serre tree for R_i , choosing finite-valence trivial vertices. Nontrivial factor vertices have injective actual object labels: a transporter in H between occurrences with label d normalizes their nontrivial intersection with S_d . Strictness puts it in S_d , hence in the factor stabilizer, so the occurrences are equal. These labels, rather than conjugacy classes, are needed in the next argument.

Finally a nonparabolic subgroup has a loxodromic element in this tree. A nontrivial elliptic element fixes a unique vertex because edge stabilizers are trivial. Two elliptic elements with disjoint fixed sets have a loxodromic product. Hence a subgroup without loxodromics fixes one vertex, and exactness identifies its nontrivial stabilizer with an actual peripheral intersection. This would make the subgroup parabolic. $\square$

Choose a fixed short-window radius large enough for Proposition 14.1. We can now verify the expansion package of Definition 8.1 for the resulting relative trees. Condition (E1) is Lemma 13.5 together with the adjacent embeddings of Lemma 13.4. For (E2), intersect the current path group with each next edge group. Lemma 13.7 preserves exact relative embeddings, and Corollary 13.8 gives finitely many nonparabolic extensions. Induction over any fixed shape gives its finite list and uniform embedding constants, also with a specified common subwindow. Proposition 14.1 gives (E3), including exact peripheral restrictions and loxodromics for nonparabolic window groups. Finally distinct continuations have distinct terminal places at the same height; Lemma 11.5 makes their common stabilizer at most cyclic, which is (E4).

Proposition 14.2 (Terminal path bound). *There is a length n_0 such that every segment of $\widehat{T}$ of length at least n_0 has trivial, cyclic, or parabolic pointwise stabilizer. Moreover there is a uniform bound on the length of a segment fixed by any nontrivial height-zero element lying outside all the S_d .*

Proof. We first rule out arbitrarily long monotone paths with nonparabolic stabilizers. Suppose they exist. Normalize a centered short window to one fixed W_0 , put $H_0 = \text{Stab}(W_0)$, and let R_0 be its relative free-product tree. Each longer window group has a loxodromic axis in R_0 . Orient such an axis and mark one edge at index zero. Translate by H_0 to make

this edge fixed. For each finite extension length there are now only finitely many actual possibilities: there are finitely many extension orbits, each relevant minimal subtree has a finite edge quotient, and a regular edge has trivial stabilizer. Passing diagonally therefore fixes every finite extension of W_0 .

At the same time, diagonalize the oriented turn labels of the axes, indexed relative to the marked edge. At each index the label is eventually constant or leaves every finite set of labels. We call the latter an escaping turn. A turn at a finite-valence trivial vertex cannot escape.

There cannot be two escaping turns. Choose consecutive escaping indices, so all intermediate turn labels are constant, and normalize the intervening edge path. Its edge markings determine actual coordinates because its edge stabilizers are trivial. Its two end vertices are then fixed peripheral vertices labeled by two distinct actual objects d, e . Distinctness here is the label assertion following Proposition 14.1, not merely distinctness of conjugacy classes. Diagonalize the window extensions in these new coordinates, retaining a regular edge of the intervening path as marking. For every fixed depth the corresponding minimal subtree contains infinitely many directions at each of these vertices. Since it has finitely many edge orbits, the stabilizer of each vertex is nontrivial: otherwise each global edge orbit could contribute at most one incident edge at that vertex. Thus every finite-depth window stabilizer intersects both S_d and S_e nontrivially. Lemma 13.2 forces the interiors of all these windows to lie on both object lines. Their limiting monotone ends coincide, contradicting the terminal system.

Consequently at least one half of the axis coordinates has no escaping turn. The constant turns and the marked edge define an actual infinite ray $\rho \subset R_0$. Every finite arc of ρ is carried on lines in the minimal subtree of every finite-window stabilizer of the limiting bilateral trajectory τ .

The first diagonal argument has produced an infinite carried ray. It has not yet produced a gate event that recurs on that ray. We obtain repeated finite arcs by a second normalization, now of disjoint blocks. Choose infinitely many disjoint blocks in ρ , each of length greater than $8R + 4$, where $2R$ is the gate-event length. Normalize a regular edge in each block by H_0 , and diagonalize its finitely many turn positions. The preceding two-escape argument applies again: together with each normalized block translate the whole trajectory, and diagonalize its finite extensions before using infinite valence. Every block is carried through every finite depth, so two escaping positions would again force distinct object lines to share an end. Removing the possible single escaping turn leaves a subblock of length at least $2R$ whose turn type is constant along a subsequence. There are therefore infinitely many disjoint H_0 -equivalent arcs of length $2R$ on ρ .

Fix a depth large enough for phase uniqueness in both signs. The corresponding window stabilizer has finite edge quotient on its minimal subtree. Among the repeated arcs, infinitely many marked initial edges belong to one orbit under this smaller stabilizer. Its transporter carries the entire arc, since the arcs already have the same H_0 -type and the marked edge has trivial stabilizer. Choose one of these arcs. Flaring supplies an expanding sign. Phase uniqueness, applied to τ and each of its translates, says that every such transporter preserves the expanding end of τ ; all required initial comparisons lie in the chosen fixed window. Bounded equivariance errors are included in the gate-event thresholds.

Each transporter has height zero and fixes that initial window. Preserving its expanding end therefore makes it fix the ray to that end pointwise. Lemma 13.1 puts every nonidentity transporter in the same S_d , because only one object line has that end. But $H_0 \cap S_d$ fixes a vertex of R_0 . Its orbit of any fixed arc has bounded distance from that vertex and cannot contain arcs going arbitrarily far along the single ray ρ . This is the required contradiction.

It follows that sufficiently long paths in $\hat{T}$ have at most cyclic, trivial or parabolic stabilizer (nonmonotone paths have a reversal). In fact a path fixed by a height-zero element outside

every S_d has bounded length: its sufficiently long blocks would all have nonparabolic cyclic stabilizers, with finite orbit dictionary for fixed block size, and repeating an oriented block would commensurate the element by a tree translation. $\square$

Corollary 14.3 (The additional root cyclic). *In the root case in which no power of the root cyclic group is claimed by the object system, its maximal cyclic overgroup may be added to the peripheral family. Its nontrivial incidences have finite orbit lists, and the relative embeddings and finite-path conclusions continue to hold with the enlarged induced systems.*

Proof. In the root case when powers of actual $U = \langle u \rangle$ are not hit by the S_d system, one can now peripheralize also the maximal cyclic containing U in G (call its translates U^*). This is a disjointly malnormal quasiconvex addition, at height zero. It is elliptic on $\hat{T}$ since a power is elliptic. The positions fixed by powers are at bounded distance as just shown, with finite branching: at a folded vertex, powers acting on its Γ either translate with the same axis (at most one preserved edge per slot), or fix a unique actual x -node, necessarily contained in any preserved K_f . Then only finitely many actual f 's there can occur, by quasiconvex double-coset finiteness between a fixed cyclic power subgroup in X_x and the corresponding $\tilde{F}$'s (finite orbits under that cyclic and each orbit finite). No power fixes an object node. Thus the finite peripheral incidence lists and exact relative embeddings still hold by the peripheral addition argument in Lemma 13.4. These are the local hypotheses for the finite-path transfer below; the terminal path bound was obtained before this addition. $\square$

14.2 The finite tests and their order of choice

For all filling assertions in this section, choose for each periodic peripheral a normal subgroup $N_d \triangleleft D_d$ with $N_d \leq S_d$ and $[S_d : N_d] < \infty$, compatibly under the actual G -action. For an added cyclic peripheral choose a finite-index subgroup. At a row incidence $P = Y_w \cap S_d$, the induced kernel is $N_P = P \cap N_d$; use the analogous restriction for a cyclic addition. Thus every row peripheral reduction P/N_P is finite, whereas D_d/N_d is finite-by-infinite-cyclic. The word *compatible* below includes these restrictions and normality conditions. Their construction, and attainment of the further finite requests below, belong to Section 15.

We record the finite-window and quotient argument more precisely. Use relative embeddings as above into one row J , with finite lists and exact induced peripheral systems, say $B, C \leq J$. Cone-quasi-isometric embeddings with this terminology have relatively finitely generated domain for the finite induced list, trivial intersections omitted. All induced peripheral subgroups inside a fixed actual peripheral are full or cyclic in the uses below, a property also preserved by taking intersections. We use the relative Dehn filling theorem and its uniform linear relative isoperimetric bound for sufficiently long fillings [55]. Thus in coned quotient graphs hyperbolicity constants are uniform and bounded avoiding paths at an apex give displacement represented by a **fixed finite original list** in its peripheral (isolated component lemma applied to the uniformly linear presentations with fixed relative relators). This holds also with any fixed finitely many ordinary letters added. Here and below fixed elements needed in ordinary connectors (even in groups only assumed relatively finitely generated) can be added as ordinary letters first. Fixed finite word equalities lift exactly, as does membership of a fixed finite element in a specified peripheral (add the tested element as an ordinary edge to use the angle bound, then finite-list injection). These are relative statements not requiring finite generation inside the peripherals. We induce standalone fillings on B, C by intersection of kernels; they are long and their reduced peripherals inject exactly there and into the ambient peripheral reductions.

The finite peripheral membership safeguards and induced subgroup fillings below have an antecedent in Groves–Manning’s H -wide filling method [25, Definitions 3.1–3.3 and Propositions 4.5–4.6]. Here the rows are only relatively finitely generated, with no finite-generation requirement inside the peripherals, and we impose additional product-membership and exact-intersection conditions. The stronger exact-intersection assertion below is therefore a separate local argument, not a theorem invoked from that source.

Definition 14.4 (Peripheral compatibility tests). Fix the rows and subgroup embeddings to be tested, and a finite list of their peripheral subgroup occurrences. For each ambient peripheral P , a filling kernel $N_P \triangleleft P$ is required to satisfy the following conditions for specified finite original lists:

- inside a peripheral, finite-list membership in each indicated intersection subgroup Q , or product QtR for a specified t and indicated subgroups Q, R , lifts from the peripheral quotient;
- $\bar{Q} \cap \bar{s}R\bar{s}^{-1} = \overline{Q \cap sRs^{-1}}$ there for specified s .

Here all letters and subgroups are in that one peripheral’s coordinates. The second condition concerns only cyclic pairs with trivial intersection (commensurate cyclics satisfy it automatically, by the gcd formula for subgroups in cyclic quotients). Full here means the entire indicated height-zero peripheral (not the whole periodic group). In Bass applications the property full-or-cyclic for new path intersections follows since at each constituent place a nontrivial object intersection is itself full or cyclic. We explain simultaneous attainment separately, including preserving the required double-coset inequalities by separability around cyclics.

The word “specified” has a fixed quantifier order. First choose the original subgroup embeddings and a path length to be tested. Next choose the geometric comparison constants and the finite original connector alphabets. Next record the resulting finite membership, product-membership and exact-intersection requests. Only then choose the kernels. The construction in Section 15 attains every such finite collection. Increasing depth later preserves previously imposed finite exclusions.

For membership, the assertion is that an element of the specified original finite list that belongs to $\bar{Q}$ after reduction already belongs to Q . For product membership it is the analogous assertion for the specified translate QtR . Exactness of peripheral intersections is a group equality, not merely a finite-element test, but only finitely many pairs and conjugators are requested.

Lemma 14.5 (Induced subgroup fillings). *For a fixed finite collection of exact relative embeddings into rows, one can choose the finite tests in Definition 14.4 so that every sufficiently long filling satisfying them has the following properties. Each standalone induced subgroup filling injects into the row filling, and the resulting coned embeddings have common quasi-isometry constants independent of the filling. Their expanded intrinsic geodesics have no repeated ambient peripheral apex.*

Proof. We prove the assertion for B ; the same argument applies to every group in the finite collection. The original embedding constants give local quasigeodesic constants (λ_0, ϵ_0) . Together with the uniform quotient hyperbolicity constant, these determine a sufficient local scale and global constants (λ_1, ϵ_1) . Choose the tested scale larger also than $\lambda_1\epsilon_1$, with the fixed offset margins, before making any finite lists.

A bounded-length intrinsic coned geodesic in $\bar{B}$ still has no repeated apices in the ambient expanded path. Otherwise take a minimal bad repetition. Internal jumps have bounded angle by the loop, hence lift by finite-list membership to bounded-list actual peripheral letters of

B preserving their values in $\bar{B}$. Equality of the endpoint ambient cosets thus already held before filling, forcing repetition of the domain apex.

Next compare an ordinary-endpoint such path, starting at identity, with any ambient geodesic shortcut. In the bounded union graph a jump not at a cut apex separating start and end has bounded angle (connect inbound to start and outbound to end, and these to each other avoiding that apex). Thus lift such letters as just above: the actual finite-list representative in an intersection subgroup after the membership test gives even the required **standalone** peripheral value, since its kernel there is the restriction and the reduced intersection subgroup injects in the ambient peripheral reduction.

Cut apices occur on both paths in the same order. Discrepancies between their ordinary inbound germs, likewise outbound germs, are from finite lists in that peripheral by avoiding paths through start or end. Lift those connectors. Between cuts all consistency equations with the connectors hold by fixed-list injection. Choose any actual lift in the indicated subgroup for the B -jump at each cut; the shortcut jump can lift compatibly via the connectors since it only has to lie in the ambient peripheral.

Thus some actual B lift of the intrinsic word has a shortcut of the same bounded cost. The intrinsic distance after reduction is no larger than that of this lift, where the original relative embedding bounds it by the shortcut cost. This proves the chosen local quasigeodesic bounds; trimming to ordinary frame endpoints costs only the fixed margins. Local-to-global now gives the uniform global constants (λ_1, ϵ_1) .

A repeated apex on the global expanded path would cut off a subpath of length at most $\lambda_1 \epsilon_1$, contradicting the local no-repetition check. The same estimate puts every potential kernel word within the tested scale. Its lifted length-zero shortcut makes the corresponding original subgroup word trivial, and the chosen peripheral lifts preserve its value in the standalone filling. Thus its standalone value is trivial. This proves injection and all the assertions simultaneously for the finite collection. $\square$

14.3 Uniform comparison and exact intersections

Lemma 14.6 (Finite original comparison data). *For the fixed embeddings and connectors of Lemma 13.6, every sufficiently long filling satisfying Lemma 14.5 has the same comparison conclusions. All discrepancy and peripheral-germ lists are images of fixed finite lists in the original groups, independent of the filling.*

Proof. The subgroup-filling lemma gives uniform quasi-isometry constants and no repeated ambient apices for expanded intrinsic geodesics. Relative filling gives a uniform hyperbolicity constant. At every fixed bypass length, the isolated-component bound places peripheral displacements in the image of a fixed finite original list. This remains true after the finitely many connector letters and offsets have been added as ordinary generators.

Apply the proof of Lemma 13.6 with these constants. Its fellow-travelling, marked-apex, and bypass arguments use only those bounds. Every bounded-angle jump in a comparison bridge consequently has a representative in its fixed original list. Products of the bounded number of such letters give fixed original checkpoint-discrepancy lists as well. The central comparison uses the same constants independently of remote endpoints, exactly as before filling. $\square$

Corollary 14.7 (Finite double-coset representatives). *For fixed original exact relative embeddings of B, C in a row, every double coset in a sufficiently long filling whose intersection contains an infinite-order loxodromic element has a representative in the image of a fixed finite original list. The list is independent of the double coset and of the filling.*

Proof. Compare paths towards the two endpoints of the common loxodromic as in Corollary 13.8, now using Lemma 14.6. A pair of central frame points gives the required representative from its fixed original discrepancy list. Row peripheral images are finite, so every infinite-order element relevant here is loxodromic. $\square$

Proposition 14.8 (Exact intersection transfer). *Fix exact relative embeddings $B, C \hookrightarrow J$ with the full-or-cyclic induced peripheral systems described above, and fix $g \in J$. There are finitely many compatibility requests of Definition 14.4, determined in the original groups, such that every sufficiently long filling satisfying them has*

$$\bar{B} \cap \bar{g}\bar{C}\bar{g}^{-1} = \overline{B \cap gCg^{-1}}.$$

The notation $\bar{B}, \bar{C}$ denotes the embedded standalone fillings given by Lemma 14.5.

Proof. Use the quotient comparison of Lemma 14.6. Its small-letter and germ-connector lists determine the finite tests before kernels are chosen. We show how their local lifts concatenate. For an element $\bar{x}$ of the intersection, compare paths from 1 to $\bar{x}$ in $\bar{B}$ and from $\bar{g}$ to $\bar{x}\bar{g}$ in $\bar{g}\bar{C}$. At paired checkpoints write

$$\bar{d}_i = \bar{b}_i \bar{u}_i, \quad \bar{u}_0 = \bar{u}_m = \bar{g}.$$

Choose once a lift u_i from the finite connector list for each checkpoint; the initial and final lifts are both the specified g . Reuse that same lift in the two strips adjacent to the checkpoint. The increments satisfy

$$\bar{\beta}_i \bar{u}_{i+1} = \bar{u}_i \bar{\gamma}_i, \quad \bar{\beta}_i \in \bar{B}, \quad \bar{\gamma}_i \in \bar{C}.$$

On a small strip, the increments have lifts $\beta_i \in B, \gamma_i \in C$ in prescribed finite lists, and injection on the resulting finite set of words gives

$$\beta_i u_{i+1} = u_i \gamma_i. \tag{14.1}$$

At a marked strip, remove the fixed entry and exit offsets. The remaining equation is exactly $qt = sr$ in one original peripheral. Its reduced solution says that $\bar{s} \in \bar{Q}\bar{t}\bar{R}$. The product-membership test for the fixed connector s supplies an original solution $q_0 t = s r_0$. For the desired reduced values $\bar{q}, \bar{r}$, one then has

$$\bar{q}\bar{q}_0^{-1} \in \bar{Q} \cap \bar{s}\bar{R}\bar{s}^{-1}.$$

Choose $a \in Q \cap sRs^{-1}$ lifting this element. The assignments

$$q = aq_0, \quad r = s^{-1}asr_0$$

solve the original equation with exactly the prescribed reduced values. Restoring the entry and exit offsets gives (14.1) on this strip too. Telescoping all strips now gives

$$(\beta_0 \cdots \beta_{m-1})g = g(\gamma_0 \cdots \gamma_{m-1}).$$

The left product belongs to $B \cap gCg^{-1}$, and its reduction is $\bar{x}$. This proves the difficult containment; the reverse containment follows from the quotient homomorphism. In particular the proof does not choose unrelated lifts of a shared connector and then assume compatibility. $\square$

14.4 Transfer along Bass–Serre paths

Theorem 14.9 (Finite-path transfer). *Fix an integer n . Under the terminal row hypotheses, there is a finite set of original compatibility requests such that every sufficiently long compatible filling has the following properties.*

- (i) *The induced row fillings and edge fillings form the quotient graph of groups for $\bar{G}$.*
- (ii) *Every reduced path of length at most n downstairs with infinite stabilizer lifts, after the permitted changes of Bass–Serre representatives, to an actual reduced path upstairs. Its stabilizer is exactly the image of the actual path stabilizer.*
- (iii) *Taking n at least the terminal path bound, the resulting Bass–Serre action is acylindrical: sufficiently long segment stabilizers are finite of uniformly bounded order. Each filled row is hyperbolic and each edge map is quasi-isometric.*

Here the ambient filled group is hyperbolic by sufficiently long filling of the original relatively hyperbolic pair: the periodic peripheral reductions are finite-by-cyclic and hence hyperbolic, and the additional cyclic reductions are finite. The conclusion about ambient quasiconvexity is proved separately below.

Proof. Before filling, Lemma 13.7 gives exact relative embeddings for finite-path stabilizers into the incident rows. Corollary 13.8 gives finitely many nonparabolic extensions of each fixed path shape. We choose the filling tests recursively, for paths of length at most n .

At each step the two groups to intersect are the current actual path stabilizer and the next edge group, both embedded in the current terminal row. Fix their original quasi-isometry constants first. Corollary 14.7 then gives a finite original list of possible extension representatives downstairs whenever the intersection is infinite. Only after fixing that list do we impose Proposition 14.8 for its representatives, and Lemma 14.5 for the resulting actual intersection groups. Repeating for the fixed length n produces finitely many requests, all before choosing kernels. Restrictions onto a path group through its two containing rows agree: both are restrictions of the same actual peripheral kernels.

Under these tests the induced row and edge fillings give a quotient graph of groups. Its Bass presentation is exactly that of $\bar{G}$. Every periodic kernel is imposed at an included place of its line, and every extra cyclic kernel at a place fixed by that cyclic. Conversely all induced row relations are restrictions of these same kernels. Thus neither presentation has additional relators.

Now let a reduced path downstairs have length at most n and infinite stabilizer. Every prefix has infinite stabilizer. Start at its first edge and induct along the path using the finite representative lists and exact intersection transfer. A left change by the stabilizer of the existing prefix preserves that prefix; a right change by the new edge stabilizer preserves the new edge. These are precisely the allowed Bass–Serre changes of representatives. At each step exact transfer makes the new stabilizer the image of the actual intersection upstairs, and the standalone subgroup filling embeds with that image. A turn reduced downstairs is reduced in its lift. This proves (ii) and explains why one cannot choose unrelated row-induced kernels at successive steps.

Each filled row is ordinarily hyperbolic. Its finite relative generating set and its finitely many finite peripheral images give a finite generating set. Insert presentations of those finite groups into a linear relative presentation. In this fixed quotient each peripheral portion has bounded linear ordinary filling cost, yielding an ordinary linear isoperimetric inequality. This argument requires no absolute finite generation of the original height kernels. The induced relative embeddings give quasi-isometric edge maps.

Take n at least the terminal path bound. An infinite stabilizer of a length- n path downstairs is the image of an actual cyclic, trivial, or parabolic stabilizer. Parabolic images

in rows are finite; hence such an infinite stabilizer is virtually cyclic. The finite lifting lists give finitely many $\bar{G}$ -orbits of oriented length- n paths with infinite stabilizer. We claim this bounds the length of every segment with infinite stabilizer in the fixed quotient.

Otherwise choose sufficiently many disjoint oriented length- n subsegments fixed by one infinite-order element c . Such an element exists in any infinite subgroup of a hyperbolic group. Two subsegments belong to the same oriented-path orbit. Their transporter preserves traversal direction, so it is hyperbolic on the Bass–Serre tree. Both virtually cyclic stabilizers contain c , so this transporter commensurates $\langle c \rangle$. Its ambient commensurator is virtually cyclic, and a virtually cyclic group containing an infinite elliptic subgroup acts elliptically on a tree. This contradiction gives a uniform segment-length bound in the fixed quotient. Hyperbolicity also bounds the orders of its finite subgroups. Thus the action is acylindrical; finite stabilizers need not be trivial.

Finally Proposition 14.11 below promotes the row and edge embeddings to ambient quasiconvexity. This is the additional conclusion needed for the hierarchy after cubulating the filled vertex groups. $\square$

Lemma 14.10 (Projection of quasiconvex cosets). *Fix finitely many quasiconvex subgroup types in a hyperbolic group J , with its ordinary Cayley metric. There are uniform constants R, C with the following property. For types H, K and a coset gK , either the projection of H to gK has diameter at most C , or there are $h \in H, k \in K$ such that*

$$w = h^{-1}gk, \quad |w| \leq R.$$

After translation by h^{-1} , that projection lies within distance C of $(H \cap wKw^{-1})w$. Thus the unbounded alternative involves only finitely many normalized intersection types. These intersections are quasiconvex in the row and in the corresponding attaching group.

Proof. Thin quadrilaterals give the first alternative when the distance between the two quasiconvex cosets is sufficiently large. More explicitly, a long interval between two projection points must, away from bounded end segments, fellow travel the side joining the two original points. Quasiconvexity puts that side near H , forcing a pair of points in H, gK at uniformly bounded distance. This fixes R and a uniform bound for the projection diameter when no such pair exists. Choose such a pair (h, gk) otherwise. Translation by h^{-1} leaves H invariant and changes gK to wK , with w in a fixed finite ball.

For the normalized close pair $(1, w)$, the same quadrilateral argument puts every projection point in $wK \cap N_L(H)$ for a uniform L . Put $I = H \cap wKw^{-1}$. Close pairs

$$(a, b) \in H \times wK, \quad a^{-1}b = d, \quad |d| \leq L$$

with a fixed discrepancy d form one I -orbit: two such pairs satisfy $a'a^{-1} = b'b^{-1} \in I$. There are finitely many possible d . Choose one pair for each nonempty class; their second coordinates lie at finite distance from w . The whole overlap, and hence the projection, lies at bounded distance from Iw . There are only finitely many choices of H, K, w , so this bound is uniform. Quasiconvex intersection gives quasiconvexity of I ; quasi-isometric subgroup inclusions give it in the attaching group as well. $\square$

Proposition 14.11 (Quasiconvexity from bounded segment stabilizers). *Let A be a hyperbolic group expressed as a finite graph of finitely generated hyperbolic groups. Suppose the edge maps are quasi-isometric embeddings and there is an integer n such that every length- n Bass–Serre segment has finite pointwise stabilizer. Then every vertex and edge group is quasiconvex in A . In particular this applies to the filled graph of groups of Theorem 14.9.*

Proof. Use the ordinary Bass–Serre tree of Cayley spaces and fix a target vertex row. Project toward it, one row at a time: take coarse nearest projection to the quasiconvex parent-edge coset, then transfer across the edge by its quasi-isometric identification. Each such map is coarsely Lipschitz, and successive maps agree up to bounded error across the attaching strips. We first work at a fixed bounded depth.

After the first transfer the projected set lies in an attaching-edge orbit. The inductive assertion is this: after j transfers the set has uniformly bounded diameter, or it lies in a uniformly bounded neighborhood of a coset of the exact stabilizer of the traversed j -edge segment, quasiconvex in the current row and belonging to a finite list of normalized types. The first transfer establishes the assertion with the finite list of edge types.

For the next transfer apply Lemma 14.10 to the controlling coset and the next attaching coset. A bounded projection gives the first alternative and remains bounded under further transfers of bounded depth. Otherwise normalize a close pair. Only finitely many bounded discrepancies occur, and the new controlling group is exactly the intersection of the previous path stabilizer with the next edge stabilizer. In the notation of that lemma, frame at w before crossing: the orbit Iw becomes the subgroup $w^{-1}Iw \leq K$ of the attaching group. Quasiconvexity is preserved by intersection and by the quasi-isometric edge map. This proves the assertion and constructs its next finite type list. The bounded neighborhood errors also remain uniform for any fixed number of transfers.

At depth $n + 1$, every surviving controlling group fixes a segment of length at least n and is therefore finite. The finite list of *normalized* groups now gives a uniform bound on their orbit diameters. This is where the normalization is essential: a bound on finite subgroup orders alone would not bound the diameter of an arbitrarily based conjugate orbit. It follows that the projection of every entire row at depth $n + 1$ has uniformly bounded diameter.

Extend over each farther branch by a constant chosen from the image of its depth- $n+1$ row. The bounded image diameter preserves coarse Lipschitz compatibility across the boundary strips. We obtain a coarse Lipschitz retraction onto the target row, equal to the identity there up to a uniform error. Summing its bound along ambient paths proves undistortion of the vertex group. Geodesic stability in the hyperbolic ambient group makes it quasiconvex. Each edge group is quasiconvex in an adjacent row, hence also in the ambient group. $\square$

15 Choice of the height-fiber kernels

We construct normal kernels that preserve height, make each height-kernel image finite, and meet the transfer tests already fixed. Their restrictions to actual incidence groups must also give the exact homological graph covers used in the later cubulation. The construction therefore combines finite-index separations with characters and their nontrivial cyclic-cover cohomology modes.

Fix one periodic peripheral D , its height kernel $S = D \cap L$, and its actual invariant subtree $Z_D \subset T$. The conclusions of Sections 5, 10, and 13 used here are the following. The group D is torsion-free hyperbolic, quasiconvex in G , and virtually compact special; $h(D) \neq 0$. Its action on Z_D is cocompact, and every vertex and edge stabilizer

$$M_\xi = D_\xi = S \cap X_\xi$$

is nontrivial, finitely generated, and of type F . These stabilizers are quasiconvex in their actual rows. The local period rule holds at every vertex and in either slot, and the subgroup division fields and the injectivity of the ambient 2-columns are those already established. In particular the argument below uses the actual rows X_ξ , not their folded replacements.

All extra requirements in this section form finite lists up to the original group action. They include prescribed finite avoidance, cyclic membership and double-coset tests from Section 14,

every actual incidence stabilizer, any additional required nontrivial groups, specified cyclic generators in S , and pairs of actual cyclic subgroups with trivial intersection. For such a pair, the maximal cyclic closures are distinct, even when they are conjugate in D . We also allow arbitrary fixed lower bounds on the girths of the incidence core graphs and on specified cyclic orders, and prescribed divisibility of those orders. All these requirements are fixed before the choices below and imposed before the normal finite-index base subgroup $H \trianglelefteq D$ is frozen. The final kernel N is then constructed with the exact restriction identity (K3), which gives the required graph covers. Intersecting N with an arbitrary further finite-index subgroup would preserve finite exclusions but need not preserve this identity.

15.1 Local cones and finite-index preparations

Lemma 15.1 (The finite family of local cones). *For a vertex $x \in Z_D$ and either slot, the incidence chain map*

$$\bigoplus_{\substack{e \text{ in the slot at } x \\ \text{mod } M_x}} \mathbb{Q}D \otimes_{\mathbb{Q}M_e} C_*(M_e) \longrightarrow \mathbb{Q}D \otimes_{\mathbb{Q}M_x} C_*(M_x)$$

has a finite based free mapping cone over $\mathbb{Q}D$ which becomes contractible over D_D . There are finitely many such cones up to D .

Proof. Choose finite classifying complexes for the stabilizers, using their already established torsion-free hyperbolicity and [46, Theorem 5], and cellular maps realizing their inclusions, with the actual attaching conjugations. The orbit list at each vertex and the list of vertex types are finite. Induction of their cellular chains therefore gives the asserted finite free complexes with rational group-ring entries.

In degree one the induced map is an isomorphism by the local period rule, because every incident row in the slot participates. In degree zero all terms vanish over D_D : every stabilizer contains a nonidentity element g , and $g - 1$ is invertible in its subgroup division field. In degree two the homology vanishes by the induced chain computation and injectivity of the ambient 2-columns. Higher homology vanishes by the dimension-two subgroup resolutions. Thus the cone is acyclic over D_D . A bounded acyclic complex of vector spaces over a division ring is contractible, by choosing complements to the boundaries in each degree. $\square$

Lemma 15.2 (Novikov openness after restriction). *There is a normal finite-index subgroup $H_0 \trianglelefteq D$ and a nonempty open cone*

$$\mathcal{O}_0 \subset \text{Hom}(H_0, \mathbb{R})$$

on which all the restricted local cones admit contractions over the ordinary rational group Novikov ring. This property persists after passing to any further finite-index subgroup, with a possibly new nonempty open cone.

Proof. A virtually compact special group is virtually RFRS [28, 1]. Apply the finite-complex form of the RFRS Novikov theorem [36, Theorem 5.2] to the direct sum of the cones in Lemma 15.1, after restriction to a finite-index RFRS subgroup. The subgroup-field compatibility gives the required field acyclicity after restriction. Take the normal core of the resulting subgroup.

Here are the restriction and openness details, which also justify later finite-index refinements. In finite coset coordinates, a Novikov contraction restricts to a matrix contraction over the completion for the restricted character. The finitely many coset coordinate shifts do

not affect summability. Given such a contraction s , truncate its entries to group polynomials s_0 , far enough that every term of

$$E = 1 - ds_0 - s_0d$$

has strictly positive character value. This is possible because the differentials have finite support. Positivity of these finitely many terms remains true in a neighborhood of the character. On that neighborhood, $(1 - E)^{-1} = \sum_{n \geq 0} E^n$ converges in the ordinary Novikov ring. Since E commutes with d , $s_0(1 - E)^{-1}$ is a contraction. Intersect the finitely many neighborhoods for the cones and take positive radial multiples. The same argument applies after every further restriction. Notice that this argument uses an actual RFRS subgroup; no selective filling extension of the Novikov theorem is required here. $\square$

Lemma 15.3 (Visible and independent evaluations). *After a further normal finite-index refinement of H_0 , the following conditions hold, and persist under further finite-index refinement. For each required nontrivial group M , there is $u_M \in M \cap H_0$ on which some rational character of H_0 is nonzero. For every specified pair $\langle a \rangle, \langle b \rangle$ with trivial intersection, evaluation on positive powers of a, b in H_0 defines a surjection*

$$\text{Hom}(H_0, \mathbb{Q}) \longrightarrow \mathbb{Q}^2.$$

The conditions hold for all the conjugated tests as well.

Proof. Choose one nonidentity element in each required M . Its cyclic group is quasiconvex in D . For a specified pair, the two endpoint pairs in the boundary of D are disjoint: in a torsion-free hyperbolic group, cyclic subgroups sharing an endpoint have a common nontrivial power. Their positive and negative axes consequently have a bounded cancellation constant at a change from one generator to the other. Choose powers whose translation lengths exceed twice this bound and the fixed local-to-global threshold for quasigeodesics. The path for a reduced alternating word, after removing the bounded pieces at its joins, is a uniform local quasigeodesic, hence a global quasigeodesic. No such nonempty word is trivial, and the orbit map of the resulting rank-two free group is a quasi-isometric embedding. Thus these powers generate a quasiconvex free group, as required.

Pass first to a compact special subgroup of finite index. Quasiconvex subgroups of a hyperbolic compact special group are virtual retracts [28, 29]. Apply this to suitable powers of each chosen cyclic group and each of the rank-two groups. A retraction onto a cyclic group gives its nonzero rational character; a retraction onto the rank-two group gives the two coordinate characters. Intersect the finitely many resulting subgroups and take a normal core in D . Restricting these characters still detects the required powers: a further power lying in the smaller subgroup has its former evaluation multiplied by a nonzero integer. The same observation preserves independence of the two evaluation functionals. Conjugation transports the resulting characters, proving the assertion for all conjugates. At a fixed normal finite-index stage only finitely many conjugacy tests on characters are needed, since inner conjugations by that subgroup act trivially on its characters. $\square$

We may simultaneously make the finite avoidance and girth requirements part of H_0 . Indeed D is residually finite and its quasiconvex subgroups are separable. Products and double cosets of the cyclic subgroups in question are separable by quasiconvex product separability [47]. For a failed membership test $u \notin QtR$, choose a finite quotient preserving that failure and intersect its kernel with the current subgroup. Tests with a full height-fiber factor are immediate, because all the letters in the peripheral coordinates lie in that fiber. For each actual incidence group M embedded in a free row, take its finite Stallings core immersed in a

rose for that row. Choose a base vertex and retain the incidence identification of the core fundamental group with the actual subgroup M . This core has only finitely many nonempty closed reduced paths of a prescribed bounded length, up to its vertices. Transporting each such loop to the base vertex along a chosen core path gives a nonidentity element of $M \leq D$. Excluding these elements, and using normality to exclude their conjugates, makes every induced incidence cover have the requested girth. Any further subgroup preserves these exclusions.

Lemma 15.4 (Separating actual cyclic pairs). *There is a normal finite-index subgroup $H \trianglelefteq D$, contained in the prescribed H_0 , such that every specified pair $\langle a \rangle, \langle b \rangle$ has nontrivial images with trivial intersection in D/H . Every specified generator may additionally be required to have order divisible by a prescribed integer, and larger than a prescribed bound.*

Proof. List the conjugacy types of the maximal cyclic groups containing the specified generators. In a torsion-free hyperbolic group these form a malnormal quasiconvex family. Apply the malnormal special quotient theorem [3, Corollary 2.8] with sufficiently deep power kernels in these maximal cyclic groups, retaining its designated finite-index normal subgroups $\dot{P}_i \trianglelefteq P_i$, and also apply the relative Dehn filling theorem [55, Theorem 1.1]. We obtain a hyperbolic virtually compact special quotient D' in which the chosen maximal cyclic groups have the exact prescribed finite orders. For $P_i = \langle c_i \rangle$, choose its power exponent divisible by both $[P_i : \dot{P}_i]$ and $[P_i : P_i \cap H_0]$, as well as by all prescribed order factors. Increase these multiples to meet the finite avoidance and lower-bound requirements. The power kernels then lie in both $\dot{P}_i$ and H_0 , as required by the two quotient constructions. Thus if $a = c^k$, with c a maximal cyclic generator, choosing the order of c divisible by $|k|r$ makes the order of a divisible by r .

We verify separation in D' , including when the two marked maximal cyclic groups are conjugate. Use the coned relative Cayley graph, with its uniform hyperbolicity constant and the uniform original-peripheral angle dictionaries of Section 14. The constants come from fixed relative relators and the uniform linear relative isoperimetric bound [55, Lemmas 2.7 and 5.3]; hence they are chosen before the power exponents. Preserve the distinctness of the finitely many marked apex cosets by the finite peripheral membership tests. Let $v \neq w$ be two resulting marked apices, and let I be the intersection of their stabilizers. For $u \in I$, compare a geodesic γ from v to w with $u\gamma$. Thin bigons bound the angle between their first germs by a constant independent of u and of the filling exponents. For a short geodesic use the path through w ; otherwise join the two geodesics a fixed distance from v . This gives an avoiding path of uniformly bounded length.

Normalize v to the apex of a maximal cyclic group $P = \langle c \rangle$. Its first ordinary neighbor has coordinate $x \in \bar{P}$, so the first-neighbor displacement for $u\gamma$ is $x^{-1}ux = u$, since $\bar{P}$ is cyclic. The uniform angle dictionary therefore represents every $u \in I$ by one of a fixed finite list of powers c^e , say $|e| \leq B$. If $|\bar{P}| = n > 3B$, no nontrivial subgroup of $\bar{P}$ is contained in these residues: every nontrivial subgroup of a cyclic group of order n has an element at cyclic distance at least $n/3$ from the identity. Thus $I = 1$. This proves the required pairwise separation in D' .

Finally use residual finiteness of D' . For each of the finitely many marked finite subgroup pairs $A, B \leq D'$, separate from the identity every nonidentity element xy^{-1} , $x \in A$, $y \in B$; also separate the nonidentity elements of each marked finite subgroup. The resulting finite quotient preserves both the exact orders and the trivial pairwise intersections. Take its product with the existing quotient D/H_0 , which factors through D' because the power-filling kernel lies in H_0 . The kernel of the resulting map from D is the required H . $\square$

Freeze H after imposing all these finite requirements. By Lemma 15.2, let $\mathcal{O} \subset \text{Hom}(H, \mathbb{R})$ be a nonempty open cone of simultaneous contraction characters. We may include its D -

translates: the local cone family was induced from D , so conjugation preserves the contraction property. Write

$$S_0 = H \cap S, \quad M^0 = M \cap H.$$

15.2 Integral characters and scalar local systems

Lemma 15.5 (A finite generic character family). *There is a nonempty finite family of integral characters $\chi_j : H \rightarrow \mathbb{Z}$, each giving all the local Novikov contractions, such that:*

- (i) *every χ_j is nonzero on each required M^0 and on the required power in H of every specified cyclic generator;*
- (ii) *for each specified pair a, b , if m_a, m_b are their orders modulo H , then*

$$(\chi_j(a^{m_a}))_j, \quad (\chi_j(b^{m_b}))_j$$

are linearly independent over $\mathbb{Q}$;

- (iii) *the family is closed under conjugation by D .*

There are arbitrarily large odd primes ℓ for which all these nonzero evaluations remain nonzero and all these pairs remain independent modulo ℓ .

Proof. The forbidden zero evaluations are finitely many proper rational hyperplanes by Lemma 15.3. For each pair, the evaluation map to $\mathbb{R}^2$ is onto. Thus the condition that the evaluations of a tuple of at least two characters have rank less than two is a proper algebraic condition. A nonzero real polynomial cannot vanish on a nonempty open set. Consequently a rational tuple in a sufficient finite power of $\mathcal{O}$ avoids all the hyperplanes and all the rank-drop conditions simultaneously. Scale its members by positive integers to obtain integral characters.

Impose the same conditions on the finitely many conjugated tests, then add all D -conjugates of the characters. These still have the contraction property; the orbit of an integral character is finite because H is normal and its inner conjugations fix every character. Adding coordinates cannot destroy the existing rank-two conditions. For reduction modulo ℓ , exclude the finitely many primes dividing any of the chosen nonzero evaluations or a chosen nonzero 2-by-2 minor for each pair. $\square$

Lemma 15.6 (Scalar specialization). *One can choose such an odd prime ℓ , followed by a finite field $\mathbb{F}$ of arbitrarily large characteristic different from ℓ , containing all ℓ -th roots of unity, so that for every j and every $\lambda^\ell = 1$, $\lambda \neq 1$, the local restriction maps*

$$H^1(M_x^0; \mathbb{F}_{\lambda^{\chi_j}}) \longrightarrow \bigoplus_{\substack{e \text{ in the chosen slot at } x \\ \text{mod } M_x^0}} H^1(M_e^0; \mathbb{F}_{\lambda^{\chi_j}})$$

are isomorphisms. Actual attaching conjugations are included in this notation. All these local groups have zero H^0 for these coefficients.

Proof. For an integral character $\chi : H \rightarrow \mathbb{Z}$, there is a ring map from the ordinary group Novikov ring to the Laurent-series field:

$$\widehat{\mathbb{Q}H}^\chi \longrightarrow \mathbb{Q}((t)), \quad \sum_{g \in H} r_g g \mapsto \sum_{n \in \mathbb{Z}} \left(\sum_{\chi(g)=n} r_g \right) t^n.$$

Each inner sum is finite and the occurring heights are bounded below. The same support condition makes every coefficient of a product a finite sum, so this is indeed a ring homomorphism. Apply it to a Novikov contraction of each restricted cone. The resulting finite

complex is contractible over $\mathbb{Q}((t))$. Its differentials already lie in $\mathbb{Q}[t, t^{-1}]$; their ranks over $\mathbb{Q}(t)$ and over $\mathbb{Q}((t))$ agree, as can be read from their minors. Hence the complex is acyclic over $\mathbb{Q}(t)$.

Choose a rational-function contraction and clear all its denominators. For the finitely many cones and characters this gives a nonzero Laurent polynomial $f(t) \in \mathbb{Z}[t, t^{-1}]$ such that all the contractions specialize wherever $f(t) \neq 0$. Include $f(t^{-1})$ in this list as well, to accommodate the dual coefficient convention in cohomology. Only finitely many complex roots of unity are roots of these polynomials. Choose the prime ℓ from Lemma 15.5 so large that no nonidentity ℓ -th root is among those roots. Equivalently, each relevant polynomial is relatively prime over $\mathbb{Q}$ to the cyclotomic polynomial $\Phi_\ell(t)$. Clearing a Bezout identity shows that this remains true modulo every prime outside a finite set. Exclude also the finitely many rational coefficient denominators and ℓ . Choose any remaining sufficiently large characteristic and a finite extension containing the ℓ -th roots. All the specialized contractions survive in this field.

For completeness, the local maps asserted in the statement are exactly the maps controlled by these cones. Restricting an induced complex from D to H decomposes it over the double cosets $H \backslash D / M_x$. Within one endpoint summand, the edge terms decompose over the incident-edge orbits of $H \cap M_x$. Thus each restricted cone is a finite direct sum of the actual local cones at the corresponding H -orbit representatives. Scalar acyclicity gives acyclicity in every summand. Dualizing finite complexes over $\mathbb{T}$, with inverse characters if required, gives precisely the displayed cohomology isomorphism. Finally each stabilizer contains an element with nonzero χ_j -value modulo ℓ . Its scalar action λ^{χ_j} is nontrivial, since ℓ is prime, so there are no invariant vectors. $\square$

Proposition 15.7 (Restriction to one actual level). *For every j, λ as in Lemma 15.6, and every vertex or edge level r of Z_D , restriction is an isomorphism*

$$H^1(S_0; \mathbb{T}_{\lambda^{\chi_j}}) \longrightarrow \prod_{\substack{\xi \text{ at level } r \\ \text{mod } S_0}} H^1(M_\xi^0; \mathbb{T}_{\lambda^{\chi_j}}). \quad (\text{K1})$$

The product on the right is finite. In particular the group on the left is finite-dimensional, and restriction to any individual actual stabilizer is onto.

Proof. The H - and S_0 -stabilizers of an actual position coincide, because actual stabilizers have height zero. Each level has finite quotient by S_0 . To see this directly, in a fixed D -orbit two positions at the same height differ by an element of S ; therefore D -cocompactness gives finitely many S -orbits per level, and S_0 has finite index in S .

Take products of the local restriction isomorphisms over all vertex orbits at one level. On either slot the incident-edge indexing is exactly the S_0 -orbit indexing at the neighboring edge level. The finite-index induction decomposition in Lemma 15.6 ensures that the coefficient transport and the attaching frames agree in every summand. Thus restriction identifies the product of the vertex-level cohomologies isomorphically with the product at either neighboring edge level.

The cohomological Mayer–Vietoris sequence for the action on the tree Z_D uses products over all vertex and edge orbits [14, 59]. All its local degree-zero terms vanish. Its degree-one part therefore identifies $H^1(S_0; \mathbb{T}_{\lambda^{\chi_j}})$ with the kernel of

$$\prod_{\xi \text{ vertex mod } S_0} H^1(M_\xi^0; \mathbb{T}_{\lambda^{\chi_j}}) \longrightarrow \prod_{\xi \text{ edge mod } S_0} H^1(M_\xi^0; \mathbb{T}_{\lambda^{\chi_j}}),$$

where the map is the difference of the two endpoint restrictions. Prescribing the data at one level determines the data at successive levels uniquely, in both directions, by the preceding

isomorphisms. Every such prescription gives a compatible product. This proves (K1); it also explains why infinite products over the whole height line cause no extra degrees of freedom. $\square$

15.3 The exact kernel and its graph covers

For every j , define

$$S_{\ell j} = \ker(S_0 \xrightarrow{x_j} \mathbb{Z} \longrightarrow \mathbb{Z}/\ell), \quad M_{\ell j} = M^0 \cap S_{\ell j}.$$

The selected nonzero evaluations imply that both S_0 and every required M^0 map onto this same cyclic deck group. Conjugation therefore defines a deck action on their first cohomology with trivial $\mathbb{T}$ -coefficients. If δ is a deck generator, put

$$\begin{aligned} e_0 &= \frac{1}{\ell} \sum_{r=0}^{\ell-1} \delta^r, \\ V_j &= \ker(e_0 : H^1(S_{\ell j}; \mathbb{T}) \rightarrow H^1(S_{\ell j}; \mathbb{T})), \\ V_{Mj} &= \ker(e_0 : H^1(M_{\ell j}; \mathbb{T}) \rightarrow H^1(M_{\ell j}; \mathbb{T})). \end{aligned}$$

Thus these spaces contain all, and only, the nontrivial deck modes.

Proposition 15.8 (Exact local annihilators). *Each V_j is finite-dimensional and restriction maps it onto V_{Mj} for every actual stabilizer M . With $\ker V_j = \bigcap_{v \in V_j} \ker v$, the subgroup*

$$N = \bigcap_j (S_{\ell j} \cap \ker V_j) \tag{K2}$$

is normal in D , has finite index in S , and satisfies

$$M \cap N = \bigcap_j \left(M_{\ell j} \cap \bigcap_{v \in V_{Mj}} \ker v \right). \tag{K3}$$

The equality uses no simultaneous surjectivity across different j .

Proof. Finite-index Shapiro and the decomposition of the regular $\mathbb{T}[\mathbb{Z}/\ell]$ -module into its characters give

$$H^1(S_{\ell j}; \mathbb{T}) \cong \bigoplus_{\lambda^\ell=1} H^1(S_0; \mathbb{T}_{\lambda^{x_j}}).$$

This remains valid when S_0 is not finitely generated; the index is finite and the decomposition has only ℓ summands. The nontrivial summands form V_j , up to the harmless inverse indexing convention for the deck characters. They are finite-dimensional by Proposition 15.7.

Since M^0 surjects onto the full deck group, the corresponding Shapiro decomposition for $M_{\ell j}$ has the same character indexing. Naturality identifies restriction on each summand with

$$H^1(S_0; \mathbb{T}_{\lambda^{x_j}}) \longrightarrow H^1(M^0; \mathbb{T}_{\lambda^{x_j}}),$$

which is onto by Proposition 15.7. Restriction is deck-equivariant (a deck generator may be represented inside M^0), so its image on V_j is exactly V_{Mj} .

With trivial coefficients, first cohomology is $\text{Hom}(S_{\ell j}, \mathbb{T}^+)$. A basis of V_j thus gives a homomorphism to a finite additive group, and $\ker V_j$ has finite index in $S_{\ell j}$. There are finitely many j , proving $[S : N] < \infty$. Conjugation by D permutes the characters, the subgroups $S_{\ell j}$, and their nontrivial-mode spaces. Consequently it preserves N . Finally surjectivity $V_j \rightarrow V_{Mj}$ says, for each j separately, that the annihilator of the restricted global classes is the annihilator of all local nontrivial classes. Intersecting these individual equalities proves (K3). $\square$

Corollary 15.9 (The outgoing graph-cover property). *Let M be an incidence group embedded in a free actual row and represented by its finite Stallings core. A component L_H of the cover determined by M^0 may be required to have arbitrarily large prescribed girth. It has a nonempty finite family of connected cyclic ℓ -covers $L_{\ell j}$, corresponding to $M_{\ell j}$. The component L_N determined by $M \cap N$ is exactly the cover whose fundamental group is the intersection of the covering kernels and the kernels of every additive first-cohomology class in the nontrivial deck part of each $L_{\ell j}$, over the finite field $\mathbb{F}$ with $\text{char } \mathbb{F} \neq \ell$.*

In particular, if e_1, e_2 are two lifts to $L_{\ell j}$ of the same oriented edge of L_H , the pullback to L_N of $e_1^ - e_2^*$ has zero period on every closed path. All these statements are invariant under the required coset and conjugation changes.*

Proof. Connectedness of $L_{\ell j}$ is the surjectivity of $\chi_j : M^0 \rightarrow \mathbb{Z}/\ell$. The exact description of its further cover is (K3), using $H^1(L_{\ell j}; \mathbb{F}) = \text{Hom}(M_{\ell j}, \mathbb{F}^+)$. The graph girth was imposed at the H stage. The deck average of $e_1^* - e_2^*$ is zero: each lifted edge in the deck orbit occurs once with each sign. Since a graph has no 2-cells, this cochain is a cocycle and its class belongs to the nontrivial deck part. Equation (K3) therefore annihilates all its periods on L_N . Transport of core graphs and local systems preserves every step. $\square$

Proposition 15.10 (Orders and exact cyclic intersections). *For every specified generator $a \in S$, let m_a be its order in D/H . Its order in S/N is exactly $m_a \ell$. For every specified pair of actual noncommensurate cyclics, their images in S/N have trivial intersection. Thus the prescribed evenness, largeness, and divisibility of cyclic orders survive the construction.*

Proof. Put $z = a^{m_a} \in S_0$. Every $\chi_j(z)$ is nonzero modulo ℓ , so z represents a deck generator for $S_0/S_{\ell j}$. The element z^ℓ lies in $S_{\ell j}$ and is fixed by conjugation by z . For a nontrivial deck eigenvector $v \in V_j$, this implies $v(z^\ell) = \lambda v(z^\ell)$ for some $\lambda \neq 1$, hence $v(z^\ell) = 0$. Decomposing V_j into eigenspaces gives $z^\ell \in \ker V_j$ for every j , so $a^{m_a \ell} \in N$. Conversely $a^q \in N \subset H$ implies $q = m_a r$, and the condition $a^q \in S_{\ell j}$, for any j , gives $r \chi_j(z) = 0 \pmod{\ell}$. Therefore $\ell \mid r$, proving the exact order.

Suppose next that $a^q N = b^r N$ for a specified pair. The images of their cyclic groups in D/H have trivial intersection by Lemma 15.4; hence $q = m_a u$ and $r = m_b v$. Since $a^q b^{-r} \in S_{\ell j}$ for every j ,

$$u(\chi_j(a^{m_a}))_j - v(\chi_j(b^{m_b}))_j = 0 \quad \text{over } \mathbb{F}_\ell.$$

The two vectors are independent, so $\ell \mid u, v$. The exact order calculation then gives $a^q, b^r \in N$. Their common image was therefore trivial. $\square$

Remark 15.11 (Additional maximal cyclic peripherals). If an added peripheral is itself $P = \langle c \rangle$ at height zero, choose $H_P = \langle c^r \rangle$ with r as large and as divisible as all its finite tests require. Use the character $\chi(c^r) = 1$ and choose ℓ avoiding the finitely many indices $[H_P : M \cap H_P]$ for its required nontrivial incidence subgroups. Then

$$N_P = \langle c^{r\ell} \rangle.$$

The character is surjective modulo ℓ on each such incidence. The deck action on the first cohomology of a cyclic kernel is trivial, so its nontrivial mode part is zero; nevertheless the exact cover description in Corollary 15.9 still holds. In particular edge-lift differences can be nonzero exact cochains on these cyclic covers. Taking r divisible by each relevant generator exponent and by the requested order factors gives all the same order and girth requirements. This applies to the cyclic root incidences of the no-hit root expansion.

Proposition 15.12 (Simultaneous compatible kernels). *For the finite orbit list of periodic peripherals and any added maximal cyclic peripherals, all the preceding choices can be made*

simultaneously and equivariantly. The resulting $N_D \trianglelefteq D$ have finite index in S_D , satisfy every prescribed finite membership, product, avoidance, girth and cyclic-order requirement, and give the exact local graph covers of Corollary 15.9. Specified noncommensurate cyclic intersections remain exact; commensurate cyclic intersections satisfy the usual intersection identity in a finite cyclic quotient.

Proof. Make every finite requirement at the H_0 and H stages, including all its finitely many orbit representatives and transporters. Perform the character choices for each periodic peripheral. There are finitely many evaluations, minors and specialization polynomials over the entire list, so one prime ℓ and then one finite field $\mathbb{F}$ can serve all of them, including Remark 15.11. Extend the kernels by conjugation. This is well-defined because each kernel is normal in its full peripheral stabilizer.

Every final kernel lies in the corresponding frozen H , so a membership or product exclusion holding modulo H also holds modulo that kernel: an equality in the finer quotient would project to the forbidden equality. The exact noncommensurate intersections and the cyclic orders are Proposition 15.10. For commensurate cyclics, write their subgroups as $m\mathbb{Z}, n\mathbb{Z}$ in their common maximal cyclic group, and let the kernel there be $k\mathbb{Z}$. The intersection of their images has generator $\text{lcm}(\gcd(m, k), \gcd(n, k))$, which equals $\gcd(\text{lcm}(m, n), k)$; prime by prime, this is the distributive identity for minimum and maximum. It is therefore exactly the image of their original intersection. Full-factor intersection tests are immediate. The graph-cover conclusion is Corollary 15.9 and its cyclic case above. $\square$

16 Filled vertex groups and central panels

Fix a terminal folded vertex row $J = Y_v$. We construct the first pieces of the walls used to cubulate its induced filling $\bar{J}$. The relative subgroups of J are denoted by P : they comprise the included height kernels $P_d = S_d$, the external cyclic intersections, and the intersections with the additional maximal cyclic family in the root case. All subgroup notation includes the specified conjugations and coset occurrences. The relative-row results of Section 13 and the kernels of Section 15 are used throughout. In particular, the standalone filling has exact peripheral reductions, its relative hyperbolicity constants are uniform, and all finitely many prescribed injection, membership, and coset-avoidance tests can be imposed before filling. There are finitely many row and incidence types. Every choice in this section is made simultaneously for those types.

Here is the local construction we seek. Free-group Cayley trees supply horizontal edges, whose original vertices are called ordinary sites. A vertex representing a peripheral coset is called an apex. After filling, its central incidences form a finite link graph: link vertices index radial edges to ordinary sites, and link edges record horizontal edges between those sites. Binary labels on the link vertices prescribe radial cochain values. Where the labels change, the intervening horizontal edge must have mod-two cochain total one. We match these cut marks at copies of shared horizontal midpoints. Large angles between the corresponding radial edges will make the matched components quasiconvex trees. We first identify the central incidences and their cyclic attachments, then build the fan model in which this matching takes place. The exact covers from the preceding section supply the binary labels; the next section extends their cochains across the cyclic attachments by cancelling odd boundary totals.

16.1 The central components before filling

Lemma 16.1 (The component presentation). *The developed presentation of J can be arranged as a tree of central components with the following properties.*

- (i) A central component has a bipartite tree of groups, with finite-rank free groups F at one class of nodes and internal peripheral groups P_d at the other. Each incidence group is a nontrivial finitely generated subgroup $M \leq P_d$ embedded in its neighboring F .
- (ii) Each edge between central components has infinite cyclic group $\langle c \rangle$. At one end it attaches to an internal node d ; at the other it attaches in a different central component. Distinct outgoing incidences at the same d have distinct target components.
- (iii) The unused relative occurrences are cyclic. We call them outer occurrences. Internal and outer nodes together label the actual peripheral cosets, without duplication. The developed data have finite quotient by J .

Every included S_d has a central incidence. The added cyclic root nodes have their root incidences. Elementary filled rows may be omitted from the wall construction.

Proof. There are three cases.

For a free actual row X_x , use the incidence tree Γ_v itself. Its actual nodes carry free groups, its object nodes carry the included S_d , and its edges carry M_x^d . These are the central data; the remaining relative groups are outer cyclics.

Suppose next that the actual row has a nonfree root, and choose one of its slots. Write the root splitting, with all translates understood, as

$$E *_{\langle z \rangle = \langle u^a \rangle} U, \quad U = \langle u \rangle,$$

where E is free. Its Bass–Serre tree is the root tree used below. First assume that powers of the root group are hit by the strict object system. The root case in the proof of Proposition 11.6 gives $\tilde{F} = X_x \cap Y_f = X_x$: all actual edges of the chosen slot above x go to one folded edge f . Each object node d likewise belongs to a unique K_f . Consequently the subtrees $K_f \subset \Gamma_v$ partition its vertices into connected subtrees. An incidence (x, d) belongs to its partition subtree precisely when M_x^d meets an E -node of the root tree nontrivially. This uses the identification of K_f with the actual incidence image of Γ_f , together with Lemma 11.1, which puts every nontrivial actual incidence in the enlarged Z_d .

A missing incidence therefore has cyclic group M_x^d acting hyperbolically on the root tree. Indeed, the local homology equality in Lemma 11.1, inherited from Proposition 5.11, forces a noncyclic incidence group to meet an E -node, whereas an elliptic cyclic incidence meets such a node after passing to a nonzero power. Collapse the K_f . The component stabilizer is Y_f , whose developed presentation Γ_f has free groups E_e , full object groups $P_d = S_d$, and incidence groups M_e^d . A missing edge joins different components: it injects into P_d at one end and into $X_x \leq Y_f$ at the other. Collapsing disjoint subtrees of a tree gives a tree, so two missing edges at a fixed d cannot lead to the same component.

Finally suppose that root powers are unclaimed. The additional maximal cyclic family U^* is then disjoint from the old object system. For the action of M_x^d on the root tree, edge-group and cyclic-node intersections are trivial. The no-hit case of Proposition 11.6 identifies the direct sum of the first homologies of its free-node intersections with $H_1(M_x^d)$, over the ambient division field. The tree Mayer sequence now has the following consequence: either M_x^d is cyclic and acts hyperbolically, or it is contained in one free-node stabilizer. If all free-node intersections are trivial, the action is free and M_x^d is a finitely generated free group. Its first division-field homology has dimension $\text{rk}(M_x^d) - 1$; the period rule makes this zero, so its nontriviality forces rank one. To see the other alternative explicitly, the quotient graph of groups has trivial edge groups. If it had two nontrivial vertex-group orbits, or a nontrivial free graph factor in addition to one such orbit, the degree-zero edge term would contribute additional first homology. Thus, when a free-node intersection is nontrivial, there is one such

orbit, no additional free factor, and the whole group fixes that free node. The fixed free node is unique because intervening root edges have trivial M_x^d -stabilizer.

Cut Γ_v along the cyclic hyperbolic incidences and expand the remaining actual nodes by their root trees. Attach every other incidence to its unique free node. Both the S_d -nodes and the cyclic U -nodes are internal. A U -node represents a distinct U^* -occurrence whose intersection with J is exactly U : a nonzero root power fixes only this actual x in Γ_v , since it fixes no object node, and the root cyclic stabilizers inside X_x are distinct maximal cyclics. An element in the same maximal cyclic group must preserve this unique x , so it belongs to the indicated root cyclic stabilizer. The same argument shows that different developed U -nodes cannot represent the same peripheral occurrence. The root incidences have group $\langle u^a \rangle$; the incidences at S_d are the actual row intersections. The cut edges again join an S_d -node to a different component, with their other map landing in X_x .

In all three cases the preceding cocompactness results give finite quotient data. Positive period dimension at a free level supplies a central incidence at every included S_d . The incidences at cyclic root nodes are displayed in the root splitting. This proves the assertions. $\square$

16.2 A tagged fan complex

Build a J -complex $\mathcal{X}$ from the presentation in Lemma 16.1. Its edge graph is given unit edge lengths. The following tags are part of the construction, even when two cells have the same image or the same endpoints.

- (1) At a free node use its Cayley tree for a finite free basis. Subdivide each positive oriented edge by a midpoint. The original vertices are *ordinary sites*; the two subdivided edges are *horizontal half-edges*. For an incidence group M , use its minimal invariant subtree, namely the lift of an unbased finite Stallings core immersion [62, Sections 3, 5 and 7]. Cone this subtree from its internal apex d , with spokes to ordinary vertices and midpoints and with a triangle over every horizontal half-edge. The resulting central complex may have several connected components.
- (2) For a component edge with group $\langle c \rangle \leq P_d$, choose a central word path in its target component from an ordinary site to its c -translate. Repeat it on an abstract indexed line and cone that line from d . This is a *satellite arm*. Choices are made on representatives of developed component-edge orbits and then extended equivariantly. The cyclic group acts on the indexed line by translation; consequently no further compatibility inside the target X_x is required. The boundary path is allowed to repeat vertices or edges, but each spoke and each fan triangle retains its occurrence tag. We choose the path so that every apex neighbor on it is ordinary and every midpoint neighbor is a true ordinary endpoint of its horizontal edge. Horizontal edges and these radial edges already connect the central components. The complex after this step is called internal.
- (3) For an outer occurrence p , with $P_p = \langle c_p \rangle$, choose in the internal complex a word path from an ordinary site to its c_p -translate. Cone the repeated indexed path from a new apex p , again retaining all tags. Use the same ordinary-neighbor convention at internal apices and midpoints.

Each central component is simply connected: it is a tree of simply connected free-node spaces, point spaces, and incidence subtrees. The satellite construction joins these components along their component tree. Each joining slab is the mapping cylinder of its indexed line's boundary map, with the other end collapsed to its source point. Van Kampen's theorem therefore preserves simple connectivity. Coning the remaining outer indexed lines preserves it as well.

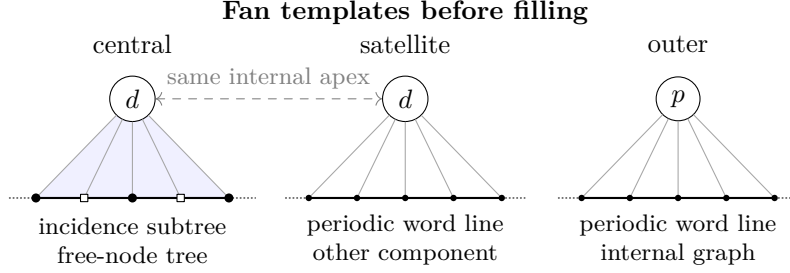


Figure 4: The three fan types before filling. The two d -symbols are separate templates for one physical internal apex. Central fans cone incidence subtrees in free-node Cayley trees; dots are ordinary sites and small squares are horizontal midpoints. Satellite fans cone indexed word lines mapping into other central components. Distinct satellite arms at d target distinct components in the developed model. An outer fan has its own apex p and an indexed word line mapping into the internal graph. The word-line maps may repeat physical vertices and edges; their tags retain the occurrences. The drawings are schematic and impose no planar geometry.

There are finitely many cell types modulo J . Apex vertices are exactly the marked peripheral cosets; all other vertices and all oriented edges have free J -action. In particular an oriented edge of a fixed type has a regular frame $g \in J$, and its endpoints have forms

$$gt \quad \text{or} \quad gtP \tag{16.1}$$

for fixed offsets t depending only on the endpoint and edge type. For satellite and outer spokes, freeness follows from the indexed-line tags, rather than from injectivity of the boundary path. The three fan types and their attachment spaces are shown in Figure 4.

Let K_J be the total kernel of the standalone induced filling of J , and write

$$\overline{\mathcal{X}} = K_J \backslash \mathcal{X}, \quad \mathcal{A} = \overline{\mathcal{X}}^{(1)}.$$

Lemma 16.2 (The filled model). *For sufficiently deep choices, $\overline{\mathcal{X}}$ is simply connected and $\mathcal{A}$ is uniformly hyperbolic. The action of $\bar{J} = J/K_J$ is cocompact, has free regular vertices and oriented edges, and has exact finite apex stabilizers $\bar{P} = P/N_P$. For each fixed filling, $\mathcal{A}$ is locally finite and proper, and $\bar{J}$ is hyperbolic.*

The central link $L_N(d)$ at an internal apex is the disjoint union of its incidence subtrees modulo N_P . Its components are exactly the Stallings covers determined by $M \cap N_P$. Satellite arms become indexed path cycles of period $n = |\bar{c}|$; an outer apex has its one indexed boundary cycle. Period is measured in generator steps, with bounded fractional offsets within the chosen word path.

Proof. The kernel K_J is generated by conjugates of peripheral kernel elements, each of which fixes an apex of $\mathcal{X}$. A loop in the quotient lifts edge by edge to a path upstairs with endpoint discrepancy in K_J . Express that discrepancy as a product of vertex-fixing elements. For each factor insert a path to its fixed apex followed by the translated reverse path. Its projection is a backtrack through that apex. The resulting closed path upstairs contracts because $\mathcal{X}$ is simply connected. This proves simple connectivity of the quotient.

Exact peripheral reduction gives $K_J \cap P = N_P$. Two tags at a specified apex are consequently identified precisely by N_P , proving the link assertion. In particular there is one apex for each marked coset, even when several arms have that source. Regular frame labels and oriented edge labels become free $\bar{J}$ -labels. The arm stabilizer acts on its abstract indexed line by its specified generator, so its quotient has exactly $|\bar{c}|$ periods.

Formula (16.1) maps every edge to a bounded path in the coned Cayley graph of the induced peripheral structure. Conversely, connectivity of $\mathcal{X}$ supplies fixed paths implementing its finite relative generators, changing regular site types, and reaching representatives of every apex orbit. These paths descend, with bounds independent of the filling. Thus $\mathcal{A}$ and that coned Cayley graph are uniformly quasi-isometric. The relative filling bounds from Section 14 give uniform hyperbolicity. There are finitely many edge orbits, free regular stabilizers, and finite apex stabilizers, so a fixed quotient graph is locally finite and the action is proper and cocompact. It follows that $\bar{J}$ is hyperbolic; its ordinary Cayley-metric comparison constants need not be uniform as the filling varies. $\square$

16.3 Angles, tags, and exact covers

For germs e, e' at an apex s , define $\angle_s(e, e')$ to be the shortest path length between their other endpoints in $\mathcal{A} \setminus \{s\}$, with value ∞ if no such path exists. Parallel tagged germs are distinguished even when this angle is zero.

Lemma 16.3 (The finite angle dictionary). *All of the following conditions can be imposed for any finite list of bounds before choosing the filling.*

- (i) *There are no self-spokes. Two distinct central spokes from a fixed apex do not end at the same physical horizontal midpoint, and likewise do not end at the same ordinary site.*
- (ii) *For each R , after the first germ is normalized to a fixed representative, every second germ at angle at most R is obtained by a translator from a fixed finite list in the original peripheral group. The list includes the two germ types and does not depend on the filling. Consequently angular balls have uniformly bounded cardinality.*
- (iii) *On one indexed arm, source germs at bounded angle have sites whose cyclic period coordinates differ by a uniformly bounded amount.*
- (iv) *Every bounded closed walk in which each apex is visited at most once cyclically has a closed labeled lift to the original fan graph, with its given edge types and projected incidences.*
- (v) *All arm orders used here can be chosen even and arbitrarily large.*

Proof. The first assertion holds upstairs. Satellite targets belong to components different from their source component. Within the central bipartite tree, at most one incidence joins a specified pair of nodes, and its incidence subtree is embedded in the free tree. A violation after filling would give an equality of endpoint frames, or membership of a fixed offset in a specified peripheral coset. There are finitely many edge types, so the finite-element and coset-avoidance tests from Section 14 exclude these violations.

For the second assertion, expand an avoiding path in $\mathcal{A}$ using (16.1). Add its finitely many offsets to the ordinary generating letters before setting the depth. Each expanded edge uses ordinary offsets and, when necessary, the cone legs at its endpoints. Neither endpoint is s . Even when a germ's other endpoint is a different apex, its connection to the designated ordinary germ coordinate uses only fixed offsets and that other apex's cone leg. Thus the expanded bypass still avoids the marked coset apex s . Closing it across the two germs gives an isolated peripheral component. The isolated-component bound for the uniformly linear relative presentations, as used in Section 14, places its translator in a fixed finite set of actual peripheral elements [53, 55]. Tags and their free edge frames bound the number of germs represented by that list.

For two sites on the same arm, normalize both word phases and write the relative rotation as a power of its generator c . The finite list just obtained must lie in the image of $\langle c \rangle$.

Finite-list membership avoidance inside the ambient hyperbolic VCS periodic peripheral ensures that such membership already holds before filling. Since N_P lies inside the chosen base finite-index subgroup, that test remains valid for the final kernel. Each actual list element then has a fixed exponent in $\langle c \rangle$, proving the period bound. For an outer cyclic peripheral this is immediate. Word phases only change the bound by a fixed amount.

For the fourth assertion, the rest of the closed walk is an avoiding path at each of its apex visits. Replace the successive normalized peripheral transporters by elements of the finite lists in the second assertion. At regular vertices the transitions are fixed offsets. After normalizing an initial oriented edge, the closing equation is one of finitely many actual word equations. Injection on that finite set makes it hold upstairs. This constructs one closed labeled lift; it does not assert compatible lifts of arbitrarily many independent translates. That stronger assertion will not be needed.

Finally use the divisible order choices and exact order conclusions of Proposition 15.10. Only finitely many bounds and element lists have been imposed, so all the choices are simultaneous. $\square$

Lemma 16.4 (Exact graph-cover data). *Let H_P be the intersection of a periodic peripheral's normal base choice H with P_d , and let $L_H(d)$ be its central link modulo H_P . Its finitely many components can be required to have arbitrarily large girth. Over each connected component B there is a nonempty finite collection of connected cyclic ℓ -covers $B_j \rightarrow B$, where $\ell > 1$, and a finite field $\mathbb{F}$ with $\text{char } \mathbb{F} \nmid \ell$. The corresponding component of $L_N(d)$ is exactly the cover associated to the intersection of*

- (i) *the kernels of all these cyclic covering maps; and*
- (ii) *inside each covering kernel, the kernels of all additive first cohomology classes in the nontrivial deck part of $H^1(B_j; \mathbb{F})$.*

The assertion holds also for the cyclic root incidences, whose nontrivial deck cohomology part is zero.

Proof. Apply Corollary 15.9 to the actual free-row incidence M . In the unclaimed-root expansion, an elliptic M_x^d is precisely the corresponding free-edge-level incidence group, so the same restriction theorem applies. Its exact restriction statement is what gives equality with the stated intersection, rather than merely a cover factoring through it. Conjugations and coset versions are included in the kernel choices. For root cyclic incidences use Remark 15.11: the character is surjective modulo ℓ on the required intersection, and a cyclic group's first cohomology has no nontrivial deck part. Finally there are finitely many original Stallings core types. Residual finiteness of the ambient periodic peripheral separates all their nonzero circuits of any fixed length, giving the girth requirement before the character choices. $\square$

16.4 Large-angle comparisons

We record the geometry used at each later wall gluing. All constants in this subsection depend on uniform hyperbolicity and the displayed quasigeodesic constants, not on the final peripheral orders.

Lemma 16.5 (Forced passage and marker comparison). *In a uniformly hyperbolic graph the following hold.*

- (i) *Geodesic legs from s with sufficiently large angle concatenate to a geodesic, and every geodesic between their other endpoints passes through s .*
- (ii) *Two uniform quasigeodesics from s to the same finite or ideal endpoint, neither revisiting s , have departure germs at bounded angle. If a geodesic has a sufficiently large interior*

angle at s , a uniform comparison quasigeodesic cannot avoid s . Nor can all its germs at s have a previously prescribed bounded angular diameter.

- (iii) Suppose uniform quasi-isometrically embedded tree pieces are joined by switches whose spokes have uniformly bounded lengths. A switch maps to a marker in the graph. For every reduced incidence path, assume each connector through a piece has distinct endpoint markers and avoids them except at its endpoints. Impose the same conditions on partial connectors at marked-node endpoints. Assume also that each piece attaches at most once at a given marker, and that all distinct spoke germs at each switch have pairwise angle larger than a sufficiently large fixed threshold. Then the joined components are uniformly quasi-isometrically embedded trees. Their node projections are injective provided the piece node projections are injective and the stated endpoint conditions hold. Every interior switch on a reduced path is a large-angle turn on a geodesic between its projected endpoints. The same conclusion holds towards an ideal endpoint of a ray when the ambient graph is proper.

The quasi-isometry constants in the third assertion can be fixed after one minimum angle threshold is met. Increasing the angle threshold later increases the forced comparison angle without changing those constants.

Proof. For the first two assertions use thin triangles and quasigeodesic stability. If a comparison path avoids s , choose points a sufficiently large fixed distance out on the two relevant legs, using the endpoints when a leg is shorter. Thinness gives bounded bridges to the other leg or to the comparison path. Take the bridge points far enough from s that the bridges avoid it. The portion of the comparison path between the bridges has bounded length; for a quasigeodesic this follows from its quasigeodesic inequality. These paths give a bounded bypass between the initial germs. Thus a sufficiently large angle precludes an avoiding geodesic and forces passage through s . Its two subpaths then have the lengths of the original legs, proving the concatenation claim. The same construction compares two departures to a common endpoint. For ideal endpoints first work on long finite initial segments and use stability; properness supplies limiting geodesics when required. Finally, if the comparison path visits s , compare its first and last visits towards the two endpoints. A large geodesic angle forces large angular diameter between these two comparison germs.

For the gluing assertion, every piece connector is a uniform quasigeodesic: adding bounded end spokes to a uniformly embedded piece path changes only its additive constants. The endpoint avoidance hypotheses allow its departure and arrival germs to be compared with those of a geodesic replacement. The angle error is bounded by the second assertion. Hence consecutive replacement geodesics still make an arbitrarily prescribed large angle at their common switch if the spoke threshold is chosen sufficiently large.

Apply the first assertion inductively along a reduced incidence path. The concatenation of its geodesic replacements is a geodesic, and every interior switch remains a large-angle vertex on it. A circuit not contained in a piece would therefore give a positive-length geodesic returning to its start; a circuit within a piece is impossible because the piece is a tree. Reusing one attachment does not create an exception, since the incidence path was reduced and each piece has at most one attachment at that marker. The same argument rules out repetition of node projections across switches; within pieces use the assumed injectivity.

There are uniform constants a, b such that the intrinsic length of each connector is at most a times the length of its geodesic replacement plus b . Distinct marker vertices have distance at least one. Summing over connectors therefore absorbs their additive errors into a uniform multiplicative bound; the two end errors remain uniformly bounded. This proves quasi-isometric embedding. Bounded initial or terminal portions of spokes do not affect it. Taking limits proves the ray statement. None of these estimates improves or worsens when

the angles are increased above the fixed minimum. $\square$

16.5 Sparse cuts from exact homological covers

A *central panel* b at d is a function $\zeta_b : V(L_N(d)) \rightarrow \{0, 1\}$, defined on all ordinary link vertices. Its *cuts* are link edges with unequal endpoint labels. The corresponding cut germs are the central spokes to their midpoints.

Proposition 16.6 (Central panel tests). *Fix K , a bound q on the number of target link vertices, and a bound r on the number of prescribed directions at an internal apex. There is a bound B , depending only on these bounds and the fixed original data, with the following property after suitable kernel choices. For every such target list and direction list there is a partition of the targets into classes, each in one component of L_N and of intrinsic diameter at most B , such that every binary assignment to these classes extends to a central panel. Its cut germs are pairwise at angle greater than K , and all are at angle greater than K from the prescribed directions. The same partition works for all binary assignments.*

There are also fillers: for every occurrence of a link edge there is a panel cutting that occurrence whose cuts all lie over its one base edge in L_H . Its cut germs can be required pairwise farther than K .

Proof. We first arrange that different lifts of any one edge of L_H are angularly farther than K . Normalize their first tagged spoke by a peripheral translator. A short angle would, by Lemma 16.3, give a translator from a fixed finite actual list. Since the lifts have the same image modulo H_P , its image belongs to H_P/N_P . Normality of H , and $N_P \leq H_P$, put the actual list element in H_P . Choose H to exclude all nonidentity members of this list. The remaining identity cannot identify two different tagged spokes, whose stabilizers are trivial.

Select support base edges successively. At each stage forbid all base edges having a lift angularly near a prescribed direction or near any lift of a support edge already chosen. The number forbidden by a direction is uniformly bounded. The number forbidden by one support edge is also uniformly bounded: the deck group H_P/N_P is transitive on all its tagged lifts, so normalize one lift and apply the finite angle dictionary. We will make at most $\binom{q}{2}$ selections. There is consequently a uniform bound B for the total number of forbidden edges at every stage. Enlarge B if necessary and require every L_H -component to have girth greater than B . The forbidden-edge subgraph, including its isolated vertices, is then a forest.

Here is the separation assertion underlying the construction. Let u, v lie in one L_N -component, and let F be its forbidden base forest. If $d_{L_N}(u, v) > B$, an allowed support edge supplies a potential distinguishing them. To prove this, choose a joining path and project its chain to each cyclic cover B_j from Lemma 16.4; call the chain c_j . Put

$$A_j = \frac{1}{\ell} \sum_{g \in \text{Deck}(B_j/B)} g$$

for deck averaging on chains and cochains. For each allowed base edge, differences of the cochains on two of its individual oriented lifts have coefficients 1, -1 and lie in $\ker A_j$. Their cohomology classes belong to the nontrivial deck part. Their pullbacks to L_N are exact by the defining kernel property, hence are differentials of $\mathbb{T}$ -valued potentials.

Suppose none of these potentials distinguishes u, v . Evaluation on the projected path says that all coefficients of c_j over any allowed edge are equal. Therefore

$$(1 - A_j)c_j \text{ is supported on the inverse image of } F, \quad \partial(1 - A_j)c_j = (1 - A_j)([v_j] - [u_j]). \quad (16.2)$$

Each lift of a base tree of F is a tree, and there are ℓ such lifts. The augmentation of a boundary on each lifted tree is zero. If the base projections of u, v belonged to different base

trees, the right side of (16.2) would have nonzero augmentation on at least one lift of either tree: its endpoint mass is the difference between one point and its deck average, and $\ell > 1$ with ℓ invertible in $\mathbb{F}$. The base endpoints must therefore belong to one forest tree. Their deck-average masses on each lift cancel, so the same augmentation test forces u_j, v_j to lie in the same lift of that tree.

Let f be the unique base forest path between the two base endpoints. It lifts to a path f_j from u_j to v_j in every B_j . The chains $(1 - A_j)c_j$ and $(1 - A_j)f_j$ have the same boundary and are both supported on a forest. A forest has no nonzero one-cycles, so they are equal. Thus $c_j - f_j$ is invariant under deck averaging. The comparison loop belongs to every cyclic-cover kernel and evaluates to zero on every nontrivial deck cohomology class. Exactness in Lemma 16.4 puts it in the final covering kernel. Consequently f lifts between the original u, v in L_N , giving $d_{L_N}(u, v) \leq |F| \leq B$, a contradiction. This proves the separation assertion.

Now process all same-component target pairs at distance greater than B . Select an allowed support edge and a distinguishing potential for each pair still requiring distinction. Extend each potential constantly on the other components. Two targets are put in the same class precisely when they belong to the same component and have the same tuple of potential values. The separation assertion gives the claimed diameter bound. Any prescribed binary values on the target classes extend to a binary function of this tuple and the component identifier, assigning arbitrary values on the remaining tuples. Its gradient is supported only over the selected support edges. The forbidden-edge rule separates different support edges, while the first paragraph separates different lifts of one support edge. This proves both cut separation and avoidance of the direction list.

For a filler, choose a cyclic cover over the component in question and take the difference cochain between the lift containing the specified occurrence and a different lift of its base edge. Its pullback is exact and its value on the specified edge is nonzero. A binary separation of the two endpoint potential values therefore cuts that occurrence, with all cuts over that one base edge. This argument only requires the nontrivial covering degree. In particular, when the nontrivial cohomology itself is zero, the difference cochain is already exact but still has a nonconstant potential. The proof works unchanged in characteristic two when ℓ is odd. $\square$

16.6 Central trees and their cochains

For the actual finite links, take finite systems realizing all required panel tests and then all peripheral translates. Decorate panels by free regular $\bar{P}_d$ -multiplicities so that the data, rather than only their isomorphism classes, translate equivariantly. There are finitely many requested orbits. We next equalize their numbers of cuts.

For each central spoke orbit type, a peripheral orbit of fillers adds a positive constant number of cuts at every spoke of that type and none of the other types. Multiply the requested systems by a common multiple of these positive numbers. Choose a common larger integer l divisible by all of them, and top up each type using its fillers. Thus exactly l cutting panels occur at every central spoke to a midpoint. The same l can be used for all finitely many types. Decorate every physical horizontal midpoint by l slots, including midpoints untouched by a particular panel system. At each apex incidence match the slots bijectively with its cutting panels. Choose this matching equivariantly: the horizontal edge frames are free, and Lemma 16.3 gives only one central occurrence from this apex at the specified midpoint. Join a panel to its assigned slots by edges projecting to its cut spokes.

Lemma 16.7 (Central trees). *Every component W_0 of the panel-slot graph is a uniformly quasi-isometrically embedded tree in $\mathcal{A}$, injective on node projections. Its constants require only a minimum cut-angle threshold and remain valid when subsequent panel systems use*

larger thresholds. There are finitely many decorated node and edge orbits. Distinct systems may be retained separately as a finite labeled pool.

Proof. Apply Lemma 16.5 with slots as point pieces and panels as switches. Along a reduced path the two panel markers adjacent to one slot are distinct: from a fixed apex, that slot is matched to only one cutting panel. The two-spoke connector through the slot avoids its endpoint apices, and each point piece attaches at most once over any apex. At a panel all cut germs have the stipulated large angle. The marker lemma proves both the tree assertion and injectivity on nodes. Finiteness of the orbit data follows from finite links, finite peripheral reductions, and the finite multiplicities just chosen. $\square$

A horizontal edge is *marked by W_0* when its midpoint has a slot in W_0 , necessarily unique. At an incident central link edge, that slot belongs to W_0 precisely together with its matched cutting panel. There is at most one panel of W_0 above any given apex.

Lemma 16.8 (The central cochain). *Each W_0 carries an equivariant mod-two cochain $\tau_0^{W_0}$ on the central complex, closed on every central face. It has value one on the positive initial half of each marked horizontal edge, zero on all other horizontal halves, and value $\zeta_b(t)$ on an ordinary radial edge $d-t$ when its panel b belongs to W_0 , or zero when there is no such panel. Every edge with nonzero value has a panel or slot anchor of W_0 at one of its endpoints.*

Proof. The assertions specify the ordinary radial and horizontal values. For an edge with positive orientation $u-m-v$, let ϵ indicate whether it is marked. Put the midpoint radial value equal to $\zeta_b(u) + \epsilon = \zeta_b(v)$, taking the ordinary radial labels to be zero if b is absent. The equality follows from the matching rule: the link edge is a cut exactly when its slot and cutting panel lie in this tree. Both fan triangles therefore have zero boundary sum. These are all central faces. The formulas also give the anchor assertion and are preserved by the decorated group action. $\square$

Lemma 16.9 (Satellite arm exclusivity and bounded support). *If W_0 has a node anchored on a satellite arm of d , then it has no node over d and no anchor on a different satellite arm of the same d . All its anchor occurrences on that arm, and all nonzero occurrences of $\tau_0^{W_0}$ on its boundary cycle, lie in a period interval of uniformly bounded diameter. These bounds are independent of the arm order and of later increases in cut separation.*

If $\tau_0^{W_0}$ has odd total on the arm cycle, its stabilizer inside $\bar{P}_d$ is trivial.

Proof. An anchor on the arm is joined to d by its tagged source spoke. If W_0 contained d , its intrinsic path to that anchor would have uniformly bounded length by Lemma 16.7. Close it by this spoke. Likewise, anchors on two arms can be joined through d by two spokes, so their intrinsic W_0 path has uniformly bounded length. In the latter case d has already been excluded. These closed walks are apex-simple: W_0 is injective on node projections and its edges are central spokes, while the added source apex occurs only at the cyclic joining point. Lemma 16.3 therefore supplies a closed labeled lift upstairs.

The lift of the W_0 segment is a central path. In the first case it would join d to a target component of its satellite arm, contrary to Lemma 16.1. In the second case it would join the targets of two distinct arms in one central component. The arms remain distinct upstairs: at a fixed source, their incidence tags are cosets $P_d/\langle c \rangle$, and the quotient construction gives the corresponding cosets in $\bar{P}_d$. If two lifted incidences were the same, their projected incidences would be the same. Thus one closed polygon lift suffices for this contradiction; no simultaneous lifting of independently chosen translates is being assumed.

The path in W_0 between two anchors on the one remaining arm has bounded length, using the two-spoke shortcut through d , and avoids d . Apply the period part of Lemma 16.3 to obtain the bounded interval. A nonzero central cochain occurrence has an anchor at an

endpoint, and the word paths have fixed lengths, so a fixed enlargement of this interval contains its support.

An odd total makes the support nonempty. Any element of $\bar{P}_d$ stabilizing W_0 must preserve its unique encountered arm. That arm's stabilizer acts faithfully by rotations of its tagged period cycle. For order sufficiently large compared with the support diameter, no nonidentity subgroup of this rotation group preserves a nonempty subset contained in such a short interval. This proves the last assertion. All bounds used the minimum quasi-isometry constants and fixed word lengths, so increasing the cut threshold does not change them. $\square$

16.7 Odd seeds on satellite arms

Proposition 16.10 (Odd inner seeds). *A finite pool of central panel systems, chosen at successive fixed angle scales, can be required to contain, on every satellite arm, a central tree W_0 for which $\tau_0^{W_0}$ has odd total on the arm's boundary cycle. The choices and all support bounds are independent of the sufficiently large individual cycle orders.*

Proof. Fix an arm cycle β with source d and period n . At each internal apex s visited by its central boundary path, list all radial half-incidences, with multiplicity. Their other endpoints are ordinary by construction. The list size is uniformly bounded: all source spokes from d to the repeated occurrence of s have the same other endpoint, hence angle zero at d ; the finite angle and period dictionaries bound both their number and their period spread. At s the listed directions are mutually close in angle by paths through d .

Use Proposition 16.6 with cuts sufficiently far from these directions and with every binary assignment on its bounded diameter target classes. For the tree through such a testing panel at s , no other anchor $w \neq s$ lies on β . Indeed the arm exclusivity lemma excludes d from this tree. Its path from s to w can be compared with the path $s-d-w$ of length at most two. The first germ of the latter is close to every listed direction at s , by the adjacent ordinary site on β . Departure comparison in Lemma 16.5 contradicts the first cut's chosen distance from those directions. The cochain's total on β is therefore exactly the sum of the tested labels at s .

If any test has odd total, it supplies the seed. Otherwise every target class occurs an even number of times. Pair the radial half-incidences within each class and replace each pair, as a mod-two chain, by a horizontal link path between its two ordinary sites. The paths have a uniform length bound B_1 . Fan triangles show that each replacement preserves evaluation by every central cochain. Perform this at all visited apices. The result is a finite purely horizontal cycle chain z ; connectivity is not required.

We justify that z is nonzero by retaining its winding around the period circle $C_n = \mathbb{R}/n\mathbb{Z}$. Initially each occurrence on β has its indexed period coordinate, with the fixed fractional coordinates inside a word. Every vertex occurrence added by a replacement has a horizontal tail of length at most B_1 to one of the original ordinary sites. Two occurrences representing the same physical vertex thus have original sites joined by a bounded path avoiding d . Their source spokes have bounded angle at d , so their period coordinates differ by a fixed bounded amount. The same applies to the two occurrences of s involved in each paired replacement. Consequently each replacement polygon, and all ambiguity in assigning coordinates to one physical vertex, occupies a period interval of bounded diameter.

For completeness, let T bound horizontal replacement lengths in our subdivided metric, and let $D(R)$ be the period-coordinate bound for original occurrences joined by a path of length at most R avoiding d . Enlarge it monotonically and put

$$M = \max\{1, D(T), D(2T + 1)\}, \quad n > 8M.$$

For every physical vertex in the finite graph used by β and the replacements, choose one original occurrence reachable by a horizontal tail of length at most T , and assign its period coordinate to the vertex. Original vertices, including apices, allow the zero-length tail. Map each physical edge by the unique short arc between its assigned endpoint coordinates. The assigned coordinate at an original occurrence differs from its parameter coordinate by at most M . Along the indexed original cycle choose these errors in $[-M, M]$. Each elementary parameter increment is at most one, so its adjusted increment has absolute value at most $1 + 2M < n/2$ and is exactly the chosen short-arc increment. The errors telescope around the cycle, leaving total increment n and degree one. For a replacement polygon centered at an occurrence of s , the witness original occurrence of any of its vertices is joined to that s by an avoiding path of length at most $2T + 1$. All assigned coordinates therefore lie in one interval of length at most $2M < n/2$. Its edge images stay in that interval, so the polygon has degree zero. The image of z thus has degree one modulo two. In particular some physical horizontal edge occurs in z with coefficient one. At its midpoint the cycle equation makes the two half-edge coefficients equal.

Choose a second central panel system with cut angles larger than a threshold fixed after B_1 , and take the central tree through a slot of that edge. It has no node above d . Otherwise close its bounded path to the slot by the slot's bounded horizontal tail to an original arm site and the source spoke to d . The added tail has no apices, so the resulting bounded walk is apex-simple and lifts by Lemma 16.3. Its lifted central path would join d to the target component, the contradiction used in Lemma 16.9.

Nor can this tree contain two distinct horizontal anchors of z . Each anchor has a bounded horizontal tail to β , and hence a bounded path to d . Between two distinct slot nodes the tree path has an interior panel marker, different from d . Its angle is larger than the fixed comparison threshold. The path between the two anchors formed from their horizontal tails and the two source spokes through d has uniformly bounded length and avoids every such marker. It is a uniform comparison quasigeodesic, with constants depending only on that length bound. Lemma 16.5 gives a contradiction. Thus the chosen tree has exactly one horizontal anchor on the support of z . Its cochain evaluates to one there, and hence to one on z and on the original β .

There are only finitely many actual links and bounded-size requests for a fixed filling. Include all first-stage tests, their translates, and the second-stage filler systems. The constants determining these requests were fixed independently of n , so this is a finite pool working for every sufficiently deep filling and every arm type. $\square$

16.8 Order and uniformity of the choices

We make explicit the quantifier order used here and in the next section. First fix the original row types, their word paths, the uniform relative hyperbolicity bounds, and a minimum bend threshold in Lemma 16.5. This fixes the quasi-isometry constants of the central trees and the arm-support bounds in Lemma 16.9. Any larger cut separations preserve these constants.

Next prescribe the finitely many target-list sizes and direction avoidance thresholds that will be tested. Proposition 16.6 provides a bound for the number of forbidden base edges and for the resulting class diameters, independently of H and N_P . For odd seeds, choose the first testing threshold, compute its replacement bound B_1 , and only then choose the second cut threshold needed to exclude the bounded comparison paths. The two systems remain separately tagged: the first-stage class bound is not recomputed using the second threshold. All angle dictionaries and short-polygon lifting tests at these finitely many scales are now fixed. Subsequent applications may repeat this finite procedure, provided each new threshold depends only on already fixed bounds.

Finally choose the normal base subgroups H to meet all resulting finite-list exclusions and girth bounds. Apply the character and cohomology kernel construction, preserving those tests because $N_P \leq H_P$. Choose the cyclic orders even and larger than the finitely many period and support bounds. The graph-cover argument uses the number of target pairs, not the number of characters or the size of a final link, so enlarging a cover at this last stage does not alter any previously fixed bound. All these choices are compatible with any initial finite-path filling requirements by Proposition 15.12.

The outgoing data are therefore a finite pool of equivariant central trees W_0 , their central cocycles τ_0 , arbitrary prescribed panel tests of the stated bounded type, and odd inner seeds with uniformly bounded support and free arm stabilizers. These are the data used to attach the satellite and outer switches.

17 Switches, cocycles, and boundary separation

Our goal is to separate every pair of boundary points by a quasiconvex subgroup, so that the boundary criterion gives a cocompact cubulation of each filled row. We first extend the central trees and cocycles across the satellite fans and then the outer fans. The second half of the section proves that the resulting cocycles supply all the required separators.

We continue with a terminal vertex row $J = Y_v$ and the filled fan complex of Section 16. Its edge graph is $\mathcal{A}$ and its acting group is $\bar{J}$. In particular, $\mathcal{A}$ is uniformly hyperbolic, and, for each individual filling, it is locally finite and the action is proper and cocompact. Internal and outer apices are denoted by d and p , respectively. All the choices below are made simultaneously over the finitely many row types. The initial filling depth, and the finite tests required by Section 14, may be prescribed in advance.

All chains and cochains in this section have coefficients in $\mathbb{F}_2$. Occurrences of edges are retained with multiplicity until a pairing is made; only then do we reduce modulo two. Thus a passage through an apex supplies two radial half-incidences, even when it uses the same physical edge twice. A *pure internal* path avoids outer apices and uses true ordinary neighbors at every internal apex and horizontal midpoint. The boundary word of every satellite or outer fan has this property. Pair replacements will produce chains, and these need not be connected.

We use the central trees W_0 and their cocycles τ_0 from Lemmas 16.7 and 16.8. An *anchor* is a projected panel or slot node of such a tree, and later also a switch node. In particular, a central radial contribution has a panel anchor at its apex, whereas a horizontal contribution has a slot anchor at its midpoint. The angle dictionary of Lemma 16.3 and the marker comparison of Lemma 16.5 will always be used with bounds fixed before the final filling kernels are chosen. We give the full order of these choices at the end of the section.

17.1 Nonmixed panels and the first switches

At an internal apex d , central and satellite incidences can coexist. Central labels prescribe values on the radial edges of incidence subtrees; satellite labels must instead integrate cochains around the attached cycles. We put these two kinds of labels on separate panel vertices. Thus testing satellite labels at d introduces no central labels there, and a central panel acquires no satellite attachment at its own apex. The later separation proofs will still check the contributions transmitted through other apices.

Definition 17.1. A *side* is a pair (W_0, a) where the central tree W_0 has an anchor occurrence on the boundary of an actual satellite arm a at its source d . The tree retains its system label and, when copies are taken, its copy index. The side is *odd* if $\tau_0^{W_0}$ has total one on the

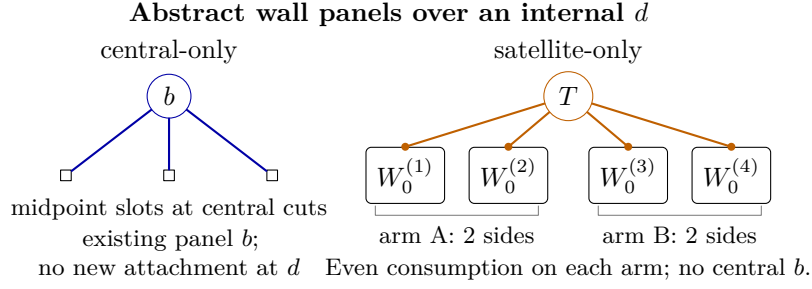


Figure 5: Nonmixed panels in the filled complex. The central-only panel b and satellite-only panel T are abstract wall vertices projecting to one internal apex d . The former joins midpoint slots at its central cuts. The latter joins chosen anchor occurrences (colored dots) in odd sides (W_0, a) , consuming an even number on each arm; two per arm are shown only as an example. Across labeled copies, every odd side is consumed exactly once. A W_0 -copy may still have sides at other source apices. Spoke angles are large in the graph metric at the physical image d , irrespective of the drawn Euclidean angles. No additional avoidance of central cuts is required. A satellite-only panel contains no central b , and its central labels are zero.

indexed boundary cycle of a . Consuming this side means assigning this particular incidence to one satellite panel and choosing an anchor occurrence on a for its attachment.

A *central-only total panel* is a central panel b , with no additional attachment at d . A *satellite-only total panel* T has no central panel and consumes finitely many odd W_0 -sides on the satellite arms at d , an even number on each arm. For every consumed side, attach a spoke from T to its chosen anchor occurrence. The spoke projects to that tagged satellite radial edge. Its germ at d is required to be far from every other spoke germ of T .

On each satellite arm, the binary labels of T are a primitive of the sum of the boundary cochains of the sides it consumes there. Such a primitive exists because the number of odd sides is even. Its additive constant is chosen independently on each arm. Central labels of T are zero. Empty satellite-only panels, including their independent constant labels, are allowed.

Figure 5 shows these two kinds of abstract panel over one physical apex. The fan spaces themselves were shown in Figure 4.

Lemma 17.2 (Satellite profile tests). *Fix bounds on the number of target ordinary satellite sites and forbidden germ directions at d , and fix an angular avoidance threshold. A finite pool containing the odd seeds of Proposition 16.10 supplies satellite-only panels whose labels realize every binary assignment to a partition of the target list into classes. Each class lies on one arm and has bounded cyclic period-coordinate diameter. Their spoke germs avoid the forbidden directions and one another by the prescribed threshold. The class-diameter bound and the required filling tests depend only on these fixed requests and the original finite data.*

Proof. On each arm, order the listed sites cyclically. Join consecutive listed sites into one cluster when the intervening gap is smaller than a fixed threshold D . There are at most as many clusters as targets, and the diameter of each cluster is at most the number of targets times D . A prescribed binary assignment changes value across an even number of the cyclically ordered gaps. For each such change, place a translate of an odd seed with its entire support cluster inside that gap. Its contribution changes the primitive by one between the neighboring target clusters.

The seed's support has uniformly bounded width. Its chosen anchor rotates freely, and the angle dictionary bounds the number of rotations whose spoke lies angularly close to a

forbidden direction or a previously placed spoke. There are only boundedly many directions and earlier placements. Choose D larger than the support width, the required end margins, and the number of forbidden positions. Every required placement is then possible. The additive constant on the arm realizes the value on the first cluster. If the assignment is constant, no seed is needed. All rotations are by integral period steps; the fixed elementary word lengths only enlarge the bounds by fixed factors. These choices involve satellite spokes and the given forbidden directions; they impose no avoidance condition against central cuts at d . $\square$

Lemma 17.3 (Equivariant consumption). *After taking finitely many labeled copies of a finite pool of central systems, every odd side can be consumed exactly once. This can be done equivariantly, with finitely many decorated panel and edge orbits, while including all the tests of Lemma 17.2 and any fixed lower bound on switch angles.*

Proof. For each of the finitely many apex types, include the full free $\bar{P}_d$ -orbits of the stipulated decorated profiles. For the actual finite links, all requests involving lists of the bounded sizes form a finite set, so this requires only finitely many profiles. Take m copies of the entire initial central systems, with trivial action on copy indices.

Consider an arm with even period n and rotation generator c . The requested profiles use any odd side and its translate by $c^{n/2}$ equally often, because they form full $\bar{P}_d$ -orbits. These two sides are distinct: the support cluster is nonempty and short compared with n . Double the requested multiplicities and choose m even and larger than all the resulting consumption counts. Each side then has an even number of unused copies, equal to the number for its antipode.

Consume the unused copies by antipodal pair panels. The pair's germs have arbitrarily large angle once the period-coordinate avoidance test is imposed. An unordered pair may have an order-two stabilizer, but a free orbit of *decorated* pair panels uses two copies of each side in such a pair. The even residual multiplicities therefore permit these free orbits; the offsets of their binary labels are transported with the decorations. There is no need to choose an offset invariant under an involution that exchanges the two sides. More explicitly, write $a = c^{n/2}$ and choose two residual indices i, j . The panel indexed by g attaches to (gs, i) and (gas, j) ; the panel indexed by ga attaches to (gas, i) and (gs, j) . Thus the free orbit uses one copy of each chosen index at every side, with no action on the copy indices themselves.

The stabilizer of an odd side at d is trivial. Indeed, arm exclusivity forces such a stabilizer to preserve its unique arm, and the short-cluster argument of Lemma 16.9 excludes a nontrivial arm rotation. Consequently copy indices can be assigned to all consumptions equivariantly at each representative apex, then transported to its orbit. At a fixed d , a given W_0 -piece encounters at most one satellite arm, so it attaches at most once there. Each central panel in each copy remains its own central-only total panel.

Assignments at distinct source apices impose separate incidence conditions. Each labeled W_0 -copy is a single piece with its original cochain. An odd side at d is consumed once, and an odd side at a different source is consumed there as well. These assignments add edges to that same piece without changing its copy index or its central cochain. Radial primitives belong to their tagged source arms, so choices at different sources prescribe values on different radial edges.

Different central systems in the pool retain their tags, slots, and cochains. Taking their disjoint union does not require rematching slot data. If satellite tests are not requested in a particular system, the antipodal construction alone suffices, without an odd-seed hypothesis. $\square$

Let W_1 be a component obtained by joining W_0 -pieces through the satellite-only total panels. A central-only panel is an existing node, rather than an additional switch.

Lemma 17.4 (The internal trees and their cocycles). *The W_1 are uniformly quasi-isometrically embedded trees, injective on projected nodes. Each contains at most one total panel over a given internal apex. They carry equivariant cocycles τ_1 on the internal graph, closed on all central and satellite fan triangles. A nonzero value on a satellite radial edge $s \rightarrow t$ has one of the following explanations:*

- (i) W_1 contains its total panel at s ;
- (ii) a constituent W_0 has an anchor on that arm within a uniformly bounded period-coordinate distance from t , accessible from t by a short internal boundary arc.

The bounds, and the quasi-isometry constants, depend only on minimum bend requirements and the basic central support bounds. Increasing subsequent angular separations does not increase these constants.

Proof. Use the satellite-only panels as markers in Lemma 16.5. By arm exclusivity, a piece attached at d avoids the physical vertex d , even at its internal vertices. Each spoke has length one. Consecutive switches through a piece are distinct, and their connectors avoid both end markers. The equivariant consumption construction attaches a piece at most once at any specified marker. The marker comparison thus gives a tree and the claimed node uniqueness and uniform embedding. Since every W_1 -edge projects to a single graph edge, a reduced path leaving a node does not revisit its projected vertex.

On the central graph define τ_1 as the sum of the τ_0 of the constituent copies. For a satellite arm a at d , start its radial labels with those of the satellite-only panel T at d , if present, and with zero otherwise, including the central-only case. For each unconsumed *even* encounter of a constituent W_0 , add the primitive of its boundary cochain on a , normalized to zero outside its short support interval. The complementary long gap makes this normalization unambiguous for large n ; a zero boundary cochain receives the zero primitive.

Every odd encounter is already accounted for: its consuming panel belongs to the same component. A side consumed on a different arm at d has no support on a , by arm exclusivity. Therefore differences of radial labels equal the full boundary cochain. This is exactly the cocycle equation on each satellite triangle; the central equations already hold for τ_0 .

A default can be nonzero only within the bounded enlargement of its support cluster, proving (ii). The direct panel labels give (i), regardless of their additive constants. The same reasoning handles nonordinary boundary sites. For a fixed edge there are only finitely many possible nearby anchors, by local finiteness of the filled graph and the finite decorated slots and links. Node uniqueness prevents infinitely many constituent pieces from using a fixed such anchor. All sums are thus defined edge by edge. Every normalization and decoration is transported equivariantly. $\square$

17.2 Outer switches and boundary sides

An *outer encounter* is similarly a pair (W_1, p) , where the outer apex p has its single indexed boundary cycle; it is odd when the τ_1 -total on that cycle is one. This indexes the pairing at each outer apex separately, even if the same W_1 encounters other outer fans.

Lemma 17.5 (Outer support clusters). *For a fixed W_1 , the nonzero positions of τ_1 on the boundary cycle of an outer apex p lie in a bounded cyclic period interval. An odd encounter has trivial stabilizer in the arm rotation group. The interval bound is independent of later increases in switch angles.*

Proof. By Lemma 17.4, each nonzero position has an anchor at uniformly bounded internal distance. Any two such anchors are at bounded ambient distance by the shortcut through p , and hence at bounded intrinsic distance in W_1 . This connecting path avoids p , as do the short

internal tails to the boundary sites. The angle and period-coordinate dictionary at p bounds the cyclic distance between any two nonzero positions. For large cycle order they consequently lie in one short interval. An odd encounter has a nonempty cluster. A nonidentity subgroup of the faithful cyclic rotation action cannot preserve a nonempty cluster of this bounded diameter when the order is sufficiently large. Only the minimum quasi-isometry and earlier cluster constants were used. $\square$

At each outer apex, pair every odd W_1 -encounter with its antipodal translate. They are distinct by Lemma 17.5. Join them through a pair switch projecting to p . Its two spokes are bounded paths: follow a radial edge to a contributing site and a short internal path to an anchor. They avoid every outer apex after departure. Choose the paths by transport from one decorated side; the opposite spoke is the antipodal translate. Individual sides have trivial stabilizer, whereas an unordered pair may have an involution exchanging its ends. This is allowed at a switch. Each W_1 attaches at most once at a specified p .

Proposition 17.6 (Completed trees and cocycles). *The resulting components $W = W_2$ are uniformly quasi-isometrically embedded trees. Their W_1 -nodes and switch nodes have distinct projections within a component. They carry cocycles τ_W on the full simply connected fan complex, supported in a uniform neighborhood of W . Under group transport the only ambiguity in these cocycles is a sum of vertex coboundaries at outer switches. There are finitely many decorated node and edge orbits.*

Proof. The period-coordinate dictionary makes the two germs of every outer switch as far apart as any prescribed fixed threshold. The W_1 -pieces avoid outer apices, and their short attachment paths avoid them except at the specified end. All hypotheses of the marker comparison hold. This proves the tree and embedding assertions and shows that changing W_1 -pieces forces a geodesic comparison with a large interior turn at an outer apex. The comparison threshold can be increased without changing the quasi-isometry constants. No assertion of simplicity is needed on the interiors of the bounded outer spokes.

On the internal graph, sum the cocycles of the constituent W_1 -pieces. At an outer cycle, all odd pieces present in this component belong to its unique pair switch there, if such a switch exists. Integrate the sum of the pair's two cochains around the cycle. Its total is zero, so it has a primitive; choose either additive constant. Integrate every even-only piece separately, with its primitive zero off its short support cluster, and add all these primitives to obtain the radial labels.

The difference of successive radial labels is the internal boundary cochain. Thus every outer triangle has cocycle sum zero. Together with Lemma 17.4, this proves closedness on every face. Even defaults are supported near anchors; the paired contribution, even when its primitive is nonzero along a long interval, lies on edges incident to its switch p . Hence the full support is uniformly near W . The same nearby-anchor argument as before proves local finiteness of the sums.

Changing the constant for the paired primitive changes precisely the values of all radial edges incident to that outer apex. Since that apex has just its one fan arm, the change is the coboundary of its vertex indicator. The possible order-two stabilizer of the unordered pair therefore creates exactly this permitted ambiguity and no further one. All even normalizations are equivariant.

Finally, the central data, finite links, decorated profiles, and labeled copies have finite orbit sets. At an outer cycle only finitely many nearby anchors contribute, so the pair-switch and chosen spoke data also have finitely many orbits. This proves the last assertion. $\square$

Lemma 17.7 (From a cocycle to boundary separation). *For each W , its stabilizer is quasiconvex in $\bar{J}$ and acts cocompactly on W . The cocycle defines a locally constant binary*

coloring on $\partial\bar{J} \setminus \Lambda W$. Its stabilizer preserves the two colors up to a global flip. If two points outside ΛW have different colors, a quasiconvex codimension-one subgroup separates them in the sense of the Bergeron–Wise criterion.

Proof. If two decorated nodes of one component are in the same ambient group orbit, any group element carrying one to the other preserves their component. Finiteness of the ambient decorated orbit sets therefore gives a finite quotient of W by its stabilizer. Node stabilizers are finite, since the projected action on $\mathcal{A}$ is proper. The stabilizer acts properly and cocompactly on the tree, and its orbit embeds quasi-isometrically in the hyperbolic graph. It is consequently a quasiconvex subgroup, and its limit set equals ΛW . Every point of that limit set is represented by a projected tree ray from any chosen node.

Simple connectivity lets us integrate τ_W to a binary vertex coloring, unique up to one additive constant. A geodesic ray to a point outside ΛW eventually escapes every fixed neighborhood of W . It has an eventual color because the cocycle is supported near W . For nearby ideal endpoints, based rays follow travel beyond a segment already far from W ; short connecting bridges also avoid the support. They have the same eventual color. This proves both independence of the ray and local constancy on the complement of the limit set.

Under an element preserving W , the cocycle changes by vertex coboundaries at switch apices. Their primitive is zero away from these apices. Thus the integrated colors away from the switch set change by one global bit. On ideal points outside ΛW this gives either preservation or a simultaneous flip. Pass to the kernel of this action on two colors, of index at most two, and call it H . It has the same limit set and remains quasiconvex. Concretely, if $\tau_W = \delta f$ and g preserves W , write $g\tau_W + \tau_W = \delta\eta_g$, where η_g is supported at its outer switch vertices. Connectedness gives

$$gf + f + \eta_g = \epsilon(g) \quad (\epsilon(g) \in \mathbb{F}_2).$$

Off the invariant switch set this is $gf + f = \epsilon(g)$. Composing two transports there shows that ϵ is a homomorphism. The possibly infinite switch set causes no summability problem: η_g is a vertex cochain, and its coboundary is evaluated on one edge at a time.

Choose a fixed neighborhood of an H -orbit containing the cocycle support. Rays of different eventual colors lie in different components of its complement: any connecting edge path would have zero cocycle sum and hence preserve the color. Both components are deep, since these rays escape every fixed orbit neighborhood. Thus H is codimension one. The two locally constant color classes also show that the given boundary points lie in different components of $\partial\bar{J} \setminus \partial H$. This is the hypothesis of [7, Theorem 1.4], including the case of a finite subgroup separating an already disconnected boundary. $\square$

17.3 Comparison paths and satellite tests

We now prove that finitely many successive systems provide the separators. There are three geometric cases: a large outer gap; a large internal turn after bounded outer turns have been replaced; and the remaining case in which all turns are bounded. In parallel, a comparison around an outer boundary cycle constructs the odd W_1 -encounters needed for the first case. This circle argument uses only the internal trees and does not assume that the completed outer switches already separate boundary points.

The two uses have different outputs. A circle comparison starts with one outer boundary cycle and produces an odd W_1 -encounter. A line comparison starts with two ideal endpoints and produces a completed W_2 whose cocycle separates them. Every W_1 , including those produced by the circle tests, already has the support bound of Lemma 17.5. That bound uses only the minimum tree geometry and is unchanged by stronger test angles. Consequently the outer-gap threshold A_p can be fixed from this bound and the basic first-direction comparison

before the line tests are chosen; the circle tests establish existence without changing the bound.

Lemma 17.8 (Replacing bounded outer turns). *Fix a bound A_p on cyclic period gaps at outer turns of a bi-infinite geodesic γ . Replacing these turns by boundary arcs and making the elementary fan detours produces a pure internal uniform quasigeodesic α . Its constants depend only on A_p and the fixed data.*

- (i) *Each physical vertex occurs a bounded number of times on α , and all its occurrences lie in a bounded parameter interval.*
- (ii) *If an internal apex d lies on γ , all new germs at d lie within bounded angular distance of one of its two original germs ν_-, ν_+ . When these are sufficiently far apart, each neighborhood contains an odd number of the new radial half-incidences.*
- (iii) *If all original internal turns have bounded angle, then the angular diameter of all germs at all visits to any one internal apex on α is bounded.*

Proof. A period gap of at most A_p gives a boundary arc of bounded elementary length; the conversion factor is the maximum of the fixed word lengths. Detour every satellite radial ending at an apex or midpoint through a neighboring ordinary site on that arm, and every central radial ending at a midpoint through a true ordinary endpoint. Fan triangles realize all these replacements. Their lengths and displacement are bounded, so the result is a uniform quasigeodesic with the same ideal endpoints. A quasigeodesic can revisit a vertex only within bounded parameter distance, which proves (i).

An elementary radial detour changes its germ at d by a bounded avoiding path. If a replaced outer arc visits a given original d , its parent outer apex is adjacent to d . Since both vertices occur on the original geodesic, they were consecutive there. The inserted directions are close via that parent apex, and the replacement polygon changes the number of half-incidences on that side by an even number. This proves (ii).

For (iii), a new internal passage in an inserted arc has a short avoiding comparison through the parent outer apex. All individual turns are therefore bounded. Between consecutive visits to an internal apex, the subpath avoids that apex and has bounded length by (i). It bounds the angle between the corresponding visit germs, proving the assertion about cumulative diameter. Arguments excluding outer switches require only avoidance of outer apices, and do not need (iii). $\square$

There will be two comparison settings. In the *line setting*, α is as in Lemma 17.8, and the tested trees are the final W_2 . In the *circle setting*, α is the specified pure internal boundary path of one outer apex p , and the tested trees are W_1 . The angle dictionary at p bounds the number and cyclic-coordinate diameter of visits to any physical site or internal apex. Directions at such an apex are close via p . No ideal-endpoint assertion is used in the circle setting.

When every test is even, we will pair radial occurrences and replace each pair by a short boundary path. The replacement is a sum of fan-triangle boundaries, so it preserves cocycle evaluation. It may disconnect the chain. The invariant that keeps the final horizontal chain nonzero is mod-two winding in the circle setting, and mod-two crossing from one end to the other in the line setting. We verify these invariants in Lemma 17.13.

Lemma 17.9 (Satellite tests and uncontaminated evaluations). *At an internal apex d visited by α , apply Lemma 17.2 to its satellite radial site list, with spoke germs far from every d -direction of α . The resulting tree through T evaluates on α exactly as the prescribed satellite labels on that list. In the line setting its limit set avoids both ideal endpoints. If all such*

evaluations are even, the satellite radial half-incidences can be paired within their classes and replaced by bounded purely central boundary arcs avoiding d .

Proof. First consider comparisons from d to other anchors or ideal endpoints. In the line setting they avoid every outer apex. Choose the outer switch turns above the marker-comparison threshold for these paths. Such a comparison forbids any change of W_1 -piece through an outer switch, even on a ray to an ideal endpoint. Thus we may work in the W_1 through T .

If another anchor $w \neq d$ lay on α , compare the tree path to it with the subpath starting at the last d -visit before w . This subpath avoids d after departure and begins with a listed direction. The tree path is uniformly quasigeodesic, never revisits d , and begins with one of the forbidden, far-away spoke germs. The first-direction comparison in Lemma 16.5 gives a contradiction. In the circle setting use the bounded shortcut through p ; its direction at d is close to the listed directions through an ordinary neighbor. The identical comparison with a tree ray excludes either ideal endpoint in the line setting.

It remains to check defaults rather than only exact anchors. A default at a satellite radial site of d needs an anchor reached along a short arc of that same arm, avoiding d . Appending this arc to the comparison again contradicts its first direction. For a satellite traversal at $s \neq d$, a default needs a W_0 -anchor w on that arm. If $w \neq d$, compare along α to s and append its direct spoke to w , starting after the last d -visit. If $w = d$, the default would require a central W_0 -node over d . There is none: T is satellite-only, and node uniqueness excludes a second node with that projection, also across W_1 -pieces of W_2 . Direct labels at $s \neq d$ and horizontal or central contributions require exact anchors already excluded. Hence only the intended satellite labels remain.

If every class has even occurrence count, pair its half-incidences. Two sites in a class lie on one arm at bounded cyclic distance, so replace the pair of radials by the intervening bounded boundary arc. These satellite boundaries are pure central paths and avoid their source d . The fan triangles show that the replacement preserves every cocycle evaluation. Write B_s for a common elementary length bound. It depends on the comparison-path and list bounds, but not on a subsequently chosen large-internal-turn threshold.

For infinite chains this is first a local chain identity. Whenever it is used for a particular separating tree, we have already excluded that tree's limit set at both ends. Its support then misses the tails and all bounded-parameter replacement patches there, so only finitely many patches affect the evaluation. This justifies the infinite-chain version without an infinite parity sum. $\square$

17.4 Large internal turns

Lemma 17.10. *In the line setting, a sufficiently large original internal turn of γ is separated by one of the constructed trees. The required turn threshold A_d is chosen after the satellite replacement bound B_s and before the regular-case tests below.*

Proof. Let d have original germs ν_-, ν_+ whose angle exceeds A_d . Perform the satellite tests only at this d . If a test is odd, the preceding lemma already gives separation. Otherwise pair and replace the satellite half-incidences there to obtain α' . They all have d as source and true ordinary sites as targets. Choose A_d larger than the angle bound supplied by a B_s -arc and the germ neighborhoods of Lemma 17.8. No pair can join the two sides, and no replacement arc passes through d . The remaining central radial counts are therefore odd near each $\nu_\pm$.

Choose a central-only panel b whose labels are zero and one, respectively, on *all* central ordinary germs in sufficiently large fixed angular neighborhoods of ν_- and ν_+ . Require its cuts to be far from both directions. These neighborhoods contain boundedly many germs by

the angle dictionary. The central panel tests of Proposition 16.6 give the required labels: after their class-diameter bound is fixed, enlarge A_d so that no class meets both neighborhoods. A class meeting both would give a bounded horizontal path avoiding d , contradicting that enlarged angle bound.

We claim that every first direction in the W_1 of b towards another node is far from both $\nu_\pm$, with any prescribed fixed comparison margin. Central edges are controlled by the cuts. The additional possibility is a spoke from b to a satellite-only panel at another apex s , because the W_0 of b has odd total on an arm at s visiting d . Suppose its germ $d \rightarrow s$ were close to a ν . An additional anchor $w \neq d$ on that arm would contradict the first cut from b in W_0 , using the shortcut through s . Thus this W_0 has anchors on the arm only at d . Every central passage at d on that arm has its two neighboring ordinary germs close to the same direction through s . Both receive the same label, so the passage contributes zero. There are no other contributions. The total on that arm is even, contrary to the supposed attachment. This proves the claim; it is the reason for using nonmixed panels.

Compare from the last d -visit along α towards either ideal endpoint. Outer changes are forbidden by avoidance of outer apices, and the claim forbids the first direction of the W_1 -ray. Neither endpoint belongs to the limit set of the resulting W_2 .

We check every possible extra contribution on α' . An exact anchor $w \neq d$ on an unchanged portion is excluded by comparison along α . On an inserted arc, use its short tail from a paired d -target; this also avoids d after departure. These comparisons forbid outer changes and then contradict the first-direction claim.

A satellite half remaining at s has $s \neq d$ and comes from α . Its default requires an anchor w on that arm. The case $w \neq d$ is excluded through s as before. If $w = d$, node uniqueness makes the original W_0 of b the only possible contributor. Here s is adjacent to d , and the quasigeodesic subpath from the last relevant d -visit to s is bounded. Its first direction, and hence the spoke $d \rightarrow s$, is close to one of $\nu_\pm$. The preceding arm argument shows that this W_0 encounters the arm only at d and has zero sum between consecutive true ordinary sites. Individual halves of a central passage need not be zero, but their pair is zero. Its even integration default is consequently constant on all true ordinary sites. The short-cluster normalization sets it to zero at distant ordinary sites, and therefore at every ordinary site. This remaining contamination also vanishes.

Only the intended central labels at d survive. Their total is one, since the counts near both original germs are odd and the two labels are zero and one. Triangle consistency transfers this total to γ . All bounds used in excluding the extra contributions depend on the already fixed substitution, quasigeodesic, and basic cluster constants. They do not depend on the size of the original large turn. $\square$

17.5 Regular comparisons and the last horizontal chain

We next treat both the circle setting and the line setting in which every original internal turn is at most A_d . Apply the satellite tests at every internal apex met by α . If none is odd, make all the paired replacements, obtaining a pure central chain α^1 . In the line setting, the quasigeodesic parameter localizes every pairing to a bounded interval. In the circle setting, the period-coordinate dictionary gives the same conclusion in the cyclic parameter.

Lemma 17.11 (Regular comparison paths). *Sites participating in α^1 have bounded connectors to true ordinary sites of α . Comparisons between them have uniformly bounded cumulative germ diameter at every internal apex. In the line setting, they avoid outer apices, are uniform quasigeodesics, and extend to either ideal endpoint. The lists needed at any tested internal apex have bounded size. These assertions persist after adding any fixed bounded horizontal tails to the connectors.*

Proof. For a site on an inserted arc, follow that arc to a paired endpoint. Each internal apex passage on this connector has a short angular bypass through the arm's source apex, which differs from the apex being traversed. For an untouched half, use that half to an ordinary neighbor when needed. From a tested apex, its connector can start with the particular germ being tested.

Join two such tails by the intervening α -path in the line setting. This is a uniform quasigeodesic with bounded end modifications, avoids outer apices, and extends in the same way towards an ideal end. In the circle setting join the ordinary ends of the tails by their two spokes through the parent outer apex p , giving a uniformly bounded comparison. All junctions between the portions are ordinary vertices. Every internal passage has bounded angle. Repeated visits to any internal apex have bounded number and bounded intervening lengths, using quasigeodesicity in the line setting and the bounded comparison length in the circle setting. Consequently the cumulative angular diameter is bounded, including when a comparison revisits its initial tested apex: its last germ there remains close to its prescribed initial one.

Finally, each substitution arc has bounded length and stays close to its assigned old positions. In the line setting quasigeodesicity bounds all possible positions assigned to one physical site or apex. In the circle setting the connector tails avoid p , so the angle dictionary at p bounds their cyclic-coordinate ambiguity. These observations also bound the number of occurrences in each tested list. Extra bounded horizontal tails change only the bounds and create no new internal turns. $\square$

Lemma 17.12 (Central tests on the regular chain). *At an internal apex s with radial halves in α^1 , central panel tests evaluate exactly the class parities of their ordinary target sites. An odd test separates the ideal endpoints in the line setting and gives an odd W_1 -total on the outer cycle in the circle setting. Otherwise all remaining radial halves can be paired and replaced by bounded horizontal link paths, giving a purely horizontal chain α^2 .*

Proof. Use a second central system, with satellite switch bends chosen after the regular comparison bounds of Lemma 17.11. At s , distinguish bounded intrinsic classes of the target list and put the central cuts far from its germ directions.

Every contribution on the pure central chain requires an actual W_0 -anchor: either its midpoint or its central apex. A comparison from the tested panel to such an anchor cannot cross an outer switch in the line setting, since the comparison avoids outer apices. It cannot change W_0 -piece through a satellite switch in either setting, since that would force an interior turn exceeding the cumulative angular bound on the comparison. Within the one W_0 , its first cut excludes every other anchor. This remains true if the comparison revisits the tested apex, because its last germ is still close to the initially prescribed one. Only the target central labels therefore contribute.

A tree ray asymptotic to either ideal endpoint is excluded in precisely the same order: it cannot use an outer switch, cannot use a satellite switch, and cannot use a first cut. Thus an odd evaluation gives the claimed separation. In the circle setting it gives the required odd total without any endpoint assertion.

If all tests are even, each bounded intrinsic class has even occurrence count. Pair within these classes and replace the radial pairs by bounded horizontal link paths. Fan triangles preserve every cocycle evaluation. Let B_h be a common bound for these new paths. The connectors of Lemma 17.11 merely acquire horizontal tails of length at most B_h . Pairing at an apex again takes place within a bounded parameter interval, by its proximity through the previous connectors to the old comparison path. $\square$

Lemma 17.13 (A surviving horizontal edge). *The chain α^2 contains a physical horizontal edge with odd multiplicity. A final central system whose cut and switch thresholds exceed the*

new regular comparison bounds has a slot tree with odd total on this chain. In the line setting its limit set avoids both ideal endpoints.

Proof. First choose the final system with its central cut separations, satellite switch bends, and, in the line setting, outer switch bends all above the bounds supplied by the connectors after the horizontal replacements. Two distinct horizontal midpoint anchors of α^2 cannot belong to one completed tree. A comparison between them forbids switches at either level. They would therefore lie in one W_0 ; its path between distinct slots has an internal panel bend, also forbidden by the same comparison. Likewise a ray asymptotic to a line endpoint can neither switch nor continue indefinitely inside W_0 , because it would have such an interior bend. This proves the limit-set assertion for every tested slot tree.

We prove nonzero physical parity without assuming that the modified chain is connected. In the circle setting, assign each participating physical vertex its position on the original parameter circle, using its occurrence or the connector to an original site. All ambiguities have bounded cyclic diameter: the connectors and replacement patches avoid the parent outer apex p , so the angle dictionary applies. Choose a single position for each physical vertex and send each edge along its short parameter arc. This is the short-arc construction used in the proof of Proposition 16.10. Every replacement polygon lies in a bounded parameter interval and has degree zero for sufficiently large cycle order. The original boundary cycle has degree one. The final chain therefore has degree one modulo two and cannot be the zero physical chain.

In the line setting, assign each participating physical vertex a single real parameter, chosen from its occurrence or connector positions on α . Quasigeodesicity and bounded tails bound the ambiguity. Cut the parameter line at a generic finite point and give each physical vertex a single binary sign according to which side its chosen parameter lies on. For an edge, count one if the endpoint signs differ. Every replacement loop has even crossing count, because its sign changes telescope modulo two. Only bounded-parameter loops can meet both signs, so only finitely many replacements affect this count. The original bi-infinite path has opposite signs at its two ends and hence odd crossing count. The final physical chain is consequently nonzero.

At a horizontal midpoint the cycle condition says that its two half-edge counts agree, in both settings; local finiteness justifies this condition in the line setting. Thus some full physical horizontal edge has odd multiplicity. Choose a slot at its midpoint in the final system. On a purely horizontal chain only marked slots contribute. No other midpoint of the chain is marked by this tree, as proved above, and node uniqueness allows just this one slot over the chosen midpoint across all constituent pieces. The evaluation is therefore odd.

All evaluations pass back through the substitutions by triangle consistency. In the line setting the previously proved limit avoidance ensures that the support misses sufficiently distant tails and their bounded-parameter patches. Thus only finitely many substitutions affect this particular evaluation, as required. $\square$

Corollary 17.14 (Odd outer encounters). *A finite pool of W_1 -systems contains, on the cycle of every outer apex, at least one W_1 with odd total. The pool and all its required bounds can be fixed before the final filling depth choices.*

Proof. Apply the circle setting. An odd satellite test suffices. If none exists, perform the satellite pairings. An odd central test then suffices. If none exists, Lemma 17.13 supplies an odd final slot tree. Only the finitely many successive systems just described are used; the bounds depend on the fixed initial circular comparisons and the preceding pairing bounds. This argument uses W_1 throughout and does not assume any separating outer W_2 switch. $\square$

17.6 Large outer turns and completion of separation

Lemma 17.15. *A sufficiently large cyclic period gap at an outer turn of a geodesic γ is separated by one of the completed trees. The gap threshold A_p depends only on the basic quasi-isometry and support-cluster bounds, not on later increased switch bends.*

Proof. Let p be that outer apex and let the two traversed sites be its entry and exit sites. Take an odd encounter from Corollary 17.14 and its antipodal partner. Rotate the pair so that one entire support cluster lies strictly inside the shorter interval between the entry and exit sites, and its antipode lies inside the complementary interval. Give both clusters bounded margins from the endpoints. The two first spoke germs must also avoid the two geodesic germs by the fixed first-direction comparison bound. Only boundedly many rotations are forbidden by this angular condition, by the angle dictionary. A sufficiently large A_p , chosen from the cluster widths, margins, and this forbidden-position bound, permits the placement.

Every tree path or ray from the switch leaves in one of these two prescribed directions. It is uniformly quasigeodesic and does not revisit p . The first-direction comparison excludes either endpoint of γ from its limit set.

We must also exclude cocycle contributions away from this traversal. A nonzero contribution on either side of γ would provide a tree anchor other than p , either exactly on that side or accessible by a uniformly bounded path from it avoiding p . For internal edges this is the support rule of Lemma 17.4. At another outer apex, either that switch itself is an anchor or an even default gives a short path through that other apex and internally to an anchor. Following the corresponding side of γ from p and then this short path contradicts the same first-direction comparison. Thus no such contribution exists.

On the traversal through p , the paired odd clusters give integrated difference one, since exactly one lies in the chosen interval. An additional even-only piece could change this difference only if an interval endpoint split its short support cluster. Such a split would supply an anchor accessible from that endpoint by a bounded internal arc and short tail, already excluded by the first-direction comparison. The full difference is therefore one. The required avoidance and margins used only the uniform minimum-threshold geometry and cluster constants. $\square$

Proposition 17.16 (Boundary separation). *There are finitely many systems of completed trees such that every pair of distinct points of $\partial\bar{J}$ lies outside the limit set of one of the trees and has different cocycle colors there.*

Proof. Choose a bi-infinite geodesic for the pair. A large outer gap is handled by Lemma 17.15. Otherwise normalize it by Lemma 17.8. A large original internal turn is handled by Lemma 17.10. In the remaining case all original internal turns are bounded. An odd satellite test separates by Lemma 17.9; failing that, an odd central test separates by Lemma 17.12; failing both, Lemma 17.13 supplies the final slot separator. Each lemma proves limit avoidance as well as odd evaluation. The finite choice of systems and its compatibility with the filling depths are verified next. $\square$

17.7 The order of the finite choices

Lemma 17.17 (No circular choice of thresholds). *All systems used above can be chosen from a finite pool by finitely many original depth, angle, girth, period-order, and avoidance tests. These tests can be imposed simultaneously with any initial filling-depth request and the finite-path tests of Section 14.*

Proof. For clarity, an angular comparison bound below is a value of the fixed bound functions supplied by the full-graph hyperbolicity and angle dictionary, for already chosen path constants

and list sizes. The number of original germ types is finite. Thus any finite collection of such requests is a finite collection of original tests, even though the final links and their periods depend on the eventual filling.

1. First impose the minimum central and internal switch bounds needed for the marker comparison, the bounded polygon tests, and the inner support clusters. They give uniform quasi-isometry constants for W_0 and W_1 , independent of any subsequent increase in sparseness. The outer support-cluster bound follows. Impose the minimum outer switch bound as well, obtaining fixed quasi-isometry constants for W_2 .
2. These basic constants determine the outer-gap threshold A_p in Lemma 17.15. They therefore determine the quasigeodesic and visit bounds for the normalized path α . Importantly, A_p uses the *basic* first-direction and cluster bounds; it does not use later regular-comparison bend requirements. Include the central odd-seed pool from Proposition 16.10 wherever satellite profile tests will be needed.
3. Fix the first satellite tests from the bounds for α and, simultaneously, from the initial circular comparisons. Their target lists, forbidden directions, and required first-direction margins are now bounded. Lemma 17.2 gives their class bound, and hence the replacement bound B_s . No bound on a possibly large original internal turn has been used.
4. Fix the angular neighborhoods and cut-avoidance requests for the large-internal-turn test, using only the preceding constants and B_s . The central tests then give their intrinsic class-diameter bound. Choose A_d larger than the resulting bound for a class straddling both neighborhoods, and larger than the bound for a satellite pair straddling them. This is the first choice of A_d ; its value is not an input to the tests that determined it.
5. Only now use the absence of turns larger than A_d in the regular line case. Lemma 17.11 gives the new cumulative angular bounds, also encompassing the circular case. A second system has internal switch bends above these bounds and central list tests whose cuts avoid the relevant directions. Its class bound gives the horizontal replacement bound B_h .
6. The extra horizontal tails of length at most B_h give the final regular comparison bounds. A final central system has all cut separations and internal switch bends above these bounds. This is a separate system; its stronger sparsity need not be fed back into the earlier tests.
7. After all these comparison requests have been recorded, choose outer antipodal switch bends above their maximum wherever needed. This is possible by the period-coordinate tests and does not alter the fixed minimum-threshold quasi-isometry constants. In systems without prescribed satellite labels, consume odd sides by antipodal pairs with the requested bend bound; no new seed or class test is needed.

The finite systems for the circle setting give the odd outer encounters before they are used in the large-outer-turn proof. There is no converse dependence: the circle setting uses only W_1 and the bounded shortcut through its parent outer apex. Its support bounds already came from the minimum-threshold geometry.

Every placement condition, polygon-lift length, cyclic-index bound, minimum period order, girth requirement, and excluded transporter list is now finite and fixed. The kernel construction of Proposition 15.12, including its graph-cover and order conclusions in Corollary 15.9 and Proposition 15.10, together with Section 16.8, imposes these tests before freezing $H, \ell, \mathbb{N}$. Membership and injectivity tests used in the angle dictionary are imposed at their corresponding fixed lengths. Taking their finite union with the prescribed initial filling and finite-path tests proves the assertion. $\square$

17.8 Good fillings and the primitive-extension conclusion

Theorem 17.18 (Virtually compact special filled vertices). *For the terminal row and peripheral data constructed in Sections 13–16, one can choose the height-fiber kernels arbitrarily deep, simultaneously satisfying the finite-path tests, so that every filled vertex group $\bar{J}$ is hyperbolic and virtually compact special.*

Proof. Hyperbolicity and the proper cocompact fan model are supplied by the row filling construction. Elementary filled groups are already virtually compact special. For every other filled row, Lemma 17.17 makes the wall constructions and Proposition 17.16 simultaneous. By Lemma 17.7, each boundary pair is separated by the limit set of a quasiconvex codimension-one subgroup. The boundary criterion [7, Theorem 1.4] gives a proper cocompact action on a CAT(0) cube complex. Agol’s theorem [2, Theorem 1.1] gives a finite-index subgroup acting specially on that complex. The quotient is compact because the original action was cocompact and the index is finite. Thus the conclusion is virtual *compact specialness*. $\square$

Theorem 17.19 (Arbitrarily deep good quotients). *Let G be the hyperbolic torsion-free primitive extension group under consideration, with the terminal finite malnormal quasiconvex virtually compact special peripheral family constructed in the preceding sections. For every prescribed sufficiently deep filling-avoidance request, there is a filling of this family to a hyperbolic virtually compact special quotient. Its peripheral images are finite-by-cyclic or finite, are hyperbolic virtually compact special, and have the inherited relative peripheral structure and filling conclusions.*

Proof. Impose the given initial request together with the finite-path transfer and the finite tests of Lemma 17.17. The terminal transfer, Theorem 14.9, gives the finite quotient splitting; Proposition 14.11 gives ambient quasiconvexity of its edge groups. Its filled vertex groups are hyperbolic virtually compact special by Theorem 17.18. Deep relative filling in G gives relative hyperbolicity of the quotient, with the stipulated finite-by-cyclic or finite peripheral images. These images are hyperbolic, so the quotient is hyperbolic. They are also virtually compact special: a finite group is virtually trivial, and a finite-by-cyclic group is virtually cyclic.

The hyperbolic quasiconvex hierarchy theorem [66], applied to the transferred finite splitting with these vertex groups, makes the whole quotient virtually compact special. Here the edge groups are quasiconvex in the ambient quotient, as required by the transfer result; this is not an inference from arbitrary embeddings of the vertex groups in right-angled Artin groups. The inherited peripheral and filling conclusions are exactly those of the deep relative filling and standalone row constructions. The initial avoidance request was arbitrary. $\square$

Theorem 17.20 (Primitive extensions). *Every hyperbolic torsion-free primitive extension group of either of Linton’s two forms specified in the introduction is virtually compact special.*

Proof. For such a group the primitive data of Section 5 and the opening reduction establish local indicability, coherence, a finite aspherical two-dimensional one-relator model, and acyclicity of its universal-cover chain complex over the rational Linnell division field. The preceding actual-row, closure, and terminal arguments construct the finite malnormal quasiconvex virtually compact special peripheral family. Theorem 17.19 supplies arbitrarily deep good quotients with hyperbolic virtually compact special peripheral images and the inherited filling conclusions. Thus all hypotheses of the filling criterion, Theorem 3.1, hold. That criterion gives virtual compact specialness of the original group G . $\square$

18 Consequences

Let G be a hyperbolic one-relator group or a hyperbolic ascending mapping torus of an injective endomorphism of a finite-rank free group. Theorems 1.1 and 1.2 give a compact-special finite-index subgroup $H \leq G$. The conclusions below also apply to a finitely generated Baumslag–Solitar-free one-relator group: the companion hyperbolicity theorem [52, Theorem 1.1] first puts it in the first class. We explain the finite-quotient and subgroup consequences before the more specific virtual-fibering result.

18.1 Linearity and finite quotients

The finite Salvetti model embeds H in a right-angled Artin group on a finite graph. Such groups are linear over $\mathbb{Z}$ [28]. Inducing a faithful integral representation from the finite-index subgroup H gives a faithful representation of G in $\mathrm{GL}_N(\mathbb{Z})$ for some finite N . Reduction of matrix entries modulo suitable integers then distinguishes every nonidentity element in a finite quotient. Thus G is linear over $\mathbb{Z}$ and residually finite. The conclusion concerns the whole group, including the one-relator groups with torsion; it is not merely a property of the chosen cover. Borisov–Sapir had already proved residual finiteness for ascending free-group tori without requiring hyperbolicity [10]; the compact-special result here also supplies integral linearity. For finite-rank free bases, this answers Druţu–Sapir’s question about the existence of hyperbolic nonlinear ascending HNN extensions in the negative [20, Problem 6].

Finite quotients also distinguish conjugacy classes. A group is *conjugacy separable* if any two nonconjugate elements have nonconjugate images in some finite quotient, and *hereditarily conjugacy separable* if every finite-index subgroup has this property. Minasyan–Zalesskii’s theorem applies to every hyperbolic virtually compact special group, so G is hereditarily conjugacy separable [49, Theorem 1.1]. For one-relator groups with torsion, they had already established this conclusion [48, Theorem 1.1].

There is also a cohomological consequence of the finite special model. Canonical completion realizes H as a virtual retract of a right-angled Artin group: it is a retract of a finite-index subgroup of that group. Proposition 3.8 of [49] therefore makes G *cohomologically good*: for every finite G -module M and every $n \geq 0$, the natural comparison map

$$H_{\mathrm{cont}}^n(\widehat{G}; M) \longrightarrow H^n(G; M)$$

is an isomorphism. Here $\widehat{G}$ is the profinite completion and the left side uses continuous cochains. Thus the finite-quotient structure recovers group cohomology with finite coefficients, not just the ability to distinguish individual elements.

18.2 Quasiconvex subgroups and largeness

If $Q \leq G$ is quasiconvex, then $Q_0 = Q \cap H$ is quasiconvex in H . Haglund–Wise’s canonical completion and retraction give a finite-index subgroup $H_0 \leq H$ containing Q_0 and a retraction $H_0 \rightarrow Q_0$ [28, Section 6 and Theorem 7.3]. In particular, Q_0 is separable in H : every element of $H \setminus Q_0$ can be excluded by a finite-index subgroup containing Q_0 . Subgroup separability passes through this finite-index change, so Q is separable in G [28, Lemma 7.5]. This permits finite covers that retain a specified quasiconvex subgroup while excluding any prescribed finite set outside it. The retraction gives the stronger structural control on $Q \cap H$ stated above. The quasiconvexity hypothesis is essential to this application; finite generation alone is not being substituted for it.

When G is *non-elementary*, meaning that it is not virtually cyclic, it is also *large*: some finite-index subgroup surjects onto a free group of rank two. Indeed, the torsion-free hyperbolic subgroup H contains a quasiconvex free subgroup of rank two; one may apply [37,

Theorem 1.11] to H with the trivial subgroup as the prescribed family. The virtual retraction just described then supplies the surjection. In particular, finite-index subgroups of G have arbitrarily large rational first Betti number: pull back finite-index subgroups of arbitrarily large rank in that free quotient. The non-elementarity condition is an additional hypothesis for these largeness and homology conclusions.

18.3 Virtual fibering and ambient embeddings

Agol’s virtual RFRS theorem for right-angled Artin groups, together with the Salvetti embedding and inheritance by subgroups, gives a finite-index RFRS subgroup of G [1, Corollary 2.3]. In this condition, a descending sequence $H' = H'_0 \geq H'_1 \geq \cdots$ of finite-index subgroups, normal in H' , has trivial intersection, and each H'_{i+1} contains the kernel of $H'_i \rightarrow H_1(H'_i; \mathbb{Q})$. It connects finite covers with rational homology and underlies the virtual-fibering results discussed in the introduction. For the ascending tori, Corollary 4.10 combines it with field acyclicity to give a finite-index epimorphism onto $\mathbb{Z}$ with finite-rank free kernel. General one-relator groups need not have that acyclicity. The theorems of Kielak–Linton instead give the following subgroup-closed conclusion and finite-rank ambient model. The one-relator conclusion is the implication they explicitly anticipated for Wise’s Conjecture 17.8 [37, Section 1.1].

Corollary 18.1. *Let G be one of the following groups.*

1. *A one-relator group*

$$G = F(X) / \langle\langle r \rangle\rangle,$$

where X is finite and $r \in F(X)$ is arbitrary, such that either G is word-hyperbolic or G contains no subgroup isomorphic to $\text{BS}(m, n)$ for any nonzero integers m, n .

2. *A word-hyperbolic ascending mapping torus*

$$G = \langle F, t \mid t^{-1}ft = \phi(f) \ (f \in F) \rangle,$$

where F is a free group of finite rank and $\phi : F \rightarrow F$ is injective.

There are a finite-index subgroup $H \leq G$, an integer $d \geq 0$, an automorphism α of the rank- d free group F_d , and an embedding

$$H \hookrightarrow F_d \rtimes_{\alpha} \mathbb{Z}.$$

In particular, G is virtually free-by-cyclic: the same H fits into an exact sequence

$$1 \longrightarrow K \longrightarrow H \longrightarrow Q \longrightarrow 1, \quad K \text{ free, } Q \leq \mathbb{Z}.$$

The rank of K may be infinite, and Q may be trivial.

Proof. If G is finite, take $H = 1$ and $d = 0$, so the ambient group is $\mathbb{Z}$. We may therefore assume that G is infinite.

For the first case, a compact connected planar surface with $|X| + 1$ boundary components has fundamental group $F(X)$; for $X = \emptyset$ this surface is a disk. If G is Baumslag–Solitar-free, Theorem 1.1 of the companion [52] first makes it hyperbolic. Theorem 1.1 then gives virtual compact specialness. Kielak–Linton’s Corollary 1.3 [37] applies to the quotient of this surface group by the normal closure of r , and makes G virtually free-by-cyclic. Its relator is arbitrary, so this application does not exclude trivial relators or proper powers.

In the second case, Corollary 4.10 already gives a finite-index free-by- $\mathbb{Z}$ subgroup with finite-rank free kernel. The weaker virtual-free-by-cyclic conclusion also follows from Theorem 1.2 and Kielak–Linton’s Corollary 1.8 [37].

For the stronger embedding, apply Theorem 1.1 of [37] to the hyperbolic virtually compact special group G . Its equivalence of (b) and (c) gives a finite-index subgroup embedded in a free-by-cyclic group with finitely generated free kernel. The proof of that theorem gives an ambient quotient $\mathbb{Z}$ in its rational cohomological dimension two case. In its dimension-at-most-one case it instead supplies a finite-index free subgroup; that subgroup has finite rank because G is finitely generated, and it embeds in its direct product with $\mathbb{Z}$. Thus in either case the ambient group has an exact sequence with finite-rank free kernel and quotient $\mathbb{Z}$. Choosing a lift of $1 \in \mathbb{Z}$ splits the sequence and gives the displayed $F_d \rtimes_{\alpha} \mathbb{Z}$.

Identify H with its image in this semidirect product and let $\pi : F_d \rtimes_{\alpha} \mathbb{Z} \rightarrow \mathbb{Z}$ be the projection. Its restriction to H has kernel $K = H \cap F_d$ and image $Q = \pi(H)$. The subgroup theorem for free groups makes K free, and Q is a subgroup of $\mathbb{Z}$. This proves the final exact sequence. Only the ambient kernel F_d is asserted to be finitely generated; finite generation does not follow for its subgroup K . $\square$

For an ascending torus, one can use the stronger finite-index subgroup from Corollary 4.10 as its own ambient finite-rank free-by- $\mathbb{Z}$ group. Its fibering character is not identified with the canonical ascending height. For injective nonsurjective ascending tori, this answers the hyperbolic case of Kielak’s virtual-fibering question recorded by Mutanguha [51, Overview, p. 3, Question (Virtual fibering)]. The separate quadratic-Dehn-function variant in that source is not covered by the hyperbolicity hypothesis here. Thus a finite-index subgroup is a mapping torus of a finite-rank free-group automorphism, even when the original ascending map is not surjective. This stronger conclusion is specific to the acyclic ascending setting; the general one-relator conclusion retains the possibly infinite-rank kernel and possibly trivial quotient stated above.

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